PART II

ODOUR THRESHOLD VALUES IN WATER

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

A

abhexone⇒5-ETHYL-3-HYDROXY-4-METHYL-2(5*H*)-FURANONE

ACENAPHTHENE *[83-32-9]*

Lillard & Powers (1975)	d	0.08

acetal⇒1,1-DIETHOXYETHANE
acetaldehyde⇒ETHANAL
acetaldehyde diethyl acetal ⇒1,1-DIETHOXYETHANE

ACETIC ACID (ethanoic acid) *[64-19-7]*

Ohrwall (1891)	r	60
Baker (1963)		24.3
Amoore *et al.* (1968); Amoore (1969)	d	30.7
Zoeteman *et al.* (1971)		200
Manning & Robinson (1973)		100
Kikuchi *et al.* (1976)		34.2
Aharoni *et al.* (1980)		200
Schnabel *et al.* (1988)	d	210 - 522
Larsen & Poll (1990)	d	100 - 1,000
Larsen & Poll (1992)	d	10 - 100
Grosch *et al.* (1993)		60
Guth (1996)		22
Buttery & Ling (1998)	d	22
Tamura *et al.* (2001); Boonbumrung *et al.* (2001)	d	25.59 - 26
Darriet *et al.* (2002)	d	50
Karagül-Yüceer *et al.* (2003)	d	22
Czerny *et al.* (2007c); Czerny *et al.* (2008)	r	180
Czerny *et al.* (2008)	d	99

acetoin⇒3-HYDROXY-2-BUTANONE
acetol ⇒HYDROXYPROPANONE
acetone⇒ PROPANONE

1-ACETONYLPYRROLE (1-(1-pyrrolyl)-2-propanone) *[4805-24-7]*

Tressl *et al.* (1985b)	d	0.1

ACETOPHENONE *[98-86-2]*

Rosen *et al.* (1963)		0.065
Baker (1963)		0.17
Pyysalo *et al.* (1977)	d	0.052
Aoki & Koizumi (1986)		0.036
Buttery *et al.* (1988a); Buttery & Ling (1995)	d	0.065
Schirack *et al.* (2006)	d	0.245
Pino & Mesa (2006)	d	0.065

acetovanillone⇒1-(4-HYDROXY-3-METHOXYPHENYL)ETHANONE

4-ACETYL-6-*tert*-BUTYL-1,1-DIMETHYLINDANE (celestolide) *[13171-00-1]*

Amoore *et al.* (1977)	d	0.0050

2-acetyl-3,4-dihydro-2*H*-azole⇒2-ACETYL-1-PYRROLINE

5-ACETYL-2,3-DIHYDRO-1,4-THIAZINE *[164524-93-0]*

Hofmann & Schieberle (1995); Schieberle & Hofmann (1996a, 1996b)		0.00125 - 0.0017
Buttery (1999)	d	0.000 6

2-ACETYLFURAN *[1192-62-7]*
Buttery *et al.* (1990a,1994a); Buttery (1993);

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Buttery & Ling (1995,1998)	d	10
Giri *et al*. (2010)	d	15.02520

7-ACETYL-1,1,3,4,4,6-HEXAMETHYL-1,2,3,4-TETRAHYDRONAPHTHALENE (fixolide, tonalide) *[1506-02-1]*

Amoore *et al.* (1977)	d	0.001 5

(-)-(2*R*,4*S*)-2-acetyl-*p*-menth-6-ene⇒(-)-1-[(1*R*,5*S*)-5-ISOPROPYL-2-METHYLCYCLOHEX-EN-1-YL]ETHANONE
(+)-(2*S*,4*R*)-2-acetyl-*p*-menth-6-ene⇒(+)-1-[(1*S*,5*R*)-5-ISOPROPYL-2-METHYLCYCLOHEX-EN-1-YL]ETHANONE

1-ACETYL-4-METHYLBENZENE (methyl acetophenone) *[122-00-9]*

Appell (1969)		0.16
Buttery *et al.* (1976b)	d	0.019
Masanetz & Grosch (1998a, 1998b)		0.024
Boonbumrung *et al.* (2001)	d	0.021

2-ACETYL-3-METHYLPYRAZINE *[23787-80-6]*

Mihara & Masuda (1988)	d	0.02

2-ACETYL-5-METHYLPYRAZINE *[22047-27-4]*

Mihara & Masuda (1988)	d	0.4
Buttery *et al.* (1999); Buttery (1999)	d	3

2-ACETYL-6-METHYLPYRAZINE *[22047-26-3]*

Mihara & Masuda (1988)	d	0.3
Buttery (1999)	d	3

ACETYLPYRAZINE *[22047-25-2]*

Teranishi *et al.* (1975); Buttery (1981,1999)	d	0.06 - 0.062

2-ACETYLPYRIDINE *[1122-62-9]*

Teranishi *et al.* (1975); Buttery (1981,1999); Buttery & Ling (1998)	d	0.019

2-ACETYLPYRROLE *[1072-83-9]*

Teranishi *et al.* (1987); Buttery *et al.* (1988a,1994a, 1999); Buttery & Ling (1998); Buttery (1999)	d	170
Giri *et al*. (2010)	d	58.58525

2-ACETYL-1-PYRROLINE (2-acetyl-3,4-dihydro-2*H*-azole) *[85213-22-5]*

Buttery *et al.* (1982a,1983,1988a,1994a,1997,1999); Buttery (1993,1999); Buttery & Ling (1995,1998)	d	0.000 1
Czerny *et al.* (2007)		0.000 1
Czerny *et al.* (2008)	d	0.000 053
Czerny *et al.* (2008)	r	0.000 12

2-ACETYL-1,4,5,6-TETRAHYDROPYRIDINE *[25343-57-1]*

Teranishi *et al.* (1975); Buttery (1981,1999); Buttery & Ling (1995,1998); Buttery *et al.* (1997,1999)	d	0.001 - 0.002

2-ACETYL-3,4,5,6-TETRAHYDROPYRIDINE *[27300-27-2]*

Buttery & Ling (1995,1998); Buttery *et al.* (1997, 1999); Buttery (1999)	d	0.001 – 0.002

6-ACETYL-1,2,3,4/2,3,4,5-TETRAHYDROPYRIDINE

Belitz & Grosch (1992a)	0.0016	

2-ACETYLTHIAZOLE *[24295-03-2]*

Teranishi & Buttery (1985); Buttery *et al.* (1994a,1999) ; Buttery (1999)	d	0.01

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Marchand *et al.* (2000)	d	0.003

4-ACETYLTHIAZOLE

Teranishi *et al.* (1975)	d	0.17

2-ACETYL-2-THIAZOLINE *[29926-41-8]*

Teranishi *et al.* (1975); Buttery (1981,1999);		
Buttery *et al.* (1994a)	d	0.001 - 0.0013
Cerny & Grosch (1993); Grosch *et al.* (1993);		
Guth & Grosch (1994); Schieberle &		
Hofmann (1996a, 1996b)		0.001
Karagül-Yüceer *et al.* (2004)	d	0.001
Limpawattana (2007)	d	0.000 12

3-(ACETYLTHIO)-2-BUTYLOCTANAL

Robert *et al.* (2004, 2005)	d	0.2

3-(ACETYLTHIO)-2-ETHYLHEXANAL

Robert *et al.* (2004, 2005)	d	0.015

3-(ACETYLTHIO)-2-METHYLPENTANAL

Robert *et al.* (2004, 2005)	d	0.005

3-(ACETYLTHIO)-2-PENTYLNONANAL

Robert *et al.* (2004, 2005)	d	0.5

3-(ACETYLTHIO)-2-PROPYLHEPTANAL

Robert *et al.* (2004, 2005)	d	0.05

acrolein⇒2-PROPENAL
acrylic acid⇒PROPENOIC ACID
acrylonitrile⇒ETHENYL CYANIDE
active amyl acetate⇒2-METHYLBUTYL ACETATE
active amyl alcohol⇒2-METHYL-1-BUTANOL
aldehyde C14⇒TETRADECANAL
aldehyde C16⇒ETHYL 3-METHYL-3-PHENYL-2,3-EPOXYPROPANOATE
aldrin⇒1,2,3,4,10,10-HEXACHLORO-1,4,4a,5,8,8a-HEXAHYDRO-1,4-*endo-exo*-5,8-DIMETHANONAPHTHALENE
allyl chloride⇒3-CHLOROPROPENE

4-ALLYL-1,2-DIMETHOXYBENZENE (methyleugenol, eugenol methyl ether, 4-allylveratrole) *[93-15-2]*

Buttery *et al.* (1974); Takeoka (2001)	d	0.068
Pyysalo *et al.* (1977)	d	0.82
Sugisawa *et al.* (1991); Yang *et al.* (1992);		
Tamura *et al.* (1993)	d	0.775 - 8.5

ALLYL ISOTHIOCYANATE (allyl mustard oil) *[57-06-7]*

Kauffmann (1907)		3.8
Buttery *et al.* (1976b); Buttery (1993)	d	0.375
Masuda *et al.* (1996)		0.046

1-ALLYL-4-METHOXYBENZENE (estragole, methylchavicol) *[140-67-0]*

Williams *et al.* (1977)	d	0.035
Zeller & Rychlik (2006)	r	0.016
Czerny *et al.* (2008)	d	0.006
Czerny *et al.* (2008)	r	0.016

5-ALLYL-1-METHOXY-2,3-(METHYLENEDIOXY)BENZENE (myristicin) *[607-91-0]*

Buttery *et al.* (1968)	d	0.025
Blank *et al.* (1992)		0.03
Zeller & Rychlik (2006)	r	0.088

4-ALLYL-2-METHOXYPHENOL (eugenol) *[97-53-0]*

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Appell (1969)		0.06
Buttery *et al.* (1971,1974,1987b,1990a);		
Buttery (1981,1993); Takeoka (2001)	d	0.006 - 0,011
Pyysalo *et al.* (1977)	d	0.030
Dietz & Traud (1978)		0.04
Sugisawa *et al.* (1991); Yang *et al.* (1992);		
Tamura *et al.* (1993)	d	0.1
Blank *et al.* (1992)		0.15
Bonfils *et al.* (2004)	d	0.47
Pino & Mesa (2006)	d	0.006
Zeller & Rychlik (2006)	r	0.0025
Czerny *et al.* (2008)	d	0.000 71
Czerny *et al.* (2008)	r	0.0025

4-ALLYL-1,2-(METHYLENEDIOXY)BENZENE (safrole) *[94-59-7]*

Helbig (1939)		0.0407
Le Magnen (1952)		0.01 - 0.033
Appell (1969)		0.16

allyl mustard oil⇒ALLYL ISOTHIOCYANATE

2-(ALLYLTHIO)ETHANAL

Rössner *et al.* (2002)		0.000 2

4-allylveratrole⇒4-ALLYL-1,2-DIMETHOXYBENZENE
ambrox⇒(3a*R*)-3a,6,6,9a-TETRAMETHYL-3aα,5aβ,9aα, 9bβ-DODECAHYDRONAPHTHO[2,1-b]FURAN
(*d*)-ambrox⇒(+)-(3a*R*)-3a,6,6,9a-TETRAMETHYL-3aα,5aβ,9aα,9bβ-DODECAHYDRONAPHTHO[2,1-b]FURAN
(*l*)-ambrox⇒(-)-(3a*S*)-3a,6,6,9a-TETRAMETHYL-3aα,5aβ,9aα,9bβ-DODECAHYDRONAPHTHO[2,1-b]FURAN
5β-ambrox⇒3aβ,6,6,9aβ-TETRAMETHYL-1,2,3a,4,5,5aβ,6,7,8,9a,9bβ-DODECAHYDRONAPHTHO[2,1-b]FURAN

2'-AMINOACETOPHENONE *[551-93-9]*

Buttery & Ling (1994,1995,1998); Buttery (1999)	d	0.000 2
Karagül-Yüceer *et al.* (2003, 2004)	d	0.000 28 – 0.000 3
Czerny *et al.* (2008)	d	0.000 27
Czerny *et al.* (2008)	r	0.000 64

4'-AMINOACETOPHENONE *[99-92-3]*

Buttery & Ling (1994); Buttery (1999)	d	100 - 108

2-AMINOBENZALDEHYDE *[529-23-7]*

Buttery (1999)	d	0.011

2-AMINO-1,5-DIHYDRO-1-METHYL-4*H*-IMIDAZOL-4-ONE (creatinine) *[60-27-5]*

Dietz & Traud (1978)		> 10

2-AMINOETHANOL (ethanolamine) *[141-43-5]*

Alexander *et al.* (1982)	d	32

2-(2-AMINOETHYLAMINO)ETHANOL (aminoethylethanolamine) *[111-41-1]*

Alexander *et al.* (1982)	d	2.8

aminoethylethanolamine⇒2-(2-AMINOETHYL)AMINOETHANOL

6-AMINOHEXANOIC ACID LACTAM (caprolactam) *[105-60-2]*

Lillard & Powers (1975)	d	59.7

(*S*)-(-)-2-AMINO-3-(4-HYDROXYPHENYL)PROPANOIC ACID (L-tyrosine) *[60-18-4]*

Dietz & Traud (1978)		> 10

(*R*)-(+)-2-AMINO-3-MERCAPTOPROPANOIC ACID (L-cysteine) *[52-90-4]*

Laska (2010)	d	24

(*S*)-(-)-2-AMINO-3-MERCAPTOPROPANOIC ACID (D-cysteine) *[921-01-7]*
Laska (2010) d 27

(*R*)-(-)-2-AMINO-4-(METHYLTHIO)BUTANOIC ACID (D-methionine) *[348-67-4]*
Laska (2010) d 1.5

(*S*)-(+)-2-AMINO-4-(METHYLTHIO)BUTANOIC ACID (L-methionine) *[63-68-3]*
Laska (2010) d 12

2-AMINO-3-PHENYLPROPANOIC ACID (phenylalanine) *[150-30-1]*
Dietz & Traud (1978) > 10

1-AMINO-2-PROPANOL (isopropanolamine) *[78-96-6]*
Alexander *et al.* (1982) d 28

N-(3-AMINOPROPYL)-1,4-BUTANEDIAMINE (spermidine) *[124-20-9]*
Wang *et al.* (1975) d 129

α-AMINOTOLUENE (benzylamine) *[100-46-9]*
Moncrieff (1957) 3,000

AMMONIA *[7664-41-7]*
Brown *et al.* (1968) 1.25
Amoore & Forrester (1976) d 23.9
Kikuchi *et al.* (1976) 110

AMMONIUM HYPOCHLORITE
Krasner & Barrett (1984) 0.35

amyl acetate⇒PENTYL ACETATE
tert-amyl acetate⇒1,1-DIMETHYLPROPYL ACETATE
amyl alcohol⇒1-PENTANOL
tert-amyl alcohol⇒2-METHYL-2-BUTANOL
amyl butyrate⇒PENTYL BUTANOATE
amyl isobutyrate⇒PENTYL 2-METHYLPROPANOATE
amyl mercaptan⇒1-PENTANETHIOL
tert-amyl mercaptan⇒2-METHYL-2-BUTANETHIOL
tert-amyl methyl ether⇒2-METHOXY-2-METHYLBUTANE
amyl salicylate⇒PENTYL SALICYLATE
amyl valerate⇒PENTYL PENTANOATE

4,16-ANDROSTADIEN-3-ONE (8*S*,9*S*,10*R*,13*R*,14*S*)-10,13-dimethyl-1,2,6,7,8,9,11,12,14,15-decahydrocyclopenta[a]phenanthren-3-one) *[4075-07-4]*
Amoore *et al.* (1977) d 0.000 98

5α-ANDROSTAN-3α-OL
Amoore *et al.* (1977) d 0.011

5α-ANDROSTAN-3β-OL *[1224-92-6]*
Amoore *et al.* (1977) d > 0.250

5α-ANDROSTAN-3-ONE *[1224-95-9]*
Amoore *et al.* (1977) d 0.000 62

5α-ANDROST-16-EN-3α-OL *[1153-51-1]*
Amoore *et al.* (1977) d 0.0062

4-ANDROSTEN-3-ONE
Amoore *et al.* (1977) d 0.002 2

5α-ANDROST-16-EN-3-ONE *[18339-16-7]*

Amoore *et al.* (1977)	d	0.000 18
Gower *et al.* (1985)		0.000 2 - 0.002

(*E*)-anethole⇒1-METHOXY-4-((*E*)1-PROPENYL)BENZENE
trans-anethole⇒1-METHOXY-4-((*E*)-1-PROPENYL)BENZENE

ANILINE *[62-53-3]*

Baker (1963)		70.1

anisaldehyde⇒4-METHOXYBENZALDEHYDE
p-anisaldehyde⇒4-METHOXYBENZALDEHYDE
anise alcohol⇒4-METHOXYBENZYL ALCOHOL
anisketone⇒1-(4-METHOXYPHENYL)-2-PROPANONE
anisole⇒METHOXYBENZENE
anisyl alcohol⇒4-METHOXYBENZYL ALCOHOL

AZACYCLOHEPTANE (hexamethyleneimine) *[111-49-9]*

Amoore *et al.* (1975b)	d	46.9

azacyclohexane⇒ PIPERIDINE

AZACYCLOOCTANE (heptamethyleneimine) *[1121-92-2]*

Amoore *et al.* (1975b)	d	82.8

AZACYCLOTRIDECANE (dodecamethyleneimine, 1,12-dodecamethylene diamine) *[295-03-4]*

Zyabbarova & Kuklina (1979)		0.1

AZETIDINE (trimethyleneimine) *[503-29-7]*

Amoore *et al.* (1975b)	d	21.9

azine⇒PYRIDINE
azole⇒PYRROLE
azolidine⇒PYRROLIDINE

B

BENZALDEHYDE *[100-52-7]*

Buttery *et al.* (1969b,1971,1987b,1988a,1994a,1997, 1999); Guadagni (1970b); Hansen *et al.* (1992);		
Buttery & Ling (1995,1998)	d	0.35
Tilgner & Ziminska (1982)		0.32
Rabin & Cain (1986)	d	1.4 - 2.2
Delfini (1987)		0.5 - 1.0
Buttery *et al.* (1990a);Buttery (1993)	d	3.5
Sugisawa *et al.* (1991); Yang *et al.* (1992);		
Tamura *et al.* (1993,1995)	d	1 - 4.6
Darriet *et al.* (2002)	d	0.3
Fabrellas *et al.* (2004)		0.024 - 0.047
Pino & Mesa (2006)	d	0.35
Giri *et al.* (2010)	d	0.75089

BENZENE *[71-43-2]*

Holluta (1960)		2
Baker (1963)		31.3
Zoeteman *et al.* (1971)		10
Alexander *et al.* (1982)	d	0.072
Sarzynski (1982)		111

benzeneacetaldehyde⇒2-PHENYLETHANAL
1,2-benzenediol⇒1,2-DIHYDROXYBENZENE
1,3-benzenediol⇒1,3-DIHYDROXYBENZENE
1,4-benzenediol⇒1,4-DIHYDROXYBENZENE

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

α-benzene hexachloride⇒α-1,2,3,4,5,6-HEXACHLOROCYCLOHEXANE
β-benzene hexachloride⇒β-1,2,3,4,5,6-HEXACHLOROCYCLOHEXANE
γ-benzene hexachloride⇒γ-1,2,3,4,5,6-HEXACHLOROCYCLOHEXANE
δ-benzene hexachloride⇒δ-1,2,3,4,5,6-HEXACHLOROCYCLOHEXANE

BENZENETHIOL (thiophenol) *[108-98-5]*

Holluta (1960)		0.001
Lane (1981)	d	0.050

1,2,3-benzenetriol⇒1,2,3-TRIHYDROXYBENZENE
1,2,4-benzenetriol⇒1,2,4-TRIHYDROXYBENZENE
1,3,5-benzenetriol⇒1,3,5-TRIHYDROXYBENZENE

BENZOTHIAZOLE *[95-16-9]*

Lillard & Powers (1975)	d	0.08
Buttery *et al.* (1988a)	d	0.08
Pino & Mesa (2006)	d	0.08

N-BENZOYLGLYCINE (hippuric acid) *[495-69-2]*

Dietz & Traud (1978)		2.5

BENZYL ACETATE *[140-11-4]*

Appell (1969)		1
Guadagni (1970b)	d	0.002
Pyysalo *et al.* (1977)	d	0.27
Rashash *et al.* (1997)		0.03
Pino & Mesa (2006)	d	0.364

benzylamine⇒α-AMINOTOLUENE

BENZYL ALCOHOL (phenylmethanol, α-hydroxytoluene) *[100-51-6]*

Pyysalo *et al.* (1977)	d	0.62
Buttery *et al.* (1987b,1988a)	d	10 - 20
Larsen & Poll (1990)	d	0.1 - 1
Yang *et al.* (1992); Tamura *et al.* (1993,1995)	d	100
Pino & Mesa (2006)	d	20
Giri *et al*. (2010)	d	2.54621

BENZYL BENZOATE *[120-51-4]*

Pino & Mesa (2006)	d	0.341

BENZYL BUTANOATE *[103-37-7]*

Pino & Mesa (2006)	d	0.376

BENZYL CYANIDE (phenylacetonitrile, α-toluenenitrile) *[140-29-4]*

Buttery *et al.* (1990a); Buttery (1993)	d	1
Freuze *et al.* (2004, 2005)	d	1.2

BENZYL ISOTHIOCYANATE (benzyl mustard oil) *[622-78-6]*

Masuda *et al.* (1996)		0.035
Pino & Mesa (2006)	d	0.000 7

3-BENZYL-2-METHOXYPYRAZINE

Murray & Whitfield (1975)		1 - 10

2-BENZYL-4-METHYLPHENOL *[716-96-1]*

Dietz & Traud (1978)		0.1

benzyl mustard oil⇒BENZYL ISOTHIOCYANATE

BENZYL SALICYLATE *[118-58-1]*

Pino & Mesa (2006)	d	0.161

BHT⇒2,6-DI-*tert*-BUTYL-4-METHYLPHENOL

BICYCLO[4.4.0]DECANE (decahydronaphthalene, decalin) *[91-17-8]*

Koppe (1965)		0.1

BIPHENYL (diphenyl, phenylbenzene) *[92-52-4]*

De Grunt (1975); De Grunt *et al.* (1977)	d	0.000 5

N,N'-BIS(3-AMINOPROPYL)-1,4-BUTANEDIAMINE (spermine) *[71-44-3]*

Wang *et al.* (1975)	d	24

bis(2-methyl-3-furyl) disulphide⇒3-(2-METHYL)FURYLDITHIO-3-(2-METHYL)FURAN
bis(methylthiomethyl) disulphide⇒2,4,5,7-TETRATHIAOCTANE
borneol⇒(1*R*,2*S*-(+)-*endo*-1,7,7,-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANOL
(1*R*,2*S*-(+)-borneol⇒(1*R*,2*S*)-(+)-*endo*-1,7,7,-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANOL
(1*S*,2*R*)-(-)-borneol⇒(1*S*,2*R*)-(-)-*endo*-1,7,7,-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANOL
bornyl acetate⇒*endo*-1,7,7,-TRIMETHYLBICYCLO[2.2.1]-2-HEPTYL ACETATE

BROMINE *[7726-95-6]*

McDonald *et al.* (2009)		0.10

BROMOCHLOROIODOMETHANE *[34970-00-8]*

Cancho *et al.* (2001)		0.0051 - 0.0084

BROMODICHLOROMETHANE *[75-27-4]*

CIRSEE in Mackey *et al.* (2004)		0.002
Khiari *et al.* (2002)		0.04
McDonald *et al.* (2009)		0.07

BROMODIIODOMETHANE *[557-95-9]*

Cancho *et al.* (2001)		0.000 1 - 0.000 2

bromoform⇒TRIBROMOMETHANE

2-BROMOPHENOL *[95-56-7]*

Bemelmans & Den Braber (1983)	d	0.000 1
Piriou *et al.* (2007)	d	0.000 024

3-BROMOPHENOL *[591-20-8]*

Bemelmans & Den Braber (1983)	d	0.050

4-BROMOPHENOL *[106-41-2]*

Bemelmans & Den Braber (1983)	d	0.160

BUTANAL (butyraldehyde) *[123-72-8]*

Guadagni *et al.* (1963); Teranishi *et al.* (1974);		
Buttery *et al.* (1988a)	d	0.009
Amoore *et al.* (1976)	d	0.0373
Ahmed *et al.* (1978a)	d	0.0159
Pino *et al.* (1986)	d	0.0082
Schnabel *et al.* (1988)	d	0.007 - 0.017
Fabrellas *et al.* (2004)		0.002 - 0.0022

1,4-BUTANEDIAMINE (putrescine) *[110-60-1]*

Holluta (1960)		20
Amoore *et al.* (1975b)	d	9,200
Wang *et al.* (1975)	d	22

1,2-BUTANEDIOL *[584-03-2]*

Schnabel *et al.* (1988)	d	70 - 100

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

1,3-BUTANEDIOL *[107-88-0]*		
Schnabel *et al.* (1988)	d	10 - 20
1,4-BUTANEDIOL *[110-63-4]*		
Schnabel *et al.* (1988)	d	810 - 990
2,3-BUTANEDIOL *[513-85-9]*		
Schnabel *et al.* (1988)	d	20 - 50
Takahashi (1990)		4,500
Buttery & Ling (1998); Buttery *et al.* (1999)	d	> 100
levo-2,3-BUTANEDIOL *[24347-58-8]*		
Giri *et al*. (2010)	d	0.0951
meso-2,3-BUTANEDIOL *[5341-95-7]*		
Buttery & Ling (1998)	d	> 100
Giri *et al*. (2010)	d	0.0951
2,3-BUTANEDIONE (diacetyl) *[431-03-8]*		
Sega *et al.* (1967)	d	0.002 61
Appell (1969)		0.01
Mulders (1972, 1973)	d	0.0065 - 0.007
Land (1968)	d	2.3
Pietrzak & Barylko-Pikielna (1976)		0.0086
Herrmann & Abd El Salam (1980b)		0.0043
Tuorila *et al.* (1982)	d	0.000 3 - 0.000 9
Boidron (1984)	d	0.015
Blank *et al*. (1991)		0.015
Buttery *et al.* (1994a,1997); Buttery & Ling (1995,1998)	d	0.003
Guth & Grosch (1994); Milo & Grosch (1997); Heiler & Schieberle (1997); Buhr *et al.* (2010)		0.015
Lawless *et al.* (1994)	r	0.005
Boonbumrung *et al.* (2001)	d	0.0011
Diaz *et al.* (2004a); Martin-Alonso *et al*. (2007)	d	0.000 05 - 0.000 08
Diaz *et al.* (2004a); Martin-Alonso *et al*. (2007)	r	0.000 18 - 0.000 20
Czerny *et al.* (2008)	d	0.001
Czerny *et al.* (2008)	r	0.0029
Leksrisompong (2008); Leksrisompong *et al.* (2010)	d	0.006
Giri *et al*. (2010)	d	0.000 059
1-BUTANETHIOL (butyl mercaptan) *[109-79-5]*		
Baker (1963)		0.006
Pietrzak & Barylko-Pikielna (1976)		0.0011
BUTANOIC ACID (butyric acid) *[107-92-6]*		
Moncrieff (1957)		10
Holluta (1960)		0.05
Amoore *et al.* (1968); Amoore (1976)	d	1.11
Guadagni (1966,1968,1969,1970a,c); Buttery & Ling (1998)	d	0.24
Zoeteman *et al.* (1971)		40
Manning & Robinson (1973)		7
Kikuchi *et al.* (1976)		3
Schnabel *et al.* (1988)	d	1.0 - 4.8
Schieberle (1991); Grosch *et al.* (1993); Gassenmeier & Schieberle (1995)	d	1
Larsen & Poll (1992)	d	0.1 - 1.0
Boonbumrung *et al.* (2001)	d	1.40
Karagül-Yüceer *et al.* (2003, 2004)	d	1.274

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Pino & Mesa (2006)	d	0.204
Czerny *et al*. (2007c); Czerny *et al.* (2008)	r	7.7
Czerny *et al.* (2008)	d	2.4

1-BUTANOL (butyl alcohol) *[71-36-3]*

Moncrieff (1957)		50
Baker (1963)		2.5
Flath *et al.* (1967); Buttery *et al.* (1971,1988a, 1994a); Buttery & Ling (1998)	d	0.5
Herrmann & Poeschel (1973)		51.0
Lillard & Powers (1975)	d	2.77
Hertz *et al.* (1975)	d	27 - 148
De Grunt (1975)	d	2.0
Amoore & Buttery (1978)	d	6.5
Alexander *et al.* (1982)	d	0.27
Sarzynski (1982)		10
Cain (1982)	d	6.7
Cain *et al.* (1983); Cain (1989)	d	< 18
Tatchell *et al.* (1985)	d	160 - 490
Eskenazi *et al.* (1986)	d	5 - 15
Granzer *et al.* (1986)		10 - 200
Heywood & Costanzo (1986)	d	< 18
Rabin & Cain (1986)	d	20 - 52
Cain *et al.* (1988)	d	6
Jones-Gotman & Zatorre (1988)	d	296 - 511
Schnabel *et al.* (1988)	d	0.8 - 2.0
Herbener *et al.* (1989)		0.2 - 0.5
Kopala *et al.* (1989)	d	600 - 1,780
Murphy *et al.* (1990)		55 - 165
Sugisawa *et al.* (1991)	d	28
Zatorre & Jones-Gotman (1991)	d	294 - 509
Martinez *et al.* (1993)		15 - 45
Lehrner *et al.* (1995)		2 - 6.1
Murphy & Jinich (1996)		6.1 - 18.3
Lehrner *et al.* (1999)		18 - 160
Boonbumrung *et al*. (2001)	d	4.33
Linschoten *et al.* (2001)		24 - 74
Jacquot *et al.* (2004)		152
Pino & Mesa (2006)	d	0.5
Chalé-Rush *et al.* (2007)	d	200
Czerny *et al.* (2008)	d	0.59
Czerny *et al.* (2008)	r	4.3
Giri *et al*. (2010)	d	0.4592

2-BUTANOL (*sec*-butyl alcohol) *[78-92-2]*

Moncrieff (1957)		50
Schnabel *et al.* (1988)	d	16 - 40
Boonbumrung *et al.* (2001)	d	3.30

BUTANONE (methyl ethyl ketone) *[78-93-3]*

Sarzynski (1982)		7
Granzer *et al.* (1986)		10 - 100
Schnabel *et al.* (1988)	d	17 - 32
Giri *et al*. (2010)	d	35.4002

(*E*)-2-BUTENAL (crotonaldehyde) *[123-73-9]*

Guadagni *et al.* (1963); Teranishi *et al.* (1974)	d	0.525
Giri *et al.* (2010)	d	0.000 3

(6*E*,2*E*)-6-(BUT-2-ENYLIDENE)-1,5,5-TRIMETHYL-1-CYCLOHEXENE

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Whitfield *et al.* (1977) 0.1

(6*Z*,2*E*)-6-(BUT-2-ENYLIDENE)-1,5,5-TRIMETHYL-1-CYCLOHEXENE
Whitfield *et al.* (1977) 0.1

2-BUTENYL ISOTHIOCYANATE (2-butenyl mustard oil) *[2253-93-2]*
Buttery *et al.* (1976b) d 0.380

3-BUTENYL ISOTHIOCYANATE (3-butenyl mustard oil) *[3386-97-8]*
Masuda *et al.* (1996) 0.017

2-(3-BUTENYL)-3-METHOXYPYRAZINE *[115652-12-5]*
Mihara & Masuda (1988) d 0.000 03

2-butenyl mustard oil⇒2-BUTENYL ISOTHIOCYANATE
3-butenyl mustard oil⇒3-BUTENYL ISOTHIOCYANATE

2-(2-BUTENYLTHIO)ETHANAL
Rössner *et al.* (2002) 0.000 04

2-(3-BUTENYLTHIO)ETHANAL
Rössner *et al.* (2002) 0.000 1

4-BUTOXY-2,5-DIMETHYL-3(2*H*)-FURANONE
Buttery (1999) d 0.25

2-BUTOXYETHANOL (butyl cellosolve, ethylene glycol *n*-butyl ether) *[111-76-2]*
Alexander *et al.* (1982) d 0.88

BUTYL ACETATE *[123-86-4]*
Flath *et al.* (1967);Guadagni (1970b);
Buttery (1981); Teranishi *et al.* (1987);
Takeoka *et al.* (1996) d 0.066
Sega *et al.* (1967) d 0.043
Cain (1974) 38
Tuorila *et al.* (1982) d 0.067 - 0.145
Sarzynski (1982) 7
Piggott & Findlay (1984) d 0.46
Granzer *et al.* (1986) 0.01 - 0.5
Schnabel *et al.* (1988) d 0.044 - 0.088
Laren & Poll (1992) d 0.01 - 0.1
Pino & Mesa (2006) d 0.066
Giri *et al.* (2010) d 0.058

2-BUTYL ACETATE *[105-46-4]*
Takeoka *et al.* (1996) d 0.006

tert-BUTYL ACETATE (2-methylpropyl-2 acetate, dimethylethyl ethanoate) *[540-88-5]*
Piggott & Findlay (1984) d 0.82
Schnabel *et al.* (1988) d 0.081 - 0.43

butyl acrylate⇒BUTYL 2-PROPENOATE
butyl alcohol⇒1-BUTANOL
sec-butyl alcohol⇒2-BUTANOL
tert-butyl alcohol⇒2-METHYL-2-PROPANOL

BUTYLAMINE *[109-73-9]*
Amoore *et al.* (1975b) d 50

butyl angelate⇒ BUTYL (*Z*)-2-METHYL-2-BUTENOATE

BUTYLBENZENE *[104-51-8]*

Zoeteman *et al.* (1971)		0.1

tert-BUTYLBENZENE *[98-06-6]*

Zoeteman *et al.* (1971)		0.05

BUTYL BUTANOATE (butyl butyrate) *[109-21-7]*

Teranishi *et al.* (1987); Takeoka *et al.* (1990a)	d	0.1
Schnabel *et al.* (1988)	d	0.087 - 0.43
Sugisawa *et al.* (1991); Yang *et al.* (1992); Tamura *et al.* (1993)	d	0.55 - 1
Tamura *et al.* (1996)	d	0.4
Tamura *et al.* (1996)	r	1.089
Boonbumrung *et al.* (2001)	d	0.11
Pino & Mesa (2006)	d	0.4

butyl butyrate⇒BUTYL BUTANOATE
butyl caproate⇒BUTYL HEXANOATE
butyl carbitol⇒3,6-DIOXA-1-DECANOL
butyl cellosolve⇒2-BUTOXYETHANOL

4-*tert*-BUTYL-2-CHLOROPHENOL *[98-28-2]*

Dietz & Traud (1978)		0.3

BUTYL CYANIDE (valeronitrile) *[110-59-8]*

Fabrellas *et al.* (2004)		0.013 - 0.049

2-*tert*-BUTYLCYCLOHEXANOL *[13491-79-7]*

Pelosi & Pisanelli (1981)	d	0.11

2-*sec*-BUTYLCYCLOHEXANONE *[1728-46-7]*

Pelosi & Viti (1978)	d	0.26

4-*tert*-butylcyclohexathiazole⇒ 4-*tert*-BUTYL-5,6,7-TRIHYDROBENZOTHIAZOLE

4(4'-*cis*-*tert*-BUTYLCYCLOHEXYL)-4-METHYL-2-PENTANONE (*cis*-ketone, pemenone)

Amoore *et al.* (1977)	d	0.0012

4-*tert*-BUTYL-1,2-DIHYDROXYBENZENE (4-*tert*-butylpyrocatechol) *[98-29-3]*

Dietz & Traud (1978)		1

4-*tert*-BUTYL-2,6-DIMETHYL-3,5-DINITROACETOPHENONE (musk ketone) *[81-14-1]*

Amoore *et al.* (1977)	d	0.0082
Geddes *et al.* (1991)	d	0.01

2-*tert*-BUTYL-4,6-DIMETHYLPHENOL *[1879-09-0]*

Dietz & Traud (1978)		0.3

5-BUTYL-2,3-DIMETHYLPYRAZINE *[15834-78-3]*

Shibamoto (1986)	d	0.77

5-*sec*-BUTYL-2,3-DIMETHYLPYRAZINE *[3226-30-0]*

Masuda & Mihara (1986); Shibamoto (1986)	d	0.68

1-*tert*-BUTYL-3,5-DIMETHYL-2,4,6-TRINITROBENZENE (musk xylene) *[81-15-2]*

Amoore *et al.* (1977)	d	0.0023

2-*sec*-BUTYL-4,6-DINITROPHENOL *[88-85-7]*

Alexander *et al.* (1982)	d	0.056

2-*tert*-BUTYL-4,6-DINITROPHENOL *[1420-07-1]*

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Dietz & Traud (1978) 0.5

5-*sec*-BUTYL-2-ETHOXY-3-METHYLPYRAZINE *[99784-15-3]*
Masuda & Mihara (1986); Shibamoto (1986) d 0.012

butyl ethyl ketone⇒3-HEPTANONE

5-BUTYL-4-ETHYLTHIAZOLE
Buttery *et al.* (1976a) d 0.000 12

4-BUTYL-5-ETHYLTHIAZOLE
Buttery *et al.* (1976a) d 0.000 02

5-*sec*-BUTYL-2-(ETHYLTHIO)-3-METHYLPYRAZINE *[99784-26-6]*
Masuda & Mihara (1986); Shibamoto (1986) d 0.061

2-BUTYL-3-(ETHYLTHIO)PYRAZINE *[113685-91-9]*
Masuda & Mihara (1988) d 0.004

BUTYL FORMATE *[592-84-7]*
De Grunt (1975) d 6.0
Pyysalo *et al.* (1977) d 0.80

2-BUTYLFURAN *[4466-24-4]*
Giri *et al*. (2010) d 0.00500

BUTYL HEXANOATE (butyl caproate) *[626-82-4]*
Takeoka *et al.* (1990a) d 0.7
Sugisawa *et al.* (1991) d 10

3-*tert*-BUTYL-4-HYDROXYANISOLE *[121-00-6]*
Dietz & Traud (1978) 3

3-*tert*-BUTYL-2-HYDROXY-5-METHYLBENZOIC ACID
Dietz & Traud (1978) 0.1

5-BUTYL-3-HYDROXY-4-METHYL-2(5*H)*-FURANONE
Rödel & Hempel (1974) r 1.4

butyl isobutyrate⇒BUTYL 2-METHYLPROPANOATE
sec-butyl isobutyrate⇒*sec*-BUTYL 2-METHYLPROPANOATE

sec-BUTYL ISOTHIOCYANATE (*sec*-butyl mustard oil) *[4426-79-3]*
Masuda *et al.* (1996) 0.09

butyl mercaptan⇒1-BUTANETHIOL
butyl methacrylate⇒BUTYL 2-METHYL-2-PROPENOATE

1-*tert*-BUTYL-2-METHOXY-4-METHYL-3,5-DINITROBENZENE (musk ambrette) *[83-66-9]*
Appell (1969) 0.3
Amoore *et al.* (1977) d 0.0036

3-*sec*-BUTYL-2-METHOXY-5-METHYLPYRAZINE
Iwabuchi *et al.* (1984) 0.002

5-*sec*-BUTYL-2-METHOXY-3-METHYLPYRAZINE *[99784-13-1]*
Masuda & Mihara (1986); Shibamoto (1986) d 0.032

2-BUTYL-3-METHOXYPYRAZINE *[32737-11-4]*
Masuda & Mihara (1988) d 0.000 05

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

2-*sec*-BUTYL-3-METHOXYPYRAZINE *[24168-70-5]*

Buttery (1981,1993)	d	0.000 002
Mihara & Masuda (1988); Mihara *et al.* (1990)	d	0.000 04
Masanetz & Grosch (1998a, 1998b)		0.000 03
Neta *et al.* (2008)	d	0.000 001

(*R*)-(-)-2-*sec*-BUTYL-3-METHOXYPYRAZINE *[124151-14-0]*

Mihara *et al.* (1990)	d	0.000 01

(*S*)-(+)-2-*sec*-BUTYL-3-METHOXYPYRAZINE *[12451-13-9]*

Mihara *et al.* (1990)	d	0.000 1

BUTYL 2-METHYLBUTANOATE *[15706-73-7]*

Schnabel *et al.* (1988)	d	0.042 - 0.087
Takeoka *et al.* (1990a)	d	0.017

2-BUTYL 2-METHYLBUTANOATE

Schnabel *et al.* (1988)	d	0.085 - 0.43

BUTYL (*E*)-2-METHYL-2-BUTENOATE (butyl tiglate) *[7785-66-2]*

Takeoka *et al.* (1998)	d	0.0042

BUTYL (*Z*)-2-METHYL-2-BUTENOATE (butyl angelate) *[7785-64-0]*

Takeoka *et al.* (1998)	d	0.013

sec-butyl methyl ketone⇒3-METHYL-2-PENTANONE

5-*sec*-BUTYL-3-METHYL-2-(METHYLTHIO)PYRAZINE *[99784-23-3]*

Masuda & Mihara (1986); Shibamoto (1986)	d	0.019

4-*tert*-BUTYL-2-METHYLPHENOL *[98-27-1]*

Dietz & Traud (1978)		0.4

2-*tert*-BUTYL-4-METHYLPHENOL *[2409-55-4]*

Dietz & Traud (1978)		0.03

2-*tert*-BUTYL-5-METHYLPHENOL *[88-60-8]*

Dietz & Traud (1978)		0.5

2-*tert*-BUTYL-6-METHYLPHENOL *[2219-82-1]*

Dietz & Traud (1978)		0.05

5-*sec*-BUTYL-3-METHYL-2-PHENOXYPYRAZINE *[99784-19-7]*

Masuda & Mihara (1986); Shibamoto (1986)	d	0.072

5-*sec*-BUTYL-3-METHYL-2-(PHENYLTHIO)PYRAZINE *[99784-31-3]*

Masuda & Mihara (1986); Shibamoto (1986)	d	0.061

BUTYL 2-METHYLPROPANOATE (butyl isobutyrate) *[97-87-0]*

Schnabel *et al.* (1988)	d	0.087 - 0.17
Takeoka *et al.* (1990a)	d	0.08
Pino & Mesa (2006)	d	0.0936

sec-BUTYL 2-METHYLPROPANOATE (*sec*-butyl isobutyrate)

Schnabel *et al.* (1988)	d	0.043 - 0.087

BUTYL 2-METHYL-2-PROPENOATE (butyl methacrylate) *[97-88-1]*

Piringer & Granzer (1984); Granzer *et al.* (1986)		0.01 - 0.1

4-BUTYL-5-METHYLTHIAZOLE *[57246-60-3]*

Buttery *et al.* (1976a)	d	0.000 01

sec-butyl mustard oil⇒*sec*-BUTYL ISOTHIOCYANATE

(*Z*)-2-BUTYL-2-OCTENAL
Robert *et al.* (2004) d 0.02

5-BUTYL-4-PENTYLTHIAZOLE
Buttery *et al.* (1976a) d 0.000 19

2-*tert*-BUTYLPHENOL *[88-18-6]*
Dietz & Traud (1978) 0.05

3-*tert*-BUTYLPHENOL *[585-34-2]*
Dietz & Traud (1978) 0.08

4-*tert*-BUTYLPHENOL *[98-54-4]*
Dietz & Traud (1978) 0.8

3-BUTYLPHTHALIDE *[6066-49-5]*
Buttery (1993) d 0.01

BUTYL PROPANOATE (butyl propionate) *[590-01-2]*
Flath *et al.* (1967) d 0.025
Schnabel *et al.* (1988) d 0.091 - 0.44
Takeoka *et al.* (1990a) d 0.2

tert-BUTYL PROPANOATE (dimethylethyl propanoate) *[20487-40-5]*
Schnabel *et al.* (1988) d 0.0086 - 0.043

BUTYL 2-PROPENOATE (butyl acrylate) *[141-32-2]*
Boiron & Pons (1978) d 0.009
Piringer & Granzer (1984);Granzer *et al.* (1986) 0.002 - 0.02

butyl propionate⇒BUTYL PROPANOATE

4-BUTYL-5-PROPYLTHIAZOLE *[57246-59-0]*
Buttery *et al.* (1976a) d 0.000 003
Pelosi *et al.* (1983) d 0.000 005

BUTYLPYRAZINE *[29460-91-1]*
Masuda & Mihara (1988) d 0.4

4-*tert*-butylpyrocatechol⇒4-BUTYL-1,2-DIHYDROXYBENZENE

2-(BUTYLTHIO)ETHANAL
Rössner *et al.* (2002) 0.000 6

butyl tiglate⇒BUTYL (*E*)-2-METHYL-2-BUTENOATE

4-*tert*-BUTYL-5,6,7-TRIHYDROBENZOTHIAZOLE (4-*tert*-butylcyclohexathiazole) *[84648-07-7]*
Pelosi *et al.* (1983) d 0.0015

butyraldehyde⇒BUTANAL
butyric acid⇒BUTANOIC ACID
γ-butyrolactone⇒4-HYDROXYBUTANOIC ACID LACTONE
butyronitrile⇒PROPYL CYANIDE

C

cadaverine⇒1,5-PENTANEDIAMINE
(+)-camphene⇒(1*R*)-(+)-2,2-DIMETHYL-3-METHYLENEBICYCLO[2.2.1]HEPTANE

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

(-)-camphene⇒(1*S*)-(-)-2,2-DIMETHYL-3-METHYLENEBICYCLO[2.2.1]HEPTANE

2-CAMPHOROXYMETHYLPYRAZINE
Calabretta (1975,1978) d > 0.5

2-CAMPHOROXY-3-METHYLPYRAZINE
Calabretta (1975,1978) d 0.3

2-CAMPHOROXY-5-METHYLPYRAZINE
Calabretta (1975,1978) d 0.3

camphor⇒1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANONE
(1*R*)-(+)-camphor⇒(1*R*)-(+)-1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANONE
(1*S*)-(-)-camphor⇒(1*S*)-(-)-1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANONE
(*d*)-camphor⇒(1*R*)-(+)-1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANONE
capric acid⇒DECANOIC ACID
caproic acid⇒HEXANOIC ACID
caprolactam⇒6-AMINOHEXANOIC ACID LACTAM
caprylaldehyde⇒OCTANAL
caprylic acid⇒OCTANOIC ACID
carbaryl⇒1-NAPHTHYL N-METHYLCARBAMATE
carbitol⇒3,6-DIOXA-1-OCTANOL

3-CARBOETHOXY-3-ETHYLTETRAHYDROFURAN-2,3-DIONE
(4-hydroxy-3-ethoxycarbonyl-3-ethyl-2-oxopentanoic acid lactone)
Rödel & Hempel (1974) r 0.2

CARBON DISULPHIDE *[75-15-0]*
ETS Laboratories (2002) 0.005

carbon tetrachloride⇒TETRACHLOROMETHANE
(+)-2-carene⇒3,7,7-TRIMETHYLBICYCLO[4.1.0]-2-HEPTENE
3-carene⇒3,7,7-TRIMETHYLBICYCLO[4.1.0]-3-HEPTENE
(1*S*)-(+)-3-carene⇒(1*S*)-(+)-3,7,7-TRIMETHYLBICYCLO[4.1.0]-3-HEPTENE
δ3-carene⇒3,7,7-TRIMETHYLBICYCLO[4.1.0]-3-HEPTENE

2-CARENE EPOXIDE *[62413-92-7]*
Buttery *et al.* (1987b) d 0.5

carotol⇒2,3,4,5,8,8a-HEXAHYDRO-3-ISOPROPYL-6,8a-DIMETHYL-3a(1*H*)-AZULENOL
carvacrol⇒5-ISOPROPYL-2-METHYLPHENOL
(*d*)-carvenone⇒(*d*)-3-ISOPROPYL-5-METHYL-2-CYCLOHEXEN-1-ONE
(*l*)-carvenone⇒(*l*)-3-ISOPROPYL-5-METHYL-2-CYCLOHEXEN-1-ONE
carveol 1⇒*cis*-4-ISOPROPENYL-1-METHYL-6-CYCLOHEXEN-2-OL
cis-carveol⇒*cis*-4-ISOPROPENYL-1-METHYL-6-CYCLOHEXEN-2-OL
trans-carveol⇒*trans*-4-ISOPROPENYL-1-METHYL-6-CYCLOHEXEN-2-OL
(*d*)-carvomenthone⇒(*d*)-5-ISOPROPYL-2-METHYLCYCLOHEXANONE
(*l*)-carvomenthone⇒(*l*)-5-ISOPROPYL-2-METHYLCYCLOHEXANONE
carvone⇒5-ISOPROPENYL-2-METHYL-2-CYCLOHEXEN-1-ONE
(*d*)-carvone⇒(*S*)-(+)-5-ISOPROPENYL-2-METHYL-2-CYCLOHEXEN-1-ONE
(*l*)-carvone⇒(*R*)-(-)-5-ISOPROPENYL-2-METHYL-2-CYCLOHEXEN-1-ONE
(4*R*)-(-)-carvone⇒(*R*)-(-)-5-ISOPROPENYL-2-METHYL-2-CYCLOHEXEN-1-ONE
(4*S*)-(+)-carvone⇒(*S*)-(+)-5-ISOPROPENYL-2-METHYL-2-CYCLOHEXEN-1-ONE
carvyl acetate⇒(1*RS*,5*R*)-5-ISOPROPENYL-2-METHYL-2-CYCLOHEXENYL ACETATE
caryophyllene⇒4,11,11-TRIMETHYL-8-METHYLENEBICYCLO[7.2.0]-4-UNDECENE
β-caryophyllene⇒4,11,11-TRIMETHYL-8-METHYLENEBICYCLO[7.2.0]-4-UNDECENE
caryophyllene epoxide⇒[1*R*-(1*R**,4*R**,6*R**,10*S**)]-4,12,12-TRIMETHYL-9-METHYLENE-5-OXATRICYCLO[8.2.0^{4,6}]DODECANE
caryophyllene oxide⇒[1*R*-(1*R**,4*R**,6*R**,10*S**)]-4,12,12-TRIMETHYL-9-METHYLENE-5-OXATRICYCLO[8.2.0^{4,6}]DODECANE
catechol⇒1,2-DIHYDROXYBENZENE
celestolide⇒4-ACETYL-6-*tert*-BUTYL-1,1-DIMETHYLINDANE
cellosolve⇒2-ETHOXYETHANOL
cellosolve acetate⇒2-ETHOXYETHYL ACETATE

CHLORAMINE (monochloramine) *[10599-90-3]*
Krasner & Barrett (1984) 0.47

chlordane⇒1,2,4,5,6,7,8,8-OCTACHLORO-2,3,3a,4,7,7a-HEXAHYDRO-4,7-METHANOINDENE

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

chloretone⇒2-METHYL-1,1,1-TRICHLORO-2-PROPANOL
chlorfenvinphos⇒2-CHLORO-1-(2,4-DICHLOROPHENYL)VINYL DIETHYL PHOSPHATE

CHLORINE *[7782-50-5]*

Klut (1913)		0.2 - 0.3
Sperry & Billings (1921)	r	1.0
Pangborn *et al.* (1970)	r	< 0.8
Piriou & Perrelle (1999)		0.05
Mackey *et al.* (2004)		0.07 - 0.13
McDonald *et al.* (2009)		0.10

CHLORINE DIOXIDE *[7758-19-2]*

Widemann (1957)		0.07 - 0.08

chloroacetyl chloride⇒1,2-DICHLOROOXOETHANE

2-CHLOROANILINE *[95-51-2]*

Zoeteman (1978)	d	0.003

3-CHLOROANILINE *[108-42-9]*

Khamuev (1965,1967)		2.6 - 3
Zoeteman (1978)	d	0.1

4-CHLOROANILINE *[106-47-8]*

Khamuev (1965,1967)		1.5
Zoeteman (1978)	d	0.01

4-CHLOROANISOLE *[623-12-1]*

Young *et al.* (1996)		0.02

CHLOROBENZENE *[108-90-7]*

Holluta (1960)		0.1
Alexander *et al.* (1982)	d	0.08

1-CHLORO-2-(1-CHLOROETHOXY)ETHANE (2,2'-dichloroethyl ether) *[111-44-4]*

Rosen *et al.* (1963)		0.36

1-CHLORO-2-(1-CHLOROISOPROPOXY)PROPANE (2,2'-dichloroisopropyl ether) *[108-60-1]*

Rosen *et al.* (1963)		0.2
Baker (1963)		0.32
Kölle *et al.* (1972)		0.3

4-chloro-*m*-cresol⇒4-CHLORO-3-METHYLPHENOL
6-chloro-*m*-cresol⇒2-CHLORO-5-METHYLPHENOL
4-chloro-*o*-cresol⇒4-CHLORO-2-METHYLPHENOL
6-chloro-*o*-cresol⇒6-CHLORO-2-METHYLPHENOL
2-chloro-*p*-cresol⇒2-CHLORO-4-METHYLPHENOL

4-CHLORO-2-CYCLOPENTYLPHENOL *[13347-42-7]*

Dietz & Traud (1978)		0.3

chlorodibromomethane⇒DIBROMOCHLOROMETHANE

2-CHLORO-1-(2,4-DICHLOROPHENYL)VINYL DIETHYL PHOSPHATE (chlorfenvinphos) *[470-90-6]*

Young *et al.* (1996)		0.34

4-CHLORO-3,5-DIMETHYLANISOLE *[6267-34-1]*

Gee *et al.* (1974)	d	0.23

CHLORODIIODOMETHANE *[638-73-3]*

Cancho *et al.* (2001)		0.000 2

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

chloroform⇒TRICHLOROMETHANE
4-chloroguaiacol⇒4-CHLORO-2-METHOXYPHENOL

5-CHLORO-4-HYDROXY-3-METHOXYBENZALDEHYDE (5-chlorovanillin) *[19463-48-0]*
Kovacs *et al.* (1984) d 3.8

6-CHLORO-4-HYDROXY-3-METHOXYBENZALDEHYDE (6-chlorovanillin) *[18268-76-3]*
Kovacs *et al.* (1984) d 0.37

CHLOROIODOMETHANE *[593-71-5]*
Cancho *et al.* (2001) 0.0058 - 0.0089

N-CHLOROISOBUTYRALDIMINE
Freuze *et al.* (2005) d 0.000 2

N-CHLOROISOVALERALDIMINE
Freuze *et al.* (2005) d 0.000 25

4-CHLORO-2-ISOPROPYL-5-METHYLPHENOL (*o*-chlorothymol) *[89-68-9]*
Dietz & Traud (1978) 0.6

4-CHLORO-2-METHOXYPHENOL (4-chloroguaiacol) *[16766-30-6]*
Kovacs *et al.* (1984) d 0.0042

5-CHLORO-2-METHYLANILINE (5-chloro-*o*-toluidine) *[95-79-4]*
Zoeteman (1978) d 0.005

4-CHLORO-3-METHYLANISOLE *[13334-71-9]*
Gee *et al.* (1974) d 0.11

2-CHLORO-6-METHYLANISOLE
Griffiths & Fenwick (1977) d 0.003

1-CHLORO-2-METHYLBENZENE (2-chlorotoluene) *[95-49-8]*
De Grunt (1975); De Grunt *et al.* (1977) d 0.01
Pisko *et al.* (1981) 0.04

1-CHLORO-3-METHYLBENZENE (3-chlorotoluene) *[108-41-8]*
Young *et al.* (1996) 0.5

1-CHLORO-4-METHYLBENZENE (4-chlorotoluene) *[106-43-4]*
Pisko *et al.* (1981) 0.04
Young *et al.* (1996) 0.15

4-CHLORO-2-METHYLPHENOL (4-chloro-*o*-cresol) *[1570-64-5]*
Griffiths & Land (1973) d 0.12
Dietz & Traud (1978) 1.8
Young *et al.* (1996) 0.2

6-CHLORO-2-METHYLPHENOL (2-chloro-6-methylphenol, 6-chloro-*o*-cresol) *[87-64-9]*
Patterson (1972) 0.000 1
Griffiths & Land (1973) d 0.000 08

4-CHLORO-3-METHYLPHENOL (4-chloro-*m*-cresol) *[59-50-7]*
Dietz & Traud (1978) 3
Young *et al.* (1996) 0.005

2-CHLORO-4-METHYLPHENOL (2-chloro-*p*-cresol) *[6640-27-3]*
Young *et al.* (1996) 0.000 3

2-CHLORO-5-METHYLPHENOL (6-chloro-*m*-cresol) *[615-74-7]*

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Dietz & Traud (1978)		0.02

2-chloro-6-methylphenol⇒6-CHLORO-2-METHYLPHENOL

1-CHLORO-2-NITROBENZENE (*o*-chloronitrobenzene) *[88-73-3]*

Duguet *et al.* (1990)		0.02

o-chloronitrobenzene⇒1-CHLORO-2-NITROBENZENE

4-CHLORO-2-NITROPHENOL *[89-64-5]*

Dietz & Traud (1978)		0.8

2-CHLORO-4-NITROPHENOL *[619-08-9]*

Dietz & Traud (1978)		5

1-CHLORO-1-NITROSOCYCLOHEXANE

Cherkinski (1961)	d	0.005

2-CHLOROPHENOL *[95-57-8]*

Hoak (1956)		0.000 33
Burttschell *et al.* (1959)		0.002
De Grunt (1975); De Grunt *et al.* (1977)	d	0.000 2 - 0.002
Dietz & Traud (1978)		0.01
Kovacs *et al.* (1984)	d	0.000 12
Young *et al.* (1996)		0.000 36

3-CHLOROPHENOL *[108-43-0]*

Hoak (1956)		0.2
Dietz & Traud (1978)		0.05

4-CHLOROPHENOL *[106-48-9]*

Helbig (1939)		0.1204
Hoak (1956)		0.033
Burttschell *et al.* (1959)		0.25
Baker (1963)		1.24
De Grunt (1975); De Grunt *et al.* (1977)	d	0.000 5
Dietz & Traud (1978)		0.06
Young *et al.* (1996)		0.02

N-CHLOROPHENYLACETALDIMINE

Freuze *et al.* (2004, 2005)	d	0.003

2-CHLORO-6-PHENYLANISOLE

Gee *et al.* (1974)	d	0.097

N'-(4-CHLOROPHENYL)-N,N-DIMETHYLUREA (monuron) *[150-68-5]*

Mikhailov (1967)	5	

o-chlorothymol⇒2-CHLORO-6-ISOPROPYL-3-METHYLPHENOL
2-chlorotoluene⇒1-CHLORO-2-METHYLBENZENE
3-chlorotoluene⇒1-CHLORO-3-METHYLBENZENE
4-chlorotoluene⇒1-CHLORO-4-METHYLBENZENE
5-chloro-*o*-toluidine⇒5-CHLORO-2-METHYLANILINE
5-chlorovanillin⇒5-CHLORO-4-HYDROXY-3-METHOXYBENZALDEHYDE
6-chlorovanillin⇒6-CHLORO-4-HYDROXY-3-METHOXYBENZALDEHYDE
chlorpyrifos⇒O,O-DIETHYL O-3,5,6-TRICHLORO-2-PYRIDYL PHOSPHOROTHIOATE
choline chloride⇒(2-HYDROXYETHYL)TRIMETHYLAMMONIUM CHLORIDE
1,8-cineole⇒1,3,3-TRIMETHYL-2-OXABICYCLO[2.2.2]OCTANE
cinnamaldehyde⇒3-PHENYL-2-PROPENAL
(*E*)-cinnamaldehyde⇒3-PHENYL-(E)-2-PROPENAL
cinnamic acid⇒3-PHENYL-2-PROPENOIC ACID
cinnamic alcohol⇒3-PHENYL-2-PROPEN-1-OL
cinnamyl alcohol⇒3-PHENYL-2-PROPEN-1-OL

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

cis-cinnamic alcohol⇒3-PHENYL-(*Z*)-2-PROPEN-1-OL
trans-cinnamic alcohol⇒3-PHENYL-(*E*)-2-PROPEN-1-OL
citral⇒3,7-DIMETHYL-2,6-OCTADIENAL
citral a⇒3,7-DIMETHYL-(*E*)-2,6-OCTADIENAL
citral b⇒3,7-DIMETHYL-(*Z*)-2,6-OCTADIENAL
citronellal⇒3,7-DIMETHYL-6-OCTENAL
(3*R*)-(+)-citronellal⇒(3*R*)-(+)-3,7-DIMETHYL-6-OCTENAL
(3*S*)-(-)-citronellal⇒(3*S*)-(-)-3,7-DIMETHYL-6-OCTENAL
citronellol⇒3,7-DIMETHYL-6-OCTEN-1-OL
(3*R*)-(+)-citronellol⇒(*R*)-(+)-3,7-DIMETHYL-6-OCTEN-1-OL
(3*S*)-(-)-citronellol⇒(*S*)-(-)-3,7-DIMETHYL-6-OCTEN-1-OL
(*l*)-citronellol⇒(*S*)-(-)-3,7-DIMETHYL-6-OCTEN-1-OL
citronellyl acetate⇒3,7-DIMETHYL-6-OCTEN-1-YL ACETATE
civetone⇒9-*cis*-CYCLOHEPTADECEN-1-ONE
co-ral⇒O,O-DIETHYL O-(3-CHLORO-4-METHYL-7-COUMARINYL)PHOSPHOROTHIOATE

COUMARIN *[91-64-5]*

Helbig (1939)		0.138
Cartwright & Kelley (1952)	d	0.05
Cartwright & Kelley (1952)	r	0.25
Appell (1969)		0.3
De Grunt (1975); De Grunt *et al.* (1977)	d	0.05
Buttery *et al.* (1978a)	d	0.034
Masanetz & Grosch (1998b)		0.025

creatinine⇒2-AMINO-1,5-DIHYDRO-1-METHYL-4*H*-IMIDAZOL-4-ONE
creosol⇒2-METHOXY-4-METHYLPHENOL
m-cresol⇒3-METHYLPHENOL
o-cresol⇒2-METHYLPHENOL
p-cresol⇒4-METHYLPHENOL
p-cresyl acetate⇒4-METHYLPHENYL ACETATE
crotonaldehyde⇒(*E*)-2-BUTENAL
cumene⇒ISOPROPYLBENZENE
cuminaldehyde⇒4-ISOPROPYLBENZALDEHYDE

1-CYANO-2,3-EPITHIOPROPANE *[58130-93-1]*

Chin *et al.* (1996)	d	180

CYCLOBUTANOL *[2919-23-5]*

Schnabel *et al.* (1988)	d	4.6 - 9.4

CYCLOBUTANONE *[1191-95-3]*

Schnabel *et al.* (1988)	d	0.9 - 4.7

CYCLOBUTYLAMINE *[2516-34-9]*

Amoore *et al.* (1975b)	d	32.5

β-cyclocitral⇒2,6,6-TRIMETHYL-1-CYCLOHEXEN-1-CARBOXALDEHYDE

CYCLODECANONE *[1502-06-3]*

Pelosi & Viti (1978)	d	0.44
Pelosi & Pisanelli (1981)	d	0.42
Schnabel *et al.* (1988)	d	0.29 - 0.48

cyclodecathiazole⇒4,5,6,7,8,9,10,11-OCTAHYDROTHIAZOLE

CYCLODODECANONE *[830-13-7]*

Pelosi & Viti (1978)	d	0.19
Pelosi & Pisanelli (1981)	d	0.43

cyclododecathiazole⇒4,5,6,7,8,9,10,11,12,13-DECAHYDROTHIAZOLE

CYCLO-1,3-ETHYLENEDIOXY-1,3-TRIDECANEDIONE (ethylene brassylate, musk T) *[105-95-3]*

Amoore *et al.* (1977)	d	0.033

CYCLOHEPTADECANONE *[3661-77-6]*

Amoore *et al.* (1977)	d	0.0043

9-*cis*-CYCLOHEPTADECEN-1-ONE (civetone) *[542-46-1]*

Amoore *et al.* (1977)	d	0.0031

CYCLOHEPTANOL *[502-41-0]*

Schnabel *et al.* (1988)	d	4.8 - 9.5

CYCLOHEPTANONE *[502-42-1]*

Pelosi & Viti (1978)	d	4.50
Schnabel *et al.* (1988)	d	0.090 - 0.47

CYCLOHEPTYLAMINE *[5452-35-7]*

Amoore *et al.* (1975b)	d	65.8

CYCLOHEXADECANONE *[2550-52-9]*

Amoore *et al.* (1977)	d	0.0073

CYCLOHEXANOL *[108-93-0]*

Cherkinski (1961)	d	3.5
De Grunt (1975); De Grunt *et al.* (1977)	d	0.4
Schnabel *et al.* (1988)	d	0.090 - 0.47

CYCLOHEXANONE *[108-94-1]*

Pelosi & Viti (1978)	d	14.4
Sarzynski (1982)		20
Schnabel *et al.* (1988)	d	0.28 - 0.67

cyclohexathiazole⇒4,5,6,7-TETRAHYDROBENZOTHIAZOLE

CYCLOHEXYLACETIC ACID *[5292-21-7]*

Amoore (1976)	d	19.1

CYCLOHEXYLAMINE *[108-91-8]*

Amoore *et al.* (1975b)	d	28.5

CYCLOHEXYL 3-METHYL-3-PHENYL-2,3-EPOXYPROPANOATE

Tekin & Karaman (1992)	d	7

3-CYCLOHEXYL-5,6-TRIMETHYLENE URACIL (hexylure) *[2164-08-1]*

Avdyushkina (1979)		0.15

CYCLONONANONE *[3350-30-9]*

Pelosi & Viti (1978)	d	0.10
Pelosi & Pisanelli (1981)	d	0.52
Schnabel *et al.* (1988)	d	0.098 - 0.48

CYCLOOCTADECANONE *[6907-37-5]*

Amoore *et al.* (1977)	d	0.023

CYCLOOCTANOL *[696-71-9]*

Schnabel *et al.* (1988)	d	10 - 19

CYCLOOCTANONE *[502-49-8]*

Pelosi & Viti (1978)	d	7.8
Pelosi & Pisanelli (1981)	d	13
Schnabel *et al.* (1988)	d	0.96 - 2.9

cyclooctathiazole⇒4,5,6,7,8,9-HEXAHYDROCYCLOOCTATHIAZOLE

cyclopentadecanolide⇒15-HYDROXYPENTADECANOIC ACID LACTONE

CYCLOPENTADECANONE (exaltone) *[502-72-7]*
Amoore *et al.* (1977) d 0.015

CYCLOPENTANOL *[96-41-3]*
Schnabel *et al.* (1988) d 9.5 - 19
Giri *et al*. (2010) d 0.1251

CYCLOPENTANONE *[120-92-3]*
Pelosi & Viti (1978) d 51.1
Schnabel *et al.* (1988) d 9.3 - 47

cyclopentathiazole⇒5,6-DIHYDRO-4*H*-CYCLOPENTATHIAZOLE

CYCLOPENTYL ACETATE *[933-05-1]*
Teranishi *et al.* (1966) d 0.021

CYCLOPENTYLAMINE *[1003-03-8]*
Amoore *et al.* (1975b) d 190

CYCLOPROPYLAMINE *[765-30-0]*
Amoore *et al.* (1975b) d 38.9

cyclotene⇒2-HYDROXY-3-METHYL-2-CYCLOPENTEN-1-ONE

CYCLOTETRADECANONE *[3603-99-4]*
Amoore *et al.* (1977) d 0.024

CYCLOTRIDECANONE *[832-10-0]*
Pelosi & Viti (1978) d 0.31

CYCLOUNDECANONE *[878-13-7]*
Pelosi & Viti (1978) d 0.12
Pelosi & Pisanelli (1981) d 0.10
Schnabel *et al.* (1988) d 0.009 - 0.047

m-cymene⇒1-ISOPROPYL-3-METHYLBENZENE
p-cymene⇒1-ISOPROPYL-4-METHYLBENZENE

D

2,4-D⇒2,4-DICHLOROPHENOXYACETIC ACID
β-damascenone⇒1-(2,6,6-TRIMETHYL-1,3-CYCLOHEXADIEN-1-YL)-(*E*)-2-BUTEN-1-ONE
α-damascone⇒1-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-2-BUTEN-1-ONE
(*d*)-α-damascone⇒(*R*)-(+)-1-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-2-BUTEN-1-ONE
(*l*)-α-damascone⇒(*S*)-(-)-1-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-2-BUTEN-1-ONE
β-damascone⇒(*E*)-1-(2,6,6-TRIMETHYL-1-CYCLOHEXEN-1-YL)-2-BUTEN-1-ONE
(*E*)-β-damascone⇒(*E*)-1-(2,6,6-TRIMETHYL-1-CYCLOHEXEN-1-YL)-2-BUTEN-1-ONE
γ-damascone⇒1-(2,2-DIMETHYL-6-METHYLENECYCLOHEXYL)-2-BUTEN-1-ONE
DDT⇒1,1,1-TRICHLORO-2,2-BIS(4-CHLOROPHENYL)ETHANE

2,4-DECADIENAL *[2363-88-4]*
Kossa *et al.* (1979); Tressl *et al.* (1979b,1980) 0.000 3

(*E*,*E*)-2,4-DECADIENAL *[25152-84-5]*
Buttery *et al.* (1969b,1971,1973,1988a,1997,1999);
Guadagni *et al.* (1972); Teranishi *et al.* (1974);
Buttery (1981); Buttery & Ling (1995,1998) d 0.000 07
Sugisawa *et al.* (1991); Yang *et al.* (1992);
Tamura *et al.* (1993,1995) d 0.01

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Grosch *et al.* (1993); Guth & Grosch (1994); Konopka *et al.* (1995); Milo & Grosch (1997); Kerscher & Grosch (2000)		0.000 2
Jensen *et al.* (1999)	d	0.000 045
Ong & Acree (1999)		0.000 2
Takahashi *et al.* (2002)		0.000 1
Karagül-Yüceer *et al.* (2004)	d	0.000 2
Schuh & Schieberle (2006)	d	0.000 16
Czerny *et al.* (2008)	d	0.000 027
Czerny *et al.* (2008)	r	0.000 077

(*E*,*Z*)-2,4-DECADIENAL *[25152-83-4]*

Buttery & Ling (1995)	d	0.000 07
Kerler & Grosch (1997); Kerscher & Grosch (2000)		0.000 04

(*Z*,*Z*)-4,7-DECADIENAL *[22644-09-3]*

Van Lier *et al.* (1986)		0.0042

decahydronaphthalene⇒BICYCLO[4.4.0]DECANE

4,5,6,7,8,9,10,11,12,13-DECAHYDROTHIAZOLE (cyclodecathiazole) *[84648-04-4]*

Pelosi *et al.* (1983)	d	0.7

γ-decalactone⇒4-HYDROXYDECANOIC ACID LACTONE
δ-decalactone⇒5-HYDROXYDECANOIC ACID LACTONE
decalin⇒BICYCLO[4.4.0]DECANE

DECALOL *[529-32-8]*

Koppe (1965)		0.01

DECANAL (decyl aldehyde) *[112-31-2]*

Guadagni *et al.* (1963); Guadagni (1966,1968,1969, 1970c); Teranishi *et al.* (1974); Buttery *et al.* (1978b,1988a); Buttery (1981); Takeoka *et al.* (1995a)	d	0.000 1 - 0.002
Ahmed *et al.* (1978a)	d	0.00197
Aoki & Koizumi (1986)		0.0062
Pino *et al.* (1986)	d	0.0049
Schnabel *et al.* (1988)	d	0.0041 - 0.0058
Kobayashi *et al.* (1990)	r	0.009
Sugisawa *et al.* (1991); Yang *et al.* (1992); Tamura *et al.* (1993,1995)	d	0.07 - 0.46
Belitz & Grosch (1992b)		0.005
Tamura *et al.* (1996)	d	0.036
Tamura *et al.* (1996)	r	0.245
Padrayuttawat *et al.* (1997)	d	0.03
Fabrellas *et al.* (2004)		0.000 08 - 0.000 19
Averbeck & Schierberle (2009, 2011)	d	0.003

DECANE *[124-18-5]*

Zoeteman *et al.* (1971)	d	10

DECANOIC ACID (capric acid) *[334-48-5]*

Cherkinski (1961)	d	10.0
Amoore *et al.* (1968)	d	2.19
Schnabel *et al.* (1988)	d	1.0 - 5.0
Boonbumrung *et al.* (2001)	d	0.13
Karagül-Yüceer *et al.* (2003)	d	10

1-DECANOL (decyl alcohol) *[112-30-1]*

Ahmed *et al.* (1978a)	d	0.047

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Schnabel *et al.* (1988)	d	0.0066 - 0.0082
Sugisawa *et al.* (1991); Yang *et al.* (1992); Tamura *et al.* (1993,1995)	d	0.775 - 2.8

2-DECANOL *[1120-06-5]*

Schnabel *et al.* (1988)	d	0.025 - 0.041

3-DECANOL *[1565-81-7]*

Schnabel *et al.* (1988)	d	0.079 - 0.41

4-decanolide⇒4-HYDROXYDECANOIC ACID LACTONE
5-decanolide⇒5-HYDROXYDECANOIC ACID LACTONE

2-DECANONE (methyl octyl ketone) *[693-54-9]*

Teranishi *et al.* (1974)	d	0.003
Schnabel *et al.* (1988)	d	0.0083 - 0.041

3-DECANONE *[928-80-3]*

Schnabel *et al.* (1988)	d	0.025 - 0.041

2,4,7-DECATRIENAL *[51325-37-2]*

Kossa *et al.* (1979); Tressl *et al.* (1979b,1980)		0.000 01 - 0.000 1

(*E*,*E*,*Z*)-2,4,7-DECATRIENAL *[66642-86-2]*

Seifert & Buttery (1980); Buttery (1989)	d	0.000 9 - 0.000 93

2-DECENAL *[3913-71-1]*

Kossa *et al.* (1979); Tressl *et al.* (1979b,1980)		0.001

(*E*)-2-DECENAL *[3913-81-3]*

Buttery *et al.* (1968,1988a); Guadagni *et al.* (1972); Teranishi *et al.* (1974); Buttery (1981) ; Buttery & Ling (1998)	d	0.000 3 - 0.000 4
Sugisawa *et al.* (1991); Yang *et al.* (1992); Tamura *et al.* (1993,1995)	d	0.017 - 0.25

(*Z*)-4-DECENAL *[21662-09-9]*

Seideneck & Schieberle (2011)		0.000 004

(*Z*)-6-DECENAL *[105683-99-6]*

Masanetz & Grosch (1998a, 1998b)		0.000 34

(*Z*)dec-7-en-5-olide⇒5-HYDROXY-(*Z*)-7-DECENOIC ACID LACTONE

DECYL ACETATE *[112-17-4]*

Sugisawa *et al.* (1991); Yang *et al.* (1992); Tamura *et al.* (1993)	d	0.225 - 1

decyl alcohol⇒1-DECANOL
decyl aldehyde⇒DECANAL

2-DECYL-3-ETHOXYPYRAZINE *[113685-84-0]*

Masuda & Mihara (1988)	d	0.06

2-DECYL-3-(ETHYLTHIO)PYRAZINE *[113685-94-2]*

Masuda & Mihara (1988)	d	0.12

2-DECYL-3-METHOXYPYRAZINE *[113685-81-7]*

Masuda & Mihara (1988)	d	0.04

2-DECYL-3-(METHYLTHIO)PYRAZINE *[113685-89-5]*

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Masuda & Mihara (1988) d 0.02

2-DECYL-3-PHENOXYPYRAZINE *[113685-86-2]*
Masuda & Mihara (1988) d 0.07

2-DECYL-3-(PHENYLTHIO)PYRAZINE *[113685-98-6]*
Masuda & Mihara (1988) d 0.3

2-DECYLPYRAZINE *[113685-78-2]*
Masuda & Mihara (1988) d 1.1

dihydro-β-ionone⇒4-(2,6,6-TRIMETHYL-1,3-CYCLOHEXADIEN-1-YL)-3-BUTEN-2-ONE
delvan⇒2,3-*p*-DIOXANEDITHIOL S,S-BIS(O,O-DIETHYLPHOSPHORODITHIOATE)
diacetone alcohol⇒4-HYDROXY-4-METHYL-2-PENTANONE
diacetyl⇒2,3-BUTANEDIONE
diallyl disulphide⇒2-PROPENYLDITHIO-2-PROPENE
diallyl sulphide⇒2-PROPENYLTHIO-2-PROPENE
1,3-diaminopropane⇒1,3-PROPANEDIAMINE
diamyl ketone⇒6-UNDECANONE
diazinon⇒*O,O*-DIETHYL *O*-2-ISOPROPYL-6-METHYL(PYRIMIDINE-4-YL) PHOSPHOROTHIOATE

DIBENZOFURAN *[132-64-9]*
Lillard & Powers (1975) d 0.12
Alexander *et al.* (1982) d 0.0033

2,3-DIBROMO-6-CHLOROANISOLE
Diaz *et al.* (2005) 0.000 014

2,5-DIBROMO-6-CHLOROANISOLE
Diaz *et al.* (2005) 0.000 006

2,6-DIBROMO-3-CHLOROANISOLE
Diaz *et al.* (2004b, 2005) 0.000 002

2,6-DIBROMO-4-CHLOROANISOLE
Diaz *et al.* (2004b, 2005) 0.000 002

DIBROMOCHLOROMETHANE (chlorodibromomethane) *[124-48-1]*
Faust & Aly (1983) in Mackey *et al.* (2004) 0.3
Khiari *et al.* (2002) 0.05
McDonald *et al.* (2009) 0.06

DIBROMOIODOMETHANE *[593-94-2]*
Cancho *et al.* (2001) 0.0029 - 0.0064

2,6-DIBROMOPHENOL *[608-33-3]*
Piriou *et al.* (2007) d 0.0013

1,3-DI-*tert*-BUTYL-4,6-DIHYDROXBENZENE (4,6-di-*tert*-butylresorcinol) *[5374-06-1]*
Dietz & Traud (1978) 0.3

dibutyl ketone⇒5-NONANONE

2,6-DI-*tert*-BUTYL-4-METHYLPHENOL (BHT) *[128-37-0]*
Dietz & Traud (1978) 1

2,4-DI-*tert*-BUTYLPHENOL *[96-76-4]*
Dietz & Traud (1978) 0.5

2,6-DI-*tert*-BUTYLPHENOL *[128-39-2]*
Dietz & Traud (1978) 0.2

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

4,6-di-*tert*-butylresorcinol⇒1,3-DI-*tert*-BUTYL-4,6-DIHYDROXYBENZENE

cis-3,5-DIBUTYL-1,2,4-TRITHIOLANE

Rödel & Kruse (1982)		0.0113

trans-3,5-DIBUTYL-1,2,4-TRITHIOLANE

Rödel & Kruse (1982)		0.00512

dichlobenil⇒2,6-DICHLOROBENZONITRILE

DICHLORAMINE *[3400-09-7]*

Krasner & Barrett (1984)		0.36

3,4-DICHLOROANILINE *[95-76-1]*

Zoeteman (1978)	d	0.003

2,4-DICHLOROANISOLE *[553-82-2]*

Griffiths (1974)	d	0.000 4
Nijssen (1988)	d	0.0019
Young *et al.* (1996)		0.000 5

2,6-DICHLOROANISOLE *[1984-65-2]*

Griffiths (1974)	d	0.000 04

1,2-DICHLOROBENZENE *[95-50-1]*

Kölle *et al.* (1972)		0.01
De Grunt *et al.* (1977)	d	0.15
Zoeteman *et al.* (1978)	d	0.45

1,3-DICHLOROBENZENE *[541-73-1]*

Kölle *et al.* (1972)		0.02
Young *et al.* (1996)		0.17

1,4-DICHLOROBENZENE *[106-46-7]*

Kölle *et al.* (1972)		0.03
De Grunt (1975); De Grunt *et al.* (1977)	d	0.000 3
Young *et al.* (1996)		0.018

2,6-DICHLOROBENZONITRILE (dichlobenil, 2,6-dichloro-1-cyanobenzene) *[1194-65-6]*

Young *et al.* (1996)		0.2

2,3-DICHLORO-6-BROMOANISOLE

Diaz *et al.* (2004b, 2005)		0.000 005

2,5-DICHLORO-6-BROMOANISOLE

Diaz *et al.* (2005)		0.000 002

2,6-DICHLORO-3-BROMOANISOLE

Diaz *et al.* (2005)		0.000 003

2,6-DICHLORO-4-BROMOANISOLE

Diaz *et al.* (2004b, 2005)		0.000 0035 - 0.000 004

4,5-dichlorocatechol⇒4,5-DICHLORO-1,2-BENZENEDIOL
2,6-dichloro-1-cyanobenzene⇒2,6-DICHLOROBENZONITRILE

4,5-DICHLORO-1,2-DIHYDROXYBENZENE (4,5-dichlorocatechol) *[3428-24-8]*

Kovacs *et al.* (1984)	d	3.5

2,4-DICHLORO-3,5-DIMETHYLANISOLE *[23400-00-2]*

Gee *et al.* (1974)	d	0.39

3,5-DICHLORO-*N*-(1,1-DIMETHYL-2-PROPYNYL)BENZAMIDE (propyzamide) *[23950-58-5]*

Young *et al.* (1996)		3

1,2-DICHLOROETHANE (ethylene chloride, ethylene dichloride) *[107-06-2]*

Amoore & Venstrom (1966); Amoore (1969)	d	29
Alexander *et al.* (1982)	d	2.8

1,1-DICHLOROETHENE (vinylidene chloride) *[156-60-5]*

Alexander *et al.* (1982)	d	1.6

2,2'-dichloroethyl ether⇒1-CHLORO-2-(1-CHLOROETHOXY)ETHANE
3,4-dichloroguaiacol⇒3,4-DICHLORO-2-METHOXYPHENOL
4,5-dichloroguaiacol⇒4,5-DICHLORO-2-METHOXYPHENOL
4,6-dichloroguaiacol⇒4,6-DICHLORO-2-METHOXYPHENOL

2,3-DICHLORO-4-HYDROXY-5-METHOXYBENZALDEHYDE (5,6-dichlorovanillin) *[18268-69-4]*

Kovacs *et al.* (1984)	d	7.5

2,2'-dichloroisopropyl ether⇒1-CHLORO-2-(1-CHLOROISOPROPOXY)PROPANE

DICHLOROMETHANE (methylene chloride) *[75-09-2]*

Alexander *et al.* (1982)	d	5.6

3,4-DICHLORO-2-METHOXYPHENOL (3,4-dichloroguaiacol) *[77102-94-4]*

Kovacs *et al.* (1984)	d	0.070

4,5-DICHLORO-2-METHOXYPHENOL (4,5-dichloroguaiacol) *[2460-49-3]*

Kovacs *et al.* (1984)	d	0.0025

4,6-DICHLORO-2-METHOXYPHENOL (4,6-dichloroguaiacol) *[16766-31-7]*

Kovacs *et al.* (1984)	d	0.0115

2,5-DICHLORO-4-METHYLANISOLE

Griffiths & Fenwick (1977)	d	0.1

2,6-DICHLORO-4-METHYLANISOLE

Griffiths & Fenwick (1977)	d	0.000 09

2,4-DICHLORO-6-METHYLANISOLE

Griffiths & Fenwick (1977)	d	0.000 1

2,3-DICHLORO-1,4-NAPHTHOQUINONE *[117-80-6]*

Lisovskaya *et al.* (1967)		20

1,2-DICHLOROOXOETHANE (chloroacetyl chloride) *[79-04-9]*

Alexander *et al.* (1982)	d	25

2,3-DICHLOROPHENOL *[576-24-9]*

Dietz & Traud (1978)		0.03

2,4-DICHLOROPHENOL *[120-83-2]*

Hoak (1956)		0.000 65
Burttschell *et al.* (1959)		0.002
Baker (1963)		0.21
Dietz & Traud (1978)		0.04
Kovacs *et al.* (1984)	d	0.056
Van Gemert (1984)	d	0.004
Young *et al.* (1996)		0.029
Czerny *et al.* (2008)	d	0.0014
Czerny *et al.* (2008)	r	0.0028

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

2,5-DICHLOROPHENOL *[583-78-8]*

Hoak (1956)		0.000 45
Dietz & Traud (1978)		0.03

2,6-DICHLOROPHENOL *[87-65-0]*

Burttschell *et al.* (1959)		0.003
De Grunt (1975); De Grunt *et al.* (1977)	d	0.0075 - 0.008
Dietz & Traud (1978)		0.2
Young *et al.* (1996)		0.022
Piriou *et al.* (2007)	d	0.002 - 0.007

3,4-DICHLOROPHENOL *[95-77-2]*

Dietz & Traud (1978)		0.1

3,5-DICHLOROPHENOL *[591-35-5]*

Van Gemert (1984)	d	0.25

2,4-DICHLOROPHENOXYACETIC ACID (2,4-D) *[94-75-7]*

Sigworth (1965)	d	3.13

N'-(3,4-DICHLOROPHENYL)-N,N-DIMETHYLUREA (diuron) *[330-54-1]*

Mikhailov (1967)	1	

1,2-DICHLOROPROPANE (propylene chloride) *[78-87-5]*

Alexander *et al.* (1982)	d	0.10

2,2-DICHLOROPROPANOIC ACID *[75-99-0]*

Sigworth (1965)	d	23.2

1,3-DICHLOROPROPENE *[542-75-6]*

Alexander *et al.* (1982)	d	0.12

5,6-dichlorovanillin⇒2,3-DICHLORO-4-HYDROXY-5-METHOXYBENZALDEHYDE
dicyclopentadiene⇒3a,4,7,7a-TETRAHYDRO-4,7-METHANOINDENE
dieldrin⇒1,2,3,4,10,10-HEXACHLORO-6,7-EPOXY-1,4,4a,5,6,7,8,8a-OCTAHYDRO-1,4-*endo*-*exo*-5,8-DIMETHANONAPHTHALENE

1,1-DIETHOXYETHANE (acetaldehyde diethyl acetal, acetal) *[105-57-7]*

Mulders (1972,1973)	d	0.04 - 0.042
Fritsch & Schieberle (2005)		0.0049

DIETHYLAMINE *[109-89-7]*

Amoore *et al.* (1975b)	d	12.1
Amoore & Forrester (1976)	d	21.2

O,O-DIETHYL O-(3-CHLORO-4-METHYL-7-COUMARINYL)PHOSPHOROTHIOATE (co-ral) *[56-72-4]*

Sigworth (1965)	d	0.02

diethyl disulphide⇒ETHYLDITHIOETHANE
diethylene glycol⇒3-OXA-1,5-PENTANEDIOL
diethylene glycol *n*-butyl ether⇒3,6-DIOXA-1-DECANOL
diethylene glycol ethyl ether⇒3,6-DIOXA-1-OCTANOL

O,O-DIETHYL *O*-2-ISOPROPYL-6-METHYL(PYRIMIDINE-4-YL) PHOSPHOROTHIOATE (diazinon) *[333-41-5]*

Young *et al.* (1996)		0.17

diethyl ketone⇒3-PENTANONE

N,N-DIETHYLMETHYLAMINE (methyldiethylamine) *[616-39-7]*

Amoore & Forrester (1976)	d	0.198

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

2,3-DIETHYL-5-METHYLPYRAZINE *[18138-04-0]*

Cerny & Grosch (1993); Grosch *et al.* (1993)		0.001
Czerny & Wagner (1995); Czerny *et al.* (1996a, 1996b)		0.000 09
Czerny *et al.* (2008)	d	0.000 031
Czerny *et al.* (2008)	r	0.000 084

O,O-DIETHYL O-(4-NITROPHENYL)PHOSPHOROTHIOATE (parathion) *[56-38-2]*

Sigworth (1965)	d	0.04

2,5-DIETHYLPYRAZINE *[13238-84-1]*

Guadagni *et al.* (1972); Teranishi *et al.* (1974)	d	0.02

2,6-DIETHYLPYRAZINE *[13067-27-1]*

Guadagni *et al.* (1972); Teranishi *et al.* (1974)	d	0.006

α,α'-DIETHYL-4,4'-STILBENEDIOL (diethylstilbestrol) *[56-53-1]*

Dietz & Traud (1978)		2.3

diethylstilbestrol⇒α,α'-DIETHYL-4,4'-STILBENEDIOL
diethyl sulphide⇒(ETHYLTHIO)ETHANE

2,4-DIETHYLTHIAZOLE *[32272-49-4]*

Teranishi *et al.* (1975)	d	0.0065

2,5-DIETHYLTHIAZOLE *[15729-76-7]*

Teranishi *et al.* (1975)	d	0.000 09

4,5-DIETHYLTHIAZOLE

Buttery *et al.* (1976a)	d	0.0051

O,O-DIETHYL O-3,5,6-TRICHLORO-2-PYRIDYL PHOSPHOROTHIOATE (chlorpyrifos) *[2921-88-2]*

Alexander *et al.* (1982)	d	0.0012

cis-3,5-DIETHYL-1,2,4-TRITHIOLANE *[38348-25-3]*

Rödel & Kruse (1982)		0.000 0323

trans-3,5-DIETHYL-1,2,4-TRITHIOLANE *[38348-26-4]*

Rödel & Kruse (1982)		0.000 0154

cis-3,5-DIHEPTYL-1,2,4-TRITHIOLANE

Rödel & Kruse (1982)		0.0263

trans-3,5-DIHEPTYL-1,2,4-TRITHIOLANE

Rödel & Kruse (1982)		0.0617

cis-3,5-DIHEXYL-1,2,4-TRITHIOLANE

Rödel & Kruse (1982)		0.0292

trans-3,5-DIHEXYL-1,2,4-TRITHIOLANE

Rödel & Kruse (1982)		0.0240

2,5-dihydro-1H-azole⇒3-PYRROLINE
3,4-dihydro-2H-azole⇒1-PYRROLINE
dihydrocarvone⇒5-ISOPROPENYL-2-METHYL-1-CYCLOHEXANONE
(*d*)-dihydrocarvone⇒(*d*)-5-ISOPROPENYL-2-METHYL-1-CYCLOHEXANONE
(*l*)-dihydrocarvone⇒ (*l*)-5-ISOPROPENYL-2-METHYL-1-CYCLOHEXANONE

5,6-DIHYDRO-4*H*-CYCLOPENTATHIAZOLE (cyclopentathiazole) *[5661-10-9]*

Pelosi *et al.* (1983)	d	0.55

2,3-DIHYDRO-3,5-DIHYDROXY-6-METHYL-4(4*H*)-PYRANONE *[28564-83-2]*

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Buttery *et al.* (1999) d 35

2,5-DIHYDRO-3,4-DIMETHYL-2,5-DIOXOTHIOPHENE
Albrand *et al.* (1980) d 0.007

dihydro-6-hydroxytheaspirane⇒2,6,10,10-TETRAMETHYL-1-OXASPIRO[4,5]DECAN-6-OL

2,3-DIHYDRO-5-HYDROXY-6-METHYL-4*H*-PYRAN-4-ONE (5-hydroxy-6-methyl-2*H*-pyran-4(3*H*)-one, dihydromaltol)
Preininger *et al.* (2009, 2010) d 0.25 - 1

3a,4-dihydro-3-isobutylenephthalide⇒3a,4-DIHYDRO-3-(2-METHYLPROPYLIDENE)PHTHALIDE
3a,4-dihydro-3-isovalidenephthalide⇒3a,4-DIHYDRO-3-(3-METHYLBUTYLIDENE)PHTHALIDE
dihydromaltol⇒2,3-DIHYDRO-5-HYDROXY-6-METHYL-4H-PYRAN-4-ONE

3a,4-DIHYDRO-3-(3-METHYLBUTYLIDENE)PHTHALIDE (3a,4-dihydro-3-isovalidenephthalide)
Gold & Wilson (1963) r < 0.1

3a,4-DIHYDRO-3-(2-METHYLPROPYLIDENE)PHTHALIDE (3a,4-dihydro-3-isobutylenephthalide) *[66983-88-8]*
Gold & Wilson (1963) r < 0.1

5,7-DIHYDRO-2-METHYLTHIENO-(3,4D)-PYRIMIDINE *[36267-71-7]*
Teranishi *et al.* (1975) d 0.000 9

1,10-dihydronootkatone⇒8-ISOPROPENYL-5,6-DIMETHYLBICYCLO[4.4.0]-3-DECANONE
11,12-dihydronootkatone⇒8-ISOPROPYL-5,6-DIMETHYLBICYCLO[4.4.0]-1-DECEN-3-ONE

3,7-DIHYDRO-1*H*-PURINE-2,6-DIONE (xanthine) *[69-89-6]*
Dietz & Traud (1978) > 10

7,9-DIHYDRO-1*H*-PURINE-2,6,8(3*H*)-TRIONE (uric acid) *[69-93-2]*
Dietz & Traud (1978) > 10

DIHYDRO-2,4,6-TRIMETHYL-1,3,5(4*H*)-DITHIAZINE (thialdine) *[638-17-5]*
Buttery (1981) d 0.001

1,2-DIHYDRO-1,1,6-TRIMETHYLNAPHTHALENE *[30364-38-6]*
Buttery *et al.* (1990b) d 0.0025

1,2-DIHYDROXYBENZENE (1,2-benzenediol, catechol, pyrocatechol) *[120-80-9]*
Dietz & Traud (1978) 8

1,3-DIHYDROXYBENZENE (1,3-benzenediol, resorcinol) *[108-46-3]*
Nesmeyanova (1953) 40
Dietz & Traud (1978) 6

1,4-DIHYDROXYBENZENE (1,4-benzenediol, hydroquinone) *[123-31-9]*
Dietz & Traud (1978) 5

3,4-DIHYDROXY-3-HEXEN-2,5-DIONE
Engel *et al.* (2001) > 1,000

2,4-dihydroxy-3-methyl-2-hexenoic acid lactone⇒5-ETHYL-3-HYDROXY-4-METHYL-2(5*H*)-FURANONE
diisobutylcarbinol⇒2,6-DIMETHYL-4-HEPTANOL

DIISOPROPANOLAMINE *[110-97-4]*
Alexander *et al.* (1982) d 10

2,5-DIISOPROPYL-3,6-DIMETHOXYPYRAZINE
Takken *et al.* (1975) d 0.67

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

2,2-DIISOPROPYL-1,3-DIOXOLANE *[4421-10-7]*

Pelosi & Pisanelli (1981)	d	13

diisopropyl disulphide⇒2-(ISOPROPYLDITHIO)PROPANE
diisopropyl ketone⇒2,4-DIMETHYL-3-PENTANONE
dill ether⇒(3*R*,3a*S*,7a*R*)-3,6-DIMETHYL-2,3,3a,4,5,7a-HEXAHYDRODIMETHYLBENZOFURAN

2,5-DIMETHOXY-3,6-DIMETHYLPYRAZINE

Takken *et al.* (1975)	d	0.18

(2,2-DIMETHOXYETHYL)BENZENE (2-phenylacetaldehyde dimethylacetal) *[101-48-4]*

Appell (1969)		0.5

2,6-DIMETHOXYPHENOL (syringol) *[91-10-1]*

Wasserman (1966)	d	1.85

(*E*)-1,2-DIMETHOXY-4-(1-PROPENYL)BENZENE (*trans*-methylisoeugenol) *[93-16-3]*

Zeller & Rychlik (2006)	r	1.6

DIMETHYLAMINE *[124-40-3]*

Baker (1963)		23.2
Amoore *et al.* (1975b)	d	17.6
Amoore & Forrester (1976)	d	34.4
Kikuchi *et al.* (1976)		30

2-DIMETHYLAMINO-3-ISOBUTYLPYRAZINE

Takken *et al.* (1975)	d	1.6

2-DIMETHYLAMINO-6-ISOBUTYLPYRAZINE

Takken *et al.* (1975)	d	5

2-DIMETHYLAMINO-3-METHYLPYRAZINE

Takken *et al.* (1975)	d	0.18

2-DIMETHYLAMINO-6-METHYLPYRAZINE

Takken *et al.* (1975)	d	1.2

2,6-DIMETHYLANTHRACENE *[613-26-3]*

Brady (1968)		0.5

2,5-DIMETHYLBENZALDEHYDE *[5779-94-2]*

Buttery *et al.* (1990b)	d	0.2

1,2-DIMETHYLBENZENE (*o*-xylene) *[95-47-6]*

Rosen *et al.* (1962)		1.8
Zoeteman *et al.* (1971)		1
Giri *et al.* (2010)	d	0.45023

1,3-DIMETHYLBENZENE (*m*-xylene) *[108-38-3]*

Rosen *et al.* (1962)		1.1
Zoeteman *et al.* (1971)		1

1,4-DIMETHYLBENZENE (*p*-xylene) *[106-42-3]*

Rosen *et al.* (1962)		0.53
Zoeteman *et al.* (1971)		1

6,6-DIMETHYLBICYCLO[3.1.1]-2-HEPTENE-2-METHANOL (myrtenol) *[515-00-4]*

Hirvi & Honkanen (1982)	d	0.007

1,3-DIMETHYLBICYCLO[3.3.0]-4-THIA-2,8-DIOXAOCTANE

Van Dort *et al.* (1984)		0.001

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

2,2-DIMETHYLBUTANAL *[2094-75-9]*

Schnabel *et al.* (1988)	d	0.24 - 0.40

2,3-DIMETHYLBUTANOIC ACID *[14287-61-7]*

Schnabel *et al.* (1988)	d	4.6 - 9.3

3,3-DIMETHYLBUTANOIC ACID *[1070-83-3]*

Schnabel *et al.* (1988)	d	0.92 - 4.5

2,2-DIMETHYL-1-BUTANOL *[1185-33-7]*

Schnabel *et al.* (1988)	d	4.2 - 8.3

2,3-DIMETHYL-1-BUTANOL *[54206-54-1]*

Schnabel *et al.* (1988)	d	0.83 - 2.5

3,3-DIMETHYL-1-BUTANOL *[624-95-3]*

Schnabel *et al.* (1988)	d	0.000 082 - 0.000 85

2,3-DIMETHYL-2-BUTANOL *[594-60-5]*

Schnabel *et al.* (1988)	d	0.082 - 0.83

3,3-DIMETHYL-2-BUTANOL *[464-07-3]*

Pelosi & Pisanelli (1981)	d	6.2
Schnabel *et al.* (1988)	d	1.6 - 4.1

2,2-DIMETHYL-3-BUTANONE (methyl *tert*-butyl ketone) *[75-97-8]*

Schnabel *et al.* (1988)	d	0.40 - 0.80

2,5-DIMETHYL-4-BUTOXY-3(2*H*)-FURANONE

Buttery & Ling (1996)	d	0.25

3,4-DIMETHYL-1,2-CYCLOPENTADIONE *[13494-06-9]*

Anon. (1986)		0.017 - 0.020

3,5-DIMETHYL-1,2-CYCLOPENTADIONE *[13494-07-0]*

Anon. (1986)		1

(8*S*,9*S*,10*R*,13*R*,14*S*)-10,13-dimethyl-1,2,6,7,8,9,11,12,14,15-decahydrocyclopenta[a]phenanthren-3-one⇒4,16-ANDROSTADIEN-3-ONE

trans-1,10-DIMETHYL-*trans*-9-DECALOL (geosmin) *[19700-21-1]*

Safferman *et al.* (1967)		0.000 16 - 0.000 2
Medsker *et al.* (1968)		0.000 05
Zoeteman & Piet (1973)		0.000 01
Lillard & Powers (1975)	d	0.000 13
Guadagni (1975)		0.000 05
Buttery *et al.* (1976c,1994a); Buttery (1981,1993)	d	0.000 02 - 0.000 021
Tyler *et al.* (1979)	d	0.000 01 - 0.000 02
Persson (1979b)	r	0.000 015
Tuorila *et al.* (1980)	d	0.000 004
Sugiura *et al.* (1983)		0.000 01
Ito *et al.* (1988)	d	0.000 022 - 0.000 055
Ito *et al.* (1988)	r	0.000 047 - 0.000 135
Sano (1988)	d	0.000 094
Sano (1988)	d	0.000 36
Tomita *et al.* (1988)		0.000 002 - 0.000 02
Burlingame (1989)		0.000 0024
Finato *et al.* (1992)	d	0.000 04
Young *et al.* (1996)		0.000 0038
Rashash *et al.* (1997)		0.000 006 - 0.000 01
Piriou *et al.* (2009)	d	0.000 000 8 - 0.000 0011

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

(+)-*trans*-1,10-DIMETHYL-*trans*-9-DECALOL ((+)-geosmin)
Piriou *et al.* (2009) d 0.000 0046 - 0.000 0091

(-)-*trans*-1,10-DIMETHYL-*trans*-9-DECALOL ((-)-geosmin)
Piriou *et al.* (2009) d 0.000 000 4 - 0.000 000 6

O,O-DIMETHYL S-(1,2-DICARBETHOXYETHYL)PHOSPHORODITHIOATE (malathion) *[121-75-5]*
Sigworth (1965) d 1.0

2,3-DIMETHYL-5-(1,2-DIMETHYLPROPYL)PYRAZINE
Shibamoto (1986) d 1.00

dimethyl disulphide⇒METHYLDITHIOMETHANE
O,O-dimethyl dithiophosphate⇒DIMETHYLDITHIOPHOSPHORIC ACID

DIMETHYLDITHIOPHOSPHORIC ACID (O,O-dimethyl dithiophosphate) *[756-80-9]*
Cherkinski (1961) d 0.1

3a,9a-DIMETHYL-3aα,5aβ,9aα,9bα-DODECAHYDRONAPHTHO[2,1-b]FURAN (9-epi-18,19-dinorambrox)
Ohloff *et al.* (1985) 0.004

3a,9a-DIMETHYL-3aα,5aβ,9aα,9bβ-DODECAHYDRONAPHTHO[2,1-b]FURAN (18,19-dinorambrox)
Ohloff *et al.* (1985) 0.0024

3a,9a-DIMETHYL-3aα,5aβ,9aβ,9bα-DODECAHYDRONAPHTHO[2,1-b]FURAN (18,19-dinorisoambrox)
Ohloff *et al.* (1985) 0.072

2,5-DIMETHYL-4-ETHOXY-3(2*H*)-FURANONE *[65330-49-6]*
Buttery & Ling (1996) d 2.9

dimethylethyl ethanoate⇒*tert*-BUTYL ACETATE
dimethylethyl propanoate⇒*tert*-BUTYL PROPANOATE

2,2-DIMETHYLHEPTANE *[1071-26-7]*
Zoeteman *et al.* (1971) 50

2,6-DIMETHYL-4-HEPTANOL (diisobutylcarbinol) *[108-82-7]*
Rosen *et al.* (1963) 1.3

2,6-DIMETHYL-5-HEPTENAL (melonal) *[106-72-9]*
Yang *et al.* (1992); Tamura *et al.* (1993) d 0.016

(3*S*,3a*S*,7a*R*)-3,6-DIMETHYL-2,3,3a,4,5,7a-HEXAHYDROBENZOFURAN ((3*R*,4*S*,8*S*)-3,9-epoxy-1-*p*-menthene, dill ether) *[70786-44-6]*
Huopalahti (1986) 0.04
Blank *et al.* (1992) 0.03

2,3-DIMETHYL-2-HEXANOL *[19550-03-9]*
Schnabel *et al.* (1988) d 0.42-0.66

2,5-DIMETHYL-2-HEXANOL *[3730-60-7]*
Schnabel *et al.* (1988) d 1.2-1.7

3,4-DIMETHYL-2-HEXANOL *[31350-88-6]*
Schnabel *et al.* (1988) d 0.85-4.2

2,2-DIMETHYL-3-HEXANOL *[4209-90-9]*
Schnabel *et al.* (1988) d 0.0013 - 0.0039

2,3-DIMETHYL-3-HEXANOL *[4166-46-5]*

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Schnabel *et al.* (1988) d 0.0078 - 0.041

2,4-DIMETHYL-3-HEXANOL *[13432-25-2]*
Schnabel *et al.* (1988) d 0.25 - 0.42

2,5-DIMETHYL-3-HEXANOL *[19550-07-3]*
Schnabel *et al.* (1988) d 0.40 - 1.7

3,5-DIMETHYL-3-HEXANOL *[4209-91-0]*
Schnabel *et al.* (1988) d 0.42 - 0.59

3,4-DIMETHYL-2-HEXANONE *[19550-10-8]*
Schnabel *et al.* (1988) d 0.24 - 0.41

2,2-DIMETHYL-3-HEXANONE *[5405-79-8]*
Schnabel *et al.* (1988) d 0.000 41 - 0.000 82

2,4-DIMETHYL-3-HEXANONE *[18641-70-8]*
Schnabel *et al.* (1988) d 0.077 - 0.024

2,5-DIMETHYL-4-((*Z*)-3-HEXENOXY)-3(2*H*)-FURANONE
Buttery & Ling (1996) d 1.9

(1S*)-1-[(1S*)-1,5-DIMETHYL-5-HEXENYL]-4-METHYL-3-CYCLOHEXENYL-1-OL
Braun *et al.* (2003) d 0.000 785

dimethylkahweofuran⇒2,4-DIMETHYL-3-OXA-8-THIABICYCLO[3.3.0]-1,4-OCTADIENE

2,5-DIMETHYL-4-METHOXY-3(2*H*)-FURANONE *[4077-47-8]*
Buttery & Ling (1996) d 3.4

2,5-DIMETHYL-4-(3-METHYLBUTOXY)-3(2*H*)-FURANONE
Buttery & Ling (1996); Buttery (1999) d 0.9

2,3-DIMETHYL-5-(1-METHYLBUTYL)PYRAZINE
Shibamoto (1986) d 0.77

2,3-DIMETHYL-5-(2-METHYLBUTYL)PYRAZINE *[75492-01-2]*
Masuda & Mihara (1986); Shibamoto (1986) d 3.20

2,5-DIMETHYL-3-(METHYLDITHIO)FURAN
Tressl & Silwar (1981); Silwar (1982) 0.000 01

(1*R*)-(+)-2,2-DIMETHYL-3-METHYLENEBICYCLO[2.2.1]HEPTANE ((+)-camphene) *[5794-03-6]*
Padrayuttawat *et al.* (1997) d 1.86

(1*S*)-(-)-2,2-DIMETHYL-3-METHYLENEBICYCLO[2.2.1]HEPTANE ((-)-camphene) *[5794-04-7]*
Padrayuttawat *et al.* (1997) d 1.98

6,6-DIMETHYL-2-METHYLENEBICYCLO[3.1.1]HEPTANE (β-pinene) *[127-91-3]*
Guadagni *et al.* (1966); Guadagni (1969);
Buttery *et al.* (1974); Takeoka (2001) d 0.006 - 0.14
Tamura *et al.* (2001) d 1.5
Pino & Mesa (2006) d 0.14

(1*S*,5*S*)-(-)-6,6-DIMETHYL-2-METHYLENEBICYCLO[3.1.1]HEPTANE ((1*S*,5*S*)-(-)-β-pinene) *[18172-67-3]*
Padrayuttawat *et al.* (1997);
Boonbumrung *et al.* (2001) d 4.16

(1*R*,5*R*)-(+)-6,6-DIMETHYL-2-METHYLENEBICYCLO[3.1.1]HEPTANE ((1*R*,5*R*)-(+)-β-pinene)
[19902-08-0]

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Padrayuttawat *et al.* (1997);
Boonbumrung *et al.* (2001) d 2.54

1-(2,2-DIMETHYL-6-METHYLENECYCLOHEXYL)-2-BUTEN-1-ONE (γ-damascone) *[35087-49-1]*
Pickenhagen & Furrer (1990) 0.0092

7,11-DIMETHYL-3-METHYLENE-1,6,10-DODECATRIENE (farnesene) *[77129-48-7]*
Tamura *et al.* (1999) d 0.087

(*E*,*E*)-2,6-DIMETHYL-10-METHYLENE-2,6,11-DODECATRIENAL (β-sinensal) *[3779-62-2]*
Guadagni (1965,1969,1970c);
Teranishi *et al.* (1981); Buttery (1981) d 0.000 05 - 0.000 2
Ahmed *et al.* (1978a) d 0.0038
Sugisawa *et al.* (1991); Yang *et al.* (1992);
Tamura *et al.* (1993) d 0.082

8,8-DIMETHYL-2-METHYLENE-6-OXABICYCLO[3.2.1]OCTANE (karahana ether) *[19901-96-3]*
Friese *et al.* (1979); Tressl *et al.* (1979a) 0.001 - 0.005

[1a*R*-(1aα,5β,5aβ,6aα)]-5,5a-DIMETHYL-6a-(1-METHYLETHYL)-1a,4,5,5a,6,6a-HEXAHYDROCYCLOPROP[a]INDEN-3(1*H*)-ONE ((*d*)-tricycloketone)
Haring *et al.* (1972) d 40 - 80

[1a*S*-(1aα,5β,5aβ,6aα)]-5,5a-DIMETHYL-6a-(1-METHYLETHYL)-1a,4,5,5a,6,6a-HEXAHYDROCYCLOPROP[a]INDEN-3(1*H*)-ONE ((*l*)-tricycloketone)
Haring *et al.* (1972) d 80 - 120

2,5-DIMETHYL-3-(METHYLTHIO)PYRAZINE *[59021-08-8]*
Mihara *et al.* (1991) d 0.0013

1,3-DIMETHYLNAPHTHALENE *[575-41-7]*
Seifert *et al.* (1975) d 0.042

2,6-DIMETHYLNAPHTHALENE *[581-42-0]*
Brady (1968) 0.010

2,3-DIMETHYL-5-NEOPENTYLPYRAZINE
Shibamoto (1986) d 3.00

2,6-DIMETHYL-4-NITROPHENOL *[2423-71-4]*
Dietz & Traud (1978) 2.5

O,O-DIMETHYL O-(4-NITROPHENYL)PHOSPHOROTHIOATE (methyl parathion) *[298-00-0]*
Sigworth (1965) d 0.0123

(*R*)-(-)-4,8-dimethylnon-7-en-4-olide⇒(+)-(*R*)-4-HYDROXY-4,8-DIMETHYL-7-NONENOIC ACID LACTONE
(*S*)-(-)-4,8-dimethylnon-7-en-4-olide⇒(-)-(*S*)-4-HYDROXY-4,8-DIMETHYL-7-NONENOIC ACID LACTONE

3,7-DIMETHYL-2,6-OCTADIENAL (citral) *[5392-40-5]*
Balavoine (1948) 0.000 5
Appell (1969) 1
Ahmed *et al.* (1978a) d 0.0853
Pino *et al.* (1986) d 0.028
Voirol & Daget (1986) d 0.04 - 0.12

3,7-DIMETHYL-(*E*)-2,6-OCTADIENAL (citral a, geranial) *[141-27-5]*
Buttery *et al.* (1971,1989,1990a);
Buttery (1981,1993) d 0.032
Sugisawa *et al.* (1991); Yang *et al.* (1992);
Tamura *et al.* (1993) d 0.1 - 0.46
Tamura *et al.* (1996) d 0.037

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Tamura *et al.* (1996)	r	0.41
Pino & Mesa (2006)	d	0.032

3,7-DIMETHYL-(*Z*)-2,6-OCTADIENAL (citral b, neral) *[106-26-3]*

Buttery *et al.* (1989); Buttery (1993)	d	0.03
Sugisawa *et al.* (1991); Yang *et al.* (1992);		
Tamura *et al.* (1993)	d	0.1 - 0.46
Tamura *et al.* (1996)	d	0.1
Tamura *et al.* (1996)	r	0.888
Boonbumrung *et al.* (2001)	d	0.053

3,7-DIMETHYL-(*E*)-2,6-OCTADIEN-1-OL (geraniol) *[106-24-1]*

Ohloff (1977,1978; Pickenhagen & Furrer (1990)		0.075
Pyysalo *et al.* (1977)	d	0.075
De Grunt (1975); De Grunt *et al.* (1977)	d	0.01
Pino *et al.* (1986)	d	0.0041
Buttery *et al.* (1987b); Takeoka *et al.* (1990a)	d	0.04
Larsen & Poll (1990)	d	0.001 - 0.01
Sugisawa *et al.* (1991); Yang *et al.* (1992);		
Tamura *et al.* (1993)	d	0.01 - 0.1
Ong & Acree (1999)		0.005
Schuh & Schieberle (2006)	d	0.0032
Czerny *et al.* (2008)	d	0.0011
Czerny *et al.* (2008)	r	0.0025
Seideneck & Schieberle (2011)		0.0066

3,7-DIMETHYL-(*Z*)-2,6-OCTADIEN-1-OL (nerol) *[106-25-2]*

Ohloff (1977,1978); Pickenhagen & Furrer (1990)		0.29 - 0.3
Sugisawa *et al.* (1991); Yang *et al.* (1992);		
Tamura *et al.* (1993)	d	0.68 - 2.2

3,7-DIMETHYL-1,6-OCTADIEN-3-OL (linalool) *[78-70-6]*

Guadagni *et al.* (1966); Buttery *et al.* (1969b,1971, 1974,1987b,1988a,1990a); Buttery (1981,1993);		
Takeoka (2001)	d	0.006
Appell (1969)		1
Ahmed *et al.* (1978a)	d	0.0053
Pino *et al.* (1986)	d	0.0047
Larsen & Poll (1990)	d	0.001 - 0.01
Sugisawa *et al.* (1991); Yang *et al.* (1992);		
Tamura *et al.* (1993)	d	0.028
Larsen & Poll (1992)	d	0.000 01 - 0.000 1
Tamura *et al.* (1996)	d	0.05
Tamura *et al.* (1996)	r	1.082
Ong & Acree (1999)		0.0015
Tamura *et al.* (1999); Mookdasanit *et al.* (2003)	d	0.007 - 0.0074
Tamura *et al.* (2001)	d	0.001
Pino & Mesa (2006)	d	0.006
Averbeck & Schierberle (2009, 2011)	d	0.000 22

(*S*)-(+)-3,7-DIMETHYL-1,6-OCTADIEN-3-OL ((3*S*)-(+)-linalool) *[126-90-9]*

Padrayuttawat *et al.* (1997);		
Boonbumrung *et al.* (2001)	d	0.0074

(*R*)-(-)-3,7-DIMETHYL-1,6-OCTADIEN-3-OL ((3*R*)-(-)-linalool) *[126-91-0]*

Padrayuttawat *et al.* (1997);		
Boonbumrung *et al.* (2001)	d	0.000 80
Fritsch & Schieberle (2005)		0.000 14
Schuh & Schieberle (2006)	d	0.000 6
Czerny *et al*. (2008)	d	0.000 087

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Czerny *et al*. (2008) r 0.000 17

3,7-DIMETHYL-1,6-OCTADIEN-3-YL ACETATE (linalyl acetate) *[115-95-7]*
Yang *et al.* (1992); Tamura *et al.* (1993) d 1

3,7-DIMETHYL-(*E*)-2,6-OCTADIEN-1-YL ACETATE (geranyl acetate) *[105-87-3]*
Guadagni *et al.* (1966); Buttery (1981);
Buttery *et al.* (1988a); Takeoka *et al.* (1991a) d 0.009
Pino *et al.* (1986) d 0.036
Sugisawa *et al.* (1991); Yang *et al.* (1992);
Tamura *et al.* (1993) d 0.15 - 0.46
Pino & Mesa (2006) d 0.15

3,7-DIMETHYL-(*Z*)-2,6-OCTADIEN-1-YL ACETATE (neryl acetate) *[141-12-8]*
Pino *et al.* (1986) d 0.042
Sugisawa *et al.* (1991); Yang *et al.* (1992);
Tamura *et al.* (1993) d 2 - 8.5

3,7-DIMETHYL-(*E*)-1,6-OCTADIEN-1-YL FORMATE (geranyl formate) *[105-86-2]*
Yang *et al.* (1992); Tamura *et al.* (1993) d 0.2

3,7-dimethyl-2,7-octadien-5-olide⇒ 5-HYDROXY-3,7-DIMETHYL-2,7-OCTADIENOIC ACID LACTONE
3,7-dimethyl-2,7-octadien-6-olide⇒ 6-HYDROXY-3,7-DIMETHYL-2,7-OCTADIENOIC ACID LACTONE

3,7-DIMETHYL-(*E*)-2,6-OCTADIEN-1-YL 2-METHYLPROPANOATE (geranyl isobutyrate) *[2345-26-8]*
Guadagni *et al.* (1966) d 0.013

3,7-DIMETHYL-(*E*)-2,6-OCTADIEN-1-YL PROPANOATE (geranyl propanoate) *[105-90-8]*
Guadagni *et al.* (1966) d 0.010

3,7-DIMETHYL-1,7-OCTANEDIOL (hydroxycitronellol) *[107-74-4]*
Hirvi & Honkanen (1983) 5

(R*,S*)-3,7-dimethyl-4-octanolide⇒(R*,S*)-4-HYDROXY-3,7-DIMETHYLOCTANOIC ACID LACTONE
(R*,R*)-3,7-dimethyl-4-octanolide⇒(R*,R*)-4-HYDROXY-3,7-DIMETHYLOCTANOIC ACID LACTONE
(R*,S*)-3,7-dimethyl-6-octanolide⇒(R*,S*)-6-HYDROXY-3,7-DIMETHYLOCTANOIC ACID LACTONE
(R*,R*)-3,7-dimethyl-6-octanolide⇒(R*,R*)-6-HYDROXY-3,7-DIMETHYLOCTANOIC ACID LACTONE

(*E*)-3,7-DIMETHYL-1,3,6-OCTATRIENE ((*E*)-β-ocimene) *[3779-61-1]*
Tamura *et al.* (2001) d 0.034

(*Z)*-3,7-DIMETHYL-1,3,6-OCTATRIENE ((*Z*)-β-ocimene) *[3338-55-4]*
Takeoka (2001) d 0.055
Tamura *et al.* (2001) d 0.034

3,7-DIMETHYL-6-OCTENAL (citronellal) *[106-23-0]*
Appell (1969) 0.16
Ahmed *et al.* (1978a) d 0.066
Pino *et al.* (1986) d 0.031
Sugisawa *et al.* (1991); Yang *et al.* (1992);
Tamura *et al.* (1993) d 0.046 - 0.1
Tamura *et al.* (1996) d 0.006
Tamura *et al.* (1996) r 0.1

(*R*)-(+)-3,7-DIMETHYL-6-OCTENAL ((3*R*)-(+)-citronellal) *[2385-77-5]*
Padrayuttawat *et al.* (1997) d 0.03

(*S*)-(-)-3,7-DIMETHYL-6-OCTENAL ((3*S*)-(-)-citronellal) *[5949-05-3]*
Padrayuttawat *et al.* (1997) d 0.03

3,7-DIMETHYL-6-OCTEN-1-OL (citronellol) *[106-22-9]*

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Appell (1969)		1
Pino *et al.* (1986)	d	0.010 6
Sugisawa *et al.* (1991); Yang *et al.* (1992); Tamura *et al.* (1993)	d	0.062 - 2.2

(*R*)-(+)-3,7-DIMETHYL-6-OCTEN-1-OL ((3*R*)-(+)-citronellol) *[1117-61-9]*

Padrayuttawat *et al.* (1997)	d	0.04

(*S*)-(-)-3,7-DIMETHYL-6-OCTEN-1-OL ((3*S*)-(-)-citronellol, (*l*)-citronellol) *[7540-51-4]*

Ohloff (1977,1978)		0.04
Padrayuttawat *et al.* (1997)	d	0.04
Ong & Acree (1999)		0.008

3,7-DIMETHYL-6-OCTEN-1-YL ACETATE (citronellyl acetate) *[150-84-5]*

Sugisawa *et al.* (1991); Yang *et al.* (1992); Tamura *et al.* (1993)	d	1
Pino & Mesa (2006)	d	1

3,7-dimethyl-2-octen-4-olide⇒4-HYDROXY-3,7-DIMETHYL-2-OCTENOIC ACID LACTONE
(R*,S*)-3,7-dimethyl-6-octen-5-olide⇒(R*,S*)-5-HYDROXY-3,7-DIMETHYL-6-OCTENOIC ACID LACTONE
(R*,R*)-3,7-dimethyl-6-octen-5-olide⇒(R*,R*)-5-HYDROXY-3,7-DIMETHYL-6-OCTENOIC ACID LACTONE
3,7-dimethyl-2-octen-6-olide⇒6-HYDROXY-3,7-DIMETHYL-2-OCTENOIC ACID LACTONE
(R*,S*)-3,7-dimethyl-7-octen-6-olide⇒(R*,S*)-6-HYDROXY-3,7-DIMETHYL-7-OCTENOIC ACID LACTONE
(R*,R*)-3,7-dimethyl-7-octen-6-olide⇒(R*,R*)-6-HYDROXY-3,7-DIMETHYL-7-OCTENOIC ACID LACTONE

2,4-DIMETHYL-3-OXA-8-THIABICYCLO[3.3.0]-1,4-OCTADIENE (dimethylkahweofuran)

Tressl (1989,1990)		0.000 5

O,O-DIMETHYL S-(4-OXO-3H-1,2,3-BENZOTRIAZINE-3-METHYL)PHOSPHORODITHIOATE (guthion) *[86-50-0]*

Sigworth (1965)	d	0.000 2

2,2-DIMETHYLPENTANAL *[14250-88-5]*

Schnabel *et al.* (1988)	d	0.41 - 0.83

3,3-DIMETHYLPENTANAL

Schnabel *et al.* (1988)	d	0.041 - 0.083

2,2-DIMETHYLPENTANE *[590-35-2]*

Zoeteman *et al.* (1971)		100

2,2-DIMETHYL-1-PENTANOL *[2370-12-9]*

Schnabel *et al.* (1988)	d	0.0081 - 0.041

2,3-DIMETHYL-1-PENTANOL *[10143-23-4]*

Schnabel *et al.* (1988)	d	0.83 - 1.6

2,4-DIMETHYL-1-PENTANOL *[6305-71-1]*

Schnabel *et al.* (1988)	d	0.24 - 0.40

2,3-DIMETHYL-2-PENTANOL *[4911-70-0]*

Schnabel *et al.* (1988)	d	1.6 - 4.2

2,4-DIMETHYL-2-PENTANOL *[625-06-9]*

Schnabel *et al.* (1988)	d	2.4 - 4.1

4,4-DIMETHYL-2-PENTANOL *[6144-93-0]*

Schnabel *et al.* (1988)	d	0.041 - 0.081

2,2-DIMETHYL-3-PENTANOL *[3970-62-5]*

Schnabel *et al.* (1988)	d	0.041 - 0.083

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

2,3-DIMETHYL-3-PENTANOL *[595-41-5]*
Schnabel *et al.* (1988) d 0.42 - 0.84

2,4-DIMETHYL-3-PENTANOL *[600-36-2]*
Schnabel *et al.* (1988) d 0.082 - 0.42

4,4-DIMETHYL-2-PENTANONE (methyl neopentyl ketone) *[590-50-1]*
Schnabel *et al.* (1988) d 0.0081 - 0.040

2,2-DIMETHYL-3-PENTANONE *[564-04-5]*
Schnabel *et al.* (1988) d 0.024 - 0.041

2,4-DIMETHYL-3-PENTANONE (diisopropyl ketone) *[565-80-0]*
Schnabel *et al.* (1988) d 0.080 - 0.40

2,3-DIMETHYL-5-PENTYLPYRAZINE
Shibamoto (1986) d 0.09

2,3-DIMETHYLPHENOL (2,3-xylenol) *[526-75-0]*
Dietz & Traud (1978) 0.5

2,4-DIMETHYLPHENOL (2,4-xylenol) *[105-67-9]*
Dietz & Traud (1978) 0.4

2,5-DIMETHYLPHENOL (2,5-xylenol) *[95-87-4]*
Dietz & Traud (1978) 0.4

2,6-DIMETHYLPHENOL (2,6-xylenol) *[576-26-1]*
Dietz & Traud (1978) 0.4

3,4-DIMETHYLPHENOL (3,4-xylenol) *[95-65-8]*
Dietz & Traud (1978) 1.2

3,5-DIMETHYLPHENOL (3,5-xylenol) *[108-68-9]*
Dietz & Traud (1978) 5.0

2,2-DIMETHYLPROPANAL *[630-19-3]*
Schnabel *et al.* (1988) d 0.078 - 0.12

2,2-DIMETHYLPROPANOIC ACID (trimethylacetic acid) *[75-98-9]*
Amoore *et al.* (1968) d 5.25
Zoeteman *et al.* (1971) 50
Schnabel *et al.* (1988) d 5.0 - 20

2,2-DIMETHYL-1-PROPANOL (neopentyl alcohol) *[75-84-3]*
Schnabel *et al.* (1988) d 0.82 - 2.5

1,1-DIMETHYLPROPYL ACETATE (*tert*-amyl acetate) *[625-16-1]*
Teranishi *et al.* (1966) d 0.030

1,2-DIMETHYLPROPYL ACETATE (3-methylbutyl-2 acetate) *[5343-96-4]*
Teranishi *et al.* (1966) d 0.006

2,2-DIMETHYLPROPYL ACETATE (neopentyl acetate) *[926-41-0]*
Teranishi *et al.* (1966) d 0.004

2,3-DIMETHYL-5-PROPYLPYRAZINE *[32262-98-9]*
Shibamoto (1986) d 0.32

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

2,3-DIMETHYLPYRAZINE *[5910-89-4]*

Guadagni *et al.* (1972); Teranishi *et al.* (1974); Buttery *et al.* (1988a,1997,1999); Buttery & Ling (1995,1998)	d	2.5
Calabretta (1973,1975,1978)	d	0.4
Mihara & Masuda (1988)	d	0.8

2,5-DIMETHYLPYRAZINE *[123-32-0]*

Seifert *et al.* (1970); Guadagni *et al.* (1972); Teranishi *et al.* (1974); Buttery *et al.* (1988a, 1994a,1997,1999); Buttery & Ling (1995,1998)	d	1.7 - 1.8
Koehler *et al.* (1971)	d	35
Mihara & Masuda (1988)	d	0.08
Pino & Mesa (2006)	d	1.75

2,6-DIMETHYLPYRAZINE *[108-50-9]*

Koehler *et al.* (1971)	d	54
Guadagni *et al.* (1972); Teranishi *et al.* (1974); Buttery *et al.* (1988a,1997,1999); Buttery & Ling (1998)	d	1.5
Mihara & Masuda (1988)	d	0.4
Schirack *et al.* (2006)	d	0.718

2,5-DIMETHYL-4-(1-PYRROLIDINYL)-3(2*H*)-FURANONE

Ottinger *et al.* (2001)		30 - 60

α,*p*-dimethylstyrene⇒1-ISOPROPENYL-4-METHYLBENZENE
dimethyl sulphide⇒METHYLTHIOMETHANE
dimethyl tetrasulphide⇒METHYLTETRATHIOMETHANE

2,4-DIMETHYLTHIAZOLE *[541-58-2]*

Teranishi *et al.* (1975)	d	0.018

4,5-DIMETHYLTHIAZOLE *[3581-91-7]*

Buttery *et al.* (1976a)	d	0.47

dimethyl trisulphide⇒METHYLTRITHIOMETHANE

cis-3,5-DIMETHYL-1,2,4-TRITHIOLANE *[38348-23-1]*

Rödel & Kruse (1982)		0.00200

trans-3,5-DIMETHYL-1,2,4-TRITHIOLANE *[38348-24-2]*

Rödel & Kruse (1982)		0.000 609

6,10-DIMETHYL-5,9-UNDECADIEN-2-ONE (geranylacetone) *[3796-70-1]*

Buttery *et al.* (1971,1987a,1988a,1989,1990a,1997); Teranishi *et al.* (1974); Buttery (1993); Buttery & Ling (1995,1998)	d	0.06
Sugisawa *et al.* (1991); Yang *et al.* (1992); Tamura *et al.* (1993,1995)	d	6 - 6.4
Tandon *et al.* (2000)	d	0.186
Pino & Mesa (2006)	d	0.06

6,10-DIMETHYL-3,5,9-UNDECATRIEN-2-ONE (pseudoionone) *[141-10-6]*

Teranishi *et al.* (1974)	d	> 0.05
Buttery *et al.* (1989,1990a); Buttery (1993)	d	0.8

4,6-dinitro-*o*-cresol⇒2-METHYL-4,6-DINITROPHENOL

2,4-DINITROPHENOL *[51-28-5]*

Dietz & Traud (1978)		4

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

2,5-DINITROPHENOL *[329-71-5]*
Dietz & Traud (1978) 2.4

2,6-DINITROPHENOL *[573-56-8]*
Dietz & Traud (1978) 10

2,6-dinitrotoluene⇒2-METHYL-1,3-DINITROBENZENE
18,19-dinorambrox⇒3a,9a-DIMETHYL-3aα,5aβ,9aα,9bβ-DODECAHYDRONAPHTHO[2,1-b]FURAN
18,19-dinorisoambrox⇒3a,9a-DIMETHYL-3aα,5aβ,9aβ,9bα-DODECAHYDRONAPHTHO[2,1-b]FURAN

3,6-DIOXA-1-DECANOL (butyl carbitol, diethylene glycol *n*-butyl ether) *[112-34-5]*
Alexander *et al.* (1982) d 0.8

2,3-*p*-DIOXANEDITHIOL S,S-BIS(O,O-DIETHYLPHOSPHORODITHIOATE) (delvan) *[78-34-2]*
Sigworth (1965) d 0.06

4,8-DIOXA-1-NONANOL (dipropylene glycol methyl ether) *[34590-94-8]*
Alexander *et al.* (1982) d 40

3,6-DIOXA-1,8-OCTANEDIOL (triethylene glycol) *[112-27-6]*
Alexander *et al.* (1982) d 700

3,6-DIOXA-1-OCTANOL (carbitol, diethylene glycol ethyl ether) *[111-90-0]*
Alexander *et al.* (1982) d 1.6

DI-(1-OXYDECYL)-PEROXIDE
Kobayashi *et al.* (1990) r 0.0045

cis-3,5-DIPENTYL-1,2,4-TRITHIOLANE
Rödel & Kruse (1982) 0.0256

trans-3,5-DIPENTYL-1,2,4-TRITHIOLANE
Rödel & Kruse (1982) 0.0340

diphenyl⇒BIPHENYL
diphenyl disulphide⇒PHENYLDITHIOBENZENE
diphenyl ether⇒PHENOXYBENZENE
dipotassium phosphate⇒POTASSIUM PHOSPHATE, DIBASIC

DIPROPYLAMINE *[142-84-7]*
Amoore *et al.* (1975b) d 22.4
Amoore & Forrester (1976) d 24.8

dipropylene glycol⇒4-OXA-2,6-HEPTANEDIOL
dipropylene glycol methyl ether⇒4,8-DIOXA-1-NONANOL
dipropyl ketone⇒4-HEPTANONE

2,4-DIPROPYLTHIAZOLE
Teranishi *et al.* (1975) d 0.026

cis-3,5-DIPROPYL-1,2,4-TRITHIOLANE
Rödel & Kruse (1982) 0.00151

trans-3,5-DIPROPYL-1,2,4-TRITHIOLANE
Rödel & Kruse (1982) 0.000 762

3,4-dithiahexane⇒ETHYLDITHIOETHANE
3,3'-dithiobis(2-methylfuran) ⇒3-(2-METHYL)FURYLDITHIO-3-(2-METHYL)FURAN
diuron⇒N'-(3,4-DICHLOROPHENYL)-N,N-DIMETHYLUREA
DMDS⇒METHYLDITHIOMETHANE
γ-dodecalactone⇒4-HYDROXYDODECANOIC ACID LACTONE
δ-dodecalactone⇒5-HYDROXYDODECANOIC ACID LACTONE

(*R*)-(+)-δ-dodecalactone⇒(*R*)-(+)-5-HYDROXYDODECANOIC ACID LACTONE
(*S*)-(-)-δ-dodecalactone⇒(*S*)-(-)-5-HYDROXYDODECANOIC ACID LACTONE
1,12-dodecamethylene diamine⇒AZACYCLOTRIDECANE
dodecamethyleneimine⇒AZACYCLOTRIDECANE

DODECANAL (lauraldehyde) *[112-54-9]*

Guadagni *et al.* (1963)	d	0.002
Ahmed *et al.* (1978a)	d	0.000 53
Pino *et al.* (1986)	d	0.0015
Schnabel *et al.* (1988)	d	0.000 83 - 0.0017
Sugisawa *et al.* (1991); Yang *et al.* (1992);		
Tamura *et al.* (1993,1995)	d	0.055
Tamura *et al.* (1996)	d	0.014
Tamura *et al.* (1996)	r	0.063
Padrayuttawat *et al.* (1997)	d	0.01
Fabrellas *et al.* (2004)		0.000 13 - 0.000 29

DODECANE *[112-40-3]*

Zoeteman *et al.* (1971)	d	10

dodecanenitrile⇒UNDECYL CYANIDE

1-DODECANOL *[112-53-8]*

Ahmed *et al.* (1978a)	d	0.073
Schnabel *et al.* (1988)	d	0.093 - 0.41
Sugisawa *et al.* (1991)	d	0.82
Tamura *et al.* (1996)	d	0.158
Tamura *et al.* (1996)	r	2.47
Boonbumrung *et al.* (2001)	d	0.016

2-DODECANOL *[10203-28-8]*

Schnabel *et al.* (1988)	d	0.041 - 0.082

3-DODECANOL *[10203-30-2]*

Schnabel *et al.* (1988)	d	0.041 - 0.082

4-dodecanolide⇒4-HYDROXYDODECANOIC ACID LACTONE
5-dodecanolide⇒5-HYDROXYDODECANOIC ACID LACTONE

2-DODECANONE (methyl decyl ketone) *[6175-49-1]*

Schnabel *et al.* (1988)	d	0.042 - 0.083

3-DODECANONE *[1534-27-6]*

Schnabel *et al.* (1988)	d	0.0083 - 0.042

(*E*)-2-DODECENAL *[20407-84-5]*

Yang *et al.* (1992); Tamura *et al.* (1993)	d	0.0014

(*R*)-*cis*-6-γ-dodecenolactone⇒(*R*)-(*Z*)-4-HYDROXY-6-DODECENOIC ACID LACTONE
(*S*)-*cis*-6-γ-dodecenolactone⇒(*S*)-(*Z*)-4-HYDROXY-6-DODECENOIC ACID LACTONE
(*Z*)-6-dodecenyl-γ-lactone⇒(*Z*)-4-HYDROXY-6-DODECENOIC ACID LACTONE

DODECYL PROPYLENE DIAMINE (lauryl propylene diamine) *[5538-95-4]*

Rudeiko *et al.* (1982)		0.13

E

EICOSANOIC ACID *[506-30-9]*

Cherkinski (1961)	d	20.0

elemol⇒[1*R*-(1α,3α,4β)]-4-ETHENYL-α,α,4-TRIMETHYL-3-(1-METHYLETHENYL)CYCLOHEXANE-METHANOL)

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

endrin⇒1,2,3,4,10,10-HEXACHLORO-6,7-EPOXY-1,4,4a,5,6,7,8,8a-OCTAHYDRO-1,4-*endo*-*endo*-5,8-DIMETHANONAPHTHALENE
(*l*)-9-epiambrox⇒(3a*S*)-3a,6,6,9a-TETRAMETHYL-3aα,5aβ,9aα,9bβ-DODECAHYDRONAPHTHO[2,1-b]FURAN
9-epi-18,19-dinorambrox⇒3a,9a-DIMETHYL-3aα,5aβ,9aα,9bα-DODECAHYDRONAPHTHO[2,1-b]FURAN
4-epinootkatone⇒4,4a,5,6,7,8-HEXAHYDRO-4,4a-DIMETHYL-6-(1-METHYLETHENYL)-2(3*H*)-NAPHTHALENONE
(*d*)-9-epi-18-norambrox⇒(3a*R*)-3a,6,9a-TRIMETHYL-3aα,5aβ,6α,9aα,9bα-DODECAHYDRONAPHTHO[2,1-b]FURAN
9-epi-19-norambrox⇒3a,6,9a-TRIMETHYL-3aα,5aβ,9aα,9bα-DODECAHYDRONAPHTHO[2,1-b]FURAN
epn-300⇒O-ETHYL O-4-NITROPHENYL PHENYLPHOSPHONOTHIOATE
1,4-epoxy-1,3-butadiene⇒FURAN

4,5-EPOXY-(*E*)-2-DECENAL *[73528-38-8]*

Heiler & Schieberle (1997)		0.000 07

trans-4,5-EPOXY-(*E*)-2-DECENAL *[73528-44-6]*

Kerler & Grosch (1996)		0.000 12
Ong *et al.* (1998)	d	0.005
Czerny *et al.* (2008)	d	0.000 038
Czerny *et al.* (2008)	r	0.000 13
Steinhaus *et al.* (2009); Sinuco *et al.* (2010)	d	0.000 22

epoxydihydrolinalool⇒α-ETHENYL-α,3,3-TRIMETHYLOXIRANPROPANOL
epoxy-β-ionone⇒(4-(2,2,6-TRIMETHYL-7-OXABICYCLO[4.1.0]HEPT-1-YL-3-BUTEN-2-ONE
(3*R*,4*S*,8*S*)-3,9-epoxy-1-*p*-menthene⇒(3*S*,3a*S*,7a*R*)-3,6-DIMETHYL-2,3,3a,4,5,7a-HEXAHYDROBENZOFURAN

1,2-EPOXYPROPANE (propylene oxide) *[75-56-9]*

Antonova *et al.* (1981)		0.43

eremophilone⇒8-ISOPROPENYL-5,6-DIMETHYLBICYCLO[4.4.0]-1-DECEN-10-ONE
estragole⇒1-ALLYL-4-METHOXYBENZENE
ETBE⇒2-ETHOXY-2-METHYLPROPANE

ETHANAL (acetaldehyde) *[75-07-0]*

Balavoine (1948)		0.000 7
Flath *et al.* (1967); Teranishi *et al.* (1974);		
Aharoni *et al.* (1980);		
Buttery *et al.* (1988a,1994a,1997)	d	0.015
Petro-Turza (1978)		2.5
Mulders (1972,1973)	d	0.12 - 0.2
Amoore *et al.* (1976)	d	0.109
Ahmed *et al.* (1978a)	d	0.0170
Pino *et al.* (1986)	d	0.0087
Schnabel *et al.* (1988)	d	0.0079 - 0.039
Takahashi (1990)		25 - 50
Guth & Grosch (1994); Heiler & Schieberle (1997)		0.025
Milo (1995a, 1995b); Milo & Grosch (1997);		
Kerscher & Grosch (2000)		0.01
Czerny *et al.* (2008); Steinhaus *et al.* (2009)	d	0.025
Czerny *et al.* (2008)	r	0.063
Giri *et al.* (2010)	d	0.0251

1,2-ETHANEDIAMINE (ethylenediamine) *[107-15-3]*

Amoore *et al.* (1975b)	d	18,300
Alexander *et al.* (1982)	d	12

1,2-ETHANEDIOL (ethylene glycol) *[107-21-1]*

Alexander *et al.* (1982)	d	197

ETHANETHIOL (ethyl mercaptan) *[75-08-1]*

Moncrieff (1957)		0.002

ethanoic acid ⇒ACETIC ACID

ETHANOL (ethyl alcohol) *[64-17-5]*

Moncrieff (1957)		3,000

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Guadagni (1966,1968,1969,1970c); Flath *et al.* (1967); Aharoni *et al.* (1980); Teranishi *et al.* (1981); Takeoka *et al.* (1989); Hansen *et al.* (1992)	d	100
Sega *et al.* (1967)	d	8.2
Mitchell & Gregson (1968)	r	11,100 - 116,200
Williams (1972)		100 - 500
Mulders (1972,1973)	d	900 - 930
De Grunt (1975)	d	100
Sarzynski (1982)		50
Eichenbaum *et al.* (1983)	d	67,500 - 191,250
Pino *et al.* (1986)	d	1,150
Granzer *et al.* (1986)		100 - 1,000
Schnabel *et al.* (1988)	d	16 - 40
Rothe & Schrödter (1996)		3.5
Jensen *et al.* (1999)	d	> 100
Mattes & DiMeglio (2001)	d	3.63
Boonbumrung *et al.* (2001)	d	4.51
Tamura *et al.* (2001)	d	200
Pino & Mesa (2006)	d	100
Czerny *et al.* (2008)	d	990
Czerny *et al.* (2008)	r	2,000
Giri *et al*. (2010)	d	950

ethanolamine⇒2-AMINOETHANOL

ETHENYLBENZENE (styrene, vinylbenzene) *[100-42-5]*

Rosen *et al.* (1963)		0.037
Baker (1963)		0.73
Zoeteman *et al.* (1971)		0.05
Boidron & Pons (1978)	d	0.08
Alexander *et al.* (1982)	d	0.0036
Young *et al*. (1996)		0.065

ETHENYL CYANIDE (acrylonitrile, vinyl cyanide) *[107-13-1]*

Baker (1963)		18.6

4-ETHENYL-2-METHOXYPHENOL (*p*-vinylguaiacol, 4-vinylguaiacol, 2-methoxy-4-vinylphenol) *[7786-61-0]*

Buttery *et al.* (1976b,1988a,1994a,1997,1999); Buttery & Ling (1995,1998); Takeoka (2001)	d	0.003
Kossa *et al.* (1979); Tressl *et al.* (1979c)		0.003 - 0.005
Schieberle (1991,1992)	d	0.1
Semmelroch *et al.* (1995)		0.020
Pino & Mesa (2006)	d	0.003
Czerny *et al.* (2007b); Czerny *et al.* (2008)	d	0.0051
Czerny *et al.* (2007b); Czerny *et al.* (2008)	r	0.019
Vanbeneden *et al.* (2008)	d	0.088
Giri *et al*. (2010)	d	0.01202

5-ETHENYL-2-METHOXYPHENOL (2-methoxy-5-vinylphenol) *[621-58-9]*

Pyysalo *et al.* (1977)	d	0.000 75
Scheidig *et al.* (2007)	d	0.000 15
Scheidig *et al.* (2007)	r	0.000 35

4-ETHENYLPHENOL (4-vinylphenol) *[2628-17-3]*

Pyysalo *et al.* (1977)	d	0.011
Buttery *et al.* (1988a,1994a,1997,1999); Buttery & Ling (1995,1998); Takeoka (2001)	d	0.01

ETHENYL PROPANOATE (vinyl propanoate, vinyl propionate) *[105-38-4]*

De Grunt (1975); De Grunt *et al.* (1977)	d	0.04

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

ETHENYLPYRAZINE (vinylpyrazine) *[4177-16-6]*
Buttery *et al.* (1997,1999); Buttery & Ling (1998) d 0.7

trans-5-ETHENYLTETRAHYDRO-α,α,5-TRIMETHYL-2-FURANMETHANOL (*trans*-furan linalool oxide, linalool oxide A) *[34995-77-2]*
Ong & Acree (1999) 0.06

[1*R*-(1α,3α,4β)]-4-ETHENYL-α,α,4-TRIMETHYL-3-(1-METHYLETHENYL)CYCLOHEXANEMETHANOL (elemol) *[639-99-6]*
Sugisawa *et al.* (1991); Yang *et al.* (1992);
Tamura *et al.* (1993) d 0.1

α-ETHENYL-α,3,3-TRIMETHYLOXIRANPROPANOL (epoxydihydrolinalool) *[15249-35-1]*
Pino *et al.* (1986) d 1.148

2-ETHOXY-3,6-DIMETHYLPYRAZINE *[90197-08-3]*
Mihara *et al.* (1991) d 0.011

4-ETHOXY-2,5-DIMETHYL-3(2*H*)-FURANONE *[65330-49-6]*
Buttery (1999) d 2.9

2-ETHOXYETHANOL (cellosolve, ethyl glycol, ethylene glycol ethyl ether) *[110-80-5]*
Moncrieff (1957) 200
Alexander *et al.* (1982) d 23.5
Sarzynski (1982) 125
Granzer *et al.* (1986) 50 - 1,000

2-ETHOXYETHYL ACETATE (cellosolve acetate, ethyl glycol acetate) *[111-15-9]*
Granzer *et al.* (1986) 5 - 50

2-ETHOXY-3-ETHYLPYRAZINE *[35243-43-7]*
Seifert *et al.* (1972); Teranishi *et al.* (1975) d 0.011
Masuda & Mihara (1988) d 0.02

3-ETHOXY-4-HYDROXYBENZALDEHYDE (ethylvanillin) *[121-32-4]*
Cartwright & Kelley (1952) d 0.1
Cartwright & Kelley (1952) r 2

2-ETHOXY-5-ISOBUTYL-3-METHYLPYRAZINE *[78246-14-7]*
Masuda & Mihara (1986); Shibamoto (1986) d 0.016

2-ETHOXY-5-ISOPROPYL-3-METHYLPYRAZINE *[99784-14-2]*
Masuda & Mihara (1986); Shibamoto (1986) d 0.12

2-ETHOXY-3-METHYL-5-(2-METHYLBUTYL)PYRAZINE *[99784-16-4]*
Masuda & Mihara (1986); Shibamoto (1986) d 0.006 - 0.0061

2-ETHOXY-3-METHYL-5-(2-METHYLPENTYL)PYRAZINE *[99784-17-5]*
Masuda & Mihara (1986); Shibamoto (1986) d 0.0027 - 0.003

2-ETHOXY-2-METHYLPROPANE (ETBE, ethyl *tert*-butyl ether) *[637-92-3]*
Vetrano (1993a) d 0.049
Vetrano (1993a) r 0.106
Van Wezel *et al.* (2009) d 0.001

2-ETHOXY-3-METHYLPYRAZINE *[32737-14-7]*
Masuda & Mihara (1986,1988) d 0.000 8

2-ETHOXY-5-METHYLPYRAZINE *[67845-34-5]*

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Mihara & Masuda (1988)	d	0.012

2-ETHOXY-6-METHYLPYRAZINE *[53163-97-6]*

Mihara & Masuda (1988)	d	0.005

2-ETHOXY-3-OCTYLPYRAZINE *[113685-83-9]*

Masuda & Mihara (1988)	d	0.002

2-ETHOXY-3-PENTYLPYRAZINE *[113685-82-8]*

Masuda & Mihara (1988)	d	0.000 08

trans-2-ETHOXY-5-(1-PROPENYL)PHENOL (propenylguaethol) *[94-86-0]*

Cartwright & Kelley (1952)	d	0.01
Cartwright & Kelley (1952)	r	0.4

ETHOXYPYRAZINE *[38028-67-0]*

Masuda & Mihara (1988)	d	0.08

2-ETHOXY THIOZOLE

Giri *et al*. (2010)	d	0.05115

ETHYL ACETATE *[141-78-6]*

Balavoine (1948)		10
Flath *et al.* (1967); Buttery (1981);		
Hansen *et al.* (1992)	d	5.0
Sega *et al.* (1967)	d	0.257
Guadagni (1970b); Teranishi *et al.* (1987);		
Takeoka *et al.* (1989)	d	0.005
Mulders (1972,1973	d	5 - 6.2
De Grunt (1975)	d	1.0
Boidron (1978)		25
Sarzynski (1982)		5
Piggott & Findlay (1984)	d	5.9
Pino *et al.* (1986)	d	0.0085
Granzer *et al.* (1986)		10 - 50
Schnabel *et al.* (1988)	d	8.8 - 45
Takahashi (1990)		25 - 50
Larsen & Poll (1992)	d	0.1 - 1.0
Tamura *et al.* (1995)		50
Takeoka *et al.* (1996)	d	13.5
Tamura *et al.* (2001) ; Boonbumrung *et al.* (2001)	d	3.28 - 3.3
Pino & Mesa (2006)	d	5
Giri *et al.* (2010)	d	0.005

ethyl acrylate⇒ETHYL 2-PROPENOATE
ethyl alcohol⇒ETHANOL

ETHYLAMINE *[75-04-7]*

Amoore *et al.* (1975b)	d	95.1
De Grunt (1975)	d	10.0

3-(ETHYLAMINO)-4-METHYLPHENOL *[120-37-6]*

Dietz & Traud (1978)		0.04

ethyl amyl ketone⇒3-OCTANONE
ethyl angelate⇒ETHYL (*Z*)-2-METHYL-2-BUTENOATE

3-ETHYLANISOLE *[10568-38-4]*

Dietz & Traud (1978)		0.02

4-ETHYLBENZALDEHYDE

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Giri *et al*. (2010)	d	0.12323

ETHYLBENZENE *[100-41-4]*

Cherkinski (1961)	d	0.2
Rosen *et al.* (1963)		0.14
Zoeteman *et al.* (1971)		0.1
Boidron & Pons (1978)	d	1.2
Alexander *et al.* (1982)	d	0.0024
Young *et al.* (1996)		0.55
Giri *et al*. (2010)	d	2.20525

ETHYL BENZOATE *[93-89-0]*

Appell (1969)		0.4
Pyysalo *et al.* (1977); Hirvi & Honkanen (1984)	d	0.10
Buttery *et al.* (1988a);Takeoka *et al.* (1991b)	d	0.06
Steinhaus *et al.* (2009); Sinuco *et al.* (2010)	d	0.053
Giri *et al*. (2010)	d	0.05556

ETHYL 3-(4-BROMOPHENYL)-3-METHYL-2,3-EPOXYPROPANOATE

Tekin & Karaman (1992)	d	0.5

2-ETHYLBUTANAL (2-ethylbutyraldehyde) *[97-96-1]*

Schnabel *et al.* (1988)	d	0.041 - 0.081

ETHYL BUTANOATE (ethyl butyrate) *[105-54-4]*

Flath *et al.* (1967); Guadagni (1970b); Buttery (1981); Teranishi *et al.* (1987); Takeoka *et al.* (1989,2008)	d	0.001
Ahmed *et al.* (1978a)	d	0.000 13
Piggott & Findlay (1984)	d	0.018
Pino *et al.* (1986)	d	0.0011
Schnabel *et al.* (1988)	d	0.0070 - 0.027
Takahashi (1990)		0.4
Schieberle (1991,1992)	d	0.001
Larsen & Poll (1992)	d	0.000 001 - 0.000 01
Boonbumrung *et al*. (2001)	d	0.000 18
Labbe *et al.* (2007)		0.01
Pino & Mesa (2006)	d	0.001
Czerny *et al.* (2008); Steinhaus *et al.* (2009); Sinuco *et al.* (2010)	d	0.000 76
Czerny *et al.* (2008)	r	0.0024
Burdack-Freitag *et al.* (2010)	d	0.000 5 (normal atmospheric pressure)
Burdack-Freitag *et al.* (2010)	d	0.0049 (low atmospheric pressure)
Burdack-Freitag *et al.* (2010)	r	0.003 (normal atmospheric pressure)
Burdack-Freitag *et al.* (2010)	r	0.0084 (low atmospheric pressure)
Giri *et al*. (2010)	d	0.000 9

2-ETHYLBUTANOIC ACID (2-ethylbutyric acid) *[88-09-5]*

Zoeteman *et al.* (1971)		10
Amoore (1976)	d	365
Schnabel *et al.* (1988)	d	0.093 - 0.46

2-ETHYL-1-BUTANOL *[97-95-0]*

De Grunt (1975)	d	0.2
Schnabel *et al.* (1988)	d	0.082 - 0.42
Giri *et al*. (2010)	d	0.0752

2-(1-ETHYLBUTYL)-5,6-DIMETHYLPYRAZINE

Shibamoto (1986)	d	0.82

ethyl *tert*-butyl ether⇒2-ETHOXY-2-METHYLPROPANE

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

2-ethylbutyraldehyde⇒2-ETHYLBUTANAL
ethyl butyrate⇒ETHYL BUTANOATE
2-ethylbutyric acid⇒2-ETHYLBUTANOIC ACID
ethyl caprate⇒ETHYL DECANOATE
ethyl caproate⇒ETHYL HEXANOATE
ethyl caprylate⇒ETHYL OCTANOATE
ethyl cinnamate⇒ETHYL 3-PHENYLPROPENOATE
ethyl (*E*)-cinnamate⇒ETHYL (*E*)-3-PHENYLPROPENOATE

ETHYL CYANIDE (propionitrile) *[107-12-0]*

Fabrellas *et al.* (2004)		0.012 - 0.068

ETHYL CYCLOBUTANECARBOXYLATE *[14924-53-9]*

Takeoka *et al.* (1991b)	d	0.000 01

ETHYL CYCLOHEPTANECARBOXYLATE *[32777-26-7]*

Takeoka *et al.* (1991b)	d	0.000 002

ETHYL CYCLOHEXANECARBOXYLATE *[3289-28-9]*

Takeoka *et al.* (1991b)	d	0.000 001

ETHYL 1-CYCLOHEXENECARBOXYLATE

Takeoka *et al.* (1991b)	d	0.000 3

ETHYL 3-CYCLOHEXENECARBOXYLATE *[15111-56-5]*

Takeoka *et al.* (1991b)	d	0.000 003

ETHYL CYCLOHEXYLACETATE *[5452-75-5]*

Takeoka *et al.* (1991b)	d	0.000 02

ETHYL CYCLOOCTANECARBOXYLATE

Takeoka *et al.* (1991b)	d	0.000 004

ETHYL CYCLOPENTANECARBOXYLATE *[5453-85-0]*

Takeoka *et al.* (1991b)	d	0.000 001

ETHYL CYCLOPROPANECARBOXYLATE *[4606-07-9]*

Takeoka *et al.* (1991b)	d	0.000 1

ETHYL (*E*,*Z*)-2,4-DECADIENOATE *[3025-30-7]*

Takeoka *et al.* (1992)	d	0.1

ETHYL DECANOATE (ethyl caprate) *[110-38-3]*

Schnabel *et al.* (1988)	d	0.0086 - 0.044
Takahashi (1990)		1.5
Takeoka *et al.* (1992)	d	0.122
Boonbumrung *et al.* (2001)	d	0.023
Pino & Mesa (2006)	d	6.3
Giri *et al.* (2010)	d	0.005

ETHYL 3-(3,4-DICHLOROPHENYL)-3-METHYL-2,3-EPOXYPROPANOATE

Tekin & Karaman (1992)	d	2

ETHYLDIMETHYLAMINE *[598-56-1]*

Amoore & Forrester (1976)	d	0.00762

ETHYL 2,2-DIMETHYLBUTANOATE

Takeoka *et al.* (1991b)	d	0.000 1

ETHYL 2,3-DIMETHYLBUTANOATE

Takeoka *et al.* (1995)	d	0.000 004

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

ETHYL 3,3-DIMETHYLBUTANOATE

Takeoka *et al.* (1991b)	d	0.000 003

2-ETHYL-5,5-DIMETHYL-1,3-DIOXANE *[768-58-1]*

Petri *et al.* (1993)	0.000 01	
Schweitzer *et al.* (1999a)		0.000 005 - 0.000 01

ETHYL 2,3-DIMETHYLPENTANOATE

Takeoka *et al.* (1995b)	d	0.000 002

ETHYL 3-(3,4-DIMETHYLPHENYL)-3-METHYL-2,3-EPOXYPROPANOATE

Tekin & Karaman (1992)	d	100

ETHYL 2,2-DIMETHYLPROPANOATE (ethyl pivalate) *[3938-95-2]*

Takeoka *et al.* (1995b)	d	0.000 2

2-ETHYL-3,5-DIMETHYLPYRAZINE *[13925-07-0]*

Koehler *et al.* (1971)	d	15
Guadagni (1973)		0.001
Cerny & Grosch (1993); Grosch *et al.* (1993)		0.002
Czerny & Wagner (1995); Czerny *et al.* (1996a, 1996b)		0.000 16
Buttery & Ling (1997b,1998); Buttery *et al.* (1997, 1999)	d	0.000 04

3-ETHYL-2,5-DIMETHYLPYRAZINE (2-ethyl-3,6-dimethylpyrazine) *[13360-65-1]*

Koehler *et al.* (1971)	d	43
Guadagni *et al.* (1972); Teranishi *et al.* (1974); Buttery (1981,1993); Buttery *et al.* (1994a); Buttery & Ling (1995)	d	0.000 4
Tressl *et al.* (1977b)		0.005
Buttery & Ling (1997b); Buttery *et al.* (1997,1999); Buttery & Ling (1998)	d	0.0086

5-ETHYL-2,3-DIMETHYLPYRAZINE *[15707-34-3]*

Shibamoto (1986); Mihara *et al.* (1991)	d	0.53

5-ETHYL-2,4-DIMETHYLTHIAZOLE *[38205-61-7]*

Teranishi *et al.* (1974); Buttery (1981)	d	0.000 09

ethyl disulphide⇒ ETHYLDITHIOETHANE

ETHYLDITHIOETHANE (ethyl disulphide, diethyl disulphide, 3,4-dithiahexane) *[110-81-6]*

De Grunt (1975); De Grunt *et al.* (1977)	d	0.000 01 - 0.000 02

ETHYL DODECANOATE (ethyl laurate) *[106-33-2]*

Takahashi (1990)		3.5
Boonbumrung *et al.* (2001)	d	0.40
Pino & Mesa (2006)	d	5.9

ethylene brassylate⇒CYCLO-1,3-ETHYLENEDIOXY-1,3-TRIDECANEDIONE
ethylene chloride⇒1,2-DICHLOROETHANE
ethylenediamine⇒1,2-ETHANEDIAMINE
ethylene dichloride⇒1,2-DICHLOROETHANE
ethylene glycol⇒1,2-ETHANEDIOL
ethylene glycol *n*-butyl ether⇒2-BUTOXYETHANOL
ethylene glycol ethyl ether⇒2-ETHOXYETHANOL
ethylene glycol methyl ether⇒2-METHOXYETHANOL

ETHYL 2-ETHYLBUTANOATE

Takeoka *et al.* (1995b)	d	0.000 007

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

ETHYL 2-ETHYLHEXANOATE *[2983-37-1]*

Takeoka *et al.* (1995b)	d	0.050

ETHYL 3-ETHYL-2-METHYL-3-PHENYL-2,3-EPOXYPROPANOATE

Tekin & Karaman (1992)	d	3.5

ETHYL 2-ETHYLPENTANOATE

Takeoka *et al.* (1995b)	d	0.000 05

ETHYL 3-ETHYL-3-PHENYL-2,3-EPOXYPROPANOATE

Tekin & Karaman (1992)	d	18

2-ETHYL-3-(ETHYLTHIO)PYRAZINE *[113685-90-8]*

Masuda & Mihara (1988)	d	0.06

ETHYL FORMATE *[109-94-4]*

Mulders (1972,1973)	d	17
Aharoni *et al.* (1980)		17
Schnabel *et al.* (1988)	d	8.9 - 89

ethyl glycol⇒2-ETHOXYETHANOL
ethyl glycol acetate⇒2-ETHOXYETHYL ACETATE
p-ethylguaiacol⇒4-ETHYL-2-METHOXYPHENOL
4-ethylguaiacol⇒4-ETHYL-2-METHOXYPHENOL

ETHYL HEPTANOATE *[106-30-9]*

Teranishi *et al.* (1987); Takeoka *et al.* (1990a)	d	0.002 - 0.0022
Pino & Mesa (2006)	d	0.002
Giri *et al*. (2010)	d	0.0019

5-ETHYL-2-HEPTANOL *[19780-40-6]*

Aoki & Koizumi (1986)		0.048

ETHYL HEXADECANOATE (ethyl palmitate) *[628-97-7]*

Teranishi *et al.* (1987)	d	2
Buttery *et al.* (1988a)	d	> 2
Pino & Mesa (2006)	d	2

2-ETHYLHEXANAL *[123-05-7]*

Schnabel *et al.* (1988)	d	0.041 - 0.082

ETHYL HEXANOATE (ethyl caproate) *[123-66-0]*

Guadagni (1970b);Buttery (1981); Teranishi *et al.* (1987); Takeoka *et al.* (1989,1990a,1992);		
Hansen *et al.* (1992)	d	0.001
Pyysalo *et al.* (1977); Hirvi & Honkanen (1984)	d	0.000 3
Boidron (1978)		0.036
Schnabel *et al.* (1988)	d	0.000 87 - 0.0043
Takahashi (1990)		0.22
Schieberle (1991,1992) ; Buhr *et al.* (2010)	d	0.005
Larsen & Poll (1992)	d	0.000 01 - 0.000 1
Boonbumrung *et al.* (2001)	d	0.0010
Pino & Mesa (2006)	d	0.001
Giri *et al*. (2010)	d	0.0022
Seideneck & Schieberle (2011)		0.005

2-ETHYLHEXANOIC ACID *[149-57-5]*

Schnabel *et al.* (1988)	d	27 - 63

2-ETHYL-1-HEXANOL *[104-76-7]*

Rosen *et al.* (1963)		0.27

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Lillard & Powers (1975)	d	1.28
Schnabel *et al.* (1988)	d	0.83 - 1.5
Suffet (1992)		0.198
Takahashi *et al.* (2002)		0.3
Giri *et al*. (2010)	d	25.4822

4-ETHYL-3-HEXANOL

Schnabel *et al.* (1988)	d	0.078 - 0.17

4-ETHYL-3-HEXANONE *[6137-12-8]*

Schnabel *et al.* (1988)	d	0.0082 - 0.041

(*Z*)-2-ETHYL-2-HEXENAL

Robert *et al.* (2004)	d	0.125

2-ETHYLHEXYL ACETATE *[103-09-3]*

Schnabel *et al.* (1988)	d	0.043 - 0.26

2-ethylhexyl acrylate⇒2-ETHYLHEXYL 2-PROPENOATE
2-ethylhexyl methacrylate⇒2-ETHYLHEXYL 2-METHYL-2-PROPENOATE

2-ETHYLHEXYL 2-METHYL-2-PROPENOATE (2-ethylhexyl methacrylate) *[688-84-6]*

Piringer & Granzer (1984); Granzer *et al.* (1986)	0.02 - 0.5

2-ETHYLHEXYL 2-PROPENOATE (2-ethylhexyl acrylate) *[103-11-7]*

Piringer & Granzer (1984); Granzer *et al.* (1986)	0.005 - 0.2

ETHYL 2-HYDROXYBUTANOATE *[52089-54-0]*

Takeoka *et al.* (1995b)	d	0.8

ETHYL 3-HYDROXYBUTANOATE *[5405-41-4]*

Takeoka *et al.* (1995b)	d	2.5

ETHYL 4-HYDROXYBUTANOATE *[999-10-0]*

Atika *et al.* (1988); Takahashi (1990)	40 – 50

(*R*)-ETHYL 3-HYDROXYHEXANOATE

Buettner & Schieberle (2001a, 2001b)	0.27

4-ETHYL-3-HYDROXY-5-METHYL-2(5*H)*-FURANONE

Rödel & Hempel (1974)	r	0.45

5-ETHYL-3-HYDROXY-4-METHYL-2(5*H)*-FURANONE (abhexone, 2,4-dihydroxy-3-methyl-2-hexenoic acid lactone, ethylsotolon) *[698-10-2]*

Pittet *et al.* (1970)		0.5 - 1.0
Rödel & Hempel (1974)	r	0.07
Semmelroch *et al.* (1995)		0.0075
Buttery (1999)	d	0.000 1

2-ETHYL-4-HYDROXY-5-METHYL-3(2*H)*-FURANONE (sugarone, homofuraneol) *[27538-10-9]*

Semmelroch & Grosch (1996)		0.02
Buttery (1999)	d	0.02

5-ETHYL-4-HYDROXY-2-METHYL-3(2*H)*-FURANONE *[27538-09-6]*

Semmelroch *et al.* (1995)	0.00115

ETHYL 2-HYDROXYPROPANOATE (ethyl lactate) *[97-64-3]*

Takahashi (1990)	50 – 250

6-ETHYL-3-ISOBUTYL-2*H*-2-PYRANONE (mucidone) *[17654-99-8]*

Aguilar (2004)	0.003

ethyl isobutyrate⇒ETHYL 2-METHYLPROPANOATE
ethyl isohexanoate⇒ETHYL 4-METHYLPENTANOATE
ethyl isovalerate⇒ETHYL 3-METHYLBUTANOATE
ethylkahweofuran⇒2-ETHYL-3-OXA-8-THIABICYCLO[3.3.0]-1,4-OCTADIENE
ethyl lactate⇒ETHYL 2-HYDROXYPROPANOATE
ethyl laurate⇒ETHYL DODECANOATE
ethyl mercaptan⇒ETHANETHIOL

2-ETHYL-3-MERCAPTO-1-PENTANOL

Widder *et al.* (2001, 2002)		0.0014

ETHYL 2-MERCAPTOPROPANOATE *[19788-49-9]*

Takeoka *et al.* (1995b)	d	0.001

ethyl methacrylate⇒ETHYL 2-METHYL-2-PROPENOATE

5-ETHYL-3-METHOXY-4-METHYL-2(5*H*)-FURANONE

Rödel & Hempel (1974)	r	4.8

5-ETHYL-3-METHOXY-2-METHYLPYRAZINE *[134079-38-2]*

Mihara *et al.* (1991)	d	0.000 2

4-ETHYL-2-METHOXYPHENOL (*p*-ethylguaiacol, 4-ethylguaiacol) *[2785-89-9]*

Semmelroch *et al.* (1995)		0.050
Czerny *et al.* (2008)	d	0.0044
Czerny *et al.* (2008)	r	0.016
Giri *et al*. (2010)	d	0.08925

ETHYL 3-(2-METHOXYPHENYL)-3-METHYL-2,3-EPOXYPROPANOATE

Tekin & Karaman (1992)	d	2

ETHYL 3-(4-METHOXYPHENYL)-3-METHYL-2,3-EPOXYPROPANOATE

Tekin & Karaman (1992)	d	2

2-ETHYL-3-METHOXYPYRAZINE *[25680-58-4]*

Seifert *et al.* (1970); Teranishi *et al.* (1974,1975)	d	0.000 4 - 0.000 425
Masuda & Mihara (1988)	d	0.01

N-ETHYLMETHYLAMINE (methylethylamine) *[624-78-2]*

Amoore *et al.* (1975b)	d	9.02

ETHYL 2-METHYLBUTANOATE (ethyl 2-methylbutyrate) *[7452-79-1]*

Flath *et al.* (1967); Guadagni (1969,1970b); Buttery (1981); Teranishi *et al.* (1987);		
Takeoka *et al.* (1989,1990a)	d	0.000 1 - 0.000 3
Jakob *et al.* (1969)	d	0.000 08
Schnabel *et al.* (1988)	d	0.000 091 - 0.000 26
Ong & Acree (1999)		0.000 5
Pino & Mesa (2006)	d	0.000 006
Takeoka *et al.* (2008)	d	0.000 007
Czerny *et al.* (2008)	d	0.000 013
Czerny *et al.* (2008)	r	0.000 063

ETHYL (*S*)-(+)-2-METHYLBUTANOATE *[10307-61-6]*

Takeoka *et al.* (1990b,1991a,1991b,1992)	d	0.000 006
Averbeck & Schierberle (2009, 2011)	d	0.000 001

ETHYL 3-METHYLBUTANOATE (ethyl isovalerate, ethyl 3-methylbutyrate) *[108-64-5]*

Schnabel *et al.* (1988)	d	0.000 091 - 0.000 43
Takeoka *et al.* (1991b)	d	0.000 01
Pino & Mesa (2006)	d	0.000 1

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Czerny *et al.* (2008)	d	0.000 023
Czerny *et al.* (2008)	r	0.000 11
Giri *et al*. (2010)	d	0.000 01

ETHYL (*E*)-2-METHYL-2-BUTENOATE (ethyl tiglate) *[5837-78-5]*

Takeoka *et al.* (1992,2008)	d	0.063 - 0.065

ETHYL (*Z*)-2-METHYL-2-BUTENOATE (ethyl angelate) *16509-44-7]*

Takeoka *et al.* (1998)	d	0.000 06

ETHYL 3-METHYL-2-BUTENOATE *[638-10-8]*

Takeoka *et al.* (1998)	d	0.025

ETHYL 3-METHYL-3-BUTENOATE *[1617-19-2]*

Takeoka *et al.* (1998)	d	0.000 22

ethyl 2-methylbutyrate⇒ETHYL 2-METHYLBUTANOATE
ethyl 3-methylbutyrate⇒ETHYL 3-METHYLBUTANOATE

2-ETHYL-4-METHYL-1,3-DIOXOLANE *[4359-46-0]*

Lillard & Powers (1975)	d	0.38
Schweitzer *et al.* (1999b)		0.000 005 - 0.000 01

5-ETHYL-4-METHYL-2(5*H)*-FURANONE

Rödel & Hempel (1974)	r	20

ETHYL 2-METHYLHEXANOATE

Takeoka *et al.* (1995b)	d	0.0025

ETHYL (*E*)-2-METHYL-2-HEXENOATE

Takeoka *et al.* (1998)	d	0.000 3

ETHYL 3-METHYL-3-(4-METHYLPHENYL)-2,3-EPOXYPROPANOATE

Tekin & Karaman (1992)	d	17

ETHYL 2-METHYL-3-(4-METHYLPHENYL)-3-PROPYL-2,3-EPOXYPROPANOATE

Tekin & Karaman (1992)	d	8.5

5-ETHYL-2-METHYL-3-(METHYLTHIO)PYRAZINE *[134079-39-3]*

Mihara *et al.* (1991)	d	0.002

ETHYL 2-METHYL-3,4-PENTADIENOATE *[60523-21-9]*

Takeoka *et al.* (1998)	d	0.000 05

2-ETHYL-3-METHYLPENTANAL *[42347-74-0]*

Schnabel *et al.* (1988)	d	0.077 - 0.42

ETHYL 2-METHYLPENTANOATE *[39255-32-8]*

Takeoka *et al.* (1995b)	d	0.000 003

ETHYL 3-METHYLPENTANOATE *[5870-68-8]*

Takeoka *et al.* (1995b)	d	0.000 008

ETHYL 4-METHYLPENTANOATE (ethyl isohexanoate) *[25415-67-2]*

Schnabel *et al.* (1988)	d	0.000 87 - 0.0043
Takeoka *et al.* (1995b)	d	0.000 01
Ong & Acree (1999)		0.005
Fritsch & Schieberle (2005)		0.000 003

2-ETHYL-3-METHYLPENTANOIC ACID *[22414-77-3]*

Schnabel *et al.* (1988)	d	9.1 - 46

ETHYL (*E*)-2-METHYL-2-PENTENOATE *[1617-40-9]*
Takeoka *et al.* (1998) d 0.0115

ETHYL 2-METHYL-4-PENTENOATE *[53399-81-8]*
Takeoka *et al.* (1998) d 0.000 01

ETHYL 4-METHYL-4-PENTENOATE *[4911-54-0]*
Takeoka *et al.* (1998) d 0.000 03

3-ETHYL-5-METHYLPHENOL *[698-71-5]*
Dietz & Traud (1978) 1

ETHYL 2-METHYL-3-PHENYL-2,3-EPOXYPROPANOATE
Tekin & Karaman (1992) d 103

ETHYL 3-METHYL-3-PHENYL-2,3-EPOXYPROPANOATE (aldehyde C16, strawberry aldehyde, ethyl 3-methyl-3-phenylglycidate) *[77-83-8]*
Appell (1969) 0.025
Tekin & Karaman (1992) d 2

ethyl 3-methyl-3-phenylglycidate⇒ETHYL 3-METHYL-3-PHENYL-2,3-EPOXYPROPANOATE

ETHYL 3-(4-METHYLPHENYL)-3-PROPYL-2,3-EPOXYPROPANOATE
Tekin & Karaman (1992) d 78

ETHYL 2-METHYLPROPANOATE (ethyl isobutyrate) *[97-62-1]*
Appell (1969) 0.025
Guadagni (1970b); Teranishi *et al.* (1987);
Takeoka *et al.* (1989, 2008) d 0.000 1
Schnabel *et al.* (1988) d 0.000 0081 - 0.000 043
Ong & Acree (1999) 0.015
Pino & Mesa (2006) d 0.000 1
Czerny *et al.* (2008) d 0.000 089
Czerny *et al.* (2008) r 0.000 22
Seideneck & Schieberle (2011) 0.000 02

ETHYL 2-METHYL-2-PROPENOATE (ethyl methacrylate) *[97-63-2]*
Piringer & Granzer (1984); Granzer *et al.* (1986) 0.002 - 0.05

2-ETHYL-3-METHYLPYRAZINE *[15707-23-0]*
Guadagni *et al.* (1972); Teranishi *et al.* (1974) d 0.13
Calabretta (1973) d 0.13
Mihara & Masuda (1988) d 0.5

2-ETHYL-5-METHYLPYRAZINE *[13360-64-0]*
Guadagni *et al.* (1972); Teranishi *et al.* (1974);
Buttery & Ling (1995,1998);
Buttery *et al.* (1997,1999) d 0.1
Mihara & Masuda (1988) d 0.016

2-ETHYL-6-METHYLPYRAZINE *[13925-03-6]*
Mihara & Masuda (1988) d 0.04

5-ETHYL-2-METHYLPYRIDINE *[104-90-5]*
Rosen *et al.* (1963) 0.019

5-ETHYL-4-METHYLTHIAZOLE *[31883-01-9]*
Teranishi *et al.* (1974) d 0.1

ETHYL 2-(METHYLTHIO)ACETATE *[4455-13-4]*

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Guadagni (1970b); Teranishi *et al.* (1987)	d	0.025

ETHYL 4-(METHYLTHIO)BUTANOATE *[22014-48-8]*

Takeoka *et al.* (1991a)	d	0.019

ETHYL 3-(METHYLTHIO)PROPANOATE *[13327-56-5]*

Takeoka *et al.* (1989)	d	0.007
Giri *et al*. (2010)	d	0.00845

ETHYL 3-(METHYLTHIO)-(*E*)-2-PROPENOATE *[136115-65-6]*

Takeoka *et al.* (1991a)	d	0.246

2-ETHYL-3-(METHYLTHIO)PYRAZINE *[21948-70-9]*

Masuda & Mihara (1988)	d	0.04

ethyl myristate⇒ETHYL TETRADECANOATE

O-ETHYL O-4-NITROPHENYL PHENYLPHOSPHONOTHIOATE (epn-300) *[2104-64-5]*

Sigworth (1965)	d	0.018

ETHYL NONANOATE (ethyl pelargonate) *[123-29-5]*

Takahashi (1990)		12
Pino & Mesa (2006)	d	0.377

ETHYL OCTANOATE (ethyl caprylate) *[106-32-1]*

Takahashi (1990)		0.2 - 1.0
Takeoka *et al.* (1992)	d	0.092
Gassenmeier & Schieberle (1995)	d	0.005
Rychlik & Grosch (1996); Rychlik (1998)		0.07
Boonbumrung *et al.* (2001)	d	0.015
Pino & Mesa (2006)	d	0.194
Giri *et al*. (2010)	d	0.0193

4-ETHYLOCTANOIC ACID *[16493-80-4]*

Boelens *et al.* (1983)	d	0.0018

ETHYL (*E*)-4-OCTENOATE *[78989-37-4]*

Takeoka *et al.* (1992)	d	0.05

2-ETHYL-3-OXA-8-THIABICYCLO[3.3.0]-1,4-OCTADIENE (ethylkahweofuran)

Tressl (1989,1990)		0.005

ethyl palmitate⇒ETHYL HEXADECANOATE
ethyl pelargonate⇒ ETHYL NONANOATE

2-ETHYLPENTANAL

Schnabel *et al.* (1988)	d	0.83 - 2.5

ETHYL PENTANOATE (ethyl valerate) *[539-82-2]*

Flath *et al.* (1967); Teranishi *et al.* (1987);		
Takeoka *et al.* (1989)	d	0.0015 - 0.005
Schnabel *et al.* (1988)	d	0.0087 - 0.044
Giri *et al*. (2010)	d	0.0058

2-ETHYLPENTANOIC ACID *[20225-24-5]*

Schnabel *et al.* (1988)	d	0.91 - 9.1

3-ETHYL-2-PENTANOL *[609-27-8]*

Schnabel *et al.* (1988)	d	0.26 - 0.43

3-ETHYL-3-PENTANOL *[597-49-9]*

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Pelosi & Pisanelli (1981)	d	15
Schnabel *et al.* (1988)	d	0.042 - 0.084

3-ETHYL-2-PENTANONE *[6137-03-7]*

Schnabel *et al.* (1988)	d	0.25 - 0.41

2-ETHYLPHENOL *[90-00-6]*

Dietz & Traud (1978)		0.3

3-ETHYLPHENOL *[620-17-7]*

Dietz & Traud (1978)		0.8
Czerny *et al.* (2008)	d	0.000 85
Czerny *et al.* (2008)	r	0.0017

4-ETHYLPHENOL *[123-07-9]*

Pyysalo *et al.* (1977)	d	0.042
Dietz & Traud (1978)		0.6
Czerny *et al.* (2008)	d	0.013
Czerny *et al.* (2008)	r	0.021

ETHYL 2-PHENYLACETATE *[101-97-3]*

Giri *et al.* (2010)	d	0.15555

ETHYL 3-PHENYL-2,3-EPOXYPROPANOATE (ethyl phenyl glycidate) *[121-39-1]*

Tekin & Karaman (1992)	d	8.5

trans-ETHYL 3-PHENYL-2,3-EPOXYPROPANOATE

Tekin & Karaman (1992)	d	10

ethyl phenyl glycidate⇒ETHYL 3-PHENYL-2,3-EPOXYPROPANOATE

ETHYL 3-PHENYLPROPENOATE (ethyl cinnamate) *[103-36-6]*

Le Magnen (1952)		0.017 - 0.04

ETHYL (*E*)-3-PHENYLPROPENOATE (ethyl (*E*)-cinnamate) *[4192-77-2]*

Pino & Mesa (2006)	d	0.165

ethyl pivalate⇒ETHYL 2,2-DIMETHYLPROPANOATE

ETHYL PROPANOATE *[105-37-3]*

Ahmed *et al.* (1978a)	d	0.0099
Piggott & Findlay (1984)	d	1.9
Teranishi *et al.* (1987); Takeoka *et al.* (1989,1992)	d	0.01
Schnabel *et al.* (1988)	d	0.0089 - 0.045
Pino & Mesa (2006)	d	0.01

ETHYL 2-PROPENOATE (ethyl acrylate) *[140-88-5]*

Baker (1963)		0.0067
Piringer & Granzer (1984); Granzer *et al.* (1986)		0.000 1 - 0.002

ETHYLPROPYLAMINE *[20193-20-8]*

Amoore *et al.* (1975b)	d	17.4

4-ETHYL-5-PROPYLTHIAZOLE *[57246-61-4]*

Buttery *et al.* (1976a)	d	0.000 06

ETHYLPYRAZINE *[13925-00-3]*

Koehler *et al.* (1971)	d	22
Guadagni *et al.* (1972); Teranishi *et al.* (1974); Buttery & Ling (1995,1998); Buttery *et al.* (1999)	d	6
Masuda & Mihara (1988)	d	4

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Buttery *et al.* (1997) d 4

ETHYL SALICYLATE *[118-61-6]*
Pino & Mesa (2006) d 0.084

ethylsotolon⇒5-ETHYL-3-HYDROXY-4-METHYL-2(5*H*)-FURANONE

ETHYL TETRADECANOATE (ethyl myristate) *[124-06-1]*
Teranishi *et al.* (1987) d 4
Boonbumrung *et al.* (2001) d 0.18
Pino & Mesa (2006) d 4

1-(3-ETHYL-5,6,7,8-TETRAHYDRO-5,5,8,8-TETRAMETHYL-2-NAPHTHALENYL)ETHANONE (versalide) *[88-29-9]*
Amoore *et al.* (1977) d 0.0024

2-(ETHYLTHIO)-3,6-DIMETHYLPYRAZINE *[59021-09-9]*
Mihara *et al.* (1991) d 0.021

2-(ETHYLTHIO)ETHANAL
Rössner *et al.* (2002) 0.000 8

(ETHYLTHIO)ETHANE (diethyl sulphide) *[352-93-2]*
Kauffmann (1907) r 0.08 - 0.11
Pietrzak & Barylko-Pikielna (1976) 0.0048

2-(ETHYLTHIO)ETHANOL *[110-77-0]*
Cherkinski (1961) d 0.01

2-(ETHYLTHIO)-5-ISOBUTYL-3-METHYLPYRAZINE *[99784-27-7]*
Masuda & Mihara (1986); Shibamoto (1986) d 0.032

2-(ETHYLTHIO)-5-ISOPROPYL-3-METHYLPYRAZINE *[99784-25-5]*
Masuda & Mihara (1986); Shibamoto (1986) d 0.085

2-(ETHYLTHIO)-3-METHYL-5-(2-METHYLBUTYL)PYRAZINE *[99784-28-8]*
Masuda & Mihara (1986); Shibamoto (1986) d 0.0025

2-(ETHYLTHIO)-3-METHYL-5-(2-METHYLPENTYL)PYRAZINE *[99784-29-9]*
Masuda & Mihara (1986); Shibamoto (1986) d 0.056

2-(ETHYLTHIO)-3-METHYLPYRAZINE *[32740-98-0]*
Masuda & Mihara (1988) d 0.07

2-(ETHYLTHIO)-3-OCTYLPYRAZINE *[113685-93-1]*
Masuda & Mihara (1988) d 0.002

2-(ETHYLTHIO)-3-PENTYLPYRAZINE *[113685-92-0]*
Masuda & Mihara (1988) d 0.001

(ETHYLTHIO)PYRAZINE *[65032-06-6]*
Masuda & Mihara (1988) d 0.9

ethyl tiglate⇒ETHYL (*E*)-2-METHYL-2-BUTENOATE

3-ETHYLTOLUENE *[620-14-4]*
Jüttner *et al.* (1995) r 0.8

4-ETHYLTOLUENE *[622-96-8]*
Jüttner *et al.* (1995) r 0.6

cis-1-ETHYL-2,2,6-TRIMETHYLCYCLOHEXANOL
Finato *et al.* (1992) d 0.000 12

(*E*)-(*R*)-2-ETHYL-4-(2,2,3-TRIMETHYL-3-PENTEN-1-YL)-2-BUTEN-1-OL
Aida *et al.* (1997, 2000) 0.000 000 18

(*E*)-(*S*)-2-ETHYL-4-(2,2,3-TRIMETHYL-3-PENTEN-1-YL)-2-BUTEN-1-OL
Aida *et al.* (1997, 2000) 0.000 0092

ethyl valerate⇒ETHYL PENTANOATE
ethylvanillin⇒2-ETHOXY-4-HYDROXYBENZALDEHYDE
ethyl vinyl ketone⇒1-PENTEN-3-ONE
eucalyptol⇒1,3,3-TRIMETHYL-2-OXABICYCLO[2.2.2]OCTANE
eugenol⇒4-ALLYL-2-METHOXYPHENOL
eugenol methyl ether⇒4-ALLYL-1,2-DIMETHOXYBENZENE
exaltolide⇒15-HYDROXYPENTADECANOIC ACID LACTONE
exaltone⇒CYCLOPENTADECANONE

F

farnesene⇒7,11-DIMETHYL-3-METHYLENE-1,6,10-DODECATRIENE
farnesol⇒3,7,11-TRIMETHYL-2,6,10-DODECATRIEN-1-OL
fenchone⇒1,3,3-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANONE
(*d*)-fenchone⇒(1*S*)-(+)-1,3,3-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANONE
(+)-fenchone⇒(1*S*)-(+)-1,3,3-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANONE
(*l*)-fenchone,⇒(1*R*)-(-)-1,3,3-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANONE
(-)-fenchone⇒(1*R*)-(-)-1,3,3-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANONE
fichtelite⇒1,2,3,4,4a,4b.α ,5,6,7,8,8a.β,9,10,10a.α-TETRADECAHYDRO-7α-ISOPPROPYL-1β,4a.β-DIMETHYLPHENANTHRENE
(*E*)-filbertone⇒5-METHYL-(*E*)-2-HEPTEN-4-ONE
fixolide⇒7-ACETYL-1,1,3,4,4,6-HEXAMETHYL-1,2,3,4-TETRAHYDRONAPHTHALENE

2-FLUOROPHENOL *[367-12-4]*
Dietz & Traud (1978) 1.5

3-FLUOROPHENOL *[372-20-3]*
Dietz & Traud (1978) 2

4-FLUOROPHENOL *[371-41-5]*
Dietz & Traud (1978) 2.3

formaldehyde⇒METHANAL

FORMIC ACID (methanoic acid) *[64-18-6]*
Guadagni (1966,1968,1969); Aharoni *et al.* (1980) d 400 - 450
Amoore & Venstrom (1966); Amoore (1969) d 1,360 - 1,500
Amoore *et al.* (1968) d 3,650
Brown *et al.* (1968) 25,000
Zoeteman *et al.* (1971) 8,000
Kikuchi *et al.* (1976) 1,000
Rosen *et al.* (1979) d 46
Schnabel *et al.* (1988) d 1,240 - 2,440

2-FORMYL-1-FURFURYLPYRROLE (N-furfuryl-2-formylpyrrole)
Buttery *et al.* (1997) d 0.097

4-formyl-2-methoxyphenol⇒4-HYDROXY-3-METHOXYBENZALDEHYDE

2-FORMYL-1-METHYLPYRROLE (N-methyl-2-formylpyrrole) *[1192-58-1]*
Buttery *et al.* (1997); Buttery (1999) d 0.037

FURAN (1,4-epoxy-1,3-butadiene) *[110-00-9]*
Mulders (1972,1973) d 4.5 - 6

furaneol⇒4-HYDROXY-2,5-DIMETHYL-3(2*H*)-FURANONE
trans-furan linalool oxide⇒*trans*-5-ETHENYLTETRAHYDRO-α,α,5-TRIMETHYL-2-FURANMETHANOL
2-furanmethanethiol⇒2-FURYLMETHANETHIOL

2-FURANMETHANOL (furfuryl alcohol) *[98-00-0]*

Buttery *et al.* (1994a,1997,1999);		
Buttery & Ling (1995,1998)	d	1.9 - 2.0
Giri *et al*. (2010)	d	4.50050

[2-(FURANYLMETHYL)THIO]PYRAZINE (2-(methylthiofurfurane)pyrazine)

Calabretta (1978)	d	< 0.001

FURFURAL *[98-01-1]*

Appell (1969)		1
Buttery *et al.* (1969b,1971,1978b,1990a,1997,1999);		
Guadagni *et al.* (1972); Buttery (1981,1993);		
Buttery & Ling (1995,1998)	d	2 - 3
Buttery *et al.* (1988a,1994a)	d	23
Pino *et al.* (1986)	d	0.282
Takahashi (1990)		50 – 150
Tamura *et al.* (2001); Boonbumrung *et al.* (2001)	d	0.77
Pino & Mesa (2006)	d	3
Giri *et al*. (2010)	d	9.56200

furfuryl alcohol⇒2-FURANMETHANOL
N-furfuryl-2-formylpyrrole⇒2-FORMYL-1-FURFURYLPYRROLE
furfuryl mercaptan⇒2-FURYLMETHANETHIOL
furfuryl methyl disulphide⇒2-[(METHYLDITHIO)METHYL]FURAN

1-FURFURYLPYRROLE *[1438-94-4]*

Tressl *et al.* (1979c,1985b); Silwar (1982)	d	0.1
Buttery *et al.* (1997)	d	0.1

furfurylthiol⇒2-FURYLMETHANETHIOL

2-FURYLACETIC ACID *[2745-26-8]*

Davídek *et al.* (1979)		34.6

2-FURYLMETHANETHIOL (furfuryl mercaptan, furfurylthiol, 2-furanmethanethiol) *[98-02-2]*

Tressl *et al.* (1979c); Tressl & Silwar (1981);		
Silwar (1982); Tressl (1989,1990)		0.000 005 - 0.000 01
Grosch *et al.* (1993); Guth & Grosch (1994)		0.000 12 - 0.000 13
Semmelroch *et al.* (1995); Schieberle &		
Hofmann (1996a, 1996b);		
Kerscher & Grosch (2000); Binggeli *et al.* (2003)		0.000 01
Buttery *et al.* (1997); Buttery & Ling (1998)	d	0.000 006
Marchand *et al.* (2000)	d	0.000 001
Czerny *et al.* (2008)	d	0.000 036
Czerny *et al.* (2008)	r	0.000 08

3-(2-FURYL)PROPANOIC ACID *[935-13-7]*

Davídek *et al.* (1979)		15.6

G

galaxolide⇒1,3,4,6,7,8-HEXAHYDRO-4,6,6,7,8,8-HEXAMETHYLCYCLOPENTA[g]-2-BENZOPYRAN
geosmin⇒*trans*-1,10-DIMETHYL-*trans*-9-DECALOL
geranial⇒3,7-DIMETHYL-*trans*-2,6-OCTADIENAL
geraniol⇒3,7-DIMETHYL-*trans*-2,6-OCTADIEN-1-OL
geranyl acetate⇒3,7-DIMETHYL-(*E*)-1,6-OCTADIEN-1-YL ACETATE

geranylacetone⇒6,10-DIMETHYLUNDECA-5,9-DIEN-2-ONE
geranyl formate⇒3,7-DIMETHYL-(*E*)-1,6-OCTADIEN-1-YL FORMATE
geranyl isobutyrate⇒3,7-DIMETHYL-(*E*)-2,6-OCTADIEN-1-YL 2-METHYLPROPANOATE

GERANYLOXYMETHYLPYRAZINE

Calabretta (1975,1978)	d	> 0.5

3-GERANYLOXY-2-METHYLPYRAZINE

Calabretta (1975)	d	0.5
Calabretta (1978)	d	0.1

5-GERANYLOXY-2-METHYLPYRAZINE

Calabretta (1975,1978)	d	0.1

geranyl propanoate⇒3,7-DIMETHYL-(*E*)-2,6-OCTADIEN-1-YL PROPANOATE
glutaraldehyde⇒1,5-PENTANEDIAL
glycerin⇒1,2,3-PROPANETRIOL
glycerol⇒1,2,3-PROPANETRIOL
guaiacol⇒2-METHOXYPHENOL
guthion⇒O,O-DIMETHYL S-(4-OXO-3H-1,2,3-BENZOTRIAZINE-3-METHYL)PHOSPHORODITHIOATE

H

heliotropine⇒3,4-(METHYLENEDIOXY)BENZALDEHYDE
heptachlor⇒3,4,5,6,7,8,8a-HEPTACHLORODICYCLOPENTADIENE

3,4,5,6,7,8,8a-HEPTACHLORODICYCLOPENTADIENE (heptachlor) *[76-44-8]*

Sigworth (1965)	d	0.02

HEPTADECANOIC ACID *[506-12-7]*

Cherkinski (1961)	d	20.0

1-HEPTADECENE *[6765-39-5]*

Tamura *et al.* (1995)		8

7-HEPTADECENE *[54290-12-9]*

Tamura *et al.* (1995)		5.6

2,4-HEPTADIENAL *[5910-85-0]*

Tamura *et al.* (2001)	d	0.15

(*E*,*E*)-2,4-HEPTADIENAL *[4313-02-4]*

Jensen *et al.* (1999)	d	0.056
Tamura *et al.* (1995)		0.8
Giri *et al*. (2010)	d	0.0154

(*E*,*Z*)-2,4-HEPTADIENAL *[4313-02-4]*

Giri *et al*. (2010)	d	0.0948

γ-heptalactone⇒ 4-HYDROXYHEPTANOIC ACID LACTONE
heptaldehyde⇒HEPTANAL
heptamethyleneimine⇒ AZACYCLOOCTANE

HEPTANAL (heptaldehyde) *[111-71-7]*

Guadagni *et al.* (1963,1972); Buttery *et al.* (1968, 1978b,1988a,1994a,1997,1999); Teranishi *et al.* (1974); Buttery & Ling (1995,1998);		
Takeoka *et al*. (1995a)	d	0.003
Amoore *et al.* (1976)	d	0.0195
Schnabel *et al.* (1988)	d	0.0041 - 0.0082
Sugisawa *et al.* (1991); Yang *et al.* (1992);		

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Tamura *et al.* (1993)	d	0.06 - 0.55
Fabrellas *et al.* (2004)		0.000 25 - 0.000 58
Limpawattana (2007)	d	0.00206
Giri *et al.* (2010)	d	0.0028
HEPTANE *[142-82-5]*		
Zoeteman *et al.* (1971)		50
(*R*,*S*)-2-HEPTANETHIOL *[628-00-2]*		
Simian *et al*. (2004); Robert *et al*. (2005)	d	0.010
(*R*)-2-HEPTANETHIOL		
Simian *et al*. (2004)	d	0.010
(*S*)-2-HEPTANETHIOL		
Simian *et al*. (2004)	d	0.010
HEPTANOIC ACID (oenanthic acid) *[111-14-8]*		
Cherkinski (1961)	d	3.0
Amoore *et al.* (1968)	d	10.4
Schnabel *et al.* (1988)	d	0.64 - 0.91
1-HEPTANOL *[111-70-6]*		
Aoki & Koizumi (1986)		0.094
Schnabel *et al.* (1988)	d	0.33 - 0.58
Sugisawa *et al.* (1991); Yang *et al.* (1992);		
Tamura *et al.* (1993,1995)	d	0.4 - 4.6
Pino & Mesa (2006)	d	0.425
Giri *et al*. (2010)	d	0.0054
2-HEPTANOL *[543-49-7]*		
Pyysalo *et al.* (1977,1979)	d	0.070
Schnabel *et al.* (1988)	d	0.041 - 0.081
Giri *et al*. (2010)	d	0.065235
3-HEPTANOL *[589-82-2]*		
Schnabel *et al.* (1988)	d	0.24 - 0.41
4-HEPTANOL *[589-55-9]*		
Schnabel *et al.* (1988)	d	0.41 - 0.81
2-HEPTANONE (methyl amyl ketone) *[110-43-0]*		
Land (1968)	r	3
Buttery *et al.* (1969b,1988a,1994a,1997,1999);		
Teranishi *et al.* (1974); Buttery & Ling (1995,1998)	d	0.14
Pyysalo *et al.* (1977)	d	0.37
Barylko-Pikielna *et al.* (1983)	d	0.75 - 1.33
Barylko-Pikielna *et al.* (1983)	r	2.66 - 3.73
Aoki & Koizumi (1986)		0.024
Schnabel *et al.* (1988)	d	0.021 - 0.041
Sugisawa *et al.* (1991)	d	4.6
Larsen & Poll (1992)	d	0.001 - 0.01
Pino & Mesa (2006)	d	0.14
3-HEPTANONE (butyl ethyl ketone) *[106-35-4]*		
De Grunt (1975); De Grunt *et al.* (1977)	d	0.0075 - 0.008
Schnabel *et al.* (1988)	d	0.080 - 0.16
4-HEPTANONE (dipropyl ketone) *[123-19-3]*		
Schnabel *et al.* (1988)	d	0.0082 - 0.041

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

(*E*)-2-HEPTENAL *[18829-55-5]*

Guadagni *et al.* (1963,1972); Buttery *et al.* (1971, 1987a,1988a,1989,1990a); Teranishi *et al.* (1974); Buttery (1993); Buttery & Ling (1995,1998)	d	0.013
Eriksson *et al.* (1976)	d	0.051
Rashash *et al.* (1997)		0.04

4-HEPTENAL *[62238-34-0]*

Kossa *et al.* (1979); Tressl *et al.* (1979b,1980)		0.000 4
Giri *et al*. (2010)	d	0.0042

(*E*)-4-HEPTENAL *[929-22-6]*

McGill *et al.* (1974)		0.01

(*Z*)-4-HEPTENAL *[6728-31-0]*

McGill *et al.* (1974)		0.000 8
Milo (1995a, 1995b)		0.000 2
Schuh & Schieberle (2006)	d	0.000 06
Czerny *et al.* (2008)	d	0.000 0087
Czerny *et al.* (2008)	r	0.000 025

(*E*)-2-HEPTEN-1-OL *[33467-76-4]*

Eriksson *et al.* (1976)	d	4.172

1-HEPTEN-3-ONE *[2918-13-0]*

Buettner & Schieberle (2001a)		0.000 04

(*E*)-3-HEPTEN-2-ONE *[5609-09-6]*

Buttery *et al.* (1969b); Teranishi *et al.* (1974); Buttery (1989)	d	0.056

6-HEPTENYL ISOTHIOCYANATE

Masuda *et al.* (1996)		0.075

HEPTYL ACETATE *[112-06-1]*

Yang *et al.* (1992); Tamura *et al.* (1993)	d	0.32
Takeoka *et al.* (1996)	d	0.42

2-HEPTYL ACETATE *[5921-82-4]*

Schnabel *et al.* (1988)	d	0.087 - 0.43
Takeoka *et al.* (1996)	d	0.89

HEPTYLAMINE *[111-68-2]*

Amoore *et al.* (1975b)	d	13.8

HEPTYL CYANIDE (octanenitrile) *[124-12-9]*

Fabrellas *et al.* (2004)		0.000 13 - 0.000 5

(*R*)-(-)-2-HEPTYLCYCLOPENTANONE

Yamamoto *et al.* (2002)		0.07

(*S*)-(+)-2-HEPTYLCYCLOPENTANONE

Yamamoto *et al.* (2002)		0.07

heptyl isobutyrate⇒HEPTYL 2-METHYLPROPANOATE

2-HEPTYL-3-METHOXYPYRAZINE *[113685-79-3]*

Masuda & Mihara (1988)	d	0.000 026

HEPTYL 2-METHYLPROPANOATE (heptyl isobutyrate) *[2349-13-5]*

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Guadagni *et al.* (1966); Teranishi *et al.* (1987) d 0.012 - 0.013

HEPTYL PROPANOATE *[2216-81-1]*
Guadagni *et al.* (1966); Teranishi *et al.* (1987) d 0.004

HEPTYLPYRAZINE *[84005-70-9]*
Masuda & Mihara (1988) d 0.1

hexachloran⇒1,2,3,4,5,6-HEXACHLOROCYCLOHEXANE

HEXACHLOROBENZENE *[118-74-1]*
Kölle *et al.* (1972) 3

1,2,3,4,5,6-HEXACHLOROCYCLOHEXANE (hexachloran) *[58-89-9]*
Kölle *et al.* (1972) 0.2 - 4.0

α-1,2,3,4,5,6-HEXACHLOROCYCLOHEXANE (α-benzene hexachloride) *[319-84-6]*
Sigworth (1965) d 0.088

β-1,2,3,4,5,6-HEXACHLOROCYCLOHEXANE (β-benzene hexachloride) *[319-85-7]*
Sigworth (1965) d 0.000 32

γ-1,2,3,4,5,6-HEXACHLOROCYCLOHEXANE (lindane, γ-benzene hexachloride) *[58-89-9]*
Sigworth (1965) d 12.0

δ-1,2,3,4,5,6-HEXACHLOROCYCLOHEXANE (δ-benzene hexachloride) *[319-86-8]*
Sigworth (1965) d 0.000 13

1,2,3,4,10,10-HEXACHLORO-6,7-EPOXY-1,4,4a,5,6,7,8,8a-OCTAHYDRO-1,4-*endo*-*endo*-5,8-DIMETHANONAPHTHALENE (endrin) *[72-20-8]*
Sigworth (1965) d 0.018

1,2,3,4,10,10-HEXACHLORO-6,7-EPOXY-1,4,4a,5,6,7,8,8a-OCTAHYDRO-1,4-*endo*-*exo*-5,8-DIMETHANONAPHTHALENE (dieldrin) *[60-57-1]*
Sigworth (1965) d 0.041

HEXACHLOROETHANE *[67-72-1]*
Cherkinski (1961) d 0.010

1,2,3,4,10,10-HEXACHLORO-1,4,4a,5,8,8a-HEXAHYDRO-1,4-*endo*-*exo*-5,8-DIMETHANONAPHTHALENE (aldrin) *[309-00-2]*
Sigworth (1965) d 0.017

hexachlorophene⇒2,2'-METHYLENEBIS(3,4,6-TRICHLORPHENOL)
ω-hexadecalactone⇒16-HYDROXYHEXADECANOIC ACID LACTONE
δ9-hexadecenolactone⇒5-HYDROXY-9-HEXADECENOIC ACID LACTONE

(*E*,*E*)-2,4-HEXADIENAL (sorbic aldehyde) *[80466-34-8]*
Teranishi *et al.* (1974); Hansen *et al.* (1992) d 0.06
Eriksson *et al.* (1976) d 0.476

2,4-HEXADIEN-1-OL *[111-28-4]*
Kossa *et al.* (1979) 0.050

(*E*,*E*)-2,4-HEXADIEN-1-OL *[17102-64-6]*
Eriksson *et al.* (1976) d 23.618

hexahydroazine⇒PIPERIDINE

4,5,6,7,8,9-HEXAHYDROCYCLOOCTATHIAZOLE (cyclooctathiazole) *[7140-68-3]*
Pelosi *et al.* (1983) d 0.67

HEXAHYDRO-1,1-DIMETHYL-4-METHYLENE-1*H*-CYCLOPENTA[c]FURAN (hop ether) *[19901-95-2]*

Friese *et al.* (1979); Tressl *et al.* (1979a)		0.001 - 0.005

4,4a,5,6,7,8-HEXAHYDRO-4,4a-DIMETHYL-6-(1-METHYLETHENYL)-2(3*H*)-NAPHTHALENONE (4-epinootkatone) *[27621-99-4]*

Guadagni (1969); Stevens *et al.* (1970)	d	0.1

1,3,4,6,7,8-HEXAHYDRO-4,6,6,7,8,8-HEXAMETHYLCYCLOPENTA[g]-2-BENZOPYRAN (galaxolide) *[1222-05-5]*

Amoore *et al.* (1977)	d	0.0041
Young *et al.* (1996)		0.005

2,3,4,5,8,8a-HEXAHYDRO-3-ISOPROPYL-6,8a-DIMETHYL-3a(1*H*)-AZULENOL (carotol) *[465-28-1]*

Buttery *et al.* (1968)	d	0.008

3,4,6,7,8,9-HEXAHYDRO-4,6,6,9,9-PENTAMETHYL-1*H*-NAPHTHO[2,3-c]PYRAN (musk 89)

Amoore *et al.* (1977)	d	0.010

γ-hexalactone⇒4-HYDROXYHEXANOIC ACID LACTONE

HEXAMETHYL-1,3-DIOXOLANE

Pelosi & Pisanelli (1981)	d	5.4

hexamethyleneimine⇒AZACYCLOHEPTANE

HEXANAL *[66-25-1]*

Guadagni *et al.* (1963,1972); Flath *et al.* (1967); Buttery *et al.* (1969b,1971,1978b,1987a,1987b, 1988a,1989,1990a,1994a,1997,1999); Teranishi *et al.* (1974); Buttery (1981,1993); Hansen *et al.* (1992); Buttery & Ling (1995,1998); Takeoka *et al.* (1995a)	d	0.0045 - 0.005
Land (1968)	r	0.4
Herrmann & Poeschel (1973)		0.75
Amoore *et al.* (1976)	d	0.0228
Eriksson *et al.* (1976)	d	0.019
Ahmed *et al.* (1978a)	d	0.00918
Kossa *et al.* (1979)		0.350
Aoki & Koizumi (1986)		0.020
Pino *et al.* (1986)	d	0.0069
Schnabel *et al.* (1988)	d	0.0041 - 0.0081
Sugisawa *et al.* (1991); Yang *et al.* (1992); Tamura *et al.* (1993,1995)	d	0.064
Larsen & Poll (1992)	d	0.01 - 0.1
Grosch *et al.* (1993); Guth & Grosch (1994)		0.010 - 0.0105
Konopka *et al.* (1995); Milo & Grosch (1997); Kerscher & Grosch (2000)		0.0045
Jensen *et al.* (1999)	d	0.0025
Tandon *et al.* (2000)	d	0.479
Arena *et al.* (2001)		0.021
Tamura *et al.* (2001)	d	0.0091
Fabrellas *et al.* (2004)		0.000 32 - 0.000 4
Czerny *et al.* (2007c); Czerny *et al.* (2008)	r	0.01
Pino & Mesa (2006)	d	0.0045
Limpawattana (2007)	d	0.00278
Czerny *et al.* (2008); Steinhaus *et al.* (2009); Sinuco *et al.* (2010)	d	0.0024
Ömür-Özbek & Dietrich (2008)		0.0039
Burdack-Freitag *et al.* (2010)	d	0.0615 (normal atmospheric pressure)
Burdack-Freitag *et al.* (2010)	d	0.098 (low atmospheric pressure)

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Burdack-Freitag *et al.* (2010)	r	0.073 (normal atmospheric pressure)
Burdack-Freitag *et al.* (2010)	r	0.098 (low atmospheric pressure)
Giri *et al.* (2010)	d	0.005

1,5-HEXANEDIOL *[928-40-5]*

Schnabel *et al.* (1988)	d	0.97 - 9.7

2,5-HEXANEDIOL *[2935-44-6]*

Schnabel *et al.* (1988)	d	19 – 48

hexanenitrile⇒PENTYL CYANIDE

2-HEXANETHIOL *[1679-06-7]*

Robert *et al.* (2005)	d	0.06

HEXANOIC ACID (caproic acid) *[142-62-1]*

Cherkinski (1961)	d	3.0
Amoore *et al.* (1968); Amoore (1976)	d	9.2
Pyysalo *et al.* (1977)	d	3.7
Schnabel *et al.* (1988)	d	0.093 - 0.93
Larsen & Poll (1990)	d	1 - 10
Buttery *et al.* (1990a,1999); Buttery (1993)	d	3
Larsen & Poll (1992)	d	1.0 - 10
Tamura *et al.* (2001); Boonbumrung *et al.* (2001)	d	1.8 - 1.84
Karagül-Yüceer *et al.* (2003, 2004)	d	0.0356 - 0.036
Pino & Mesa (2006)	d	3
Schuh & Schieberle (2006)	d	0.89

1-HEXANOL (hexyl alcohol) *[111-27-3]*

Williams (1953)		2
Flath *et al.* (1967); Buttery *et al.* (1971,1978b,1989, 1997,1999); Hansen *et al.* (1992); Buttery (1993)	d	0.5
Johansson *et al.* (1973)	d	2.0
Eriksson *et al.* (1976)	d	4.865
De Grunt (1975); De Grunt *et al.* (1977)	d	0.2
Buttery *et al.* (1988a); Takeoka *et al.* (1990a); Buttery & Ling (1995)	d	2.5
Aoki & Koizumi (1986)		0.150
Schnabel *et al.* (1988)	d	0.82 - 1.6
Sugisawa *et al.* (1991); Yang *et al.* (1992); Tamura *et al.* (1993,1995)	d	3.125 - 10
Larsen & Poll (1992)	d	0.01 - 0.1
Tandon *et al.* (2000)	d	0.759
Tamura *et al.* (2001); Boonbumrung *et al.* (2001)	d	1.6 - 1.62
Ito & Kubota (2005)	r	0.391 - 6.25
Pino & Mesa (2006)	d	0.5
Giri *et al.* (2010)	d	0.0056

2-HEXANOL *[626-93-7]*

Pino *et al.* (1986)	d	0.0152
Schnabel *et al.* (1988)	d	0.082 - 0.41
Giri *et al*. (2010)	d	1.5082

3-HEXANOL *[623-37-0]*

Schnabel *et al.* (1988)	d	0.82 - 2.5
Giri *et al*. (2010)	d	0.8205

4-hexanolide⇒4-HYDROXYHEXANOIC ACID LACTONE

2-HEXANONE (methyl butyl ketone) *[591-78-6]*

Schnabel *et al.* (1988)	d	0.040 - 0.081

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Boonbumrung *et al*. (2001)	d	0.56
3-HEXANONE *[589-38-8]*		
Schnabel *et al.* (1988)	d	0.041 - 0.081
2-HEXENAL *[505-57-7]*		
Kossa *et al.* (1979)		0.030
(*E*)-2-HEXENAL (leaf aldehyde) *[6728-26-3]*		
Guadagni *et al.* (1963); Flath *et al.* (1967); Buttery *et al.* (1969b,1971,1978b,1987a,1987b,1988a,1989, 1990a); Teranishi *et al.* (1974); Buttery (1981,1989, 1993); Hansen *et al.* (1992); Buttery & Ling (1998)	d	0.017
Herrmann & Poeschel (1973)		0.5
Eriksson *et al.* (1976)	d	0.316
Ahmed *et al.* (1978a)	d	0.0242
Sugisawa *et al.* (1991); Yang *et al.* (1992); Tamura *et al.* (1993)	d	0.48 - 1
Larsen & Poll (1992)	d	0.01 - 0.1
Rashash *et al.* (1997)		0.05
Jensen *et al.* (1999)	d	0.096
Tandon *et al.* (2000)	d	0.123
Arena *et al.* (2001)		0.082
Tamura *et al.* (2001); Boonbumrung *et al.* (2001)	d	0.082
Czerny *et al*. (2007c); Czerny *et al*. (2008)	r	0.19
Czerny *et al.* (2008); Steinhaus *et al.* (2009)	d	0.11
Burdack-Freitag *et al.* (2010)	d	0.0887 (normal atmospheric pressure)
Burdack-Freitag *et al.* (2010)	d	0.0885 (low atmospheric pressure)
Burdack-Freitag *et al.* (2010)	r	0.4286 (normal atmospheric pressure)
Burdack-Freitag *et al.* (2010)	r	0.3981 (low atmospheric pressure)
(*Z*)-2-HEXENAL *[16635-54-4]*		
Giri *et al.* (2010)	d	0.0192
(*E*)-3-HEXENAL *[69112-21-6]*		
Tamura *et al.* (2001)	d	0.16
(*Z*)-3-HEXENAL *[6789-80-6]*		
Buttery *et al.* (1971,1987a,1987b,1989,1990a); Teranishi *et al.* (1974); Buttery (1981,1989, 1993); Hansen *et al.* (1992)	d	0.000 25
Tandon *et al.* (2000)	d	0.026
Tamura *et al.* (2001) Boonbumrung *et al.* (2001)	d	0.0017
Czerny *et al.* (2008); Steinhaus *et al.* (2009); Sinuco *et al.* (2010)	d	0.000 12
Czerny *et al.* (2008)	r	0.000 21
(*E*)-4-HEXENAL *[25166-87-4]*		
Teranishi *et al.* (1974)	d	0.000 3
(*E*)-2-HEXEN-1-OL *[928-95-0]*		
Eriksson *et al.* (1976)	d	6.7
Sugisawa *et al.* (1991); Yang *et al.* (1992); Tamura *et al.* (1993)	d	4.25 - 10
Hansen *et al.* (1992)	d	0.4
Larsen & Poll (1992)	d	0.1 - 1.0
Arena *et al.* (2001)		4.2
Tamura *et al.* (2001)	d	0.10
Giri *et al.* (2010)	d	0.2319

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

(*Z*)-2-HEXEN-1-OL *[928-94-9]*

Giri *et al.* (2010)	d	0.3593

3-HEXEN-1-OL *[544-12-7]*

Kossa *et al.* (1979)		0.070
King (1985)	d	1.63

(*E*)-3-HEXEN-1-OL *[928-97-2]*

King (1985)	d	1.55
Tamura *et al.* (2001); Boonbumrung *et al.* (2001)	d	0.11

(*Z*)-3-HEXEN-1-OL (leaf alcohol) *[928-96-1]*

Buttery *et al.* (1969b,1971,1978b,1987a,1987b, 1989,1990a,1994a); Buttery (1981,1993);		
Hansen *et al.* (1992)	d	0.07
King (1985)	d	0.36
Larsen & Poll (1990)	d	0.1 - 1
Sugisawa *et al.* (1991); Yang *et al.* (1992);		
Tamura *et al.* (1993,1995)	d	3.625 - 7
Khiari *et al.* (1995)		0.05
Masanetz & Grosch (1998a)		0.039
Tamura *et al.* (2001); Boonbumrung *et al.* (2001)	d	0.91
Darriet *et al.* (2002)	d	0.2
Ito & Kubota (2005)	r	0.195 - 1.56
Czerny *et al.* (2008)	d	0.0039
Czerny *et al.* (2008)	r	0.013
Jaeger *et al.* (2010)		1.9

1-HEXEN-3-ONE *[1629-60-3]*

Buttery *et al.* (1978b);		
Buttery (1981,1989,1993)	d	0.000 02 - 0.000 024

4-((*Z*)-3-HEXENOXY)-2,5-DIMETHYL-3-(2*H*)-FURANONE

Buttery (1999)	d	1.9

3-HEXENYL ACETATE *[1708-82-3]*

King (1985)	d	0.21

(*E*)-3-HEXENYL ACETATE *[3681-82-1]*

King (1985)	d	0.87

(*Z*)-3-HEXENYL ACETATE *[3681-71-8]*

King (1985)	d	0.056
Yang *et al.* (1992); Tamura *et al.* (1993)	d	0.32
Khiari *et al.* (1995)		0.001 - 0.002
Masanetz & Grosch (1998a)		0.008
Tamura *et al.* (2001)	d	0.016
Czerny *et al.* (2008)	d	0.013
Czerny *et al.* (2008)	r	0.031

(*Z*)-3-HEXENYL BUTANOATE *[16491-36-4]*

Yang *et al.* (1992); Tamura *et al.* (1993)	d	0.32
Tamura *et al.* (2001); Boonbumrung *et al.* (2001)	d	0.50

(*E*)-2-HEXENYL HEXANOATE *[53398-86-0]*

Ito & Kubota (2005)	r	0.781 - 3.13

(*Z*)-2-HEXENYL HEXANOATE *[56922-79-3]*

Ito & Kubota (2005)	r	0.195- 0.781

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

5-HEXENYL ISOTHIOCYANATE *[49776-81-0]*

Masuda *et al.* (1996)		0.021

HEXYL ACETATE *[142-92-7]*

Flath *et al.* (1967);Guadagni (1970b); Buttery (1981); Teranishi *et al.* (1987)	d	0.002
Jakob *et al.* (1969)	d	0.085
Williams *et al.* (1977)	d	0.08
Schnabel *et al.* (1988)	d	0.0043 - 0.0088
Sugisawa *et al.* (1991); Yang *et al.* (1992); Tamura *et al.* (1993)	d	0.48 - 8.2
Larsen & Poll (1992)	d	0.01 - 0.1
Takeoka *et al.* (1996)	d	0.115

2-HEXYL ACETATE *[5953-49-1]*

Takeoka *et al.* (1996)	d	0.86

hexyl alcohol⇒1-HEXANOL

HEXYLAMINE *[111-26-2]*

Amoore *et al.* (1975b)	d	3.67
Amoore & Forrester (1976)	d	4.84

4-hexyl-1,3-benzenediol⇒4-HEXYL-1,3-DIHYDROXYBENZENE

HEXYL BUTANOATE (hexyl butyrate) *[2639-63-6]*

Takeoka *et al.* (1990a)	d	0.25
Pino & Mesa (2006)	d	0.203

hexyl butyrate⇒HEXYL BUTANOATE

(*R*)-(-)-2-HEXYLCYCLOPENTANONE

Yamamoto *et al.* (2002)		0.01

(*S*)-(+)-2-HEXYLCYCLOPENTANONE

Yamamoto *et al.* (2002)		0.01

(*Z*)-2-HEXYL-2-DECENAL

Robert *et al.* (2004)	d	0.1

5-HEXYLDIHYDRO-2(3*H*)-THIOPHENONE (γ-thiodecalactone)

Roling *et al.* (1998)		0.047

4-HEXYL-1,3-DIHDROXYBENZENE (4-hexyl-1,3-benzenediol, 4-hexylresorcin) *[136-77-6]*

Dietz & Traud (1978)		2

HEXYL HEXANOATE *[6378-65-0]*

Sugisawa *et al.* (1991)	d	6.4

5-HEXYL-3-HYDROXY-4-METHYL-2(5*H*)-FURANONE

Rödel & Hempel (1974)	r	1.3

hexyl isobutyrate⇒HEXYL 2-METHYLPROPANOATE

2-HEXYL-3-METHOXYPYRAZINE *[25680-56-2]*

Seifert *et al.* (1970); Teranishi *et al.* (1974)	d	0.000 001
Masuda & Mihara (1988)	d	0.000 07

HEXYL 2-METHYLBUTANOATE *[10032-15-2]*

Jakob *et al.* (1969)	d	0.06
Schnabel *et al.* (1988)	d	0.086 - 0.43

Takeoka *et al.* (1990a) d 0.022

HEXYL 2-METHYLPROPANOATE (hexyl isobutyrate) *[2349-07-7]*
Guadagni *et al.* (1966); Teranishi *et al.* (1987);
Takeoka *et al.* (1990a) d 0.006 - 0.013
Schnabel *et al.* (1988) d 0.0086 - 0.026

HEXYL PROPANOATE *[2445-76-3]*
Guadagni *et al.* (1966); Teranishi *et al.* (1987);
Takeoka *et al.* (1990a) d 0.008

HEXYLPYRAZINE *[28217-91-6]*
Masuda & Mihara (1988) d 0.2

4-hexylresorcin⇒4-HEXYL-1,3-DIHYDROXYBENZENE

2-(HEXYLTHIO)ETHANAL
Rössner *et al.* (2002) 0.000 08

hexylure⇒3-CYCLOHEXYL-5,6-TRIMETHYLENE URACIL
hippuric acid⇒N-BENZOYLGLYCINE
homofuraneol⇒2-ETHYL-4-HYDROXY-5-METHYL-3(2*H*)-FURANONE
hop ether⇒HEXAHYDRO-1,1-DIMETHYL-4-METHYLENE-1*H*-CYCLOPENTA[c]FURAN
humulene⇒2,6,6,9-TETRAMETHYL-1,4,8-CYCLOUNDECATRIENE
α-humulene⇒2,6,6,9-TETRAMETHYL-1,4,8-CYCLOUNDECATRIENE
humulene epoxide⇒I 1,5,9,9-TETRAMETHYL-[1R-(1R*,4E,7E,11R*)]-12-0XABICYCLO[9.1.0]DODECA-4,7-DIENE
hydrocinnamonitrile⇒2-PHENYLETHYL CYANIDE

HYDROGEN CHLORIDE *[7647-01-0]*
Alexander *et al.* (1982) d 135

HYDROGEN CYANIDE (prussic acid) *[74-90-8]*
Bailey & Powell (1885) 15
Brown & Robinette (1967) 0.001 - 0.01
Brown *et al.* (1968) 0.000 65 - 0.0065

HYDROGEN SULPHIDE *[7783-06-4]*
Pippen & Mecchi (1969) 0.01
Pomeroy & Cruse (1969) 0.000 01 - 0.000 1

hydroquinone⇒1,4-DIHYDROXYBENZENE

4'-HYDROXYACETOPHENONE *[99-93-4]*
Pyysalo *et al.* (1977) d 5.5

1-hydroxyadamantane⇒1-HYDROXYTRICYCLO[3.3.1.1(3.7)]DECANE

2-HYDROXYBENZALDEHYDE (salicylaldehyde) *[90-02-8]*
Dietz & Traud (1978) 0.03
Aoki & Koizumi (1986) 0.34

4-HYDROXYBENZOIC ACID *[99-96-7]*
Dietz & Traud (1978) 5

2-hydroxybiphenyl⇒2-PHENYLPHENOL
4-hydroxybiphenyl⇒4-PHENYLPHENOL

4-HYDROXYBUTANOIC ACID LACTONE (γ-butyrolactone) *[96-48-0]*
Akita *et al.* (1988); Takahashi (1990) 20 – 50
Buttery & Ling (1998) d > 1

3-HYDROXY-2-BUTANONE (acetoin) *[513-86-0]*

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Pyysalo *et al.* (1977)	d	0.055
Boidron (1984)	d	8
Larsen & Poll (1990)	d	1 - 10
Buttery *et al.* (1990a,1994a); Hansen *et al.* (1992); Buttery (1993)	d	0.8
Buttery & Ling (1995,1998); Buttery *et al.* (1999)	d	8
Boonbumrung *et al.* (2001)	d	0.014

hydrocinnamic acid⇒3-PHENYLPROPANOIC ACID
hydroxycitronellol⇒3,7-DIMETHYL-1,7-OCTANEDIOL

4-HYDROXYDECANOIC ACID LACTONE (γ-decalactone, 4-decanolide) *[706-14-9]*

Engel *et al.* (1988); Buttery (1993)	d	0.011
Buttery (1981,1989)		0.000 7
Tóth-Markus *et al.* (1989)	d	0.6
Larsen & Poll (1992)	d	0.001 - 0.01
Pino & Mesa (2006)	d	0.005
Czerny *et al*. (2007c); Czerny *et al.* (2008)	r	0.0026
Czerny *et al.* (2008)	d	0.0011

5-HYDROXYDECANOIC ACID LACTONE (δ-decalactone, 5-decanolide) *[705-86-2]*

Engel *et al.* (1988); Buttery (1993)	d	0.1
Ong & Acree (1999)		0.0025
Karagül-Yüceer *et al.* (2003, 2004)	d	0.03
Pino & Mesa (2006)	d	0.41
Czerny *et al*. (2007c); Czerny *et al.* (2008)	r	0.051
Czerny *et al.* (2008)	d	0.031
Leksrisompong (2008); Leksrisompong *et al.* (2010)	d	0.066

5-HYDROXY-(*Z*)-7-DECENOIC ACID LACTONE ((*Z*)dec-7-en-5-olide)

Engel *et al.* (1988)	d	2.0

6-hydroxydihydrotheaspirane⇒2,6,10,10-TETRAMETHYL-1-OXASPIRO[4,5]DECAN-6-OL

4-HYDROXY-3,5-DIMETHOXYBENZALDEHYDE (syringaldehyde) *[134-96-3]*

Dietz & Traud (1978)		1.9

2-HYDROXY-3,4-DIMETHYL-2-CYCLOPENTEN-1-ONE *[21835-00-7]*

Nishimura & Mihara (1990)	d	0.02

3-HYDROXY-4,5-DIMETHYL-2(5*H*)-FURANONE (sotolon) *[28664-35-9]*

Rödel & Hempel (1974)	r	0.07
Takahashi *et al.* (1976); Ohba *et al.* (1983); Takahashi & Akiyama (1993)		5.804
Grosch *et al.* (1993); Guth & Grosch (1994); Schieberle & Hofmann (1996a, 1996b)		0.000 3
Semmelroch *et al.* (1995)		0.02
Blank *et al.* (1996)	r	0.03
Buttery (1999)	d	0.000 4
Schuh & Schieberle (2006)	d	0.02
Zeller & Rychlik (2006); Czerny *et al.* (2007); Czerny *et al.* (2008)	r	0.0011
Czerny *et al.* (2008)	d	0.000 49
Steinhaus *et al.* (2009); Sinuco *et al.* (2010)	d	0.0017

4-HYDROXY-2,5-DIMETHYL-3(2*H*)-FURANONE (furaneol) *[3658-77-3]*

Pittet *et al.* (1970)		0.1 - 0.2
Pyysalo & Suihko (1976b); Pyysalo *et al.* (1977); Honkanen *et al.* (1980); Buttery (1993)	d	0.000 03 - 0.000 04
Wilson *et al.* (1990)	d	1.7
Schieberle (1991,1992); Grosch *et al.* (1993)		0.1

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Larsen & Poll (1992)	d	0.001 - 0.01
Buttery *et al.* (1994b,1995,1997,1999);		
Buttery & Ling (1998); Buttery (1999)	d	0.06
Guth & Grosch (1994)		0.025
Semmelroch *et al.* (1995); Schieberle &		
Hofmann (1996a, 1996b);		
Kerscher & Grosch (2000)		0.01
Guth (1996)		0.021 - 0.06
Ong & Acree (1999)		0.025
Karagül-Yüceer *et al.* (2004)	d	0.039
Schuh & Schieberle (2006)	d	0.03
Czerny *et al.* (2008); Steinhaus *et al.* (2009);		
Sinuco *et al.* (2010)	d	0.04
Czerny *et al.* (2008)	r	0.16
Leksrisompong (2008); Leksrisompong *et al.* (2010)	d	0.0223

(+)-(*R*)-4-HYDROXY-4,8-DIMETHYL-7-NONENOIC ACID LACTONE ((*R*)-(-)-4,8-dimethylnon-7-en-4-olide)
Kitaura *et al.* (2004) 0.000 16

(-)-(*S*)-4-HYDROXY-4,8-DIMETHYL-7-NONENOIC ACID LACTONE ((*S*)-(-)-4,8-dimethylnon-7-en-4-olide)
Kitaura *et al.* (2004) 0.000 65

5-HYDROXY-3,7-DIMETHYL-2,7-OCTADIENOIC ACID LACTONE (3,7-dimethyl-2,7-octadien-5-olide)
Tateba & Mihara (1990) d 0.4

6-HYDROXY-3,7-DIMETHYL-2,7-OCTADIENOIC ACID LACTONE (3,7-dimethyl-2,7-octadien-6-olide)
Tateba & Mihara (1990) d 0.5

(R*,S*)-4-HYDROXY-3,7-DIMETHYLOCTANOIC ACID LACTONE ((R*,S*)-3,7-dimethyl-4-octanolide)
Tateba & Mihara (1990) d 0.01

(R*,R*)-4-HYDROXY-3,7-DIMETHYLOCTANOIC ACID LACTONE ((R*,R*)-3,7-dimethyl-4-octanolide)
Tateba & Mihara (1990) d 0.02

(R*,S*)-6-HYDROXY-3,7-DIMETHYLOCTANOIC ACID LACTONE ((R*,S*)-3,7-dimethyl-6-octanolide)
Tateba & Mihara (1990) d 0.3

(R*,R*)-6-HYDROXY-3,7-DIMETHYLOCTANOIC ACID LACTONE ((R*,R*)-3,7-dimethyl-6-octanolide)
Tateba & Mihara (1990) d 1.0

4-HYDROXY-3,7-DIMETHYL-2-OCTENOIC ACID LACTONE (3,7-dimethyl-2-octen-4-olide)
Tateba & Mihara (1990) d 0.1

6-HYDROXY-3,7-DIMETHYL-2-OCTENOIC ACID LACTONE (3,7-dimethyl-2-octen-6-olide)
Tateba & Mihara (1990) d 0.2

(R*,S*)-5-HYDROXY-3,7-DIMETHYL-6-OCTENOIC ACID LACTONE
((R*,S*)-3,7-dimethyl-6-octen-5-olide)
Tateba & Mihara (1990) d 0.2

(R*,R*)-5-HYDROXY-3,7-DIMETHYL-6-OCTENOIC ACID LACTONE
((R*,R*)-3,7-dimethyl-6-octen-5-olide)
Tateba & Mihara (1990) d 1.3

(R*,S*)-6-HYDROXY-3,7-DIMETHYL-7-OCTENOIC ACID LACTONE
((R*,S*)-3,7-dimethyl-7-octen-6-olide)
Tateba & Mihara (1990) d 0.2

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

(R*,R*)-6-HYDROXY-3,7-DIMETHYL-7-OCTENOIC ACID LACTONE
((R*,R*)-3,7-dimethyl-7-octen-6-olide)
Tateba & Mihara (1990) d 1.6

1-HYDROXY-2,4-DINITRONAPHTHALENE *[605-69-6]*
Dietz & Traud (1978) > 10

4-HYDROXYDODECANOIC ACID LACTONE (γ-dodecalactone, 4-dodecanolide) *[2305-05-7]*
Engel *et al.* (1988); Buttery (1993) d 0.007
Czerny *et al*. (2007c); Czerny *et al.* (2008) r 0.002
Czerny *et al.* (2008) d 0.000 43

5-HYDROXYDODECANOIC ACID LACTONE (δ-dodecalactone, 5-dodecanolide) *[713-95-1]*
Karagül-Yüceer *et al.* (2003, 2004) d 0.000 46 - 0.005
Czerny *et al.* (2008) d 0.053
Czerny *et al.* (2008) r 0.098

(*R*)-(+)-5-HYDROXYDODECANOIC ACID LACTONE ((*R*)-(+)-δ-dodecalactone)
Yamamoto *et al.* (2002) 0.5

(*S*)-(-)-5-HYDROXYDODECANOIC ACID LACTONE ((*S*)-(-)-δ-dodecalactone)
Yamamoto *et al.* (2002) 0.05

(*Z*)-4-HYDROXY-6-DODECENOIC ACID LACTONE ((*Z*)-6-dodecenyl-γ-lactone, (*R*/*S*)-γ-(*Z*)-6-dodecenolactone)
Karagül-Yüceer *et al.* (2003) d 0.000 7
Christlbauer (2005) 0.013
Czerny *et al.* (2008) d 0.0054
Czerny *et al.* (2008) r 0.0096

(*R*)-(*Z*)-4-HYDROXY-6-DODECENOIC ACID LACTONE ((*R*)-*cis*-6-γ-dodecenolactone)
Guichard *et al.* (1991) < 0.5

(*S*)-(*Z*)-4-HYDROXY-6-DODECENOIC ACID LACTONE ((*S*)-*cis*-6-γ-dodecenolactone)
Guichard *et al.* (1991) < 0.5

4-hydroxy-3-ethoxycarbonyl-3-ethyl-2-oxopentanoic acid lactone⇒3-CARBOETHOXY-3-ETHYLTETRAHYDROFURAN-2,3-DIONE
2-hydroxyethyl acrylate⇒2-HYDROXYETHYL 2-PROPENOATE

2-HYDROXYETHYL 2-PROPENOATE (2-hydroxyethyl acrylate) *[818-61-1]*
Alexander *et al.* (1982) d 0.010

(2-HYDROXYETHYL)TRIMETHYLAMMONIUM CHLORIDE (choline chloride) *[67-48-1]*
Alexander *et al.* (1982) d 0.16

4-HYDROXYHEPTANOIC ACID LACTONE (γ-heptalactone) *[105-21-5]*
Engel *et al.* (1988) d 0.4

16-HYDROXYHEXADECANOIC ACID LACTONE (ω-hexadecalactone) *[109-29-5]*
Amoore *et al.* (1977) d 0.0046

5-HYDROXY-9-HEXADECENOIC ACID LACTONE (δ9-hexadecenolactone)
Amoore *et al.* (1977) d 0.0010

4-HYDROXYHEXANOIC ACID LACTONE (γ-hexalactone, 4-hexanolide) *[695-06-7]*
Buttery *et al.* (1971); Engel *et al.* (1988) d 1.6
Buttery *et al* (1999) d 0.05
Boonbumrung *et al.* (2001) d 0.26
Ito & Kubota (2005) r 12.5 - 100

4-HYDROXY-(2-HYDROXY)-3-METHYLHEXANOIC ACID LACTONE

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Rödel & Hempel (1974)	r	7

4-(2-HYDROXYISOPROPYL)-1-METHYL-1-CYCLOHEXENE (α-terpineol) *[98-55-5]*

Helbig (1939)		0.086
Buttery *et al.* (1968,1971,1974,1990a); Takeoka *et al.* (1990a); Buttery (1993); Takeoka (2001)	d	0.33 - 0.35
Ahmed *et al.* (1978a)	d	0.280
Pino *et al.* (1986)	d	0.0046
Sugisawa *et al.* (1991); Yang *et al.* (1992); Tamura *et al.* (1993)	d	10 - 15
Tamura *et al.* (2001); Boonbumrung *et al.* (2001)	d	5.0
Averbeck & Schierberle (2011)	d	1.2

(*S*)-(-)-4-(2-HYDROXYISOPROPYL)-1-METHYL-1-CYCLOHEXENE ((4*S*)-(-)-α-terpineol) *[10482-56-1]*

Padrayuttawat *et al.* (1997)	d	9.18

(*R*)-(+)-4-(2-HYDROXYISOPROPYL)-1-METHYL-1-CYCLOHEXENE ((4*R*)-(+)-α-terpineol) *[7785-53-7]*

Padrayuttawat *et al.* (1997)	d	6.80

4-HYDROXY-3-METHOXYBENZALDEHYDE (vanillin, 4-formyl-2-methoxyphenol) *[121-33-5]*

Helbig (1939)		0.1533
Cartwright & Kelley (1952)	d	0.2
Cartwright & Kelley (1952)	r	4
Appell (1969)		1
Dietz & Traud (1978)		0.06
Herrmann & Abd El Salam (1980b,1981)		0.226 - 1.78
Tilgner & Ziminska (1982)		0.28
Tuorila *et al.* (1982)	d	0.029 - 0.107
Mozell *et al.* (1986); Schwartz *et al.* (1987)	d	1.5 - 41
Voirol & Daget (1986)	d	0.3 - 1.6
Eccles *et al.* (1989)	d	66.7
Buttery & Ling (1995); Buttery *et al.* (1997,1999)	d	0.058
Semmelroch *et al.* (1995)		0.025
Ong & Acree (1999)		1.2
Karagül-Yüceer *et al.* (2003, 2004)	d	0.064
Schuh & Schieberle (2006)	d	0.025
Czerny *et al.* (2008)	d	0.053
Czerny *et al.* (2008)	r	0.21

1-(4-HYDROXY-3-METHOXYPHENYL)ETHANONE (acetovanillone) *[498-02-2]*

Pyysalo *et al.* (1977)	d	0.78

alpha-hydroxy-4-methoxytoluene ⇒4-METHOXYBENZYL ALCOHOL

2-HYDROXY-3-METHYL-2-CYCLOPENTEN-1-ONE (3-methyl-1,2-cyclopentanedione, cyclotene, methylcyclopentenolone, maple lactone) *[80-71-7, 765-70-8]*

Wasserman (1968)		2
Nishimura & Mihara (1990)	d	0.3

4-HYDROXY-5-METHYL-3(2*H*)-FURANONE (norfuraneol) *[19332-27-1]*

Buttery *et al.* (1994b,1995); Buttery (1999)	d	23
Schieberle & Hofmann (1996a, 1996b)		8.52

3-HYDROXY-4-METHYL-2(5*H*)-FURANONE

Rödel & Hempel (1974)	r	40

5-(HYDROXYMETHYL)FURFURAL (oxymethylfurfural) *[67-47-0]*

Takahashi (1990)		1,000

4-HYDROXY-3-METHYLHEXANOIC ACID LACTONE

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Rödel & Hempel (1974)	r	15

4-HYDROXY-4-METHYL-2-PENTANONE (diacetone alcohol) *[123-42-2]*

Moncrieff (1957)		100
Lillard & Powers (1975)	d	44.12

3-HYDROXY-4-METHYL-5-PROPYL-2(5*H*)-FURANONE

Rödel & Hempel (1974)	r	0.3

3-HYDROXY-6-METHYL-2(2*H*)-PYRANONE

Schieberle & Hofmann (1996b)		0.015

5-hydroxy-6-methyl-2*H*-pyran-4(3*H*)-one⇒2,3-DIHYDRO-5-HYDROXY-6-METHYL-4*H*-PYRAN-4-ONE

3-HYDROXY-2-METHYL-4-PYRONE (maltol) *[118-71-8]*

Pittet *et al.* (1970)		35
Bingham *et al.* (1990)	d	29
Buttery & Ling (1997a,1998);		
Buttery *et al.* (1997,1999)	d	2.5
Ong *et al.* (1998)	d	10
Schieberle (1998)		9
Karagül-Yüceer *et al.* (2003, 2004)	d	0.21
Labbe *et al.* (2007)		1.24

1-HYDROXYNAPHTHALENE (1-naphthol) *[90-15-3]*

Dietz & Traud (1978)		4

2-HYDROXYNAPHTHALENE (2-naphthol) *[135-19-3]*

Nesmeyanova (1953)		7
Baker (1963)		1.29
Dietz & Traud (1978)		3

4-HYDROXYNONANOIC ACID LACTONE (γ-nonalactone, 4-nonanolide) *[104-61-0]*

Appell (1969)		0.025
Belitz & Grosch (1992h)		0.030 - 0.065
Czerny *et al*. (2007c); Czerny *et al*. (2008)	r	0.027
Czerny *et al.* (2008)	d	0.0097

4-HYDROXYOCTANOIC ACID LACTONE (γ-octalactone, 4-octanolide) *[104-50-7]*

Buttery *et al.* (1971,1999); Buttery (1981);		
Engel *et al.* (1988); Buttery & Ling (1998)	d	0.007 - 0.008
Boonbumrung *et al.* (2001)	d	0.014
Pino & Mesa (2006)	d	0.007
Czerny *et al*. (2007c); Czerny *et al*. (2008)	r	0.024
Czerny *et al.* (2008)	d	0.0065
Burdack-Freitag *et al.* (2010)	d	0.0120 (normal atmospheric pressure)
Burdack-Freitag *et al.* (2010)	d	0.0156 (low atmospheric pressure)
Burdack-Freitag *et al.* (2010)	r	0.0179 (normal atmospheric pressure)
Burdack-Freitag *et al.* (2010)	r	0.0222 (low atmospheric pressure)

5-HYDROXYOCTANOIC ACID LACTONE (δ-octalactone, 5-octanolide) *[698-76-0]*

Engel *et al.* (1988)	d	0.4
Czerny *et al.* (2008)	d	0.2
Czerny *et al.* (2008)	r	0.42

15-HYDROXYPENTADECANOIC ACID LACTONE (exaltolide, ω-pentadecalactone, cyclopentadecanolide, thibetolid) *[106-02-5]*

Le Magnen (1950,1952)		0.000 000 001 - 0.002
Amoore & Venstrom (1966); Amoore (1969)	d	0.000 7
Vierling & Rock (1967)		0.000 1
Whissell-Buechy & Amoore (1973)	d	0.001

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Amoore *et al.* (1975a)	d	0.001
Amoore *et al.* (1975b)	d	0.00398
Amoore *et al.* (1977)	d	0.0018

1-(4-HYDROXYPHENYL)-3-BUTANONE (raspberry ketone) *[5471-51-2]*

Schmidlin-Mészáros (1971)		< 0.1
Larsen & Poll (1990)	d	0.001 - 0.01
Belitz & Grosch (1992d)		0.01

L-4-hydroxyproline⇒L-4-HYDROXY-2-PYRROLIDINECARBOXYLIC ACID

HYDROXYPROPANONE (acetol) *[116-09-6]*

Buttery *et al.* (1994a,1999) ; Buttery & Ling (1998)	d	80 - 100
Takahashi *et al.* (2002)		10

3-HYDROXY-2-PYRONE *[496-64-0]*

Pyysalo *et al.* (1977)	d	22.5

L-4-HYDROXY-2-PYRROLIDINECARBOXYLIC ACID (L-4-hydroxyproline) *[51-35-4]*

Dietz & Traud (1978)		> 10

hydroxyquinol⇒1,2,4-TRIHYDROXYBENZENE

14-HYDROXYTETRADECANOIC ACID LACTONE (ω-tetradecalactone)

Amoore *et al.* (1977)	d	0.001 7

1-HYDROXY-3,7,11,15-TETRAMETHYL-2-HEXADECENE (phytol) *[150-86-7]*

Tamura *et al.* (1995)		0.64

α-hydroxytoluene⇒BENZYL ALCOHOL

1-HYDROXYTRICYCLO[3.3.1.1(3.7)]DECANE (1-hydroxyadamantane) *[768-95-6]*

Pelosi & Pisanelli (1981)	d	2.0

trans-2-HYDROXY-3,4,5-TRIMETHYL-2-CYCLOPENTEN-1-ONE

Nishimura & Mihara (1990)	d	0.4

cis-2-HYDROXY-3,4,5-TRIMETHYL-2-CYCLOPENTEN-1-ONE

Nishimura & Mihara (1990)	d	0.015

4-HYDROXYUNDECANOIC ACID LACTONE (γ-undecalactone) *[104-67-6]*

Appell (1969)		0.06
Pino & Mesa (2006)	d	0.41
Czerny *et al*. (2007c); Czerny *et al.* (2008)	r	0.0042
Czerny *et al.* (2008)	d	0.0021

(*R*)-(+)-5-HYDROXYUNDECANOIC ACID LACTONE ((*R*)-(+)-δ-undecalactone)

Yamamoto *et al.* (2002)		0.1

(*S*)-(-)-5-HYDROXYUNDECANOIC ACID LACTONE ((*S*)-(-)-δ-undecalactone)

Yamamoto *et al.* (2002)		0.03

HYPOCHLOROUS ACID *[7790-92-3]*

Krasner & Barrett (1984)		0.21

I

INDAN *[496-11-7]*

De Grunt (1975); De Grunt *et al.* (1977)	d	0.01

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

INDENE *[95-13-6]*

Zoeteman *et al.* (1971)		0.001

INDOLE *[120-72-9]*

Le Magnen (1952)		0.13 - 0.59
Holluta (1960)		0.3
Dietz & Traud (1978)		0.3
Buttery *et al.* (1988a)	d	0.14
Karagül-Yüceer *et al.* (2003)	d	0.021
Ito & Kubota (2005)	r	1.56 - 25
Czerny *et al.* (2008)	d	0.011
Czerny *et al.* (2008)	r	0.04

IODINE *[7553-56-2]*

Dietrich *et al.* (1999)		0.5

iodoform⇒TRIIODOMETHANE

1-IODOOCTANE *[629-27-6]*

Takahashi *et al.* (2002)		0.000 2

1-IODOPENTANE *[628-17-1]*

Takahashi *et al.* (2002)		1

2-IODOPHENOL *[533-58-4]*

Mirlohi (1997); Dietrich *et al.* (1999)		0.001

4-IODOPHENOL *[540-38-5]*

Mirlohi (1997); Dietrich *et al.* (1999)		0.5

α-ionone⇒4-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE
(*d*)-α-ionone⇒(*R*)-(+)-4-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE
(*l*)-α-ionone⇒(*S*)-(-)-4-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE
β-ionone⇒4-(2,6,6-TRIMETHYL-1-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE
(*E*)-β-ionone⇒(*E*)-4-(2,6,6-TRIMETHYL-1-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE
(+)-*cis*-α-irone⇒(+)-4-((1*S*,5*R*)-2,5,6,6-TETRAMETHYL-2-CYCLOHEXEN-1-YL)-*cis*-3-BUTEN-2-ONE
(-)-*cis*-α-irone⇒(-)-4-((1*R*,5*S*)-2,5,6,6-TETRAMETHYL-2-CYCLOHEXEN-1-YL)-*cis*-3-BUTEN-2-ONE
isoambrox⇒(3a*S*)-3a,6,6,9a-TETRAMETHYL-3aα,5aα,9aβ,9bα-DODECAHYDRONAPHTHO-[2,1-b]-FURAN
isoamyl alcohol ⇒3-METHYL-1-BUTANOL
isoamyl acetate⇒ISOPENTYL ACETATE
isoamyl butyrate⇒ISOPENTYL BUTANOATE
isoamyl caproate⇒ISOPENTYL HEXANOATE
isoamyl caprylate⇒ISOPENTYL OCTANOATE
isoamyl isobutyrate⇒ISOPENTYL 2-METHYLPROPANOATE
isoamyl isovalerate⇒ISOPENTYL 3-METHYLBUTANOATE
isoamyl propylate⇒ISOPENTYL PROPANOATE
isoborneol⇒*exo*-1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANOL
isobutanol⇒2-METHYL-1-PROPANOL

ISOBUTYL ACETATE *[110-19-0]*

Sega *et al.* (1967)	d	0.073
Guadagni (1970b); Takeoka *et al.* (1992)	d	0.065
Schnabel *et al.* (1988)	d	0.44 - 0.88
Takahashi (1990)		0.5 - 1.6
Giri *et al.* (2010)	d	0.025

isobutyl alcohol⇒2-METHYL-1-PROPANOL

ISOBUTYLBENZENE *[538-93-2]*

De Grunt (1975); De Grunt *et al.* (1977)	d	0.000 8 - 0.08

ISOBUTYL CYANIDE (isovaleronitrile, 3-methylbutanenitrile) *[625-28-5]*

Buttery *et al.* (1987a,1989,1990a); Buttery (1993)	d	1

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Fabrellas *et al.* (2004)		0.0023 - 0.0032
Freuze *et al.* (2005)	d	0.21

5-ISOBUTYL-2,3-DIMETHYLPYRAZINE *[32263-00-6]*

Masuda & Mihara (1986); Shibamoto (1986)	d	0.77
Shibamoto (1986)	d	4.10

2-ISOBUTYL-4,5-DIMETHYLTHIAZOLE *[53498-32-1]*

Buttery (1981)	d	0.01

3-isobutylenephthalide⇒3-(2-METHYLPROPYLIDENE)PHTHALIDE
isobutyl isobutyrate⇒ISOBUTYL 2-METHYLPROPANOATE

ISOBUTYL ISOTHIOCYANATE (isobutyl mustard oil) *[591-82-2]*

Brevard *et al.* (1992)	d	2.5
Masuda *et al.* (1996)		0.13

2-ISOBUTYL-3-METHOXY-5,6-DIMETHYLPYRAZINE

Seifert *et al.* (1972); Teranishi *et al.* (1974,1975)	d	0.315

5-ISOBUTYL-2-METHOXY-3-METHYLPYRAZINE *[78246-20-5]*

Masuda & Mihara (1986); Shibamoto (1986)	d	0.000 18

2-ISOBUTYL-3-METHOXY-5-METHYLPYRAZINE

Seifert *et al.* (1972); Teranishi *et al.* (1974,1975)	d	0.000 26

2-ISOBUTYL-3-METHOXY-6-METHYLPYRAZINE

Seifert *et al.* (1972); Teranishi *et al.* (1974,1975)	d	0.0026

2-ISOBUTYL-3-METHOXYPYRAZINE (pepper pyrazine, 3-isobutyl-2-methoxypyrazine) *[24683-00-9]*

Buttery *et al.* (1969a,1969b); Seifert *et al.* (1970, 1972); Guadagni (1969,1970c); Teranishi *et al.* (1974,1975); Buttery (1981,1993)	d	0.000 002
Takken *et al.* (1975)	d	0.000 016
Calabretta (1975,1978)	d	0.010
Mihara & Masuda (1988)	d	0.000 045
Semmelroch & Grosch (1996)		0.000 005
Young *et al.* (1996)		0.000 001
Darriet *et al.* (2002)	d	0.000 002
Czerny et al. (2008)	d	0.000 0062
Czerny et al. (2008)	r	0.000 038

3-isobutyl-2-methoxypyrazine⇒2-ISOBUTYL-3-METHOXYPYRAZINE

2-ISOBUTYL-5-METHOXYPYRAZINE

Calabretta (1975,1978)	d	0.010

3-ISOBUTYL-2-METHOXYPYRIDINE

Seifert *et al.* (1972); Teranishi *et al.* (1975)	d	0.011

ISOBUTYL 2-METHYLBUTANOATE *[2445-67-2]*

Schnabel *et al.* (1988)	d	0.043 - 0.085

5-ISOBUTYL-3-METHYL-2-(METHYLTHIO)PYRAZINE *[78246-17-0]*

Masuda & Mihara (1986); Shibamoto (1986)	d	0.015

5-ISOBUTYL-3-METHYL-2-PHENOXYPYRAZINE *[78246-16-9]*

Masuda & Mihara (1986); Shibamoto (1986)	d	0.32

5-ISOBUTYL-3-METHYL-2-PHENYLPYRAZINE

Shibamoto (1986)	d	0.10

5-ISOBUTYL-3-METHYL-2-(PHENYLTHIO)PYRAZINE *[78246-19-2]*

Masuda & Mihara (1986)	d	0.1

ISOBUTYL 2-METHYLPROPANOATE (isobutyl isobutyrate) *[97-85-8]*

Guadagni *et al.* (1966); Teranishi *et al.* (1987);		
Takeoka *et al.* (1990a)	d	0.03
Amoore *et al.* (1968)	d	4.37
Amoore & Forrester (1976)	d	2.67
Amoore *et al.* (1976)	d	2.1
Amoore *et al.* (1977)	d	2.84
Schnabel *et al.* (1988)	d	0.026 - 0.061

2-ISOBUTYL-3-METHYLPYRAZINE *[13925-06-9]*

Guadagni *et al.* (1972); Teranishi *et al.* (1974)	d	0.035
Takken *et al.* (1975)	d	0.13

2-ISOBUTYL-3-(METHYLTHIO)PYRAZINE

Takken *et al.* (1975)	d	0.000 33

isobutyl mustard oil⇒ISOBUTYL ISOTHIOCYANATE

ISOBUTYLPYRAZINE *[29460-92-2]*

Seifert *et al.* (1970)	d	0.4

2-ISOBUTYLTHIAZOLE *[18640-74-9]*

Buttery *et al.* (1971,1987a,1989,1990a);		
Buttery (1981,1993)	d	0.0035
Tandon *et al.* (2000)	d	0.029

2-(ISOBUTYLTHIO)ETHANAL

Rössner *et al.* (2002)		0.000 08

isobutyric acid⇒2-METHYLPROPANOIC ACID
isobutyraldehyde⇒2-METHYLPROPANAL
isobutyronitrile⇒ISOPROPYL CYANIDE
isocaproic acid⇒4-METHYLPENTANOIC ACID
isoeugenol⇒2-METHOXY-4-(1-PROPENYL)PHENOL

2-ISOHEXYL-3-METHOXYPYRAZINE *[68844-95-1]*

Mihara & Masuda (1988)	d	0.000 006

isonootkatone⇒(*d*)-8-ISOPROPYLIDENE-5,6-DIMETHYLBICYCLO[4.4.0]-1-DECEN-3-ONE

ISOOCTYL 2,4-DICHLOROPHENOXYACETATE *[25168-26-7]*

Sigworth (1965)	d	0.12

ISOPENTYL ACETATE (isoamyl acetate, 3-methylbutyl acetate) *[123-92-2]*

Williams (1953)		0.5
Teranishi *et al.* (1966); Guadagni (1969,1970b);		
Buttery (1981); Teranishi *et al.* (1987);		
Takeoka *et al.* (1989,1992)	d	0.002 - 0.005
Sega *et al.* (1967)	d	0.019
Appell (1969)		0.4
Schnabel *et al.* (1988)	d	0.0087 - 0.043
Takahashi (1990)		1.6 - 2.5
Schieberle (1998)		0.088
Pino & Mesa (2006)	d	0.002
Giri *et al*. (2010)	d	0.000 15

ISOPENTYL BENZOATE *[94-46-2]*

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Hirvi & Honkanen (1984) d 0.25

ISOPENTYL BUTANOATE (isoamyl butyrate) *[106-27-4]*
Pino & Mesa (2006) d 0.000 13
Giri *et al*. (2010) d 0.015

2-ISOPENTYL-5,6-DIMETHYLPYRAZINE *[18450-01-6]*
Shibamoto (1986) d 6.00

ISOPENTYL HEXANOATE (isoamyl caproate) *[2198-61-0]*
Hirvi & Honkanen (1984) d 0.32

2-ISOPENTYL-3-METHOXYPYRAZINE *[32737-13-6]*
Mihara & Masuda (1988) d 0.000 0063

ISOPENTYL 2-METHYLBUTANOATE *[27625-35-0]*
Hirvi & Honkanen (1984) d 0.14
Schnabel *et al.* (1988) d 0.0086 - 0.043

ISOPENTYL 3-METHYLBUTANOATE (isoamyl isovalerate) *[659-70-1]*
Hirvi & Honkanen (1984) d 0.02

ISOPENTYL 2-METHYLPROPANOATE (isoamyl isobutyrate) *[2050-01-3]*
Schnabel *et al.* (1988) d 0.087 - 0.43

ISOPENTYL OCTANOATE (isoamyl caprylate) *[2035-99-6]*
Hirvi & Honkanen (1984) d 0.07

ISOPENTYL PROPANOATE (isoamyl propylate) *[105-68-0]*
Schnabel *et al.* (1988) d 0.0086 - 0.043

2-(ISOPENTYLTHIO)ETHANAL
Rössner *et al.* (2002) 0.000 3

isophorone⇒3,5,5-TRIMETHYL-2-CYCLOHEXENONE
isoprene⇒2-METHYL-1,3-BUTADIENE
isopropanol⇒2-PROPANOL
isopropanolamine⇒1-AMINO-2-PROPANOL

4-ISOPROPENYL-1-CYCLOHEXEN-1-CARBOXALDEHYDE (perillaldehyde) *[2111-75-3]*
Ahmed *et al.* (1978a) d 0.0301
Sugisawa *et al.* (1991); Yang *et al.* (1992);
Tamura *et al.* (1993) d 0.062 - 0.1
Tamura *et al.* (1996) d 0.056
Tamura *et al.* (1996) r 0.29
Boonbumrung *et al.* (2001) d 0.030

(*S*)-(-)-4-ISOPROPENYL-1-CYCLOHEXEN-1-CARBOXALDEHYDE ((4*S*)-(-)-perillaldehyde) *[18031-40-8]*
Padrayuttawat *et al.* (1997) d 0.03

(*R*)-(+)-4-ISOPROPENYL-1-CYCLOHEXEN-1-CARBOXALDEHYDE ((4*R*)-(+)-perillaldehyde)
Padrayuttawat *et al.* (1997) d 0.07

(4-ISOPROPENYL-1-CYCLOHEXENYL)METHANOL (perillyl alcohol) *[536-59-4]*
Sugisawa *et al.* (1991); Yang *et al.* (1992);
Tamura *et al.* (1993) d 7

(*R*)-(+)-(4-ISOPROPENYL-1-CYCLOHEXENYL)METHANOL ((4*R*)-(+)-perillyl alcohol) *[57717-97-2]*
Padrayuttawat *et al.* (1997) d 1.66

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

(*S*)-(-)-(4-ISOPROPENYL-1-CYCLOHEXENYL)METHANOL ((4*S*)-(-)-perillyl alcohol) *[536-59-4]*

Padrayuttawat *et al.* (1997)	d	1.10

8-ISOPROPENYL-5,6-DIMETHYLBICYCLO[4.4.0]-3-DECANONE (1,10-dihydronootkatone) *[1803-39-0]*

Stevens *et al.* (1970)	d	0.05

8-ISOPROPENYL-5,6-DIMETHYLBICYCLO[4.4.0]-1-DECEN-3-ONE (nootkatone) *[4674-50-4]*

Guadagni (1969,1970c); Stevens *et al.* (1970);		
Teranishi *et al.* (1981); Buttery (1981)	d	0.17 - 0.20
Pino *et al.* (1986)	d	0.8
Sugisawa *et al.* (1991); Yang *et al.* (1992);		
Tamura *et al.* (1993)	d	0.28

(*d*)-8-ISOPROPENYL-5,6-DIMETHYLBICYCLO[4.4.0]-1-DECEN-3-ONE ((*d*)-nootkatone, (+)-nootkatone)

Haring *et al.* (1972)	d	0.6 - 1.0

(*l*)-8-ISOPROPENYL-5,6-DIMETHYLBICYCLO[4.4.0]-1-DECEN-3-ONE ((*l*)-nootkatone, (-)-nootkatone) *[38427-78-0]*

Haring *et al.* (1972)	d	400 – 800
Belitz & Grosch (1992d)		1

8-ISOPROPENYL-5,6-DIMETHYLBICYCLO[4.4.0]-1-DECEN-10-ONE (eremophilone) *[562-23-2]*

Guadagni (1969); Stevens *et al.* (1970)	d	0.045

1-ISOPROPENYL-4-METHYLBENZENE (α,*p*-dimethylstyrene) *[1195-32-0]*

Masanetz & Grosch (1998a, 1998b)		0.085

4-ISOPROPENYL-1-METHYL-1,3-CYCLOHEXADIENE (*p*-mentha-1,3,8-triene) *[18368-95-1]*

Masanetz & Grosch (1998a, 1998b)		0.015

2-ISOPROPENYL-5-METHYL-1-CYCLOHEXANOL (isopulegol) *[89-79-2]*

Sugisawa *et al.* (1991)	d	1

5-ISOPROPENYL-2-METHYL-1-CYCLOHEXANONE (dihydrocarvone) *[7764-50-3]*

Sugisawa *et al.* (1991)	d	3.25

(*d*)-5-ISOPROPENYL-2-METHYL-1-CYCLOHEXANONE ((*d*)-dihydrocarvone) *[5524-05-0]*

Pelosi & Viti (1978)	d	0.82

(*l*)-5-ISOPROPENYL-2-METHYL-1-CYCLOHEXANONE ((*l*)-dihydrocarvone) *[619-02-3]*

Pelosi & Viti (1978)	d	1.06

4-ISOPROPENYL-1-METHYL-1-CYCLOHEXENE (limonene) *[138-86-3]*

Appell (1969)		1
Guadagni (1969); Buttery *et al.* (1969b,1971,1974, 1987b,1988a); Buttery (1981);		
Takeoka *et al.* (1990a)	d	0.01
De Grunt (1975); De Grunt *et al.* (1977)	d	0.004
Tuorila *et al.* (1982)	d	0.011 - 0.072
Pino *et al.* (1986)	d	0.229
Tamura *et al.* (1995)		3
Tamura *et al.* (1996); Mookdasanit *et al.* (2003)	d	1
Tamura *et al.* (1996)	r	3.5
Buttery & Ling (1998)	d	0.3
Tamura *et al.* (2001)	d	0.20
Averbeck (2002)		0.034
Pino & Mesa (2006)	d	1
Buhr *et al.* (2010)	d	0.2

(*R*)-(+)-4-ISOPROPENYL-1-METHYL-1-CYCLOHEXENE ((*d*)-limonene, (4*R*)-(+)-limonene) *[5989-27-5]*

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Ahmed *et al.* (1978a)	d	0.060
Belitz & Grosch (1992e)		0.2
Blank *et al.* (1992)		0.15
Buttery (1993); Buttery & Ling (1995)	d	0.2
Padrayuttawat *et al.* (1997);		
Boonbumrung *et al.* (2001)	d	1.20
Averbeck & Schierberle (2009, 2011)	d	0.034

(*S*)-(-)-4-ISOPROPENYL-1-METHYL-1-CYCLOHEXENE ((*l*)-limonene, (4*S*)-(-)-limonene) *[5989-54-8]*

Buttery (1993)	d	0.5
Padrayuttawat *et al.* (1997);		
Boonbumrung *et al.* (2001)	d	1.04

cis-4-ISOPROPENYL-1-METHYL-6-CYCLOHEXEN-2-OL (carveol 1, *cis*-carveol) *[1197-06-4]*

Sugisawa *et al.* (1991); Yang *et al.* (1992);		
Tamura *et al.* (1993)	d	4
Mookdasanit *et al.* (2003)	d	0.25

trans-4-ISOPROPENYL-1-METHYL-6-CYCLOHEXEN-2-OL (*trans*-carveol) *[1197-07-5]*

Yang *et al.* (1992); Tamura *et al.* (1993)	d	4
Tamura *et al.* (2001); Boonbumrung *et al.* (2001)	d	0.25

5-ISOPROPENYL-2-METHYL-2-CYCLOHEXEN-1-ONE (carvone) *[99-49-0]*

Pino *et al.* (1986)	d	0.0067
Sugisawa *et al.* (1991)	d	0.82
Yang *et al.* (1992); Tamura *et al.* (1993)	d	0.067
Mookdasanit *et al.* (2003)	d	0.1
Pino & Mesa (2006)	d	0.027

(*S*)-(+)-5-ISOPROPENYL-2-METHYL-2-CYCLOHEXEN-1-ONE ((*d*)-carvone, (4*S*)-(+)-carvone) *[2244-16-8]*

Leitereg *et al.* (1971a,1971b)	d	0.085 - 0.13
Ahmed *et al.* (1978a)	d	0.0027
Pelosi & Viti (1978)	d	0.60
Polak *et al.* (1989)		0.001 - 4.0
Polak *et al.* (1989)		0.001 - 0.7
Padrayuttawat *et al.* (1997)	d	0.25
Averbeck & Schierberle (2009, 2011)	d	0.160

(*R*)-(-)-5-ISOPROPENYL-2-METHYL-2-CYCLOHEXEN-1-ONE ((*l*)-carvone, (4*R*)-(-)-carvone) *[6485-40-1]*

Leitereg *et al.* (1971a,1971b)	d	0.002
Ohloff (1977,1978)		0.05
Peliso & Viti (1978)	d	0.043
Pelosi & Pisanelli (1981)	d	0.058
Polak *et al.* (1989)		0.000 7 - 2.0
Polak *et al.* (1989)		0.02 - 0.15
Padrayuttawat *et al.* (1997)	d	0.01
Averbeck & Schierberle (2009, 2011)	d	0.007

(1*RS*,5*R*)-5-ISOPROPENYL-2-METHYL-2-CYCLOHEXENYL ACETATE (carvyl acetate) *[97-42-7]*

Hirvi & Honkanen (1982)	d	0.0015

ISOPROPYL ACETATE *[108-21-4]*

Sarzynski (1982)		8
Granzer *et al.* (1986)		0.5 - 5
Schnabel *et al.* (1988)	d	1.7 - 4.4

ISOPROPYLAMINE *[75-31-0]*

De Grunt (1975)	d	5.0

isopropyl alcohol⇒2-PROPANOL

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

4-ISOPROPYLBENZALDEHYDE (cuminaldehyde) *[122-03-2]*
Mookdasanit *et al.* (2003) d 0.06
Pino & Mesa (2006) d 0.4

ISOPROPYLBENZENE (cumene) *[98-82-8]*
Zoeteman *et al.* (1971) 0.1
Young *et al.* (1996) 0.07

ISOPROPYL BUTANOATE (isopropyl butyrate) *[638-11-9]*
Schnabel *et al.* (1988) d 0.043 - 0.086

isopropyl butyrate⇒ISOPROPYL BUTANOATE

ISOPROPYL CYANIDE (isobutyronitrile, 2-methylpropanenitrile) *[78-82-0]*
Fabrellas *et al.* (2004) 0.064 - 0.204
Freuze *et al.* (2005) d 0.43

ISOPROPYL 2,4-DICHLOROPHENOXYACETATE *[94-11-1]*
Sigworth (1965) d 0.0031 - 0.055

2-ISOPROPYL-3,5-DIMETHOXY-6-METHYLPYRAZINE
Takken *et al.* (1975) d 0.07

2-ISOPROPYL-3,6-DIMETHOXY-5-METHYLPYRAZINE
Takken *et al.* (1975) d 0.25

(*d*)-8-ISOPROPYL-5,6-DIMETHYLBICYCLO[4.4.0]-1,9-DECADIEN-3-ONE
Haring *et al.* (1972) d 5 - 10

(*l*)-8-ISOPROPYL-5,6-DIMETHYLBICYCLO[4.4.0]-1,9-DECADIEN-3-ONE
Haring *et al.* (1972) d 40 - 100

8-ISOPROPYL-5,6-DIMETHYLBICYCLO[4.4.0]-3-DECANONE (tetrahydronootkatone) *[38427-80-4]*
Stevens *et al.* (1970) d 0.045

(*d*)-8-ISOPROPYL-5,6-DIMETHYLBICYCLO[4.4.0]-3-DECANONE ((*d*)-tetrahydronootkatone)
Haring *et al.* (1972) d 5 - 20

(*l*)-8-ISOPROPYL-5,6-DIMETHYLBICYCLO[4.4.0]-3-DECANONE ((*l*)-tetrahydronootkatone)
Haring *et al.* (1972) d 30 - 70

8-ISOPROPYL-5,6-DIMETHYLBICYCLO[4.4.0]-1-DECEN-3-ONE (11,12-dihydronootkatone)
Stevens *et al.* (1970) d 0.5

2-ISOPROPYL-5,6-DIMETHYLPYRAZINE *[40790-21-4]*
Masuda & Mihara (1986); Shibamoto (1986) d 0.53

2-(ISOPROPYLDITHIO)PROPANE (diisopropyl disulphide) *[4253-89-8]*
Mookdasanit *et al.* (2003) d 0.043

8-ISOPROPYLIDENE-5,6-DIMETHYLBICYCLO[4.4.0]-1-DECEN-3-ONE (α-vetivone, isonootkatone) *[15764-04-2]*
Stevens *et al.* (1970) d 0.175

(*d*)-8-ISOPROPYLIDENE-5,6-DIMETHYLBICYCLO[4.4.0]-1-DECEN-3-ONE ((*d*)-α-vetivone, isonootkatone)
Haring *et al.* (1972) d 0.6 - 1.0

(*l*)-8-ISOPROPYLIDENE-5,6-DIMETHYLBICYCLO[4.4.0]-1-DECEN-3-ONE ((*l*)-α-vetivone)
Haring *et al.* (1972) d 6 - 15

(*R*)-(+)-4-ISOPROPYLIDENE-1-METHYL-3-CYCLOHEXANONE ((*d*)-pulegone) *[89-82-7]*

Pelosi & Viti (1978)	d	0.13

4-ISOPROPYLIDENE-1-METHYL-3-CYCLOHEXENE (terpinolene) *[586-62-9]*

Buttery *et al.* (1968,1987b); Buttery (1981)	d	0.2
Tamura *et al.* (2001); Boonbumrung *et al.* (2001)	d	0.041
Pino & Mesa (2006)	d	0.2

2-{[(1*S*,2*R*,5*S*)-2-isopropyl-5-methylcyclohexyl]oxy}pyrazine⇒MENTHYLOXYPYRAZINE
(+)-2-{[(1*S*,2*R*,5*S*)-2-isopropyl-5-methylcyclohexyl]oxy}pyrazine⇒(*d*)-MENTHYLOXYPYRAZINE
(-)-2-{[(1*R*,2*S*,5*R*)-2-isopropyl-5-methylcyclohexyl]oxy}pyrazine⇒(*l*)-MENTHYLOXYPYRAZINE
isopropyl isobutyrate⇒ISOPROPYL 2-METHYLPROPANOATE

ISOPROPYL ISOTHIOCYANATE (isopropyl mustard oil) *[625-59-2]*

Brevard *et al.* (1992)	d	10.0
Masuda *et al.* (1996)		0.63

4-ISOPROPYL-8-MERCAPTO-1-METHYLTHIOL-1-CYCLOHEXENE (α,α,4-trimethyl-3-cyclohexene-1-methanethiol, 1-*p*-menthene-8-thiol) *[71159-90-5]*

Belitz & Grosch (1992d)		0.000 000 02

5-ISOPROPYL-2-METHOXY-3-METHYLPYRAZINE *[38346-76-8]*

Masuda & Mihara (1986); Shibamoto (1986)	d	0.026

2-ISOPROPYL-3-METHOXY-5-METHYLPYRAZINE

Takken *et al.* (1975)	d	0.000 045

2-ISOPROPYL-3-METHOXY-6-METHYLPYRAZINE

Takken *et al.* (1975)	d	0.000 05

2-ISOPROPYL-3-METHOXYPYRAZINE (3-isopropyl-2-methoxypyrazine) *[25773-40-4]*

Seifert *et al.* (1970); Buttery (1981,1993)	d	0.000 002
Calabretta (1975,1978)	d	0.010
Mihara & Masuda (1988)	d	0.000 024
Young *et al.* (1996)		0.000 000 2
Masanetz & Grosch (1998a)		0.000 004
Zeller & Richlik (2007); Czerny *et al.* (2008)	r	0.000 013
Neta *et al.* (2008)	d	0.000 000 4
Czerny et al. (2008)	d	0.000 0039

3-isopropyl-2-methoxypyrazine⇒2-ISOPROPYL-3-METHOXYPYRAZINE

2-ISOPROPYL-5-METHOXYPYRAZINE

Calabretta (1975,1978)	d	0.010

1-ISOPROPYL-3-METHYLBENZENE (*m*-cymene) *[535-77-3]*

Tamura *et al.* (1995)		0.8

1-ISOPROPYL-4-METHYLBENZENE (*p*-cymene, 4-isopropyltoluene) *[99-87-6]*

Zoeteman *et al.* (1971)		0.1
Buttery *et al.* (1974); Takeoka (2001)	d	0.15
Ahmed *et al.* (1978a)	d	0.0114
Pino *et al.* (1986)	d	0.0062
Young *et al.* (1996)		0.4
Padrayuttawat *et al.* (1997); Tamura *et al.* (2001); Boonbumrung *et al.* (2001)	d	0.1 - 0.12
Pino & Mesa (2006)	d	0.0114
Giri *et al*. (2010)	d	0.00501

trans-5-ISOPROPYL-2-METHYLBICYCLO[3.1.0]-2-HEXANOL (*trans*-sabinene hydrate) *[17699-16-0]*

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Sugisawa *et al.* (1991); Yang *et al.* (1992);
Tamura *et al.* (1993) d 55

1-ISOPROPYL-4-METHYLBICYCLO[3.1.0]-3-HEXANONE (α/β-thujone) *[546-80-5]*
Pelosi & Pisanelli (1981) d 0.36

1-ISOPROPYL-4-METHYLBICYCLO[3.1.0]-3-HEXEN-2-ONE (umbellulone) *[24545-81-1]*
Buttery *et al.* (1974) d 0.75

ISOPROPYL 2-METHYLBUTANOATE *[66576-71-4]*
Schnabel *et al.* (1988) d 0.043-0.087

4-ISOPROPYL-1-METHYL-1,3-CYCLOHEXADIENE (α-terpinene) *[99-86-5]*
Boonbumrung *et al.* (2001);
Mookdasanit *et al.* (2003) d 0.08 - 0.085

4-ISOPROPYL-1-METHYL-1,4-CYCLOHEXADIENE (γ-terpinene) *[99-85-4]*
Tamura *et al.* (1996) d 1
Tamura *et al.* (1996) r 2.89
Padrayuttawat *et al.* (1997); Tamura *et al.* (1999) d 0.6
Tamura *et al.* (2001); Boonbumrung *et al.* (2001) d 0.26
Takeoka (2001) d 0.065
Pino & Mesa (2006) d 1

4-ISOPROPYL-1-METHYL-1,5-CYCLOHEXADIENE (α-phellandrene) *[99-83-2]*
Padrayuttawat *et al.* (1997); Tamura *et al.* (1999);
Boonbumrung *et al.* (2001) d 0.16
Pino & Mesa (2006) d 0.04

(*R*)-(-)-4-ISOPROPYL-1-METHYL-1,5-CYCLOHEXADIENE ((*l*)-α-phellandrene) *[4221-98-1]*
Belitz & Grosch (1992e); Blank *et al.* (1992) 0.5

(*S*)-(+)-4-ISOPROPYL-1-METHYL-1,5-CYCLOHEXADIENE ((*d*)-α-phellandrene) *[2243-33-6]*
Huopalahti (1986) 0.04
Belitz & Grosch (1992e); Blank *et al.* (1992) 0.2

(1*S*,2*R*,5*S*)-(+)-2-ISOPROPYL-5-METHYLCYCLOHEXANOL ((1*S*,3*S*,4*R*)-(+)-menthol) *[15356-60-2]*
Padrayuttawat *et al.* (1997) d 0.92

(1*R*,2*S*,5*R*)-(-)-2-ISOPROPYL-5-METHYLCYCLOHEXANOL ((1*R*,3*R*,4*S*)-(-)-menthol, (*l*)-menthol)
[2216-51-5]
Pelosi & Viti (1978) d 0.95
Pelosi & Pisanelli (1981) d 2.5
Padrayuttawat *et al.* (1997) d 2.28
Ottinger *et al.* (2001) 0.1 - 0.2

(1*S*,2*S*,5*R*)-(+)-2-ISOPROPYL-5-METHYLCYCLOHEXANOL ((1*R*,3*S*,4*S*)-(+)-neomenthol) *[2216-52-6]*
Padrayuttawat *et al.* (1997) d 0.81

(1*R*,2*R*,5*S*)-(-)-2-ISOPROPYL-5-METHYLCYCLOHEXANOL ((1*S*,3*R*,4*R*)-(-)-neomenthol) *[20747-49-3]*
Padrayuttawat *et al.* (1997) d 1.05

(*d*)-5-ISOPROPYL-2-METHYLCYCLOHEXANONE ((*d*)-carvomenthone)
Pelosi & Viti (1978) d 0.88

(*l*)-5-ISOPROPYL-2-METHYLCYCLOHEXANONE ((*l*)-carvomenthone)
Pelosi & Viti (1978) d 1.12

2-ISOPROPYL-5-METHYLCYCLOHEXANONE (menthone) *[10458-14-7]*
Amoore & Venstrom (1966); Amoore (1969) d 0.17

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Vickers *et al.* (2001) 2.4 - 4.9

(2*R*,5*S*)-(+)-2-ISOPROPYL-5-METHYLCYCLOHEXANONE ((2*R*,5*S*)-(+)-menthone) *[3391-87-5]*
Padrayuttawat *et al.* (1997) d 0.21

(2*S*,5*R*)-(-)-2-ISOPROPYL-5-METHYLCYCLOHEXANONE ((2*S*,5*R*)-(-)-menthone, (*l*)-menthone) *[14073-97-3]*
Pelosi & Viti (1978) d 0.17
Padrayuttawat *et al.* (1997) d 0.35

4-ISOPROPYL-7-METHYLCYCLOHEXATHIAZOLE *[84648-08-8]*
Pelosi *et al.* (1983) d 0.000 36

(*l*)-4-ISOPROPYL-7-METHYLCYCLOHEXATHIAZOLE *[84710-13-4]*
Pelosi *et al.* (1983) d 0.022

1-ISOPROPYL-4-METHYL-3-CYCLOHEXEN-1-OL (1-terpinen-4-ol) *[562-74-3]*
Buttery *et al.* (1968,1974); Takeoka (2001) d 0.34
Sugisawa *et al.* (1991); Yang *et al.* (1992);
Tamura *et al.* (1993) d 6.4
Tamura *et al.* (1996) d 0.59
Tamura *et al.* (1996) r 4.37
Tamura *et al.* (2001) d 1.2

(*S*)-(+)-1-ISOPROPYL-4-METHYL-3-CYCLOHEXEN-1-OL ((4*S*)-(+)-terpinen-4-ol) *[2438-10-0]*
Padrayuttawat *et al.* (2001) d 1.29

(*R*)-(-)-1-ISOPROPYL-4-METHYL-3-CYCLOHEXEN-1-OL ((4*R*)-(-)-terpinen-4-ol)
Padrayuttawat *et al.* (2001) d 1.19

(*d*)-3-ISOPROPYL-5-METHYL-2-CYCLOHEXEN-1-ONE ((*d*)-carvenone)
Pelosi & Viti (1978) d 1.46

(*l*)-3-ISOPROPYL-5-METHYL-2-CYCLOHEXEN-1-ONE ((*l*)-carvenone)
Pelosi & Viti (1978) d 2.00

4-ISOPROPYL-1-METHYL-1-CYCLOHEXEN-3-ONE (piperitone) *[89-81-6]*
Pelosi & Viti (1978) d 0.68

(-)-1-[(1*R*,5*S*)-5-ISOPROPYL-2-METHYLCYCLOHEX-EN-1-YL]ETHANONE ((-)-(2*R*,4*S*)-2-acetyl-*p*-menth-6-ene)
Yamamoto *et al.* (2001) 0.1

(+)-1-[(1*S*,5*R*)-5-ISOPROPYL-2-METHYLCYCLOHEX-EN-1-YL]ETHANONE ((+)-(2*S*,4*R*)-2-acetyl-*p*-menth-6-ene)
Yamamoto *et al.* (2001) 2

1-ISOPROPYL-4-METHYLENEBICYCLO[3.1.0]HEXANE (sabinene) *[3387-41-5]*
Buttery *et al.* (1968,1974); Buttery (1981) d 0.075
Pino *et al.* (1986) d 0.037
Boonbumrung *et al*. (2001);
Mookdasanit *et al.* (2003) d 0.98

4-ISOPROPYL-1-METHYLENE-2-CYCLOHEXENE (β-phellandrene) *[555-10-2]*
Buttery *et al.* (1987b) d 0.5
Masanetz & Grosch (1998a, 1998b) 0.036

5-ISOPROPYL-3-METHYL-2-(METHYLTHIO)PYRAZINE *[99784-22-2]*
Masuda & Mihara (1986); Shibamoto (1986) d 0.016

5-ISOPROPYL-2-METHYLPHENOL (carvacrol) *[499-75-2]*

Dietz & Traud (1978)		0.8
Yang *et al.* (1992); Tamura *et al.* (1993)	d	2.29
Mookdasanit *et al.* (2003)	d	0.07
Pino & Mesa (2006)	d	2.29

2-ISOPROPYL-5-METHYLPHENOL (thymol) *[89-83-8]*

Nesmeyanova (1953)		0.05
Buttery *et al.* (1974)	d	0.086
Pelosi & Viti (1978)	d	0.40
Dietz & Traud (1978)		0.5
Huopalahti (1986)		0.12
Yang *et al.* (1992); Tamura *et al.* (1993)	d	0.79
Mookdasanit *et al.* (2003)	d	0.188
Pino & Mesa (2006)	d	1.7

2-ISOPROPYL-6-METHYLPHENOL *[3228-04-4]*

Dietz & Traud (1978)		1

5-ISOPROPYL-3-METHYL-2-PHENOXYPYRAZINE *[99784-18-6]*

Masuda & Mihara (1986); Shibamoto (1986)	d	0.37

ISOPROPYL 3-METHYL-3-PHENYL-2,3-EPOXYPROPANOATE

Tekin & Karaman (1992)	d	8.5

5-ISOPROPYL-3-METHYL-2-(PHENYLTHIO)PYRAZINE *[99784-30-2]*

Masuda & Mihara (1986); Shibamoto (1986)	d	0.16

ISOPROPYL 2-METHYLPROPANOATE (isopropyl isobutyrate) *[617-50-5]*

Schnabel *et al.* (1988)	d	0.026 - 0.060

2-ISOPROPYL-3-METHYLPYRAZINE *[15986-81-9]*

Mihara & Masuda (1988)	d	0.016

2-ISOPROPYL-3-(METHYLTHIO)PYRAZINE *[67952-59-4]*

Mihara & Masuda (1988)	d	0.000 047

isopropyl mustard oil⇒ ISOPROPYL ISOTHIOCYANATE

2-ISOPROPYLPHENOL *[88-69-7]*

Dietz & Traud (1978)		1

ISOPROPYLPYRAZINE *[29460-90-0]*

Calabretta (1973)	d	0.1

2-(ISOPROPYLTHIO)ETHANAL

Rössner *et al.* (2002)		0.000 5

4-isopropyltoluene⇒1-ISOPROPYL-4-METHYLBENZENE
isopseudocumenol⇒2,3,5-TRIMETHYLPHENOL
isopulegol⇒2-ISOPROPENYL-5-METHYL-1-CYCLOHEXANOL
isovaleraldehyde⇒3-METHYLBUTANAL
isovaleric acid⇒3-METHYLBUTANOIC ACID
isovaleronitrile⇒ISOBUTYL CYANIDE
3-isovalidenephthalide⇒3-(3-METHYLBUTYLIDENE)PHTHALIDE

K

cis-ketone⇒4(4'-*cis*-*tert*-BUTYLCYCLOHEXYL)-4-METHYL-2-PENTANONE
(*d*)-ketone⇒(*d*)-2-METHYL-4[5.5.6(*exo*)-TRIMETHYL-2-(*exo*)-NORBORNYL]CYCLOHEXANONE
(*l*)-ketone⇒(*l*)-2-METHYL-4[5.5.6(*exo*)-TRIMETHYL-2-(*exo*)-NORBORNYL]CYCLOHEXANONE

kahweofuran⇒2-METHYL-3-OXA-8-THIABICYCLO[3.3.0]-1,4-OCTADIENE
karahana ether⇒8,8-DIMETHYL-2-METHYLENE-6-OXABICYCLO[3.2.1]OCTANE

L

lauraldehyde⇒DODECANAL
lauryl propylene diamine⇒DODECYL PROPYLENE DIAMINE
leaf alcohol⇒(*Z*)-3-HEXEN-1-OL
leaf aldehyde⇒(*E*)-2-HEXENAL
lenthionine⇒1,2,3,5,6-PENTATHIEPANE
linolenic acid⇒9,12,15-OCTADECATRIENOIC ACID
limonene⇒4-ISOPROPENYL-1-METHYL-1-CYCLOHEXENE
(4*R*)-(+)-limonene⇒(*R*)-(+)-4-ISOPROPENYL-1-METHYL-1-CYCLOHEXENE
(4*S*)-(-)-limonene⇒(*S*)-(-)-4-ISOPROPENYL-1-METHYL-1-CYCLOHEXENE
(*d*)-limonene⇒(*R*)-(+)-4-ISOPROPENYL-1-METHYL-1-CYCLOHEXENE
(*l*)-limonene⇒(*S*)-(-)-4-ISOPROPENYL-1-METHYL-1-CYCLOHEXENE

LIMONENE EPOXIDE (limonene oxide) *[1195-92-2]*
Buttery *et al.* (1987b) d 0.1

limonene oxide⇒LIMONENE EPOXIDE

cis-LIMONENE OXIDE *[13837-75-7]*
Yang *et al.* (1992); Tamura *et al.* (1993) d 0.25

trans-LIMONENE OXIDE *[4959-35-7]*
Yang *et al.* (1992); Tamura *et al.* (1993) d 0.25

8,9-LIMONENE OXIDE *[31684-93-2]*
Boonbumrung *et al.* (2001) d 0.20

(4*R*)-(+)-*cis*-LIMONENE-1,2-OXIDE
Padrayuttawat *et al.* (1997) d 0.26

(4*R*)-(+)-*trans*-LIMONENE-1,2-OXIDE
Padrayuttawat *et al.* (1997) d 0.20

(4*R*,8*R*)-LIMONENE-8,9-OXIDE
Padrayuttawat *et al.* (1997) d 0.1

(4*R*,8*S*)-LIMONENE-8,9-OXIDE
Padrayuttawat *et al.* (1997) d 0.1

linalool⇒3,7-DIMETHYL-1,6-OCTADIEN-3-OL
(3*S*)-(+)-linalool⇒(3*S*)-(+)-3,7-DIMETHYL-1,6-OCTADIEN-3-OL
(3*R*)-(-)-linalool⇒(3*R*)-(-)-3,7-DIMETHYL-1,6-OCTADIEN-3-OL
linalool oxide A⇒*trans*-5-ETHENYLTETRAHYDRO-α,α,5-TRIMETHYL-2-FURANMETHANOL
cis-linalooloxide⇒*cis*-α,α,5-TRIMETHYL-5-VINYLTETRAHYDROFURFURYL ALCOHOL
trans-linalool oxide⇒*trans*-α,α,5-TRIMETHYL-5-VINYLTETRAHYDROFURFURYL ALCOHOL
linalyl acetate⇒3,7-DIMETHYL-1,6-OCTADIEN-3-YL ACETATE
lindane⇒γ-1,2,3,4,5,6-HEXACHLOROCYCLOHEXANE

M

malathion⇒O,O-DIMETHYL S-(1,2-DICARBETHOXYETHYL)PHOSPHORODITHIOATE
maltol⇒3-HYDROXY-2-METHYL-4-PYRONE
maple lactone⇒2-HYDROXY-3-METHYL-2-CYCLOPENTEN-1-ONE
melonal⇒2,6-DIMETHYL-5-HEPTENAL
p-mentha-1,3,8-triene⇒4-ISOPROPENYL-1-METHYL-1,3-CYCLOHEXADIENE
1-*p*-menthene-8-thiol⇒4-ISOPROPYL-8-MERCAPTO-1-METHYLTHIOL-1-CYCLOHEXENE
(1*S*,3*S*,4*R*)-(+)-menthol⇒(1*S*,2*R*,5*S*)-(+)--2-ISOPROPYL-5-METHYLCYCLOHEXANOL
(1*R*,3*R*,4*S*)-(-)-menthol⇒(1*R*,2*S*,5*R*)-(-)-2-ISOPROPYL-5-METHYLCYCLOHEXANOL
(*l*)-menthol⇒(1*R*,2*S*,5*R*)-(-)-2-ISOPROPYL-5-METHYLCYCLOHEXANOL

menthone⇒2-ISOPROPYL-5-METHYLCYCLOHEXANONE
(2*R*,5*S*)-(+)-menthone⇒(2*R*,5*S*)-(+)-2-ISOPROPYL-5-METHYLCYCLOHEXANONE
(2*S*,5*R*)-(-)-menthone⇒(2*S*,5*R*)-(-)-2-ISOPROPYL-5-METHYLCYCLOHEXANONE
(*l*)-menthone⇒(2*S*,5*R*)-(-)-2-ISOPROPYL-5-METHYLCYCLOHEXANONE

MENTHYLOXYPYRAZINE (2-{[(1*S*,2*R*,5*S*)-2-isopropyl-5-methylcyclohexyl]oxy}pyrazine)
Masuda & Mihara (1989) d 0.003

(*d*)-MENTHYLOXYPYRAZINE ((+)-2-{[(1*S*,2*R*,5*S*)-2-isopropyl-5-methylcyclohexyl]oxy}pyrazine)
Masuda & Mihara (1989) d 0.01

(*l*)-MENTHYLOXYPYRAZINE ((-)-2-{[(1*R*,2*S*,5*R*)-2-isopropyl-5-methylcyclohexyl]oxy}pyrazine)
Masuda & Mihara (1989) d 0.002

2-MERCAPTOBENZOTHIAZOLE *[149-30-4]*
Lillard & Powers (1975) d 1.76

3-MERCAPTO-2-BUTANONE *[40789-98-8]*
Schieberle & Hofmann (1996a, 1996b) 0.003

2-MERCAPTOETHANOL *[60-24-2]*
Baker (1963) 0.64
Brown *et al.* (1968) 12.5

3-MERCAPTO-1-HEXANOL ((*R*/*S*)-3-sulfanylhexan-1-ol) *[51755-83-0]*
Steinhaus *et al.* (2009); Sinuco *et al.* (2010) d 0.000 06

3-MERCAPTOHEXYL ACETATE ((*R*/*S*)-3-sulfanylhexyl acetate) *[136954-20-6]*
Steinhaus *et al.* (2009); Sinuco *et al.* (2010) d 0.000 02

2-MERCAPTO-2-METHYL-1-BUTANOL
Sabater-Lűntzel *et al.* (2001) 0.000 1

3-MERCAPTO-2-METHYL-1-BUTANOL *[227456-33-9]*
Widder *et al.* (2001, 2002) 0.000 23

3-MERCAPTO-3-METHYL-1-BUTANOL *[34300-94-2]*
Holscher *et al.* (1992) d 0.002 - 0.006

3-MERCAPTO-3-METHYLBUTYL FORMATE *[50746-10-6]*
Belitz & Grosch (1992f) 0.000 003
Holscher *et al.* (1992) d 0.000 002 - 0.000 005

3-MERCAPTO-2-METHYL-1-HEPTANOL
Widder *et al.* (2001, 2002) 0.077

3-MERCAPTO-2-METHYL-1-HEXANOL
Widder *et al.* (2001, 2002) 0.000 31

3-MERCAPTO-2-METHYLPENTANAL *[227456-28-2]*
Widder *et al.* (2000) d 0.000 95

3-MERCAPTO-2-METHYL-1-PENTANOL *[227456-30-6]*
Widder *et al.* (2000) d 0.000 15
Widder *et al.* (2001, 2002) 0.000 63
Granvogl *et al.* (2004) d 0.000 000 7
Granvogl *et al.* (2004) r 0.000 0016

(2*R*,3*S*)-3-MERCAPTO-2-METHYL-1-PENTANOL
Sabater-Lüntzel *et al.* (2000); Pickenhagen (2001) d 0.000 04
Widder *et al.* (2001, 2002) d 0.000 09

(2*S*,3*R*)-3-MERCAPTO-2-METHYL-1-PENTANOL

Sabater-Lüntzel *et al.* (2000); Pickenhagen (2001)	d	0.000 03

(2*R*,3*R*)-3-MERCAPTO-2-METHYL-1-PENTANOL

Sabater-Lüntzel *et al.* (2000)	d	> 0.012

(2*S*,3*S*)-3-MERCAPTO-2-METHYL-1-PENTANOL

Sabater-Lüntzel *et al.* (2000)	d	> 0.03

2-MERCAPTO-2-METHYL-4-PENTANONE (4-mercapto-4-methylpentan-2-one) *[19872-52-7]*

Tressl *et al.* (1980)		0.000 005
Kendall (1980)		0.000 000 1

4-mercapto-4-methylpentan-2-one⇒2-MERCAPTO-2-METHYL-4-PENTANONE

3-MERCAPTO-2-PENTANONE *[67633-97-0]*

Schieberle & Hofmann (1996a, 1996b); Kerscher & Grosch (2000)		0.000 7

2-MERCAPTO-3-PENTANONE *[17042-24-9]*

Schieberle & Hofmann (1996a, 1996b)		0.000 7

3-MERCAPTO-2-PROPYL-1-PENTANOL

Widder *et al.* (2001, 2002)		0.0105

mesifurane⇒4-METHOXY-2,5-DIMETHYL-3(2*H*)-FURANONE
mesitol⇒2,4,6-TRIMETHYLPHENOL
mesitylene⇒1,3,5-TRIMETHYLBENZENE
methacrylic acid⇒2-METHYLPROPENOIC ACID

METHANAL (formaldehyde) *[50-00-0]*

Middleton (1956)	d	25
Baker (1963)		49.9
Schnabel *et al.* (1988)	d	990 - 2,050

METHANETHIOL (methyl mercaptan) *[74-93-1]*

Guadagni *et al.* (1963,1968); Guadagni (1966, 1969,1970c); Hansen *et al.* (1992)	d	0.000 02
Sato *et al.* (1975); Takahashi (1990)		0.004
Guth & Grosch (1994); Milo (1995b); Milo & Grosch (1997); Kerscher & Grosch (2000)		0.000 2

methanoic acid⇒FORMIC ACID

METHANOL (methyl alcohol) *[67-56-1]*

Moncrieff (1957)		2,000
Sega *et al.* (1967)	d	36.9
Mitchell & Gregson (1968)	r	6,300 - 50,400
De Grunt (1975); De Grunt *et al.* (1977)	d	10
Sarzynski (1982)		125
Pino *et al.* (1986)	d	369.828
Schnabel *et al.* (1988)	d	800 - 1,200

methional⇒3-(METHYLTHIO)PROPANAL
methionol⇒3-(METHYLTHIO)-1-PROPANOL
methionyl acetate⇒3-(METHYLTHIO)PROPYL ACETATE

4-METHOXYBENZALDEHYDE (*p*-anisaldehyde, anisaldehyde) *[123-11-5]*

Zeller & Rychlik (2006)	r	0.047
Pino & Mesa (2006)	d	0.03
Czerny *et al.* (2008)	d	0.027

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Czerny *et al.* (2008)	r	0.047

METHOXYBENZENE (anisole) *[100-66-3]*

Griffiths (1974)	d	0.05
Dietz & Traub (1978)		0.2
Burdack-Freitag *et al.* (2010)	d	0.1525 (normal atmospheric pressure)
Burdack-Freitag *et al.* (2010)	d	0.1975 (low atmospheric pressure)
Burdack-Freitag *et al.* (2010)	r	0.1775 (normal atmospheric pressure)
Burdack-Freitag *et al.* (2010)	r	0.2100 (low atmospheric pressure)

4-METHOXYBENZYL ALCOHOL (anise alcohol, anisyl alcohol, *alpha*-hydroxy-4-methoxytoluene) *[105-13-5]*

Zeller & Rychlik (2006)	r	38

2-METHOXYBICYCLO[4.4.0]DECANE (β-methoxydecalin)

Koppe (1965)		0.005

methoxychlor⇒1,1,1-TRICHLORO-2,2-BIS(4-METHOXYPHENYL)ETHANE
β-methoxydecalin⇒2-METHOXYBICYCLO[4.4.0]DECANE

4-METHOXY-2,5-DIMETHYL-3(2*H*)-FURANONE (mesifurane) *[4077-47-8]*

Pyysalo & Honkanen (1976); Pyysalo *et al.* (1977)	d	0.000 03 - 0.000 04
Pyysalo *et al.* (1979)	d	0.01
Larsen & Poll (1992)	d	0.1 - 1.0
Buttery (1999)	d	3.4
Boonbumrung *et al.* (2001)	d	0.016
Pino & Mesa (2006)	d	0.000 03
Steinhaus *et al.* (2009);Sinuco *et al.* (2010)	d	0.16

2-METHOXY-3,5-DIMETHYLPYRAZINE

Calabretta (1973)	d	0.003
Czerny & Grosch (2000)	d	0.000 000 4

2-METHOXY-3,6-DIMETHYLPYRAZINE *[19846-22-1]*

Mihara *et al.* (1991)	d	0.000 1

2-METHOXYETHANOL (ethylene glycol methyl ether, methyl cellosolve, methyl glycol) *[109-86-4]*

Alexander *et al.* (1982)	d	100

2-METHOXY-2-METHYLBUTANE (TAME, *tert*-amyl methyl ether) *[994-05-8]*

Vetrano (1993b)	d	0.194
Vetrano (1993b)	r	0.443
Van Wezel *et al.* (2009)	d	0.008

1-METHOXY-3-METHYL-3-BUTANETHIOL *[94087-83-9]*

Rigaud *et al.* (1986)		0.000 001
Belitz & Grosch (1992f)		0.000 000 02

2-METHOXY-3-(2-METHYLBUTYL)PYRAZINE *[115652-10-3]*

Mihara & Masuda (1988)	d	0.000 012

2-METHOXY-3-METHYL-5-(2-METHYLBUTYL)PYRAZINE *[78246-21-6]*

Masuda & Mihara (1986); Shibamoto (1986)	d	0.000 01

2-METHOXY-3-METHYL-5-(2-METHYLPENTYL)PYRAZINE *[78261-14-0]*

Masuda & Mihara (1986); Shibamoto (1986)	d	0.0012

2-METHOXY-3-(2-METHYLPENTYL)PYRAZINE *[115652-11-4]*

Mihara & Masuda (1988)	d	0.000 008

2-METHOXY-4-METHYLPHENOL (creosol, *p*-methylguaiacol, 4-methyl-2-methoxyphenol) *[93-51-6]*

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Wasserman (1966)	d	0.09
Dietz & Traub (1978)		0.01
Czerny *et al*. (2008)	d	0.021
Czerny *et al*. (2008)	r	0.03

2-METHOXY-2-METHYLPROPANE (MTBE, methyl *tert*-butyl ether) *[1634-04-4]*

ARCO (1993)	d	0.041 - 0.048
ARCO (1993)	r	0.044 - 0.065
CFDRA (1993)	d	0.000 36 - 0.000 48
Vetrano (1993a); API (1994)	d	0.095
Vetrano (1993a); API (1994)	r	0.193/0.174 - 0.212
Young *et al.* (1996)		0.034
Dale *et al.* (1997)		0.043 - 0.071
Shen *et al.* (1998)		0.0135 - 0.0454
DEA (2000)		0.0074
Stocking *et al.* (2001)		0.015
CCFRA (2003)		0.018
Van Wezel *et al.* (2009)	d	0.007

METHOXYMETHYLPYRAZINE *[63450-30-6]*

Calabretta (1973,1975,1978)	d	0.15

2-METHOXY-3-METHYLPYRAZINE *[2847-30-5]*

Seifert *et al.* (1970); Teranishi *et al.* (1974,1975)	d	0.004
Calabretta (1975,1978)	d	0.003
Masuda & Mihara (1988); Mihara & Masuda (1988)	d	0.007

2-METHOXY-5-METHYLPYRAZINE *[2882-22-6]*

Calabretta (1973,1975,1978)	d	0.015
Mihara & Masuda (1988)	d	0.02

2-METHOXY-6-METHYLPYRAZINE *[2882-21-5]*

Mihara & Masuda (1988)	d	0.017

2-METHOXY-3-OCTYLPYRAZINE *[113685-80-6]*

Masuda & Mihara (1988)	d	0.000 06

2-METHOXY-3-(3-(*E*)-PENTENYL)PYRAZINE *[115652-13-6]*

Mihara & Masuda (1988)	d	0.000 13

2-METHOXY-3-(3-(*Z*)-PENTENYL)PYRAZINE *[115652-14-7]*

Mihara & Masuda (1988)	d	0.000 5

2-METHOXY-3-PENTYLPYRAZINE *[32737-12-5]*

Masuda & Mihara (1988)	d	0.000 02

2-METHOXYPHENOL (guaiacol) *[90-05-1]*

Le Magnen (1952)		0.02 - 0.1
Hoak (1956)		0.002 - 0.070
Wasserman (1966)	d	0.021
Buttery *et al.* (1971,1987b,1988a,1997,1999); Buttery (1981); Hansen *et al.* (1992); Buttery & Ling (1998)	d	0.003
Dietz & Traub (1978)		0.01
Kovacs *et al.* (1984)	d	0.02
Cerny & Grosch (1993)		0.0025
Ong & Acree (1999)		0.002
Karagül-Yüceer *et al.* (2003)	d	0.0109
Eisele & Semon (2005)	d	0.000 48

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Czerny *et al*. (2008)	d	0.000 84
Czerny *et al*. (2008)	r	0.0016

1-(4-METHOXYPHENYL)-2-PROPANONE (anisketone) *[122-84-9]*

Zeller & Rychlijk (2006)	r	1.8

1-METHOXY-2-PROPANOL (propylene glycol methyl ether) *[107-98-2]*

Alexander *et al.* (1982)	d	4

1-METHOXY-4-((*E*)-1-PROPENYL)BENZENE (*trans*-anethole, (*E*)-anethole) *[4180-23-8]*

Zeller & Rychlik (2006); Czerny *et al.* (2008)	r	0.073
Pino & Mesa (2006)	d	0.086
Czerny *et al.* (2008)	d	0.015

2-METHOXY-4-(1-PROPENYL)PHENOL (isoeugenol) *[97-54-1]*

Dietz & Traud (1978)		0.1

2-METHOXY-3-PROPYLPYRAZINE *[25680-57-3]*

Seifert *et al.* (1970); Teranishi *et al.* (1974)	d	0.000 006
Masuda & Mihara (1988)	d	0.000 12

METHOXYPYRAZINE *[3149-28-8]*

Seifert *et al.* (1970); Teranishi *et al.* (1974)	d	0.7
Masuda & Mihara (1988)	d	0.4

2-METHOXYTETRACHLOROPHENOL (tetrachloroguaiacol) *[2539-17-5]*

Kovacs *et al.* (1984)	d	0.88

2-METHOXY-3,4,5-TRICHLOROPHENOL (3,4,5-trichloroguaiacol) *[57057-83-7]*

Kovacs *et al.* (1984)	d	0.074

2-METHOXY-4,5,6-TRICHLOROPHENOL (4,5,6-trichloroguaiacol) *[2668-24-8]*

Kovacs *et al.* (1984)	d	0.074

2-methoxy-4-vinylphenol⇒4-ETHENYL-2-METHOXYPHENOL
2-methoxy-5-vinylphenol⇒5-ETHENYL-2-METHOXYPHENOL

METHYL ACETATE *[79-20-9]*

Balavoine (1948)		0.002
Schnabel *et al.* (1988)	d	1.5 - 47

1-(1-METHYL)ACETONYLPYRROLE *[98612-18-1]*

Tressl *et al.* (1985b)	d	0.1

methyl acetophenone⇒1-ACETYL-4-METHYLBENZENE

METHYL 3-ACETOXYHEXANOATE *[77118-93-5]*

Takeoka *et al.* (1989)	d	0.19

methyl acrylate⇒METHYL 2-PROPENOATE
methyl alcohol⇒METHANOL

METHYLAMINE *[74-89-5]*

Cherkinski (1961)	d	1.0
Baker (1963)		3.33
Amoore & Forrester (1976)	d	182

METHYL 2-AMINOBENZOATE (methyl anthranilate) *[134-20-3]*

Appell (1969)		0.06
Hirvi & Honkanen (1982)	d	0.003
Buttery (1999)	d	0.007

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Pino & Mesa (2006)	d	0.003

methyl amyl ketone⇒2-HEPTANONE
methyl angelate⇒METHYL (*Z*)-2-METHYL-2-BUTENOATE

2-METHYLANISOLE *[578-58-5]*

Dietz & Traub (1978)		0.3

3-METHYLANISOLE *[100-84-5]*

Dietz & Traub (1978)		0.6

4-METHYLANISOLE (methyl *p*-cresol) *[104-93-8]*

Appell (1969)		0.025
Dietz & Traub (1978)		0.2

methyl anthranilate⇒METHYL 2-AMINOBENZOATE
N-methylazolidine⇒1-METHYLPYRROLIDINE
methylbenzene⇒TOLUENE

METHYL BENZOATE *[93-58-3]*

Appell (1969)		1
Pyysalo *et al.* (1977)	d	0.11
Pino & Mesa (2006)	d	0.000 52
Steinhaus *et al.* (2009); Sinuco *et al.* (2010)	d	0.073

2-METHYLBENZOTHIAZOLE *[120-75-2]*

De Grunt (1975); De Grunt *et al.* (1977)	d	0.0075 - 0.008

1-METHYLBICYCLO[3.3.0]-2,8-DIOXA-4-THIAOCTANE

Van Dort *et al.* (1984)		0.1

1-METHYLBICYCLO[3.3.0]-2,4-DITHIA-8-OXAOCTANE (2,3-(methylenedithio)-2-methyltetrahydrofuran) *[67411-25-0]*

Buttery *et al.* (1981)	d	0.000 000 4
Buttery & Seifert (1982)	d	0.006
Van Dort *et al.* (1984)		0.001
Flament *et al.* (1988)		0.006

2-METHYL-1,3-BUTADIENE (isoprene) *[78-79-5]*

Cherkinski (1961)	d	0.005

2-METHYLBUTANAL *[96-17-3]*

Amoore *et al.* (1976)	d	0.0125
Schnabel *et al.* (1988)	d	0.0024 - 0.0056
Buttery *et al.* (1994a,1997,1999);		
Buttery & Ling (1995,1998)	d	0.003
Milo & Grosch (1996a); Milo & Grosch (1997)		0.0013
Czerny *et al.* (2007a); Czerny *et al.* (2008)	r	0.0044
Czerny *et al.* (2008)	d	0.0015
Giri *et al.* (2010)	d	0.001

3-METHYLBUTANAL (isovaleraldehyde) *[590-86-3]*

Guadagni *et al.* (1963,1972); Buttery *et al.* (1987a, 1989,1990a,1994a,1997,1999); Buttery (1993);		
Buttery & Ling (1995,1998)	d	0.000 15 - 0.000 2
Land (1968)	r	13
Mulders (1972,1973)	d	0.007 - 0.009
Amoore *et al.* (1976)	d	0.00202
Schnabel *et al.* (1988)	d	0.0024 - 0.0040
Guth & Grosch (1994); Milo & Grosch (1996,1997);		
Kerscher & Grosch (2000)		0.000 35 - 0.000 4

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Tandon *et al.* (2000)	d	0.006
Fabrellas *et al.* (2004)		0.000 8 - 0.0039
Freuze *et al.* (2005)	d	0.005
Czerny *et al*. (2007a); Czerny *et al.* (2008)	r	0.0012
Czerny *et al*. (2008)	d	0.000 5
Giri *et al.* (2010)	d	0.0011

3-methylbutanenitrile⇒ISOBUTYL CYANIDE

2-METHYL-2-BUTANETHIOL (*tert*-amyl mercaptan) *[1679-09-0]*

Guadagni (1979)	d	0.000 002

METHYL BUTANOATE (methyl butyrate) *[623-42-7]*

Ahmed *et al.* (1978a)	d	0.043
Pino *et al.* (1986)	d	0.0151
Schnabel *et al.* (1988)	d	0.009 - 0.09
Takeoka *et al.* (1989,1990,2008)	d	0.06 - 0.076
Larsen & Poll (1992)	d	0.001 - 0.01
Pino & Mesa (2006)	d	0.059

2-METHYLBUTANOIC ACID (α-methylbutyric acid) *[116-53-0]*

Amoore *et al.* (1968); Amoore (1976)	d	0.60
Zoeteman *et al.* (1971)		40
Pyysalo *et al.* (1977)	d	0.18
Schnabel *et al.* (1988)	d	4.7 - 9.3
Larsen & Poll (1992)	d	0.01 - 0.1
Gassenmeier & Schieberle (1995)	d	0.1
Rychlik & Grosch (1996); Rychlik (1998); Buhr *et al.* (2010)		0.5 - 0.54
Czerny *et al*. (2008)	d	2.2
Czerny *et al*. (2008)	r	5.8

3-METHYLBUTANOIC ACID (isovaleric acid) *[503-74-2]*

Amoore (1968)	d	0.19
Amoore *et al.* (1968); Amoore (1976)	d	0.132
Zoeteman *et al.* (1971)		5
Amoore *et al.* (1975b)	d	0.142
Pyysalo *et al.* (1977); Hirvi & Honkanen (1984)	d	0.15
Boelens *et al.* (1983)	d	1
Schnabel *et al.* (1988)	d	0.92 - 2.8
Buttery *et al.* (1990a,1999); Buttery (1993); Buttery & Ling (1998)	d	0.25
Schieberle (1991,1992); Grosch *et al.* (1993); Gassenmeier & Schieberle (1995)	d	0.1
Rychlik & Grosch (1996); Rychlik (1998)		0.74
Rashash *et al.* (1997)		0.02
Ong & Acree (1999)		0.012
Pino & Mesa (2006)	d	0.07
Czerny *et al*. (2007a,2007c); Czerny *et al.* (2008)	r	1.2
Czerny *et al*. (2008)	d	0.49

2-METHYL-1-BUTANOL (active amyl alcohol) *[137-32-6]*

Sega *et al.* (1967)	d	4.15
Pyysalo *et al.* (1977)	d	1.9
Schnabel *et al.* (1988)	d	0.25 - 0.42
Takeoka *et al.* (1990a); Buttery & Ling (1995)	d	0.3
Takahashi (1990)		50 - 65
Sugisawa *et al.* (1991)	d	32.5
Czerny *et al*. (2008)	d	1.2
Czerny *et al*. (2008)	r	3.7
Giri *et al*. (2010)	d	0.0159

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

3-METHYL-1-BUTANOL (isoamyl alcohol) *[123-51-3]*

Williams (1953)		0.5
Sega *et al.* (1967)	d	0.291
Buttery *et al.* (1971,1978b,1987a,1988a,1989,1990a, 1994a); Hansen *et al.* (1992); Buttery (1993); Buttery & Ling (1995)	d	0.25 - 0.3
Mulders (1972,1973)	d	0.7 - 0.77
Pino *et al.* (1986)	d	1.005
Schnabel *et al.* (1988)	d	0.81 - 4.1
Takahashi (1990)		50 - 70
Schieberle (1991,1992); Gassenmeier & Schieberle (1995)	d	1
Sugisawa *et al.* (1991)	d	2.8
Tandon *et al.* (2000)	d	0.071
Boonbumrung *et al.* (2001)	d	3.06
Pino & Mesa (2006)	d	0.3
Czerny *et al*. (2008)	d	0.22
Czerny *et al*. (2008)	r	0.98
Giri *et al*. (2010)	d	0.004

2-METHYL-2-BUTANOL (*tert*-amyl alcohol) *[75-85-4]*

Schnabel *et al.* (1988)	d	20 - 33

3-METHYL-2-BUTANOL *[598-75-4]*

Schnabel *et al.* (1988)	d	0.41 - 0.82

3-METHYL-2-BUTANONE (methyl isopropyl ketone) *[563-80-4]*

Amoore *et al.* (1976)	d	1.67
Schnabel *et al.* (1988)	d	0.81 - 1.2

4-(2-methylbutan-2-yl)phenol⇒4-*tert*-PENTYLPHENOL

2-METHYL-2-BUTENAL *[1115-11-3]*

Buttery *et al.* (1990a); Buttery (1993)	d	0.5

(*E*)-2-METHYL-2-BUTENAL (tiglaldehyde, tiglic aldehyde) *[497-03-0]*

Giri *et al.* (2010)	d	0.4589

3-METHYL-2-BUTENE-1-THIOL (prenyl mercaptan) *[5287-45-6]*

Tressl *et al.* (1977a,1979b,1980)		0.000 002 - 0.000 005
Holscher *et al.* (1992)	d	0.000 000 2 - 0.000 000 4

3-METHYL-2-BUTENOIC ACID *[541-47-9]*

Pyysalo *et al.* (1977)	d	14.0

3-METHYL-3-BUTENOIC ACID *[1617-31-8]*

Pyysalo *et al.* (1977)	d	0.60

3-METHYL-2-BUTEN-1-OL (prenyl alcohol) *[556-82-1]*

Pyysalo *et al.* (1977)	d	0.14
Sugisawa *et al.* (1991)	d	7.75
Boonbumrung *et al.* (2001)	d	0.25

2-METHYL-3-BUTEN-1-OL *[4516-90-9]*

Tamura *et al.* (2001)	d	1.1

3-METHYL-3-BUTEN-1-OL *[763-32-6]*

Giri *et al.* (2010)	d	0.547125

2-METHYL-3-BUTEN-2-OL *[115-18-4]*

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Sugisawa *et al.* (1991); Tamura *et al.* (1995) d 100
Boonbumrung *et al*. (2001) d 1.14

2-(3-METHYL-2-BUTENYL)-3-METHYLFURAN (rose furan) *[15186-51-3]*
Ohloff (1977,1978); Pickenhagen & Furrer (1990) 0.08 - 0.2

(1-METHYLBUTOXY)PYRAZINE
Masuda & Mihara (1989) d 0.1

(*d*)-(1-METHYLBUTOXY)PYRAZINE
Masuda & Mihara (1989) d 0.1

(*l*)-(1-METHYLBUTOXY)PYRAZINE
Masuda & Mihara (1989) d 0.1

2-METHYLBUTYL ACETATE (active amyl acetate) *[624-41-9]*
Teranishi *et al.* (1966); Flath *et al.* (1967); Guadagni (1970b); Buttery (1981); Teranishi et al (1987);
Takeoka *et al.* (1992) d 0.005 - 0.011

3-methylbutyl acetate⇒ISOPENTYL ACETATE
3-methylbutyl-2 acetate⇒1,2-DIMETHYLPROPYL ACETATE

2-METHYLBUTYLAMINE *[96-15-1]*
De Grunt (1975) d 10.0

methyl tert-butyl ether⇒2-METHOXY-2-METHYLPROPANE

2-METHYLBUTYL HEXANOATE *[2601-13-0]*
Takeoka *et al.* (1990a) d 0.032

3-(3-METHYLBUTYLIDENE)PHTHALIDE (3-isovalidenephthalide)
Gold & Wilson (1963) r < 0.1

2-methylbutyl isobutyrate⇒2-METHYLBUTYL 2-METHYLPROPANOATE
methyl butyl ketone⇒2-HEXANONE
methyl *tert*-butyl ketone⇒2,2-DIMETHYL-3-BUTANONE

2-METHYLBUTYL 2-METHYLBUTANOATE *[2445-78-5]*
Guadagni *et al.* (1966); Teranishi *et al.* (1987) d 0.024
Giri *et al*. (2010) d 0.075

2-METHYLBUTYL 2-METHYLPROPANOATE (2-methylbutyl isobutyrate) *[2445-69-4]*
Guadagni *et al.* (1966); Teranishi *et al.* (1987) d 0.014

2-METHYLBUTYL PROPANOATE *[2438-20-2]*
Guadagni *et al.* (1966); Teranishi *et al.* (1987) d 0.028

6-(2-METHYLBUTYL)-2,3,5-TRIMETHYLPYRAZINE
Shibamoto (1986) d 1.12

methyl butyrate⇒ METHYL BUTANOATE
α-methylbutyric acid⇒2-METHYLBUTANOIC ACID
methyl caprate⇒METHYL DECANOATE
methyl caproate⇒METHYL HEXANOATE
methyl caprylate⇒METHYL OCTANOATE
methyl cellosolve⇒2-METHOXYETHANOL
methylchavicol⇒1-ALLYL-4-METHOXYBENZENE
methylchloroform⇒1,1,1-TRICHLOROETHANE
methyl *p*-cresol⇒4-METHYLANISOLE

2-METHYLCYCLOHEXANONE *[583-60-8]*
Schnabel *et al.* (1988) d 0.92 - 1.9

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

3-METHYLCYCLOHEXANONE *[591-24-2]*
Schnabel *et al.* (1988) d 0.46 - 0.92

4-METHYLCYCLOHEXANONE *[589-92-4]*
Schnabel *et al.* (1988) d 0.46 - 0.92

4-methylcyclohexathiazole⇒4-METHYL-4,5,6,7-TETRAHYDROBENZOTHIAZOLE
6-methylcyclohexathiazole⇒6-METHYL-4,5,6,7-TETRAHYDROBENZOTHIAZOLE

2-(4-METHYL-3-CYCLOHEXENYL)ISOPROPYL ACETATE (terpinyl acetate) *[80-26-2]*
Sugisawa *et al.* (1991); Yang *et al.* (1992);
Tamura *et al.* (1993) d 2.5

3-METHYLCYCLOPENTADECANONE (muscone) *[541-91-3]*
Amoore *et al.* (1977) d 0.0098
Pickenhagen (1989) 0.103

(*R*)-(-)-3-METHYLCYCLOPENTADECANONE ((*R*)-(-)-muscone)
Pickenhagen (1989) 0.061

(*S*)-(+)-3-METHYLCYCLOPENTADECANONE ((*S*)-(+)-muscone)
Pickenhagen (1989) 0.233

3-methyl-1,2-cyclopentanedione⇒2-HYDROXY-3-METHYL-2-CYCLOPENTEN-1-ONE

5-METHYL-5*H*-CYCLOPENTA[B]PYRAZINE *[65128-99-6]*
Schieberle & Grosch (1987);
Grosch & Schieberle (1991) 0.015 - 0.025

methylcyclopentenolone⇒2-HYDROXY-3-METHYL-2-CYCLOPENTEN-1-ONE

METHYL (*Z*,*Z*)-4,8-DECADIENOATE
Buttery (1966); Guadagni *et al.* (1966);
Teranishi *et al.* (1987) d 0.010

METHYL DECANOATE (methyl caprate) *[110-42-9]*
Schnabel *et al.* (1988) d 0.0043 - 0.0088

METHYL (*Z*)-4-DECENOATE *[7367-83-1]*
Buttery (1966) 0.018
Guadagni *et al.* (1966); Teranishi *et al.*(1987);
Takeoka *et al.* (1989) d 0.003

methyl decyl ketone⇒2-DODECANONE
methyldiethylamine⇒N,N-DIETHYLMETHYLAMINE

METHYL 2,2-DIMETHYLPROPANOATE *[598-98-1]*
Takeoka *et al.* (1995b) d 0.028

2-METHYL-1,3-DINITROBENZENE (2,6-dinitrotoluene) *[606-20-2]*
Kölle *et al.* (1972) 0.05 - 1

2-METHYL-4,6-DINITROPHENOL (4,6-dinitro-*o*-cresol) *[534-52-1]*
Popov & Vrochinskii (1976) 0.93
Dietz & Traud (1978) 1.3

METHYLDITHIOMETHANE (dimethyl disulphide, DMDS) *[624-92-0]*
Amoore & Venstrom (1966); Amoore (1969) d 0.0012
Mulders (1972,1973) d 0.000 16 - 0.000 25
Land (1968) r 0.09

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Sato *et al*. (1975); Takahashi (1990)		0.0065 - 0.007
Buttery *et al.* (1976b); Hansen *et al.* (1992);		
Buttery & Ling (1998)	d	0.012
Milo (1995a, 1995b)		0.000 3
Buttery *et al.* (1997)	d	0.000 3
Giri *et al*. (2010)	d	0.0011

2-[(METHYLDITHIO)METHYL]FURAN (furfuryl methyl disulphide) *[57500-00-2]*
Mulders *et al.* 1976 0.000 04

2,2'-METHYLENEBIS(4-CHLOROPHENOL) *[97-23-4]*
Dietz & Traud (1978) 0.5

2,2'-METHYLENEBIS(3,4,6-TRICHLORPHENOL) (hexachlorophene) *[70-30-4]*
Dietz & Traud (1978) 10

methylene chloride⇒DICHLOROMETHANE

3,4-(METHYLENEDIOXY)BENZALDEHYDE (piperonal, heliotropine) *[120-57-0]*
Helbig (1939) 0.062
Appell (1969) 1

2,3-(methylenedithio)-2-methyltetrahydrofuran⇒1-METHYLBICYCLO[3.3.0]-2,4-DITHIA-8-OXAOCTANE
(±)-methyl epijasmonate⇒METHYL 3-OXO-2-(2-(*Z*)-PENTENYL)CYCLOPENTANE-1-ACETATE
(+)-(*Z*)-methyl epijasmonate⇒(1*R*,2*S*)-(+)-METHYL 3-OXO-2-(2-(*Z*)-PENTENYL)CYCLOPENTANE-1-ACETATE
methyleugenol⇒4-ALLYL-1,2-DIMETHOXYBENZENE
methylethylamine⇒N-ETHYLMETHYLAMINE
methyl ethyl ketone⇒BUTANONE

METHYL FORMATE *[107-31-3]*
Schnabel *et al.* (1988) d 190 – 490

N-methyl-2-formylpyrrole⇒2-FORMYL-1-METHYLPYRROLE

2-METHYLFURAN *[534-22-5]*
Mulders (1972,1973) d 3.5 - 4
Trubko & Teplyakova (1981b) 0.2

5-methylfuranacetic acid⇒5-METHYL-2-FURYLACETIC ACID

2-METHYL-3-FURANTHIOL *[28588-74-1]*
Gasser & Grosch (1988) 0.000 005 - 0.000 01
Belitz & Grosch (1992c) 0.000 000 4
Schieberle & Hofmann (1996a, 1996b);
Kerscher & Grosch (2000) 0.000 007

5-METHYLFURFURAL *[620-02-0]*
Buttery *et al.* (1997,1999) d 0.5
Boonbumrung *et al.* (2001) d 1.11

(5-methylfurfuryl)mercaptan⇒2-(5-METHYL)FURYLMETHANETHIOL

1-(5-METHYLFURFURYL)PYRROLE *[13678-52-9]*
Tressl *et al.* (1979c,1981,1985b); Silwar (1982) d 0.01

5-methyl-2- ⇒2-(5-METHYL)FURYLMETHANETHIOL

5-METHYL-2-FURYLACETIC ACID (5-methylfuranacetic acid)
Davídek *et al.* (1979) 6.8

3-(2-METHYL)FURYLDITHIO-3-(2-METHYL)FURAN (bis(2-methyl-3-furyl) disulphide, 3,3'-dithiobis(2-methylfuran)) *[28588-75-2]*

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Buttery *et al.* (1984)	d	0.000 000 02
Jezussek (2002)		0.000 000 08
Czerny *et al.* (2008)	d	0.000 000 32
Czerny *et al.* (2008)	r	0.000 000 76

2-(5-METHYL)FURYLMETHANETHIOL ((5-methylfurfuryl)mercaptan, 5-methyl-2-furfurylthiol) *[59303-05-8]*

Tressl & Silwar (1981); Silwar (1982);		
Tressl (1989,1990)		0.000 05
Schieberle & Hofmann (1996b)		0.000 05

methyl glycol⇒2-METHOXYETHANOL
p-methylguaiacol⇒2-METHOXY-4-METHYLPHENOL

2-METHYL-2-(*E*)-4-HEPTADIEN-6-ONE

Buttery *et al.* (1971,1988a,1990a); Buttery (1993)	d	0.375 - 0.38

2-METHYLHEPTANE *[592-27-8]*

Zoeteman *et al.* (1971)		50

METHYL HEPTANOATE *[106-73-0]*

Guadagni *et al.* (1966); Teranishi *et al.* (1987);		
Takeoka *et al.* (1989)	d	0.004

2-METHYLHEPTANOIC ACID *[1188-02-9]*

Amoore (1976)	d	29.5

2-METHYL-2-HEPTANOL *[625-25-2]*

Schnabel *et al.* (1988)	d	0.16 - 0.40

6-METHYL-2-HEPTANOL *[4730-22-7]*

Schnabel *et al.* (1988)	d	0.039 - 0.40

2-METHYL-3-HEPTANOL *[18720-62-2]*

Schnabel *et al.* (1988)	d	0.33 - 0.57

3-METHYL-3-HEPTANOL *[5582-82-1]*

Schnabel *et al.* (1988)	d	0.0078 - 0.042

4-METHYL-3-HEPTANOL *[14979-39-6]*

Schnabel *et al.* (1988)	d	0.23 - 0.40

5-METHYL-3-HEPTANOL *[18720-65-5]*

Schnabel *et al.* (1988)	d	0.039 - 0.078

6-METHYL-3-HEPTANOL *[18720-66-6]*

Schnabel *et al.* (1988)	d	0.25 - 0.42

2-METHYL-4-HEPTANOL *[21570-35-4]*

Schnabel *et al.* (1988)	d	0.40 - 0.81

3-METHYL-4-HEPTANOL *[1838-73-9]*

Schnabel *et al.* (1988)	d	0.078 - 0.42

4-METHYL-4-HEPTANOL *[598-01-6]*

Schnabel *et al.* (1988)	d	0.57 - 0.82

6-METHYL-2-HEPTANONE *[928-68-7]*

Schnabel *et al.* (1988)	d	0.0081 - 0.024

2-METHYL-3-HEPTANONE *[13019-20-0]*

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Schnabel *et al.* (1988)	d	0.075 - 0.41

5-METHYL-3-HEPTANONE *[541-85-5]*

Schnabel *et al.* (1988)	d	0.041 - 0.082

2-METHYL-4-HEPTANONE *[626-33-5]*

Schnabel *et al.* (1988)	d	0.024 - 0.058

3-METHYL-4-HEPTANONE (5-methyl-4-heptanone) *[15726-15-5]*

Burdack-Freitag & Schieberle (2010)		0.000 05

5-methyl-4-heptanone⇒3-METHYL-4-HEPTANONE

6-METHYL-5-HEPTEN-2-OL *[1569-60-4]*

Buttery *et al.* (1990a); Buttery (1993)	d	2

methylheptenone⇒6-METHYL-5-HEPTEN-2-ONE

6-METHYL-5-HEPTEN-2-ONE (methylheptenone) *[110-93-0]*

Buttery *et al.* (1971,1987a,1988a,1989,1990a);		
Buttery (1981,1993)	d	0.05
Pyysalo *et al.* (1977)	d	0.85
Sugisawa *et al.* (1991); Yang *et al.* (1992);		
Tamura *et al.* (1993)	d	1 - 8.2
Tandon *et al.* (2000)	d	0.525
Tamura *et al.* (2001)	d	0.16
Giri *et al*. (2010)	d	0.068

5-METHYL-(*E*)-2-HEPTEN-4-ONE ((*E*)-filbertone) *[102322-83-8]*

Belitz & Grosch (1992d)		0.000 05
Burdack-Freitag & Schieberle (2010)		0.000 05

methyl heptine carbonate⇒METHYL 2-OCTYNOATE
methyl heptyl ketone⇒2-NONANONE

METHYL HEXADECANOATE (methyl palmitate) *[112-39-0]*

Buttery *et al.* (1988a)	d	> 2

N-methylhexahydroazine⇒N-METHYLPIPERIDINE

3-METHYLHEXANE *[589-34-4]*

Zoeteman *et al.* (1971)		500

METHYL HEXANOATE (methyl caproate) *[106-70-7]*

Pyysalo *et al.* (1977)	d	0.087
Schnabel *et al.* (1988)	d	0.039 - 0.43
Takeoka *et al.* (1989,1990a)	d	0.07 - 0.084
Larsen & Poll (1992)	d	0.01 - 0.1
Pino & Mesa (2006)	d	0.07

2-METHYLHEXANOIC ACID *[4536-23-6]*

Schnabel *et al.* (1988)	d	0.92 - 2.7

5-METHYLHEXANOIC ACID *[628-46-6]*

Schnabel *et al.* (1988)	d	4.6 - 27

5-METHYL-1-HEXANOL *[627-98-5]*

Schnabel *et al.* (1988)	d	0.41 - 0.58

2-METHYL-2-HEXANOL *[625-23-0]*

Schnabel *et al.* (1988)	d	1.6 - 3.3

3-METHYL-2-HEXANOL *[2313-65-7]*
Schnabel *et al.* (1988) d 0.081 - 0.41

5-METHYL-2-HEXANOL *[627-59-8]*
Schnabel *et al.* (1988) d 0.33 - 0.65

2-METHYL-3-HEXANOL *[617-29-8]*
Schnabel *et al.* (1988) d 0.046 - 0.081

3-METHYL-3-HEXANOL *[597-96-6]*
Schnabel *et al.* (1988) d 0.046 - 0.081

4-METHYL-3-HEXANOL *[615-29-2]*
Schnabel *et al.* (1988) d 0.17 - 0.43

5-METHYL-3-HEXANOL *[623-55-2]*
Schnabel *et al.* (1988) d 0.41 - 0.83

3-METHYL-2-HEXANONE *[2550-21-2]*
Schnabel *et al.* (1988) d 0.041 - 0.083

4-METHYL-2-HEXANONE *[105-42-0]*
Schnabel *et al.* (1988) d 0.000 81 - 0.0041

5-METHYL-2-HEXANONE (methyl isoamyl ketone) *[110-12-3]*
Schnabel *et al.* (1988) d 0.062 - 0.089

2-METHYL-3-HEXANONE *[7379-12-6]*
Schnabel *et al.* (1988) d 0.0081 - 0.040

5-METHYL-3-HEXANONE *[623-56-3]*
Schnabel *et al.* (1988) d 0.040 - 0.081

METHYL (*E*)-4-HEXENOATE *[14017-81-3]*
Takeoka *et al.* (1991a) d 0.147

METHYL 5-HEXENOATE *[2396-80-7]*
Takeoka *et al.* (1991a) d 0.194

(1-METHYLHEXOXY)PYRAZINE
Masuda & Mihara (1989) d 0.09

(*d*)-(1-METHYLHEXOXY)PYRAZINE
Masuda & Mihara (1989) d 0.03

(*l*)-(1-METHYLHEXOXY)PYRAZINE
Masuda & Mihara (1989) d 0.09

methyl hexyl ketone⇒2-OCTANONE

METHYL 3-HYDROXYBENZOATE *[19348-10-9]*
Pyysalo *et al.* (1977) d 2.0

METHYL 4-HYDROXYBENZOATE *[99-76-3]*
Pyysalo *et al.* (1977) d 2.6

(*R*)-METHYL 3-HYDROXYHEXANOATE
Buettner & Schieberle (2001b) 3.76

METHYL 4-HYDROXY-3-METHOXYBENZOATE (methyl vanillate) *[3943-74-6]*

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Pyysalo *et al.* (1977)	d	0.79

METHYL (2*R*,3*S*)-2-HYDROXY-3-METHYLPENTANOATE

Steinhaus *et al.* (2009); Sinuco *et al.* (2010)	d	0.013

METHYL (2*S*,3*S*)-2-HYDROXY-3-METHYLPENTANOATE

Steinhaus *et al.* (2009); Sinuco *et al.* (2010)	d	0.0024

3-METHYLINDOLE (skatole) *[83-34-1]*

Kauffmann (1907)		0.052
Dietz & Traud (1978)		0.01
Buttery *et al.* (1979); Buttery *et al.* (1997)	d	0.000 2
Heiler (1995); Heiler & Schieberle (1997)		0.000 7
Milo & Reineccius (1997)		0.15
Karagül-Yüceer *et al.* (2003)	d	0.003
Czerny *et al.* (2008)	d	0.000 13
Czerny *et al.* (2008)	r	0.000 41

methyl isoamyl ketone⇒5-METHYL-2-HEXANONE
2-methylisoborneol⇒1,2,7,7-TETRAMETHYL-*exo*-BICYCLO[2.2.1]-2-HEPTANOL
methyl isobutyl ketone⇒4-METHYL-2-PENTANONE
methyl isobutyrate⇒METHYL 2-METHYLPROPANOATE
trans-methylisoeugenol⇒(*E*)-1,2-DIMETHOXY-4-(1-PROPENYL)BENZENE
methyl isopropyl ketone⇒3-METHYL-2-BUTANONE
methyl isovalerate⇒METHYL 3-METHYLBUTANOATE

METHYL ISOTHIOCYANATE (methyl mustard oil) *[556-61-6]*

Brevard *et al.* (1992)	d	12.6

(±)-methyl jasmonate⇒METHYL 3-OXO-2-(2-(*Z*)-PENTENYL)CYCLOPENTANE-1-ACETATE
(-)-(*Z*)-methyl jasmonate⇒ (1*R*,2*R*)-(-)-METHYL 3-OXO-2-(2-(*Z*)-PENTENYL)CYCLOPENTANE-1-ACETATE
methyl mercaptan⇒METHANETHIOL
2-methylmercaptoacetaldehyde⇒2-(METHYLTHIO)ETHANAL
2-methylmercaptoethanol⇒2-(METHYLTHIO)ETHANOL
4-methylmercaptophenol⇒4-(METHYLTHIO)PHENOL
methyl methacrylate⇒METHYL 2-METHYL-2-PROPENOATE

METHYL METHANETHIOSULPHINATE *[13882-12-7]*

Chin & Lindsay (1994)	d	0.55

METHYL METHANETHIOSULPHONATE *[2949-92-0]*

Chin & Lindsay (1994)	d	5

4-methyl-2-methoxyphenol⇒2-METHOXY-4-METHYLPHENOL

METHYL N-METHYLANTHRANILATE *[85-91-6]*

Pino & Mesa (2006)	d	0.349

METHYL 2-METHYLBUTANOATE *[868-57-5]*

Teranishi *et al.* (1987); Takeoka *et al.* (1989)	d	0.000 25
Schnabel *et al.* (1988)	d	0.000 93 - 0.0044
Blank *et al.* (1992)		0.000 4
Takeoka *et al.* (1998, 2008)	d	0.000 1 - 0.000 14
Pino & Mesa (2006)	d	0.000 25

METHYL (*S*)-(+)-2-METHYLBUTANOATE *[10307-60-5]*

Blank *et al.* (1992)		0.000 3

METHYL 3-METHYLBUTANOATE (methyl isovalerate) *[556-24-1]*

Schnabel *et al.* (1988)	d	0.0044 - 0.044

METHYL (*E*)-2-METHYL-2-BUTENOATE (methyl tiglate) *[6622-76-0]*

Takeoka *et al.* (1998,2008)	d	0.13

METHYL (*Z*)-2-METHYL-2-BUTENOATE (methyl angelate)
Takeoka *et al.* (1998) d 0.0016

3-METHYL-5-(2-METHYLBUTYL)-2-(METHYLTHIO)PYRAZINE *[99784-24-4]*
Masuda & Mihara (1986); Shibamoto (1986) d 0.000 01 - 0.000 1

3-METHYL-5-(2-METHYLBUTYL)-2-PHENOXYPYRAZINE *[99784-20-0]*
Masuda & Mihara (1986); Shibamoto (1986) d 0.056 - 0.12

3-METHYL-5-(2-METHYLBUTYL)-2-(PHENYLTHIO)PYRAZINE *[99784-32-4]*
Masuda & Mihara (1986) d 0.056

2-METHYL-3-(METHYLDITHIO)FURAN *[65505-17-1]*
Tressl & Silwar (1981); Silwar (1982) 0.000 01
Schieberle (1998) 0.000 004

7-METHYL-3-METHYLENE-1,6-OCTADIENE (myrcene) *[123-35-3]*
Guadagni *et al.* (1966); Guadagni (1969,1970c);
Buttery *et al.* (1968,1974,1987b);
Teranishi *et al.* (1981); Takeoka (2001) d 0.013 - 0.015
Ahmed *et al.* (1978a) d 0.036
Pino *et al.* (1986) d 0.046
Tamura *et al.* (1996) d 0.67
Tamura *et al.* (1996) r 0.915
Masanetz & Grosch (1998a, 1998b) 0.014
Padrayuttawat *et al.* (1997); Tamura *et al.* (2001);
Boonbumrung *et al.* (2001) d 0.099 - 0.10
Pino & Mesa (2006) d 0.015
Zeller & Rychlik (2006); Czerny *et al*. (2008) r 0.0049
Czerny *et al.* (2008) d 0.0012

METHYL 4-METHYL-2-HEXENOATE *[54378-97-1]*
Teranishi *et al.* (1987) d 0.006

3-methyl-4-methylmercaptophenol⇒3-METHYL-4-(METHYLTHIO)PHENOL

3-METHYL-5-(2-METHYLPENTYL)-2-(METHYLTHIO)PYRAZINE *[78246-18-1]*
Masuda & Mihara (1986); Shibamoto (1986) d 0.012

3-METHYL-5-(2-METHYLPENTYL)-2-PHENOXYPYRAZINE *[99784-21-1]*
Masuda & Mihara (1986); Shibamoto (1986) d 0.14

3-METHYL-5-(2-METHYLPENTYL)-2-(PHENYLTHIO)PYRAZINE *[99784-33-5]*
Masuda & Mihara (1986); Shibamoto (1986) d 0.16

METHYL 3-METHYL-3-PHENYL-2,3-EPOXYPROPANOATE
Tekin & Karaman (1992) d 34

METHYL 2-METHYLPROPANOATE (methyl isobutyrate) *[547-63-7]*
Schnabel *et al.* (1988) d 0.0045 - 0.0086
Takeoka *et al.* (1989,2008) d 0.007

METHYL 2-METHYL-2-PROPENOATE (methyl methacrylate) *[80-62-6]*
Piringer & Granzer (1984); Granzer *et al.* (1986) 0.05 - 0.5

cis-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN (*cis*-rose oxide) *[876-17-5]*
Ong & Acree (1999) 0.000 1

(*l*)-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN ((*l*)-rose oxide) *[3033-23-6]*
Ohloff (1977,1978) 0.000 5

METHYL 4-(METHYLTHIO)BUTANOATE *[53053-51-3]*
Takeoka *et al.* (1991a) d 0.133

METHYL(METHYLTHIO)DITHIOMETHANE (methyl (methylthio)methyl disulphide, 2,3,5-trithiahexane) *[423474-44-2]*
Kubota *et al.* (1994) d 0.04
Spadone *et al.* (2006) d 0.000 8

2-METHYL-3-(METHYLTHIO)FURAN *[63012-97-5]*
Tressl & Silwar (1981); Silwar (1982) 0.000 05
Gasser & Grosch (1988) 0.025 - 0.030

methyl (methylthio)methyl disulphide⇒ METHYL(METHYLTHIO)DITHIOMETHANE

3-METHYL-4-(METHYLTHIO)PHENOL (3-methyl-4-methylmercaptophenol) *[3120-74-9]*
Dietz & Traud (1978) 0.2

METHYL 3-(METHYLTHIO)PROPANOATE *[13532-18-8]*
Takeoka *et al.* (1989) d 0.18

METHYL 3-(METHYLTHIO)-(*E*)-2-PROPENOATE *[15804-85-5]*
Takeoka *et al.* (1991a) d 0.095

METHYL 3-(METHYLTHIO)-(*Z*)-2-PROPENOATE *[15904-84-4]*
Takeoka *et al.* (1991a) d 0.025

2-METHYL-3-(METHYLTHIO)PYRAZINE *[2882-20-4]*
Calabretta (1973,1975,1978) d 0.001
Masuda & Mihara (1986,1988) d 0.004

2-METHYL-5-(METHYLTHIO)PYRAZINE *[2884-14-2]*
Calabretta (1973,1978) d 0.004
Mihara & Masuda (1988) d 0.06

2-METHYL-6-(METHYLTHIO)PYRAZINE *[2884-13-1]*
Mihara & Masuda (1988) d 0.02

methyl mustard oil⇒METHYL ISOTHIOCYANATE

1-METHYLNAPHTHALENE *[90-12-0]*
Lillard & Powers (1975) d 0.02
De Grunt (1975); De Grunt *et al.* (1977) d 0.0075 - 0.008

2-METHYLNAPHTHALENE *[91-57-6]*
Lillard & Powers (1975) d 0.01
De Grunt *et al.* (1977) d 0.003

methyl neopentyl ketone⇒4,4-DIMETHYL-2-PENTANONE

1-METHYL-2-NITROBENZENE (2-nitrotoluene) *[88-72-2]*
De Grunt (1975) d 0.13

1-METHYL-4-NITROBENZENE (4-nitrotoluene) *[99-99-0]*
De Grunt (1975); De Grunt *et al.* (1977) d 0.003

3-METHYL-1-NITROBUTANE *[627-67-8]*
Buttery *et al.* (1989,1990a); Buttery (1993) d 0.15

4-METHYL-2-NITROPHENOL (2-nitro-*p*-cresol) *[119-33-5]*
Dietz & Traud (1978) 0.8

5-METHYL-2-NITROPHENOL (6-nitro-*m*-cresol) *[700-38-9]*
Dietz & Traud (1978) 0.2

4-METHYL-3-NITROPHENOL (3-nitro-*p*-cresol) *[2042-14-0]*
Dietz & Traud (1978) 2

3-METHYL-4-NITROPHENOL (4-nitro-*m*-cresol) *[2581-34-2]*
Dietz & Traud (1978) 5

2-METHYL-5-NITROPHENOL (5-nitro-*o*-cresol) *[5428-54-6]*
Dietz & Traud (1978) 0.3

2-METHYL-1-NITROPROPANE *[625-74-1]*
Buttery *et al.* (1990b) d 5

2-METHYLNONANE *[871-83-0]*
Zoeteman *et al.* (1971) 10

3-METHYL-2,4-NONANEDIONE *[113486-29-6]*
Masanetz & Grosch (1998b) 0.000 03
Schuh & Schieberle (2006) d 0.000 01

methyl nonyl ketone⇒2-UNDECANONE

3-METHYLOCTANE *[2216-33-3]*
Zoeteman *et al.* (1971) 100

METHYL OCTANOATE (methyl caprylate) *[111-11-5]*
Schnabel *et al.* (1988) d 0.27 - 0.87
Takeoka *et al.* (1989,1992) d 0.2
Pino & Mesa (2006) d 0.2

METHYL (*E*)-3-OCTENOATE *[35234-16-3]*
Takeoka *et al.* (1989) d 0.15

methyl octyl ketone⇒2-DECANONE

METHYL 2-OCTYNOATE (methyl heptine carbonate) *[111-12-6]*
Appell (1969) 0.025

2-METHYL-3-OXA-8-THIABICYCLO[3.3.0]-1,4-OCTADIENE (kahweofuran) *[26693-24-3]*
Tressl & Silwar (1981); Silwar (1982) 0.005
Tressl (1989,1990) 0.000 5

METHYL 3-OXO-2-(2-(*Z*)-PENTENYL)CYCLOPENTANE-1-ACETATE ((±)-methyl jasmonate) *[20073-13-6]*
Acree *et al.* (1984,1985) d 5.7

(1*R*,2*R*)-(-)-METHYL 3-OXO-2-(2-(*Z*)-PENTENYL)CYCLOPENTANE-1-ACETATE ((-)-(*Z*)-methyl jasmonate) *[1211-29-6]*
Acree *et al.* (1985) d > 0.07

METHYL 3-OXO-2-(2-(*Z*)-PENTENYL)CYCLOPENTANE-1-ACETATE ((±)-methyl epijasmonate)
Acree *et al.* (1984,1985) d 0.013

(1*R*,2*S*)-(+)-METHYL 3-OXO-2-(2-(*Z*)-PENTENYL)CYCLOPENTANE-1-ACETATE ((+)-(*Z*)-methyl epijasmonate) *[95722-42-2]*
Acree *et al.* (1985) d 0.003

methyl palmitate⇒METHYL HEXADECANOATE

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

methyl parathion⇒O,O-DIMETHYL O-(4-NITROPHENYL)PHOSPHOROTHIOATE

2-METHYLPENTANAL *[123-15-9]*
Schnabel *et al.* (1988) d 0.0016 - 0.003 2

3-METHYLPENTANAL *[15877-57-3]*
Schnabel *et al.* (1988) d 0.000 41 - 0.000 81

4-METHYLPENTANAL *[1119-16-0]*
Amoore *et al.* (1976) d 0.39

METHYL PENTANOATE (methyl valerate) *[624-24-8]*
Schnabel *et al.* (1988) d 0.044 - 0.089
Takeoka *et al.* (1989) d 0.02

2-METHYLPENTANOIC ACID *[97-61-0]*
Amoore (1976) d 10.0
Schnabel *et al.* (1988) d 9.3 - 46

3-METHYLPENTANOIC ACID *[105-43-1]*
Schnabel *et al.* (1988) d 0.046 - 0.28

4-METHYLPENTANOIC ACID (isocaproic acid) *[646-07-1]*
Amoore *et al.* (1968) d 0.81

2-METHYL-1-PENTANOL *[105-30-6]*
Schnabel *et al.* (1988) d 0.83 - 1.2

3-METHYL-1-PENTANOL *[589-35-5]*
Schnabel *et al.* (1988) d 0.83 - 4.1
Giri *et al*. (2010) d 0.0075

4-METHYL-1-PENTANOL *[626-89-1]*
Schnabel *et al.* (1988) d 0.82 - 4.1

2-METHYL-2-PENTANOL *[590-36-3]*
Schnabel *et al.* (1988) d 4.1 - 8.2

3-METHYL-2-PENTANOL *[565-60-6]*
Schnabel *et al.* (1988) d 0.83 - 4.2

4-METHYL-2-PENTANOL *[108-11-2]*
Schnabel *et al.* (1988) d 2.5 - 4.1

2-METHYL-3-PENTANOL *[565-67-3]*
Schnabel *et al.* (1988) d 0.42 - 0.83

3-METHYL-3-PENTANOL *[77-74-7]*
Schnabel *et al.* (1988) d 2.5 - 4.1

3-METHYL-2-PENTANONE (*sec*-butyl methyl ketone) *[565-61-7]*
Schnabel *et al.* (1988) d 0.041 - 0.081

4-METHYL-2-PENTANONE (methyl isobutyl ketone) *[108-10-1]*
Sarzynski (1982) 7
Schnabel *et al.* (1988) d 0.24 - 0.64

2-METHYL-3-PENTANONE *[565-69-5]*
Schnabel *et al.* (1988) d 0.04 - 0.08

2-METHYL-2-PENTENAL *[623-36-9]*

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Persson & Jüttner (1983)	r	0.29
(*E*)-2-METHYL-2-PENTENAL		
Robert *et al.* (2004)	d	0.25
(1-METHYLPENTOXY)PYRAZINE		
Masuda & Mihara (1989)	d	0.1
(*d*)-(1-METHYLPENTOXY)PYRAZINE		
Masuda & Mihara (1989)	d	0.07
(*l*)-(1-METHYLPENTOXY)PYRAZINE		
Masuda & Mihara (1989)	d	0.2
4-METHYL-2-PENTYLAMINE		
De Grunt (1975)	d	150
2-METHYLPHENOL (*o*-cresol) *[95-48-7]*		
Rosen *et al.* (1962)		0.26
Baker (1963)		0.65
Dietz & Traud (1978)		1.4
3-METHYLPHENOL (*m*-cresol) *[108-39-4]*		
Rosen *et al.* (1962)		0.25
Baker (1963)		0.68
Dietz & Traud (1978)		0.8
Czerny *et al.* (2008)	d	0.015
Czerny *et al.* (2008)	r	0.031
4-METHYLPHENOL (*p*-cresol) *[106-44-5]*		
Rosen *et al.* (1962)		0.055
De Grunt (1975); De Grunt *et al.* (1977)	d	0.1
Buttery *et al.* (1988a)	d	0.055
Dietz & Traud (1978)		0.2
Kerscher (2000)		0.001
Karagül-Yüceer *et al.* (2003)	d	0.0027
Czerny *et al.* (2008)	d	0.0039
Czerny *et al.* (2008)	r	0.01
2-METHYL-3-PHENOXYPYRAZINE *[91137-78-9]*		
Masuda & Mihara (1988)	d	0.2
4-METHYLPHENYL ACETATE (*p*-cresyl acetate, 4-tolyl acetate) *[140-39-7]*		
Appell (1969)		0.025

methyl phenyl carbinol⇒1-PHENYLETHANOL

(2*R*,4*S*)-*cis*-2-METHYL-4-PHENYL-1,3-DIOXOLANE		
Pickenhagen & Schatkowski (2001, 2007)		0.125
(2*S*,4*R*)-*cis*-2-METHYL-4-PHENYL-1,3-DIOXOLANE		
Pickenhagen & Schatkowski (2001, 2007)		0.384
(2*R*,4*R*)-*trans* -2-METHYL-4-PHENYL-1,3-DIOXOLANE		
Pickenhagen & Schatkowski (2001, 2007)		0.384
(2*S*,4*S*)-*trans*-2-METHYL-4-PHENYL-1,3-DIOXOLANE		
Pickenhagen & Schatkowski (2001, 2007)		0.589
METHYL 3-PHENYL-2,3-EPOXYPROPANOATE		
Tekin & Karaman (1992)	d	15

trans-METHYL 3-PHENYL-2,3-EPOXYPROPANOATE

Tekin & Karaman (1992)	d	17

3-METHYL-1-PHENYL-3-PENTANOL (β-phenylethylmethylethylcarbinol) *[10415-87-9]*

Amoore & Venstrom (1966); Amoore (1968)	d	6.4
Rosen *et al.* (1979)	d	0.018

2-METHYL-3-(PHENYLTHIO)PYRAZINE *[91091-02-0]*

Masuda & Mihara (1988)	d	0.3

N-METHYLPIPERIDINE (N-methylhexahydroazine) *[626-67-5]*

Amoore & Forrester (1976)	d	1.17

2-METHYLPROPANAL (isobutyraldehyde) *[78-84-2]*

Guadagni *et al.* (1963,1972); Buttery *et al.* (1997)	d	0.000 9 - 0.001
Amoore *et al.* (1968)	d	0.0435
Amoore (1969)	d	0.009
Mulders (1972,1973)	d	0.01
Amoore *et al.* (1976)	d	0.0023
Schnabel *et al.* (1988)	d	0.000 40 - 0.0024
Milo & Grosch (1996,1997);		
Kerscher & Grosch (2000)		0.000 7
Fabrellas *et al.* (2004)		0.000 32 - 0.0015
Freuze *et al.* (2005)	d	0.004
Czerny *et al.* (2008)	d	0.000 49
Czerny *et al.* (2008)	r	0.0019
Giri *et al.* (2010)	d	0.0015

2-methylpropanenitrile⇒ISOPROPYL CYANIDE

METHYL PROPANOATE (methyl propionate) *[554-12-1]*

Ahmed *et al.* (1978a)	d	0.100
Schnabel *et al.* (1988)	d	4.6 - 8.8

2-METHYLPROPANOIC ACID (isobutyric acid) *[79-31-2]*

Holluta (1960)		0.05
Amoore *et al.* (1968); Amoore (1976)	d	8.5
Zoeteman *et al.* (1971)		30
Pyysalo *et al.* (1977)	d	0.3
Schnabel *et al.* (1988)	d	0.97 - 9.5
Larsen & Poll (1992)	d	0.01 - 0.1
Buttery & Ling (1998)	d	0.05
Pino & Mesa (2006)	d	5.3
Czerny *et al.* (2008)	d	16
Czerny *et al.* (2008)	r	29
Giri *et al*. (2010)	d	6.5505

2-METHYL-1-PROPANOL (isobutyl alcohol, isobutanol) *[78-83-1]*

Moncrieff (1957)		200
Sega *et al.* (1967)	d	0.565
Amoore *et al.* (1968)	d	22.5
Mulders (1972,1973)	d	3.2 - 3.3
Amoore *et al.* (1976)	d	12.5
Alexander *et al.* (1982)	d	0.36
Sarzynski (1982)		13
Schnabel *et al.* (1988)	d	8.2 - 20
Takahashi (1990)		16
Guth (1996)		1
Czerny *et al.* (2008)	d	0.55
Czerny *et al.* (2008)	r	2.3

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Giri *et al*. (2010)	d	6.5052

2-METHYL-2-PROPANOL (*tert*-butyl alcohol) *[75-65-0]*

Moncrieff (1957)		1,000
Schnabel *et al.* (1988)	d	8.2 - 19

METHYL 2-PROPENOATE (methyl acrylate) *[96-33-3]*

Piringer & Granzer (1984); Granzer *et al.* (1986)		0.005 - 0.01

2-METHYLPROPENOIC ACID (methacrylic acid) *[79-41-4]*

Piringer & Granzer (1984); Granzer *et al.* (1986)		2 – 100

methyl propionate⇒METHYL PROPANOATE

(1-METHYLPROPOXY)PYRAZINE

Masuda & Mihara (1989)	d	0.1

(*d*)-(1-METHYLPROPOXY)PYRAZINE

Masuda & Mihara (1989)	d	0.1

(*l*)-(1-METHYLPROPOXY)PYRAZINE

Masuda & Mihara (1989)	d	0.1

2-methylpropyl-2 acetate⇒*tert*-BUTYL ACETATE

3-(2-METHYLPROPYLIDENE)PHTHALIDE (3-isobutylenephthalide)

Gold & Wilson (1963)	r	< 0.1

methyl propyl ketone⇒ 2-PENTANONE

(*d*)-*cis*-2-METHYL-4-PROPYL-1,3-OXATHIANE

Pickenhagen & Brönner-Schindler (1984); Pickenhagen (2001)	d	0.002

(*l*)-*cis*-2-METHYL-4-PROPYL-1,3-OXATHIANE

Pickenhagen & Brönner-Schindler (1984); Pickenhagen (2001)	d	0.004

2-METHYL-3-PROPYLPYRAZINE *[15986-80-8]*

Mihara & Masuda (1988)	d	0.06

2-METHYL-6-PROPYLPYRAZINE *[29444-46-0]*

Sizer *et al.* (1975)		< 0.1

METHYLPYRAZINE *[109-08-0]*

Koehler *et al.* (1971)	d	105
Guadagni *et al.* (1972); Teranishi *et al.* (1974); Buttery *et al.* (1994a,1999); Buttery & Ling (1995,1998)	d	60
Buttery *et al.* (1988a)	d	0.06
Calabretta (1973,1975,1978)	d	100
Masuda & Mihara (1988)	d	30

1-METHYLPYRROLIDINE (N-methylazolidine) *[120-94-5]*

Amoore *et al.* (1975b)	d	0.0114
Amoore & Forrester (1976)	d	0.0167

3-METHYL-2-(1-PYRROLIDINYL)-2-CYCLOPENTEN-1-ONE

Ottinger *et al.* (2001)		43.5 - 72.5

5-METHYL-2-(1-PYRROLIDINYL)-2-CYCLOPENTEN-1-ONE *[41357-00-0]*

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Ottinger *et al.* (2001)		2.6 – 5.2

(*S*)-(-)-3-(1-METHYL-2-PYRROLIDINYL)PYRIDINE (*S*-(-)-nicotine) *[54-11-5]*

Hummel *et al.* (1991)		47 - 191
Hummel *et al.* (1992)	d	16 - 260
Rosenblatt *et al.* (1998)		470 - 2,000

(*R*)-(+)-3-(1-METHYL-2-PYRROLIDINYL)PYRIDINE (*R*-(+)-nicotine)

Hummel *et al.* (1992)	d	32 - 130

METHYL SALICYLATE *[119-36-8]*

Helbig (1939)		0.0349
Buttery *et al.* (1969b,1971,1987a,1987b,1989,1990a); Hansen *et al.* (1992); Buttery (1993); Buttery & Ling (1995)	d	0.04
Pino & Mesa (2006)	d	0.04

3-(methylsulfanyl)propanal⇒3-(METHYLTHIO)PROPANAL

4-METHYL-4,5,6,7-TETRAHYDROBENZOTHIAZOLE (4-methylcyclohexathiazole) *[84648-05-5]*

Pelosi *et al.* (1983)	d	0.024

6-METHYL-4,5,6,7-TETRAHYDROBENZOTHIAZOLE (6-methylcyclohexathiazole) *[84648-06-6]*

Pelosi *et al.* (1983)	d	0.049

METHYLTETRATHIOMETHANE (dimethyl tetrasulphide) *[5756-24-1]*

Granvogl & Schieberle (2010)	r	0.000 02

4-METHYLTHIAZOLE *[693-95-8]*

Marchand *et al.* (2000)	d	0.055

5-METHYL-2-THIENYLACETIC ACID

Davídek *et al.* (1979)		4.4

2-(METHYLTHIO)BENZOTHIAZOLE *[615-22-5]*

Kölle *et al.* (1972)		0.005
De Grunt (1975)	d	0.0075

3-(METHYLTHIO)BUTANAL *[16630-52-7]*

Kubota (1981)		0.0025

4-(METHYLTHIO)BUTYL ISOTHIOCYANATE *[4430-36-8]*

Buttery *et al.* (1976b); Buttery (1981)	d	0.003 - 0.0034

METHYL THIOCYANATE *[556-64-9]*

Brevard *et al.* (1992)	d	10.0
Hansen *et al.* (1992)	d	0.010

2-(METHYLTHIO)ETHANAL (2-methylmercaptoacetaldehyde) *[23328-62-3]*

Buttery *et al.* (1971)	d	0.016
Rössner *et al.* (2002)		0.000 1

1-(METHYLTHIO)ETHANETHIOL *[31331-53-0]*

Brinkman *et al.* (1972)	r	< 0.001

2-(METHYLTHIO)ETHANOL (2-methylmercaptoethanol) *[5271-38-5]*

Buttery *et al.* (1971); Hansen *et al.* (1992)	d	0.12

2-(methylthiofurfurane)pyrazine⇒[2-(FURANYLMETHYL)THIO]PYRAZINE

2-METHYL-3-THIOFURFURYLPYRAZINE

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Calabretta (1978)	d	< 0.001
2-METHYL-5-THIOFURFURYLPYRAZINE		
Calabretta (1978)	d	< 0.001
METHYL THIOHEPTANOATE		
Guadagni *et al.* (1966); Teranishi *et al.* (1987)	d	0.002
7-(METHYLTHIO)HEPTYL ISOTHIOCYANATE *[4430-38-0]*		
Masuda *et al.* (1996)		0.28
3-(METHYLTHIO)HEXANAL *[38433-74-8]*		
Buttery *et al.* (1990b)	d	0.003
METHYL THIOHEXANOATE *[2432-77-1]*		
Guadagni *et al.* (1966); Teranishi *et al.* (1987)	d	0.000 3
6-(METHYLTHIO)HEXYL ISOTHIOCYANATE *[4430-39-1]*		
Masuda *et al.* (1996)		0.49
METHYLTHIOMETHANE (dimethyl sulphide) *[75-18-3]*		
Guadagni *et al.* (1963); Guadagni (1966,1968,1969);		
Buttery *et al.* (1971,1990a,1994a);		
Buttery (1981,1993)	d	0.000 3 - 0.000 33
Koptyaev (1967)		0.03
Mulders (1972,1973)	d	0.001
Sato *et al.* (1975)		0.010
De Grunt (1975)	d	0.010
Spedding & Raut (1982)		0.000 08
Belitz & Grosch (1992c)		0.001
Milo & Grosch (1997); Kerscher & Grosch (2000)		0.000 3
Karagül-Yüceer *et al.* (2004)	d	0.000 3
Czerny *et al.* (2008)	d	0.000 84
Czerny *et al.* (2008)	r	0.0011
Averbeck & Schierberle (2009, 2011)	d	0.000 12
2-(METHYLTHIOMETHANE)PYRAZINE		
Calabretta (1973,1978)	d	0.02
2-(METHYLTHIOMETHYL)ETHANAL		
Rössner *et al.* (2002)		0.000 03
1-(METHYLTHIO)-3-OCTANONE		
Buttery *et al.* (1990b)	d	0.0055
2-(METHYLTHIO)-3-OCTYLPYRAZINE *[113685-88-4]*		
Masuda & Mihara (1988)	d	0.000 7
1-(METHYLTHIO)-3-PENTANONE *[66735-69-1]*		
Buttery *et al.* (1990b)	d	0.15
5-(METHYLTHIO)PENTYL ISOTHIOCYANATE *[4430-42-6]*		
Masuda *et al.* (1996)		0.80
2-(METHYLTHIO)-3-PENTYLPYRAZINE *[113685-87-3]*		
Masuda & Mihara (1988)	d	0.000 12
2-METHYL-3-THIOPHENETHIOL *[2527-76-6]*		
Schieberle & Hofmann (1996a, 1996b)		0.000 02

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

4-(METHYLTHIO)PHENOL (4-methylmercaptophenol) *[1073-72-9]*

Dietz & Traud (1978)		0.8

3-(METHYLTHIO)PROPANAL (methional, 3-(methylsulfanyl)propanal, 4-thiapentanal) *[3268-49-3]*

Buttery *et al.* (1971,1990a,1994a,1997,1999);		
Guadagni *et al.* (1972); Buttery (1981, 1993);		
Buttery & Ling (1995,1998)	d	0.000 2
Cerny & Grosch (1993); Grosch *et al.* (1993)		0.0018
Guth & Grosch (1994); Milo (1995a);		
Milo & Grosch (1997); Kerscher & Grosch (2000)		0.000 2
Karagül-Yüceer *et al.* (2004)	d	0.000 4
Czerny *et al*. (2007a); Czerny *et al.* (2008)	r	0.0014
Czerny *et al*. (2008); Steinhaus *et al.* (2009);		
Sinuco *et al.* (2010)	d	0.000 43
Giri *et al*. (2010)	d	0.000 45

3-(METHYLTHIO)-1-PROPANOL (methionol) *[505-10-2]*

Fritsch & Schieberle (2005)		0.25
Czerny *et al.* (2008)	d	0.036
Czerny *et al.* (2008)	r	0.069
Giri *et al*. (2010)	d	0.12323

3-(METHYLTHIO)PROPYL ACETATE (methionyl acetate) *[16630-55-0]*

Wyllie & Leach (1992)		0.03

3-(METHYLTHIO)PROPYL CYANIDE

Buttery *et al.* (1976b)	d	0.082

3-(METHYLTHIO)PROPYL ISOTHIOCYANATE *[505-79-3]*

Buttery *et al.* (1976b); Buttery (1981)	d	0.005
Masuda *et al.* (1996)		0.65

2-(METHYLTHIO)-3-PROPYLPYRAZINE *[68650-83-8]*

Mihara & Masuda (1988)	d	0.001

(METHYLTHIO)PYRAZINE *[21948-70-9]*

Masuda & Mihara (1988)	d	0.2

methyl tiglate⇒METHYL (*E*)-2-METHYL-2-BUTENOATE

4-METHYL-2,3,6-TRICHLOROANISOLE

Griffiths & Fenwick (1977)	d	0.000 8

2-METHYL-1,1,1-TRICHLORO-2-PROPANOL (chloretone) *[57-15-8]*

Pelosi & Pisanelli (1981)	d	50

(2*R*,3*R*,1'*R*,2'*R*,6'*R*,7'*R*)-3-METHYL-4-(TRICYCLO[5.2.1.$0^{2,6}$]DEC-4-EN-8-YLIDENE)BUTAN-2-OL

Kappey *et al.* (2003); Hölscher *et al.* (2004)	d	0.137

(2*R*,3*R*,1'*S*,2'*S*,6'*S*,7'*S*)-3-METHYL-4-(TRICYCLO[5.2.1.$0^{2,6}$]DEC-4-EN-8-YLIDENE)BUTAN-2-OL

Kappey *et al.* (2003); Hölscher *et al.* (2004)	d	0.204

(2*R*,3*S*,1'*R*,2'*R*,6'*R*,7'*R*)-3-METHYL-4-(TRICYCLO[5.2.1.$0^{2,6}$]DEC-4-EN-8-YLIDENE)BUTAN-2-OL

Kappey *et al.* (2003); Hölscher *et al.* (2004)	d	0.070

(2*R*,3*S*,1'*S*,2'*S*,6'*S*,7'*S*)-3-METHYL-4-(TRICYCLO[5.2.1.$0^{2,6}$]DEC-4-EN-8-YLIDENE)BUTAN-2-OL

Kappey *et al.* (2003); Hölscher *et al.* (2004)	d	0.168

(2*S*,3*R*,1'*R*,2'*R*,6'*R*,7'*R*)-3-METHYL-4-(TRICYCLO[5.2.1.$0^{2,6}$]DEC-4-EN-8-YLIDENE)BUTAN-2-OL

Kappey *et al.* (2003); Hölscher *et al.* (2004)	d	0.005

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

(2*S*,3*R*,1'*S*,2'*S*,6'*S*,7'*S*)-3-METHYL-4-(TRICYCLO[5.2.1.$0^{2,6}$]DEC-4-EN-8-YLIDENE)BUTAN-2-OL
Kappey *et al.* (2003); Hölscher *et al.* (2004) d 0.072

(2*S*,3*S*,1'*R*,2'*R*,6'*R*,7'*R*)-3-METHYL-4-(TRICYCLO[5.2.1.$0^{2,6}$]DEC-4-EN-8-YLIDENE)BUTAN-2-OL
Kappey *et al.* (2003); Hölscher *et al.* (2004) d 0.046

(2*S*,3*S*,1'*S*,2'*S*,*6'S*,*7'S*)-3-METHYL-4-(TRICYCLO[5.2.1.$0^{2,6}$]DEC-4-EN-8-YLIDENE)BUTAN-2-OL
Kappey *et al.* (2003); Hölscher *et al.* (2004) d 0.159

12-METHYLTRIDECANAL *[75853-49-5]*
Guth & Grosch (1993); Grosch *et al.* (1993);
Kerscher & Grosch (2000) 0.000 1

(*d*)-2-METHYL-4[5.5.6(*exo*)-TRIMETHYL-2-(*exo*)-NORBORNYL]CYCLOHEXANONE ((*d*)-ketone)
Amoore *et al.* (1977) d 0.000 28

(*l*)-2-METHYL-4[5.5.6(*exo*)-TRIMETHYL-2-(*exo*)-NORBORNYL]CYCLOHEXANONE ((*l*)-ketone)
Amoore *et al.* (1977) d 0.009

(*E*)-(*R*)-2-METHYL-4-(2,2,3-TRIMETHYL-3-PENTEN-1-YL)-2-BUTEN-1-OL
Aida *et al.* (1997, 2000) 0.000 0099

(*E*)-(*S*)-2-METHYL-4-(2,2,3-TRIMETHYL-3-PENTEN-1-YL)-2-BUTEN-1-OL
Aida *et al.* (1997, 2000) 0.000 082

METHYLTRITHIOMETHANE (dimethyl trisulphide, 2,3,4-trithiapentane) *[3658-80-8]*
Buttery *et al.* (1976b,1990b,1994a,1997,1999);
Buttery (1981,1993); Takeoka *et al.* (1991a);
Hansen *et al.* (1992); Buttery & Ling (1998);
Takeoka (2001) d 0.000 01
Wajon *et al.* (1986) 0.000 005 - 0.000 01
Milo & Grosch (1997) 0.000 006
Wright *et al.* (2006) d 0.000 000 07
Czerny *et al.* (2008) d 0.000 0099
Czerny *et al.* (2008) r 0.000 016
Giri *et al*. (2010) d 0.000 10

2-METHYLUNDECANE *[7045-71-8]*
Zoeteman *et al.* (1971) 10

methyl valerate⇒METHYL PENTANOATE
methyl vanillate⇒METHYL 4-HYDROXY-3-METHOXYBENZOATE
MIB⇒1,2,7,7-TETRAMETHYL-*exo*-BICYCLO[2.2.1]-2-HEPTANOL
monochloramine⇒CHLORAMINE
monuron⇒N'-(4-CHLOROPHENYL)-N,N-DIMETHYLUREA
MTBE⇒2-METHOXY-2-METHYLPROPANE
mucidone⇒6-ETHYL-3-ISOBUTYL-*2H*-2-PYRANONE
muscone⇒3-METHYLCYCLOPENTADECANONE
(*R*)-(-)-muscone⇒(*R*)-(-)-3-METHYLCYCLOPENTADECANONE
(*S*)-(+)-muscone⇒(*S*)-(+)-3-METHYLCYCLOPENTADECANONE
musk ambrette⇒1-*tert*-BUTYL-2-METHOXY-4-METHYL-3,5-DINITROBENZENE
musk ketone⇒4-*tert*-BUTYL-2,6-DIMETHYL-3,5-DINITROACETOPHENONE
musk T⇒CYCLO-1,3-ETHYLENEDIOXY-1,3-TRIDECANEDIONE
musk xylene⇒1-*tert*-BUTYL-3,5-DIMETHYL-2,4,6-TRINITROBENZENE
musk 89⇒3,4,6,7,8,9-HEXAHYDRO-4,6,6,9,9-PENTAMETHYL-1*H*-NAPHTHO[2,3-c]PYRAN
myrcene⇒7-METHYL-3-METHYLENE-1,6-OCTADIENE
myristaldehyde⇒TETRADECANAL
myristic acid⇒TETRADECANOIC ACID
myristicin⇒5-ALLYL-1-METHOXY-2,3-(METHYLENEDIOXY)BENZENE
myrtenol⇒6,6-DIMETHYLBICYCLO[3.1.1]-2-HEPTENE-2-METHANOL

N

NAPHTHALENE *[91-20-3]*

Holluta (1960)		0.5
Rosen *et al.* (1963)		0.0068
Koppe (1965)		0.001
Zoeteman *et al.* (1971)		0.005
Young *et al*. (1996)		0.006

1-naphthol⇒1-HYDROXYNAPHTHALENE
2-naphthol⇒2-HYDROXYNAPHTHALENE

1-NAPHTHYL N-METHYLCARBAMATE (carbaryl) *[63-25-2]*

Young *et al.* (1996)		0.28

(1*R*,3*S*,4*S*)-(+)-neomenthol⇒(1*S*,2*S*,5*R*)-(+)-2-ISOPROPYL-5-METHYLCYCLOHEXANOL
(1*S*,3*R*,4*R*)-(-)-neomenthol⇒(1*R*,2*R*,5*S*)-(-)-2-ISOPROPYL-5-METHYLCYCLOHEXANOL
neopentyl acetate⇒2,2-DIMETHYLPROPYL ACETATE
neopentyl alcohol⇒2,2-DIMETHYL-1-PROPANOL
neral⇒3,7-DIMETHYL-*cis*-2,6-OCTADIENAL
nerol⇒3,7-DIMETHYL-*cis*-2,6-OCTADIEN-1-OL
nerolidol⇒3,7,11-TRIMETHYL-1,6,10-DODECATRIEN-3-OL
neryl acetate⇒3,7-DIMETHYL-(*Z*)-2,6-OCTADIEN-1-YL ACETATE
S-(-)-nicotine⇒(*S*)-(-)-3-(1-METHYL-2-PYRROLIDINYL)PYRIDINE
R-(+)-nicotine⇒(*R*)-(+)-3-(1-METHYL-2-PYRROLIDINYL)PYRIDINE

NITROBENZENE *[98-95-3]*

De Grunt (1975)	d	0.2

4-nitro-*m*-cresol⇒3-METHYL-4-NITROPHENOL
6-nitro-*m*-cresol⇒5-METHYL-2-NITROPHENOL
5-nitro-*o*-cresol⇒2-METHYL-5-NITROPHENOL
2-nitro-*p*-cresol⇒4-METHYL-2-NITROPHENOL
3-nitro-*p*-cresol⇒4-METHYL-3-NITROPHENOL

NITROCYCLOHEXANE *[1122-60-7]*

Cherkinski (1961)	d	0.15

nitrogen trichloride⇒TRICHLORAMINE

2-NITROPHENOL *[88-75-5]*

Dietz & Traud (1978)		1.7

3-NITROPHENOL *[554-84-7]*

Dietz & Traud (1978)		0.6

4-NITROPHENOL *[100-02-7]*

Dietz & Traud (1978)		2.5

1-NITRO-2-PHENYLETHANE *[6125-24-2]*

Buttery *et al.* (1989,1990a); Buttery (1993)	d	0.002

2-NITROPROPANE *[79-46-9]*

Amoore *et al.* (1976)	d	24.9
Sarzynski (1982)		22

N-NITROSODIPHENYLAMINE *[62-75-9]*

Korolev *et al.* (1980)		0.1

2-nitrotoluene⇒1-METHYL-2-NITROBENZENE
4-nitrotoluene⇒1-METHYL-4-NITROBENZENE

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

NONADECANOIC ACID *[646-30-0]*

Cherkinski (1961)	d	20.0

2,4-NONADIENAL *[6750-03-4]*

Tressl *et al.* (1979b,1980)		0.000 05

(*E*,*E*)-2,4-NONADIENAL *[5910-87-2]*

Teranishi *et al.* (1974)	d	0.000 09
Konopka *et al.* (1995)		0.000 09
Tamura *et al.* (1995)		0.008
Milo & Grosch (1997)		0.000 1
Jensen *et al.* (1999)	d	0.000 0017
Buttery & Ling (1998)	d	0.000 01
Schuh & Schieberle (2006)	d	0.000 16
Czerny *et al.* (2008)	d	0.000 062
Czerny *et al.* (2008)	r	0.000 19
Giri *et al*. (2010)	d	0.000 1

2,6-NONADIENAL *[17587-33-6]*

Kossa *et al.* (1979); Tressl *et al.* (1979b,1980)		0.000 5

(*E*,*Z*)-2,6-NONADIENAL *[557-48-2]*

Buttery *et al.* (1969b); Teranishi *et al.* (1974);		
Buttery (1981,1989,1993)	d	0.000 01
Burlingame *et al.* (1992)	d	0.000 0018 - 0.000 0022
Burlingame *et al.* (1992)	r	0.000 0067 - 0.000 01
Tamura *et al.* (1995)		0.000 108
Milo & Grosch (1997); Kerscher & Grosch (2000)		0.000 02
Rashash *et al.* (1997)		0.000 004
Boonbumrung *et al.* (2001)	d	0.000 14
Pino & Mesa (2006)	d	0.000 02
Schuh & Schieberle (2006)	d	0.000 03
Czerny *et al.* (2008)	d	0.000 0045
Czerny *et al.* (2008)	r	0.000 0091
Giri *et al*. (2010)	d	0.000 8

(*E*,*Z*)-2,6-NONADIEN-1-OL *[28069-72-9]*

Heiler (1995); Heiler & Schieberle (1997)		0.000 2
Arena *et al.* (2001)		0.001
Takahashi *et al.* (2002)		0.001

(*E*,*Z*)-3,6-NONADIEN-1-OL *[56805-23-3]*

Buttery *et al.* (1982b)	d	0.003

(*Z*,*Z*)-3,6-NONADIEN-1-OL *[53406-97-2]*

Buttery (1981)	d	0.01

(*E*,*E*)-2,5-NONADIEN-4-ONE *[61759-51-1]*

Buttery *et al.* (1969b); Buttery (1989)	d	0.004

(*E*,*Z*)-3,6-NONADIENYL ACETATE *[76649-26-8]*

Buttery *et al.* (1982b); Teranishi *et al.* (1987)	d	0.011 - 0.015

γ-nonalactone⇒4-HYDROXYNONANOIC ACID LACTONE

NONANAL *[124-19-6]*

Guadagni *et al.* (1963,1972); Teranishi *et al.* (1974);		
Buttery *et al.* (1978b,1988a,1999); Buttery (1981);		
Buttery & Ling (1995,1998); Takeoka *et al*. (1995a)	d	0.001
Ahmed *et al.* (1978a)	d	0.00253
Aoki & Koizumi (1986)		0.0058

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Pino *et al.* (1986)	d	0.0044
Schnabel *et al.* (1988)	d	0.0041 - 0.008 2
Sugisawa *et al.* (1991); Yang *et al.* (1992); Tamura *et al.* (1993,1995)	d	0.1
Belitz & Grosch (1992b)		0.005
Tamura *et al.* (1996)	d	0.04
Tamura *et al.* (1996)	r	0.26
Mookdasanit *et al.* (2003)	d	0.049
Fabrellas *et al.* (2004)		0.000 32 - 0.000 55
Limpawattana (2007)	d	0.00206
Czerny *et al.* (2008)	d	0.0028
Czerny *et al.* (2008)	r	0.008
Giri *et al*. (2010)	d	0.0011

NONANE *[111-84-2]*

Zoeteman *et al.* (1971)		10

2-NONANETHIOL *[13281-11-3]*

Robert *et al.* (2005)	d	0.09

NONANOIC ACID (pelargonic acid) *[112-05-0]*

Cherkinski (1961)	d	3.0
Amoore *et al.* (1968)	d	8.8
Schnabel *et al.* (1988)	d	4.6 - 9.0

1-NONANOL *[143-08-8]*

Pyysalo & Suihko (1976a)	d	0.09
Aoki & Koizumi (1986)		0.034
Schnabel *et al.* (1988)	d	0.042 - 0.082
Buttery *et al.* (1988a); Takeoka *et al.* (1991a)	d	0.05
Sugisawa *et al.* (1991); Yang *et al.* (1992); Tamura *et al.* (1993)	d	1
Giri *et al*. (2010)	d	0.0455

2-NONANOL *[628-99-9]*

Schnabel *et al.* (1988)	d	0.058 - 0.082

3-NONANOL *[624-51-1]*

Schnabel *et al.* (1988)	d	0.087 - 0.25

4-NONANOL *[5932-79-6]*

Schnabel *et al.* (1988)	d	0.087 - 0.42

5-NONANOL *[623-93-8]*

Schnabel *et al.* (1988)	d	0.087 - 0.25

4-nonanolide⇒4-HYDROXYNONANOIC ACID LACTONE

2-NONANONE (methyl heptyl ketone) *[821-55-6]*

Teranishi *et al.* (1974)	d	0.005
Hirvi & Honkanen (1982)	d	0.038
Aoki & Koizumi (1986)		0.0109
Buttery *et al.* (1988a)	d	0.2
Schnabel *et al.* (1988)	d	0.041 - 0.082

3-NONANONE *[925-78-0]*

Schnabel *et al.* (1988)	d	0.017 - 0.033

4-NONANONE *[4485-09-0]*

Schnabel *et al.* (1988)	d	0.0082 - 0.041

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

5-NONANONE (dibutyl ketone) *[502-56-7]*		
Schnabel *et al.* (1988)	d	0.0082 - 0.041
(*E,E,Z*)-2,4,6-NONATRIENAL		
Schuh & Schieberle (2006)	d	0.000 026
2-NONENAL *[2463-53-8]*		
Kossa *et al.* (1979); Tressl *et al.* (1979b,1980)		0.000 1
(*E*)-2-NONENAL *[18829-56-6]*		
Guadagni *et al.* (1963,1972); Buttery *et al.* (1968, 1971,1978b,1988a); Teranishi *et al.* (1974); Buttery (1981,1989,1993); Buttery & Ling (1995,1998)	d	0.000 08
Heiler (1995)		0.000 15
Milo (1995a, 1995b); Milo & Grosch (1996b); Kerscher & Grosch (2000)		0.000 25
Tamura *et al.* (1995)		0.000 125
Jensen *et al.* (1999)	d	0.000 23
Ong & Acree (1999)		0.000 1
Tamura *et al.* (2001)	d	0.000 4
Pino & Mesa (2006)	d	0.000 08
Schuh & Schieberle (2006)	d	0.000 4
Czerny *et al.* (2007c); Czerny *et al.* (2008)	r	0.000 69
Limpawattana (2007)	d	0.000 09
Czerny *et al.* (2008)	d	0.000 19
(*Z*)-2-NONENAL *[60784-31-8]*		
Heiler (1995)		0.000 004
Kerscher & Grosch (2000); Kerscher (2001)		0.000 02
(*Z*)-6-NONENAL *[2277-19-2]*		
Guadagni (1970b); Seifert (1981); Buttery (1989)	d	0.000 005
Buttery (1981)	d	< 0.001
(*E*)-2-NONEN-1-OL *[31502-14-4]*		
Yang *et al.* (1992); Tamura *et al.* (1993)	d	0.13
Arena *et al.* (2001)		0.209
(*Z*)-6-NONEN-1-OL *[35854-86-5]*		
Buttery *et al.* (1982b)	d	0.001
3-NONEN-2-ONE *[14309-57-0]*		
Takeoka *et al.* (1990a)	d	0.8
1-NONEN-3-ONE *[24415-26-7]*		
Ott *et al.* (1997)	d	0.000 000 008
Pons *et al.* (2011)	d	0.000 001
1-NONEN-4-ONE *[61168-10-3]*		
Buttery *et al.* (1969b); Buttery (1981,1989)	d	0.000 4
(*E*)-2-NONEN-4-ONE *[27743-70-0]*		
Buttery *et al.* (1969b,1971); Teranishi *et al.* (1974); Buttery (1981,1989)	d	0.000 9
(*E*)-3-NONENYL ACETATE		
Buttery *et al.* (1982b)	d	0.060
(*Z*)-6-NONENYL ACETATE *[76238-22-7]*		

Buttery *et al.* (1982b); Teranishi *et al.* (1987);
Buttery (1989) d 0.002

NONYL ACETATE *[143-15-5]*
Guadagni (1970b); Teranishi *et al.* (1987) d 0.2
Ahmed *et al.* (1978a) d 0.057
Yang *et al.* (1992); Tamura *et al.* (1993) d 0.6

2-NONYL ACETATE *[14936-66-4]*
Takeoka *et al.* (1996) d 0.31

4-NONYLPHENOL *[104-40-5]*
Dietz & Traud (1978) 1

nootkatone⇒8-ISOPROPENYL-5,6-DIMETHYLBICYCLO[4.4.0]-1-DECEN-3-ONE
(*d*)-nootkatone⇒(*d*)-8-ISOPROPENYL-5,6-DIMETHYLBICYCLO[4.4.0]-1-DECEN-3-ONE
(*l*)-nootkatone⇒(*l*)-8-ISOPROPENYL-5,6-DIMETHYLBICYCLO[4.4.0]-1-DECEN-3-ONE
(+)-nootkatone⇒(*d*)-8-ISOPROPENYL-5,6-DIMETHYLBICYCLO[4.4.0]-1-DECEN-3-ONE
(-)-nootkatone⇒(*l*)-8-ISOPROPENYL-5,6-DIMETHYLBICYCLO[4.4.0]-1-DECEN-3-ONE
(*l*)-18-norambrox⇒[3aR-(3aα,5aβ,6α,9aα,9bβ)]-3a,6,9a-TRIMETHYLNAPHTHO[2,1-b]FURAN
19-norambrox⇒(3aα,5aβ,6β,9aα,9bβ)-3a,6,9a-TRIMETHYLNAPHTHO[2,1-b]FURAN
norfuraneol⇒4-HYDROXY-5-METHYL-3(2*H)*-FURANONE
(*d*)-18-norisoambrox⇒[3a*R*-(3aα ,5aβ,6α,9aβ,9bα)]-3a,6,9a-TRIMETHYLNAPHTHO[2,1-b]FURAN
(*l*)-19-norisoambrox⇒[3a*S*-(3aα,5aβ,6β,9aβ,9bα)]-3a,6,9a-TRIMETHYLNAPHTHO[2,1-b]FURAN

O

(*E*)-β-ocimene⇒(*E*)-3,7-DIMETHYL-1,3,6-OCTATRIENE
(*Z*)-β-ocimene⇒(*Z*)-3,7-DIMETHYL-1,3,6-OCTATRIENE

1,2,4,5,6,7,8,8-OCTACHLORO-2,3,3a,4,7,7a-HEXAHYDRO-4,7-METHANOINDENE (chlordane) *[5103-71-9]*
Sigworth (1965) d 0.000 5 - 0.0025

OCTADECANOIC ACID (stearic acid) *[57-11-4]*
Cherkinski (1961) d 20.0

9,12,15-OCTADECATRIENOIC ACID (linolenic acid) *[463-40-1]*
Rashash *et al.* (1997) 0.005

(*E*,*E*)-2,4-OCTADIENAL *[30361-28-5]*
Tamura *et al.* (1995) 0.01

1,3-OCTADIENE *[1002-33-1]*
Persson & Jüttner (1983) r 5.6

(*Z*)-1,5-OCTADIEN-3-OL *[50306-18-8]*
Whitfield *et al.* (1982) r 0.01

1,5-OCTADIEN-2-ONE
Tressl *et al.* (1979b,1980) 0.000 002

(*E*,*E*)-3,5-OCTADIEN-2-ONE *[30086-02-3]*
Buttery (1989); Hansen *et al.* (1992);
Buttery & Ling (1995,1998) d 0.10 - 0.15

(*E*,*Z*)-3,5-OCTADIEN-2-ONE *[4173-41-5]*
Buttery & Ling (1995) d 0.15

1,5-OCTADIEN-3-ONE *[65213-86-7]*
Kossa *et al.* (1979) 0.000 002

(*Z*)-1,5-OCTADIEN-3-ONE *[65767-22-8]*

Swoboda & Peers (1977)	d	0.000 0012
Whitfield *et al.* (1982)	r	0.000 001
Buttery (1989)		0.000 001
Belitz & Grosch (1992b); Milo & Grosch (1997)		0.000 0012
Darriet *et al.* (2002)	d	0.000 000 7 - 0.000 000 9

1,2,3,4,5,6,7,8-OCTAHYDROANTHRACENE (octanthrene)

Koppe (1965)		0.001

4,5,6,7,8,9,10,11-OCTAHYDROTHIAZOLE (cyclodecathiazole) *[84648-03-3]*

Pelosi *et al.* (1983)	d	0.58

γ-octalactone⇒4-HYDROXYOCTANOIC ACID LACTONE
δ-octalactone⇒5-HYDROXYOCTANOIC ACID LACTONE

OCTANAL (caprylaldehyde) *[124-13-0]*

Guadagni *et al.* (1963); Buttery *et al.* (1968, 1978b,1988a,1999); Teranishi *et al.* (1974); Buttery (1981,1989); Buttery & Ling (1998)	d	0.000 7
Ahmed *et al.* (1978a)	d	0.00141
Aoki & Koizumi (1986)		0.0031
Pino *et al.* (1986)	d	0.0064
Schnabel *et al.* (1988)	d	0.000 41 - 0.000 82
Sugisawa *et al.* (1991); Yang *et al.* (1992); Tamura *et al.* (1993,1995)	d	0.08 - 0.082
Belitz & Grosch (1992b)		0.008
Tamura *et al.* (1996)	d	0.03
Tamura *et al.* (1996)	r	0.248
Padrayuttawat *et al.* (1997)	d	0.000 9
Fabrellas *et al.* (2004)		0.000 32 - 0.000 46
Pino & Mesa (2006)	d	0.000 7
Limpawattana (2007)	d	0.000 98
Czerny *et al.* (2008)	d	0.0034
Czerny *et al.* (2008)	r	0.0069
Averbeck & Schierberle (2009, 2011)	d	0.000 8
Giri *et al*. (2010)	d	0.000 587

OCTANE *[111-65-9]*

Zoeteman *et al.* (1971); De Grunt (1975)		10

octanenitrile⇒HEPTYL CYANIDE

2-OCTANETHIOL *[3001-66-9]*

Robert *et al.* (2005)	d	0.05

OCTANOIC ACID (caprylic acid) *[124-07-2]*

Cherkinski (1961)	d	3.0
Amoore *et al.* (1968); Amoore (1976)	d	19.0
Schnabel *et al.* (1988)	d	0.91 - 1.9
Karagül-Yüceer *et al.* (2003, 2004)	d	1.405
Pino & Mesa (2006)	d	3

1-OCTANOL *[111-87-5]*

Baker (1963)		0.13
Pyysalo & Suihko (1976a)	d	0.48
Ahmed *et al.* (1978a)	d	0.190
Aoki & Koizumi (1986)		0.0775
Schnabel *et al.* (1988)	d	0.042 - 0.083
Buttery *et al.* (1988a); Takeoka *et al.* (1992)	d	0.11

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Sugisawa *et al.* (1991); Yang *et al.* (1992);		
Tamura *et al.* (1993,1995)	d	0.875 - 1
Padrayuttawat *et al.* (1997)	d	0.1
Pino & Mesa (2006)	d	0.19
Giri *et al*. (2010)	d	0.1258
2-OCTANOL *[123-96-6]*		
Aoki & Koizumi (1986)		0.0715
Schnabel *et al.* (1988)	d	0.0078 - 0.042
3-OCTANOL *[589-98-0]*		
Pyysalo & Suihko (1976a)	d	0.018
Schnabel *et al.* (1988)	d	0.078 - 0.25
4-OCTANOL *[589-62-8]*		
Schnabel *et al.* (1988)	d	0.40 - 0.82

4-octanolide⇒4-HYDROXYOCTANOIC ACID LACTONE
5-octanolide⇒5-HYDROXYOCTANOIC ACID LACTONE

2-OCTANONE (methyl hexyl ketone) *[111-13-7]*		
Aoki & Koizumi (1986)		0.0145
Schnabel *et al.* (1988)	d	0.041 - 0.062
Buttery *et al.* (1988a); Buttery & Ling (1998)	d	0.05
Pino & Mesa (2006)	d	0.005
Giri *et al*. (2010)	d	0.0502
3-OCTANONE (ethyl amyl ketone) *[106-68-3]*		
Pyysalo & Suihko (1976a)	d	0.050
Schnabel *et al.* (1988)	d	0.021 - 0.041
Buttery *et al.* (1988a)	d	0.028
Takahashi *et al.* (2002)		0.07
Darriet *et al.* (2002)	d	0.023
Giri *et al*. (2010)	d	0.0214
4-OCTANONE *[589-63-9]*		
Schnabel *et al.* (1988)	d	0.041 - 0.082

octanthrene⇒1,2,3,4,5,6,7,8-OCTAHYDROANTHRACENE

2-OCTENAL *[2363-89-5]*		
Kossa *et al.* (1979)		0.000 2
(*E*)-2-OCTENAL *[2548-87-0]*		
Buttery *et al.* (1971,1988a,1999); Guadagni *et al.* (1972);		
Teranishi *et al.* (1974); Buttery & Ling (1998)	d	0.003
Eriksson *et al.* (1976)	d	0.004
Pyysalo & Suihko (1976a)	d	0.003
Belitz & Grosch (1992b)		0.004
Tamura *et al.* (1995)		0.02
Rashash *et al.* (1997)		< 0.006
Jensen *et al.* (1999)	d	0.000 34
Pino & Mesa (2006)	d	0.003
1-OCTENE *[111-66-0]*		
De Grunt (1975); De Grunt *et al.* (1977)	d	0.000 5
2-OCTEN-1-OL *[22104-78-5]*		
Kossa *et al.* (1979)		0.075

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

(*E*)-2-OCTEN-1-OL *[18409-17-1]*

Kaminski *et al.* (1972)	r	0.1
Eriksson *et al.* (1976)	d	0.84
Pyysalo & Suihko (1976a)	d	0.040
Takahashi *et al.* (2002)		0.02

(*Z*)-5-OCTEN-1-OL *[64275-73-6]*

Seifert (1981)	d	0.006

1-OCTEN-3-OL *[3391-86-4]*

Cronin & Ward (1971)		0.1
Kaminski *et al.* (1972)		0.01
Buttery *et al.* (1974,1988a,1999); Buttery (1981); Takeoka *et al.* (1991a); Buttery & Ling (1995,1998)	d	0.001 - 0.0014
Kossa *et al.* (1979)		0.000 5
Pyysalo & Suihko (1976a)	d	0.010
Whitfield *et al.* (1982)	r	0.025
Buttery (1981,1989)		0.000 005
Tamura *et al.* (1995)		0.01
Ong & Acree (1999)		0.018
Darriet *et al.* (2002)	d	0.007
Giri *et al*. (2010)	d	0.0015

(*Z*)-5-OCTEN-2-ONE *[22610-86-2]*

Buttery (1989); Hansen *et al.* (1992)	d	0.030

(*Z*)-5-OCTEN-3-ONE

Darriet *et al.* (2002)	d	0.001

1-OCTEN-3-ONE *[4312-99-6]*

Cronin & Ward (1971)		0.01
Pyysalo & Suihko (1976a)	d	0.004
Swoboda & Peers (1977)	d	0.000 089
Buttery *et al.* (1978b,1990b); Buttery (1981,1989,1993)	d	0.000 005
Kossa *et al.* (1979); Tressl *et al.* (1979b,1980)		0.000 5
Whitfield *et al.* (1982)	r	0.000 1
Konopka *et al.* (1995); Milo & Grosch (1997); Kerscher & Grosch (2000)		0.000 05
Buttery *et al.* (1997,1999); Buttery & Ling (1998)	d	0.000 05
Darriet *et al.* (2002)	d	0.000 007
Czerny *et al*. (2007c); Czerny *et al*. (2008)	r	0.000 036 - 0.000 04
Czerny *et al.* (2008)	d	0.000 016
Pons *et al.* (2011)	d	0.000 003

1-OCTEN-3-YL ACETATE *[2442-10-6]*

Pyysalo & Suihko (1976a)	d	0.09

1-OCTEN-3-YL PROPANOATE *[63156-02-5]*

Pyysalo & Suihko (1976a)	d	0.022

OCTYL ACETATE *[112-14-1]*

Guadagni *et al.* (1966); Teranishi *et al.* (1987); Takeoka *et al.* (1992)	d	0.012
Ahmed *et al.* (1978a)	d	0.047
Sugisawa *et al.* (1991); Yang *et al.* (1992); Tamura *et al.* (1993)	d	0.45 - 4.6
Takeoka *et al.* (1996)	d	0.19
Pino & Mesa (2006)	d	0.047

2-OCTYL ACETATE *[2051-50-5]*
Takeoka *et al.* (1996) d 0.1

OCTYLAMINE *[111-86-4]*
Amoore *et al.* (1975b) d 2.64

OCTYL BUTANOATE (octyl butyrate) *[110-39-4]*
Sugisawa *et al.* (1991) d 8.2
Boonbumrung *et al.* (2001) d 0.25

octyl butyrate⇒OCTYL BUTANOATE
octyl isobutyrate⇒OCTYL 2-METHYLPROPANOATE

OCTYL 3-METHYL-3-PHENYL-2,3-EPOXYPROPANOATE
Tekin & Karaman (1992) d 720

OCTYL 2-METHYLPROPANOATE (octyl isobutyrate) *[109-15-9]*
Guadagni *et al.* (1966); Teranishi *et al.* (1987) d 0.006
Pino & Mesa (2006) d 0.006

2-OCTYL-3-(PHENYLTHIO)PYRAZINE *[113685-97-5]*
Masuda & Mihara (1988) d 0.08

OCTYLPYRAZINE *[110823-72-8]*
Masuda & Mihara (1988) d 0.4

oenanthic acid⇒HEPTANOIC ACID

4-OXA-2,6-HEPTANEDIOL (dipropylene glycol) *[110-98-5]*
Alexander *et al.* (1982) d 240

11-OXAHEXADECALACTONE *[3391-83-1]*
Amoore *et al.* (1977) d 0.019

12-OXAHEXADECALACTONE *[6707-60-4]*
Amoore *et al.* (1977) d 0.045

3-OXA-1,5-PENTANEDIOL (diethylene glycol) *[111-46-6]*
Alexander *et al.* (1982) d 240

4-OXODECANAL *[43160-78-7]*
Takeoka *et al.* (1995a) d 0.012

4-OXOHEPTANAL
Takeoka *et al.* (1995a) d 0.8

4-OXOHEXANAL
Takeoka *et al.* (1995a) d 3

4-OXONONANAL
Takeoka *et al.* (1995a) d 0.15

oxymethylfurfural⇒5-(HYDROXYMETHYL)FURFURAL

P

parathion⇒O,O-DIETHYL O-(4-NITROPHENYL)PHOSPHOROTHIOATE
pelargonic acid⇒NONANOIC ACID
pemenone⇒4(4'-*cis*-*tert*-BUTYLCYCLOHEXYL)-4-METHYL-2-PENTANONE

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

PENTABROMOANISOLE *[1825-26-9]*

Diaz *et al.* (2005)		0.000 043

PENTACHLOROANISOLE *[1825-21-4]*

Curtis *et al.* (1972)		0.004
Nijssen (1988)	d	0.000 41
Diaz *et al.* (2005)		0.000 24

PENTACHLOROBENZENE *[608-93-5]*

Kölle *et al.* (1972)		0.06

PENTACHLOROPHENOL *[87-86-5]*

Hoak (1956)		0.3
Dietz & Traud (1978)		1.6
Young *et al.* (1996)		0.023

6,9-PENTADECADIEN-2-ONE

Buttery *et al.* (1967)		0.001

ω-pentadecalactone⇒15-HYDROXYPENTADECANOIC ACID LACTONE

PENTADECANAL *[2765-11-9]*

Von Ranson & Belitz (1992)	r	0.43
Tamura *et al.* (1995)		1

PENTADECANOIC ACID *[1002-84-2]*

Cherkinski (1961)	d	10.0

1-PENTADECENE *[13360-61-7]*

Tamura *et al.* (1995)		3.6

pentamethyleneimine⇒PIPERIDINE

PENTANAL (valeraldehyde) *[110-62-3]*

Guadagni *et al.* (1963); Teranishi *et al.* (1974); Buttery *et al.* (1978b,1988a,1997);		
Buttery & Ling (1995)	d	0.012
Johansson *et al.* (1973)	d	0.100
Amoore *et al.* (1976)	d	0.0606
Schnabel *et al.* (1988)	d	0.016 - 0.024
Jensen *et al.* (1999)	d	0.022
Fabrellas *et al.* (2004)		0.0027 - 0.005
Pino & Mesa (2006)	d	0.012

1,5-PENTANEDIAL (glutaraldehyde) *[111-30-8]*

Nichiporenko *et al.* (1987)		100

1,5-PENTANEDIAMINE (cadaverine) *[462-94-2]*

Holluta (1960)		20
Wang *et al.* (1975)		190

1,4-PENTANEDIOL *[626-95-9]*

Schnabel *et al.* (1988)	d	9.9 - 49

1,5-PENTANEDIOL *[111-29-5]*

Schnabel *et al.* (1988)	d	500 - 790

2,4-PENTANEDIOL *[625-69-4]*

Schnabel *et al.* (1988)	d	9.6 - 29

2,3-PENTANEDIONE *[600-14-6]*

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Buttery *et al.* (1994a,1997,1999); Buttery & Ling (1995,1998)	d	0.020
Blank *et al.* (1991); Milo & Grosch (1996,1997)		0.029 - 0.030
1-PENTANETHIOL (amyl mercaptan) *[110-66-7]*		
Pietrzak & Barylko-Pikielna (1976)		0.000 8
PENTANOIC ACID (valeric acid) *[109-52-4]*		
Cherkinski (1961)	d	3.0
Amoore *et al.* (1968); Amoore (1976)	d	1.37
Zoeteman *et al.* (1971)		10
Schnabel *et al.* (1988)	d	0.94 - 4.7
Buttery *et al.* (1999)	d	3
Boonbumrung *et al.* (2001)	d	0.28
Karagül-Yüceer *et al.* (2003)	d	1.207
Czerny *et al.* (2007c)		0.024
Czerny *et al.* (2008)	d	11
Czerny *et al.* (2008)	r	17
1-PENTANOL (amyl alcohol) *[71-41-0]*		
Sega *et al.* (1967)	d	2.54
Rabin & Cain (1986)	d	28 - 70
Schnabel *et al.* (1988)	d	1.6 - 4.1
Buttery *et al.* (1988a,1989,1990a,1994a,1997,1999); Hansen *et al.* (1992); Buttery (1993); Buttery & Ling (1995,1998)	d	4
Boonbumrung *et al.* (2001)	d	0.12
Pino & Mesa (2006)	d	4
Giri *et al.* (2010)	d	0.1502
2-PENTANOL *[6032-29-7]*		
Schnabel *et al.* (1988)	d	8.1 - 16
3-PENTANOL *[584-02-1]*		
Schnabel *et al.* (1988)	d	5.7 - 8.2
Giri *et al.* (2010)	d	4.125
2-PENTANONE (methyl propyl ketone) *[107-87-9]*		
Manning & Robinson (1973)		70
Schnabel *et al.* (1988)	d	1.6 - 4.0
Larsen & Poll (1992)	d	0.01 - 0.1
Boonbumrung *et al.* (2001)	d	1.38
3-PENTANONE (diethyl ketone) *[96-22-0]*		
Schnabel *et al.* (1988)	d	0.040 - 0.082
1,2,3,5,6-PENTATHIEPANE (lenthionine) *[292-46-6]*		
Wada *et al.* (1967)		0.27 - 0.53
2-PENTENAL *[764-39-6]*		
Tandon *et al.* (2000)	d	0.055
(*E*)-2-PENTENAL *[1576-87-0]*		
Buttery *et al.* (1989); Buttery (1993)	d	1.5
Jensen *et al.* (1999)	d	0.31
Tamura *et al.* (2001)	d	0.98
2-PENTEN-1-OL *[20273-24-9]*		
Tamura *et al.* (1995)	d	1.1

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

(*E*)-2-PENTEN-1-OL *[1576-96-1]*

Giri *et al*. (2010)	d	0.0892

(*Z*)-2-PENTEN-1-OL *[1576-95-0]*

Tamura *et al.* (2001); Boonbumrung *et al.* (2001)	d	0.72

1-PENTEN-3-OL *[616-25-1]*

Buttery *et al.* (1971,1989,1990a); Buttery (1993); Buttery & Ling (1995,1998)	d	0.4
Giri *et al*. (2010)	d	0.3581

3-PENTEN-2-ONE *[625-33-2]*

Buttery *et al.* (1988a)	d	0.0015
Boonbumrung *et al.* (2001)	d	1.20

1-PENTEN-3-ONE (ethyl vinyl ketone) *[1629-58-9]*

Buttery *et al.* (1971,1987a,1989,1990a); Guadagni *et al.* (1972); Buttery (1989,1993)	d	0.001 - 0.0013
Ahmed *et al.* (1978a)	d	0.000 9
Tandon *et al.* (2000)	d	0.023

4-PENTENYL ISOTHIOCYANATE *[18060-79-2]*

Masuda *et al.* (1996)		0.22

PENTYL ACETATE (amyl acetate) *[628-63-7]*

Balavoine (1948)		0.000 01
Baker (1963)		0.08
Guadagni (1966,1968,1969,1970c); Flath *et al.* (1967); Teranishi *et al.* (1966,1981,1987); Buttery (1981)	d	0.005
Sega *et al.* (1967)	d	0.009
Rausch & Serafetinides (1975a,b)	d	0.0041
Rausch & Serafetinides (1975b)	r	8.2
Delaney *et al.* (1977); Bosma *et al.* (1981)	d	0.013
Delaney *et al.* (1977); Bosma *et al.* (1981)	r	130
Schnabel *et al.* (1988)	d	0.0089 - 0.089
Takeoka *et al.* (1996)	d	0.043

2-PENTYL ACETATE *[626-38-0]*

Teranishi *et al.* (1966)	d	0.002
Schnabel *et al.* (1988)	d	0.000 87 - 0.0087
Takeoka *et al.* (1996)	d	0.015

3-PENTYL ACETATE *[620-11-1]*

Teranishi *et al.* (1966)	d	0.009

PENTYLAMINE *[110-58-7]*

Amoore *et al.* (1975b)	d	15.9
Amoore & Forrester (1976)	d	9.21

4-*tert*-PENTYLANISOLE

Gee *et al.* (1974)	d	0.031

PENTYL BUTANOATE (amyl butyrate) *[540-18-1]*

Takeoka *et al.* (1990a)	d	0.21

2-PENTYL BUTANOATE (2-pentyl butyrate) *[60415-61-4]*

Schnabel *et al.* (1988)	d	0.044 - 0.087

2-pentyl butyrate⇒2-PENTYL BUTANOATE

PENTYL CYANIDE (hexanenitrile) *[628-73-9]*

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Fabrellas *et al.* (2004)		0.0032 - 0.004
2-PENTYLFURAN *[3777-69-3]*		
Buttery *et al.* (1969b,1971,1978b,1990a,1997,1999);		
Buttery (1993); Buttery & Ling (1995,1998)	d	0.006
Aoki & Koizumi (1986)		0.0145
Giri *et al*. (2010)	d	0.00580
2-PENTYL HEXANOATE *[88164-61-8]*		
Schnabel *et al.* (1988)	d	0.043 - 0.089
PENTYL 2-METHYLBUTANOATE *[68039-26-9]*		
Schnabel *et al.* (1988)	d	0.0086 - 0.043
PENTYL 3-METHYL-3-PHENYL-2,3-EPOXYPROPANOATE		
Tekin & Karaman (1992)	d	59
PENTYL 2-METHYLPROPANOATE (amyl isobutyrate) *[2445-72-9]*		
Schnabel *et al.* (1988)	d	0.043 - 0.085
(*Z*)-2-PENTYL-2-NONENAL		
Robert *et al.* (2004)	d	0.2
PENTYL PENTANOATE (amyl valerate) *[2173-56-0]*		
Schnabel *et al.* (1988)	d	0.026 - 0.060
4-*tert*-PENTYLPHENOL (4-(2-methylbutan-2-yl)phenol) *[80-46-6]*		
Dietz & Traud (1978)		0.8
2-PENTYL-3-PHENOXYPYRAZINE *[113685-72-9]*		
Masuda & Mihara (1988)	d	0.05
2-PENTYL-3-(PHENYLTHIO)PYRAZINE *[113685-96-4]*		
Masuda & Mihara (1988)	d	0.01
PENTYLPYRAZINE *[6303-75-9]*		
Koehler *et al.* (1971)	d	1
Masuda & Mihara (1988)	d	0.005
2-PENTYLPYRIDINE *[2294-76-0]*		
Buttery *et al.* (1977); Buttery (1993)	d	0.000 6
6-PENTYL-2-PYRONE *[27593-23-3]*		
Engel *et al.* (1988)	d	0.15
PENTYL SALICYLATE (amyl salicylate) *[2050-08-0]*		
Le Magnen (1952)		0.000 8 - 0.0029
2-(PENTYLTHIO)ETHANAL		
Rössner *et al.* (2002)		0.000 1
6-PENTYLTETRAHYDRO-2*H*-THIOPYRAN-2-ONE (δ-thiodecalactone)		
Roling *et al.* (1998)		0.006

pepper pyrazine⇒2-ISOBUTYL-3-METHOXYPYRAZINE
perchloroethylene⇒TETRACHLOROETHENE
perillaldehyde⇒4-ISOPROPENYL-1-CYCLOHEXEN-1-CARBOXALDEHYDE
(4*S*)-(-)-perillaldehyde⇒(4*S*)-(-)-4-ISOPROPENYL-1-CYCLOHEXEN-1-CARBOXALDEHYDE
(4*R*)-(+)-perillaldehyde⇒(4*R*)-(+)-4-ISOPROPENYL-1-CYCLOHEXEN-1-CARBOXALDEHYDE
perillyl alcohol⇒(4-ISOPROPENYL-1-CYCLOHEXENYL)METHANOL
(4*R*)-(+)-perillyl alcohol⇒(4*R*)-(+)-(4-ISOPROPENYL-1-CYCLOHEXENYL)METHANOL

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

(4*S*)-(-)-perillyl alcohol⇒(4*S*)-(-)-(4-ISOPROPENYL-1-CYCLOHEXENYL)METHANOL
α-phellandrene⇒4-ISOPROPYL-1-METHYL-1,5-CYCLOHEXADIENE
β-phellandrene⇒4-ISOPROPYL-1-METHYLENE-2-CYCLOHEXENE

PHENANTHRENE *[85-01-8]*

Koppe (1965)		1.0

phenethyl acetate⇒2-PHENYLETHYL ACETATE
phenethyl alcohol⇒2-PHENYLETHANOL

PHENOL *[108-95-2]*

Helbig (1939)		7.13
Le Magnen (1952)		8.7 - 28.6
Nesmeyanova (1953)		25
Hoak (1956)		0.5
Burttschell *et al.* (1959)		> 1
Holluta (1960)		1
Rosen *et al.* (1962)		4.2
Baker (1963)		5.9
Dietz & Traud (1978)		4
Persson *et al.* (1987)		5.6
Young *et al.* (1996)		0.031
Mirlohi (1997); Dietrich *et al.* (1999)		5
Giri *et al*. (2010)	d	58.58525

PHENOXYBENZENE (diphenyl ether) *[101-84-8]*

Appell (1969)		0.5
Kölle *et al.* (1972)		0.015

3-PHENOXYBENZYL CHLORIDE *[53874-66-1]*

Shtabskii *et al.* (1991)		0.03

PHENOXYPYRAZINE *[107697-82-5]*

Masuda & Mihara (1988)	d	0.03

phenylacetaldehyde⇒2-PHENYLETHANAL
2-phenylacetaldehyde dimethylacetal⇒(2,2-DIMETHOXYETHYL)BENZENE

PHENYLACETIC ACID *[103-82-2]*

Appell (1969)		1
Münch & Schieberle (1998); Schieberle (1998)		1
Ong & Acree (1999)		2
Carunchia Whetstine *et al.* (2005)	d	0.464
Czerny *et al.* (2008)	d	6.1
Czerny *et al.* (2008)	r	12

phenylacetonitrile⇒BENZYL CYANIDE
phenylalanine⇒2-AMINO-3-PHENYLPROPANOIC ACID
phenylbenzene⇒BIPHENYL

1-PHENYL-1-CYCLOHEXENE *[771-98-2]*

Koszinowski *et al.* (1980)	r	0.010

3-PHENYL-1-CYCLOHEXENE *[15232-96-9]*

Koszinowski *et al.* (1980)	r	0.010

4-PHENYL-1-CYCLOHEXENE *[4994-16-5]*

Koszinowski *et al.* (1980)	r	0.010

PHENYLDITHIOBENZENE (diphenyl disulphide) *[882-33-7]*

Brady (1968)		0.005

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

2-PHENYLETHANAL (benzeneacetaldehyde, phenylacetaldehyde) *[122-78-1]*

Buttery *et al.* (1969b,1971,1978b,1988a,1989, 1990a,1997,1999); Guadagni *et al.* (1972); Buttery (1981,1993); Hansen *et al.* (1992); Buttery & Ling (1995,1998)	d	0.004
Fabrellas *et al.* (2004)		0.000 3 - 0.000 43
Freuze *et al.* (2004, 2005)	d	0.03
Carunchia Whetstine *et al.* (2005)	d	0.002
Pino & Mesa (2006)	d	0.004
Schuh & Schieberle (2006)	d	0.0063

1-PHENYLETHANOL (phenylmethylcarbinol, methyl phenyl carbinol, styralyl alcohol) *[98-85-1]*

Rosen *et al.* (1963)		1.45
Pyysalo *et al.* (1977)	d	0.70
Tandon *et al.* (2000)	d	0.479

2-PHENYLETHANOL (phenyl ethyl alcohol, phenethyl alcohol, β-phenylethyl alcohol) *[60-12-8]*

Sega *et al.* (1967)	d	0.00317
Appell (1969)		1
Rausch & Serafetinides (1975a,b)	d	0.000 015
Rausch & Serafetinides (1975b)	r	1.2
Pyysalo *et al.* (1977)	d	0.086
Buttery *et al.* (1987a,1987b,1988a,1989,1990a); Buttery (1993)	d	1 - 1.1
Zatorre & Jones-Gotman (1990)	d	1.5 - 3.5
Takahashi (1990)		40 - 75
Schieberle (1991,1992); Gassenmeier & Schieberle (1995)	d	1
Graham *et al.* (1995)	d	1.5
Rashash *et al.* (1997)		0.06
Ong & Acree (1999)		2
Bonfils *et al.* (2004)	d	7.2
Jacquot *et al.* (2004)		472
Carunchia Whetstine *et al.* (2005)	d	0.122
Pino & Mesa (2006)	d	1.1
Schuh & Schieberle (2006)	d	1
Czerny *et al*. (2007a); Czerny *et al.* (2008)	r	0.39
Czerny *et al*. (2008)	d	0.14
Giri *et al*. (2010)	d	0.56423

1-PHENYLETHYL ACETATE (styralyl acetate) *[93-92-5]*

Appell (1969)		0.16

2-PHENYLETHYL ACETATE (phenethyl acetate) *[103-45-7]*

Takahashi (1990)		3 - 5
Ong & Acree (1999)		0.02
Carunchia Whetstine *et al.* (2005)	d	0.019
Pino & Mesa (2006)	d	0.48
Giri *et al*. (2010)	d	0.24959

phenyl ethyl alcohol⇒2-PHENYLETHANOL
β-phenylethyl alcohol ⇒2-PHENYLETHANOL

2-PHENYLETHYL CYANIDE (hydrocinnamonitrile) *[645-59-0]*

Buttery *et al.* (1976b)	d	0.015

2-PHENYLETHYL FORMATE *[104-62-1]*

Pyysalo *et al.* (1977)	d	0.27

2-PHENYLETHYL ISOTHIOCYANATE (2-phenylethyl mustard oil) *[2257-09-2]*

Buttery *et al.* (1976b); Buttery (1981,1993)	d	0.006

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Masuda *et al.* (1996) 0.24

β-phenylethylmethylethylcarbinol⇒ 3-METHYL-1-PHENYL-3-PENTANOL
2-phenylethyl mustard oil ⇒2-PHENYLETHYL ISOTHIOCYANATE

PHENYL ISOTHIOCYANATE (phenyl mustard oil) *[103-72-0]*
Kauffmann (1907) 2.1

phenylmethanol⇒BENZYL ALCOHOL
phenylmethylcarbinol⇒1-PHENYLETHANOL
phenyl mustard oil⇒ PHENYL ISOTHIOCYANATE

2-PHENYLPHENOL (2-hydroxybiphenyl) *[90-43-7]*
Dietz & Traud (1978) 0.4

4-PHENYLPHENOL (4-hydroxybiphenyl) *[92-69-3]*
Dietz & Traud (1978) 0.7

3-PHENYLPROPANOIC ACID (hydrocinnamic acid) *[501-52-0]*
Ong & Acree (1999) 5

3-PHENYL-2-PROPENAL (cinnamaldehyde) *[104-55-2]*
Amoore (1969) 0.05
Appell (1969) 0.16
Pyysalo *et al.* (1977) d 0.75

3-PHENYL-(*E*)2-PROPENAL ((*E*)-cinnamaldehyde) *[14371-10-9]*
Pino & Mesa (2006) d 6

(*E*)-3-PHENYL-2-PROPENOIC ACID (cinnamic acid) *[140-10-3]*
Ong *et al.* (1998) d 5
Ong & Acree (1999) 2

3-PHENYL-2-PROPEN-1-OL (cinnamic alcohol, cinnamyl alcohol) *[104-54-1]*
Appell (1969) 1
Steinhaus *et al.* (2009); Sinuco *et al.* (2010) d 0.077

3-PHENYL-(*Z*)-2-PROPEN-1-OL (*cis*-cinnamic alcohol) *[4510-34-3]*
Pyysalo *et al.* (1977) d 0.081

3-PHENYL-(*E*)-2-PROPEN-1-OL (*trans*-cinnamic alcohol) *[4407-36-7]*
Pyysalo *et al.* (1977) d 2.8

3-PHENYL-2-PROPEN-1-YL ACETATE (cinnamyl acetate) *[103-54-8]*
Steinhaus *et al.* (2009); Sinuco *et al.* (2010) d 0.15

2-(PHENYLTHIO)-3-PROPYLPYRAZINE *[113685-95-3]*
Masuda & Mihara (1988) d 0.09

(PHENYLTHIO)PYRAZINE *[107697-83-6]*
Masuda & Mihara (1988) d 0.4

phloroglucinol⇒1,3,5-TRIHYDROXYBENZENE
phytol⇒1-HYDROXY-3,7,11,15-TETRAMETHYL-2-HEXADECENE
picric acid⇒2,4,6-TRINITROPHENOL
α-pinene⇒2,6,6-TRIMETHYLBICYCLO[3.1.1]-2-HEPTENE
(1*S*,5*S*)-(-)-α-pinene⇒(1*S*,5*S*)-(-)-2,6,6-TRIMETHYLBICYCLO[3.1.1]-2-HEPTENE
(1*R*,5*R*)-(+)-α-pinene⇒(1*R*,5*R*)-(+)-2,6,6-TRIMETHYLBICYCLO[3.1.1]-2-HEPTENE
(*R*)-α-pinene⇒(1*R*,5*R*)-(+)-2,6,6-TRIMETHYLBICYCLO[3.1.1]-2-HEPTENE
β-pinene⇒6,6-DIMETHYL-2-METHYLENEBICYCLO[3.1.1]HEPTANE
(1*S*,5*S*)-(-)-β-pinene⇒(1*S*,5*S*)-(-)-6,6-DIMETHYL-2-METHYLENEBICYCLO[3.1.1]HEPTANE
(1*R*,5*R*)-(+)-β-pinene⇒(1*R*,5*R*)-(+)-6,6-DIMETHYL-2-METHYLENEBICYCLO[3.1.1]HEPTANE
1-piperideine⇒1,2,3,4-TETRAHYDROPYRIDINE

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

3-piperideine⇒1,2,3,6-TETRAHYDROPYRIDINE

PIPERIDINE (azacyclohexane, hexahydroazine, pentamethyleneimine) *[110-89-4]*

Amoore *et al.* (1975b)	d	65.8
Amoore & Forrester (1976)	d	70.6

piperitone⇒4-ISOPROPYL-1-METHYL-1-CYCLOHEXEN-3-ONE
piperonal⇒3,4-(METHYLENEDIOXY)BENZALDEHYDE

POTASSIUM ETHYL XANTHATE *[140-89-6]*

Trofimovich *et al.* (1976)		0.1

potassium isoamyl xanthate⇒POTASSIUM ISOPENTYL XANTHATE

POTASSIUM ISOBUTYL XANTHATE *[13001-46-2]*

Trofimovich *et al.* (1976)		0.005

POTASSIUM ISOPENTYL XANTHATE (potassium isoamyl xanthate) *[928-70-1]*

Trofimovich *et al.* (1976)		0.005

POTASSIUM ISOPROPYL XANTHATE *[140-92-1]*

Trofimovich *et al.* (1976)		0.05

POTASSIUM PHOSPHATE, DIBASIC (dipotassium phosphate) *[7758-11-4]*

Alexander *et al.* (1982)	d	> 40,000

prenyl alcohol⇒3-METHYL-2-BUTEN-1-OL
prenyl mercaptan⇒3-METHYL-2-BUTENE-1-THIOL

2-(PRENYLTHIO)ETHANAL

Rössner *et al.* (2002)		0.000 05

PROPANAL (propional, propionaldehyde) *[123-38-6]*

Guadagni *et al.* (1963); Teranishi *et al.* (1974);		
Buttery *et al.* (1988a) ; Buttery *et al.* (1997)	d	0.0095 - 0.01
Amoore *et al.* (1976)	d	0.145
Herrmann & Abd El Salam (1980a)		0.067
Schnabel *et al.* (1988)	d	0.081 - 0.16
Milo (1995a, 1995b); Milo & Grosch (1996a)		0.010
Fabrellas *et al.* (2004)		0.011 - 0.012
Giri *et al.* (2010)	d	0.0151

1,3-PROPANEDIAMINE (trimethylenediamine, 1,3-diaminopropane) *[109-76-2]*

Amoore *et al.* (1975b)	d	12,400

1,2-PROPANEDIOL (propylene glycol) *[57-55-6]*

Alexander *et al.* (1982)	d	340

1-PROPANETHIOL (propyl mercaptan) *[107-03-9]*

Pietrzak & Barylko-Pikielna (1976)		0.0031

1,2,3-PROPANETRIOL (glycerin, glycerol) *[56-81-5]*

Alexander *et al.* (1982)	d	> 20,000

PROPANOIC ACID (propionic acid) *[79-09-4]*

Amoore *et al.* (1968); Amoore (1976)	d	40.3
Zoeteman *et al.* (1971)		200
Kikuchi *et al.* (1976)		32.8
Schnabel *et al.* (1988)	d	0.96 - 5.0
Larsen & Poll (1992)	d	0.1 - 1.0
Buttery & Ling (1998); Buttery *et al.* (1999)	d	2

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Karagül-Yüceer *et al.* (2003)	d	2.19

1-PROPANOL (propyl alcohol) *[71-23-8]*

Williams (1953)		6 - 45
Moncrieff (1957)		50
Guadagni (1966,1968,1969); Flath *et al.* (1967)	d	9
Sega *et al.* (1967)	d	6.08
Mitchell & Gregson (1968)	r	600 - 6,300
Mulders (1972,1973)	d	40
Cain (1974)		337
Halpin (1978)		200,000 - 650,000
Sarzynski (1982)		67
Schnabel *et al.* (1988)	d	5.7 - 7.8
Takahashi (1990)		50
Pino & Mesa (2006)	d	9
Giri *et al*. (2010)	d	8.5056

2-PROPANOL (isopropyl alcohol, isopropanol) *[67-63-0]*

Moncrieff (1957)		70
Sega *et al.* (1967)	d	601
Antonova & Salmina (1981)		0.25
Sarzynski (1982)		8,333
Granzer *et al.* (1986)		100 - 2,000
Schnabel *et al.* (1988)	d	40 - 78
Giri *et al*. (2010)	d	9.7879

PROPANONE (acetone) *[67-64-1]*

Cheesman & Mayne (1953)		187 - 263
Moncrieff (1957)		300
Holluta (1960)		5
Baker (1963)		40.9
Blondheim & Reznik (1971)		10,000
Mulders (1972,1973)	d	300 - 335
Manning & Robinson (1973)		500
De Grunt (1975); De Grunt *et al.* (1977)	d	40
Sarzynski (1982)		476
Eichenbaum *et al.* (1983)	d	59,375
Pino *et al.* (1986)	d	437.119
Granzer *et al.* (1986)		50 - 1,000
Schnabel *et al.* (1988)	d	120 - 160
Odeigah (1994)		20,000
Rashash *et al.* (1997)		0.04 - 0.08
Tandon *et al.* (2000)	d	0.832

2-propanoyl-1-pyrroline⇒2-PROPIONYL-1-PYRROLINE

2-PROPENAL (acrolein) *[107-02-8]*

Guadagni *et al.* (1963); Teranishi *et al.* (1974)	d	0.11

PROPENOIC ACID (acrylic acid) *[79-10-7]*

Piringer & Granzer (1984); Granzer *et al.* (1986)		0.5-10

2-PROPENYLDITHIO-2-PROPENE (diallyl disulphide) *[2179-57-9]*

Teleky-Vamossy & Petro-Turza (1986)	d	0.03
Teleky-Vamossy & Petro-Turza (1986)	r	0.08

propenylguaethol⇒*trans*-2-ETHOXY-5-(1-PROPENYL)PHENOL

2-PROPENYLTHIO-2-PROPENE (diallyl sulphide) *[592-88-1]*

Pietrzak & Barylko-Pikielna (1976)		0.0325
Teleky-Vamossy & Petro-Turza (1986)	d	0.1

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Teleky-Vamossy & Petro-Turza (1986)	r	0.3

propional⇒PROPANAL
propionaldehyde⇒PROPANAL
propionic acid⇒PROPANOIC ACID
propionitrile⇒ETHYL CYANIDE

2-PROPIONYL-1-PYRROLINE (2-propanoyl-1-pyrroline) *[133447-37-7]*

Belitz & Grosch (1992a)		0.000 1
Buttery *et al.* (1997) ; Buttery (1999)	d	0.000 1

PROPYL ACETATE *[109-60-4]*

Sarzynski (1982)		3
Piggott & Findlay (1984)	d	11.0
Schnabel *et al.* (1988)	d	2.7 - 6.2
Takeoka *et al.* (1996)	d	2

2-PROPYL ACETATE *[108-21-4]*

Takeoka *et al.* (1996)	d	1.99

propyl alcohol⇒1-PROPANOL

PROPYLAMINE *[107-10-8]*

Amoore *et al.* (1975b)	d	90.1
Amoore & Forrester (1976)	d	62.4

PROPYLBENZENE *[103-65-1]*

Giri *et al.* (2010)	d	0.17712

PROPYL BUTANOATE (propyl butyrate) *[105-66-8]*

Flath *et al.* (1967); Teranishi *et al.* (1987)	d	0.018
Takeoka *et al.* (1990a,1992)	d	0.124
Schnabel *et al.* (1988)	d	0.000 87 - 0.0043
Boonbumrung *et al.* (2001)	d	0.16
Pino & Mesa (2006)	d	0.018

propyl butyrate⇒PROPYL BUTANOATE

PROPYL CYANIDE (butyronitrile) *[109-74-0]*

Fabrellas *et al.* (2004)		0.032 - 0.070

propylene chloride⇒1,2-DICHLOROPROPANE
propylene glycol⇒1,2-PROPANEDIOL
propylene glycol methyl ether⇒1-METHOXY-2-PROPANOL
propylene oxide⇒1,2-EPOXYPROPANE

(*Z*)-2-PROPYL-2-HEPTENAL

Robert *et al.* (2004)	d	0.15

PROPYL 4-HYDROXYBENZOATE *[94-13-3]*

Dietz & Traud (1978)		0.02

propyl isobutyrate⇒PROPYL 2-METHYLPROPANOATE
propyl isovalerate⇒PROPYL 3-METHYLBUTANOATE
propyl mercaptan⇒1-PROPANETHIOL

PROPYL 2-METHYLBUTANOATE *[37064-20-3]*

Schnabel *et al.* (1988)	d	0.000 087 - 0.000 43
Takeoka *et al.* (1998)	d	0.000 2

PROPYL 3-METHYLBUTANOATE (propyl isovalerate) *[557-00-6]*

Schnabel *et al.* (1988)	d	0.0087 - 0.043

PROPYL (*E*)-2-METHYL-2-BUTENOATE *[61692-83-9]*

Takeoka *et al.* (1998)	d	0.012

PROPYL (*Z*)-2-METHYL-2-BUTENOATE *[53082-57-8]*

Takeoka *et al.* (1998)	d	0.07

PROPYL 2-METHYLPROPANOATE (propyl isobutyrate) *[644-49-5]*

Schnabel *et al.* (1988)	d	0.000 086 - 0.000 43

2-PROPYLPENTANAL *[18295-59-5]*

Schnabel *et al.* (1988)	d	0.42 - 0.83

2-PROPYLPHENOL *[644-35-9]*

Czerny & Buettner (2009)	d	0.039
Czerny & Buettner (2009)	r	0.064

3-PROPYLPHENOL *[621-27-2]*

Czerny & Buettner (2009)	d	0.000 29
Czerny & Buettner (2009)	d	0.000 62

4-PROPYLPHENOL *[645-56-7]*

Dietz & Traud (1978)		0.5
Czerny & Buettner (2009)	d	0.107
Czerny & Buettner (2009)	r	0.149

PROPYL PROPANOATE *[106-36-5]*

Flath *et al.* (1967); Teranishi *et al.* (1987)	d	0.057
Schnabel *et al.* (1988)	d	0.44 - 0.88

PROPYLPYRAZINE *[18138-03-9]*

Masuda & Mihara (1988)	d	0.3

4-PROPYLTHIAZOLE *[41981-60-6]*

Teranishi *et al.* (1974)	d	0.3

2-(PROPYLTHIO)ETHANAL

Rössner *et al.* (2002)		0.000 07

propyzamide⇒3,5-DICHLORO-*N*-(1,1-DIMETHYL-2-PROPYNYL)BENZAMIDE
prussic acid⇒HYDROGEN CYANIDE
pseudocumene⇒1,2,4-TRIMETHYLBENZENE
pseudoionone⇒6,10-DIMETHYL-3,5,9-UNDECATRIEN-2-ONE
(*d*)-pulegone⇒(*R*)-(+)-4-ISOPROPYLIDENE-1-METHYL-3-CYCLOHEXANONE
putrescine⇒1,4-BUTANEDIAMINE

PYRAZINE *[290-37-9]*

Teranishi *et al.* (1971,1974); Buttery *et al.* (1994a, 1999); Buttery & Ling (1995,1998)	d	75 - 180
Calabretta (1973,1975,1978)	d	500
Masuda & Mihara (1988)	d	300

PYRIDINE (azine) *[110-86-1]*

Kauffmann (1907)		4
Le Magnen (1952)		0.83 - 1.66
Moncrieff (1957)		0.5
Holluta (1960)		0.1
Baker (1963)		0.82
Henkin & Bartter (1966); Henkin & Hoye (1966); Henkin (1967a,1967b,1968)	d	0.000 79
Henkin & Bartter (1966); Henkin & Hoye (1966);		

Henkin (1967b,1968)	r	7.9 - 40
Henkin *et al.* (1968); Hoye *et al.* (1970)	d	0.0079
Henkin *et al.* (1968); Hoye *et al.* (1970)	r	790
Henkin *et al.* (1971,1972); Henkin & Smith (1971); Marshall & Henkin (1971); Henkin & Larson (1972); Schechter & Henkin (1974)	d	0.0079
Henkin *et al.* (1971,1972); Henkin & Smith (1971); Marshall & Henkin (1971); Henkin & Larson (1972); Schechter & Henkin (1974)	r	79
Baker & Luh (1973)	d	0.000 3 - 2.9
Amoore *et al.* (1975b)	d	4.33
Filla *et al.* (1975); De Michele *et al.* (1976); Campanella *et al.* (1978)	d	0.000 079 - 0.79
Rausch & Serafetinides (1975a,b)	d	0.20
Rausch & Serafetinides (1975b)	r	25
Henkin *et al.* (1976); Henkin (1976); Delaney *et al.* (1977); Bosma *et al.* (1981)	d	0.000 079 - 0.79
Henkin *et al.* (1976); Henkin (1976); Delaney *et al.* (1977); Bosma *et al.* (1981)	r	7.9 - 790
Sullivan & Burch (1976); Burch *et al.* (1978)	d	0.27
Sullivan & Burch (1976); Burch *et al.* (1978)	r	11
Sherman *et al.* (1979)	d	3.8
Lawas *et al.* (1979)	d	19
Lawas *et al.* (1979)	r	302
Rosen *et al.* (1979)	d	0.0079
Perry *et al.* (1980)	d	0.000 03
Vreman *et al.* (1980)	d	9 - 39
Garrett-Laster *et al.* (1984)	d	0.0079
Garrett-Laster *et al.* (1984)	r	0.079
Amoore (1986a,b)	d	2 - 8
Amoore (1986a)	r	12
Rabin & Cain (1986)	d	3 - 9
Buttery *et al.* (1988a,1994a,1999); Buttery & Ling (1998)	d	2
Pino & Mesa (2006)	d	2

pyrocatechol⇒1,2-DIHYDROXYBENZENE
pyrogallol⇒1,2,3-TRIHYDROXYBENZENE

PYRROLE (azole) *[109-97-7]*

Holluta (1960)		10
Amoore *et al.* (1975b)	d	49.6
Buttery & Ling (1995)	d	20

2-PYRROLECARBALDEHYDE *[1003-29-8]*

Pyysalo *et al.* (1977)	d	65.0

PYRROLIDINE (azolidine) *[123-75-1]*

Amoore *et al.* (1975b)	d	20.2

(*R*)-(+)-2-PYRROLIDINECARBOXYLIC ACID (D-proline) *[344-25-2]*

Laska (2010)	d	8,630

(*S*)-(-)-2-PYRROLIDINECARBOXYLIC ACID (L-proline) *[147-85-3]*

Laska (2010)	d	11,510

PYRROLIDINO[1,2-e]-4H-2,4-DIMETHYL-1,3,5-DITHIAZINE *[116505-60-3]*

Kubota *et al.* (1991)	d	0.000 000 000 000 01

1-PYRROLINE (3,4-dihydro-2H-azole) *[5724-81-2]*

Amoore *et al.* (1975b)	d	0.0220

Teranishi *et al.* (1975) d 0.0023

3-PYRROLINE (2,5-dihydro-1H-azole) *[109-96-6]*
Amoore *et al.* (1975b) d 62.8

1-(1-pyrrolyl)-2-propanone⇒1-ACETONYLPYRROLE

Q

QUINOLINE *[91-22-5]*
Baker (1963) 0.71

R

raspberry ketone⇒1-(4-HYDROXYPHENYL)-3-BUTANONE
resorcinol⇒1,3-DIHYDROXYBENZENE
rose furan⇒2-(3-METHYLBUTEN-2-YL)-3-METHYLFURAN
(*l*)-rose oxide⇒(*l*)-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN
cis-rose oxide⇒*cis*-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN

ROTUNDONE
Wood *et al.* (2008, 2010) d 0.000 008

S

sabinene⇒1-ISOPROPYL-4-METHYLENEBICYCLO[3.1.0]HEXANE
trans-sabinene hydrate⇒*trans*-5-ISOPROPYL-2-METHYLBICYCLO[3.1.0]-2-HEXANOL
safrole⇒4-ALLYL-1,2-(METHYLENEDIOXY)BENZENE
salicylaldehyde⇒2-HYDROXYBENZALDEHYDE

SELINENE
Koppe (1965) 0.001

α-sinensal⇒2,6,10-TRIMETHYL-2,6,9,11-DODECATETRAENAL
β-sinensal⇒(*E*,*E*)-2,6-DIMETHYL-10-METHYLENE-2,6,11-DODECATRIENAL
skatole⇒3-METHYLINDOLE
sorbic aldehyde⇒(*E*,*E*)-2,4-HEXADIENAL
sotolon⇒3-HYDROXY-4,5-DIMETHYL-2(5*H*)-FURANONE
spermidine⇒N-(3-AMINOPROPYL)-1,4-BUTANEDIAMINE
spermine⇒N,N'-BIS(3-AMINOPROPYL)-1,4-BUTANEDIAMINE
stearic acid⇒OCTADECANOIC ACID
strawberry aldehyde⇒ETHYL 3-METHYL-3-PHENYL-2,3-EPOXYPROPANOATE
styralyl acetate⇒1-PHENYLETHYL ACETATE
styralyl alcohol⇒1-PHENYLETHANOL
styrene⇒ETHENYLBENZENE
sugarone⇒ 2-ETHYL-4-HYDROXY-5-METHYL-3(2*H*)-FURANONE
((*R*/*S*)-3-sulfanylhexan-1-ol ⇒3-MERCAPTO-1-HEXANOL
((*R*/*S*)-3-sulfanylhexyl acetate⇒3-MERCAPTOHEXYL ACETATE

SULPHUR DIOXIDE *[7446-09-5]*
Brenner *et al.* (1955) r 50 - 100

syringaldehyde⇒4-HYDROXY-3,5-DIMETHOXYBENZALDEHYDE
syringol⇒2,6-DIMETHOXYPHENOL

T

TAME⇒2-METHOXY-2-METHYLBUTANE
γ-terpinene⇒4-ISOPROPYL-1-METHYL-1,4-CYCLOHEXADIENE
1-terpinen-4-ol⇒1-ISOPROPYL-4-METHYL-3-CYCLOHEXEN-1-OL
(4*S*)-(+)-terpinen-4-ol⇒(*S*)-(+)-1-ISOPROPYL-4-METHYL-3-CYCLOHEXEN-1-OL

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

(4*R*)-(-)-terpinen-4-ol⇒(*R*)-(-)-1-ISOPROPYL-4-METHYL-3-CYCLOHEXEN-1-OL
α-terpineol⇒4-(2-HYDROXYISOPROPYL)-1-METHYL-1-CYCLOHEXENE
(4*S*)-(-)-α-terpineol⇒(*S*)-(-)-4-(2-HYDROXYISOPROPYL)-1-METHYL-1-CYCLOHEXENE
(4*R*)-(+)-α-terpineol⇒(*R*)-(+)-4-(2-HYDROXYISOPROPYL)-1-METHYL-1-CYCLOHEXENE
terpinolene⇒4-ISOPROPYLIDENE-1-METHYL-1-CYCLOHEXENE
terpinyl acetate⇒2-(4-METHYL-3-CYCLOHEXENYL)ISOPROPYL ACETATE
tetra ⇒TETRACHLOROMETHANE

2,3,4,6-TETRACHLOROANISOLE *[938-22-7]*

Curtis *et al.* (1972)		0.000 004
Nijssen (1988)	d	0.000 0011

1,2,3,4-TETRACHLOROBENZENE *[634-66-2]*

Kölle *et al.* (1972)		0.02

1,2,3,5-TETRACHLOROBENZENE *[634-90-2]*

Kölle *et al.* (1972)		0.4

1,2,4,5-TETRACHLOROBENZENE *[95-94-3]*

Kölle *et al.* (1972)		0.13

tetrachlorocatechol⇒TETRACHLORO-1,2-DIHYDROXYBENZENE

TETRACHLORO-1,2-DIHYDROXYBENZENE (tetrachlorocatechol) *[1198-55-6]*

Kovacs *et al.* (1984)	d	> 3.5

TETRACHLOROETHENE (perchloroethylene, tetrachloroethylene) *[127-18-4]*

Kölle *et al.* (1972)		0.3
Alexander *et al.* (1982)	d	0.24

tetrachloroethylene⇒TETRACHLOROETHENE

TETRACHLOROMETHANE (tetra, carbon tetrachloride) *[56-23-5]*

Alexander *et al.* (1982)	d	1.08

tetrachloroguaiacol⇒2-METHOXYTETRACHLOROPHENOL

2,3,4,6-TETRACHLOROPHENOL *[58-90-2]*

Hoak (1956)		47
Dietz & Traud (1978)		0.6
Kovacs *et al.* (1984)	d	10

1,2,3,4,4a,4b.α,5,6,7,8,8a.β,9,10,10a.α-TETRADECAHYDRO-7α-ISOPPROPYL-1β,4a.β-DIMETHYLPHENANTHRENE (fichtelite) *[2221-95-6]*

Koppe (1965)		1.0

ω-tetradecalactone⇒14-HYDROXYTETRADECANOIC ACID LACTONE

TETRADECANAL (myristaldehyde, aldehyde C14) *[124-25-4]*

Von Ranson & Belitz (1992)	r	0.064
Yang *et al.* (1992); Tamura *et al.* (1993)	d	0.067
Tamura *et al.* (1996)	d	0.053
Bonfils *et al.* (2004)	d	0.11

TETRADECANE *[629-59-4]*

Tamura *et al.* (1995)		1

TETRADECANOIC ACID (myristic acid) *[544-63-8]*

Cherkinski (1961)	d	10.0

(*Z*)-7-TETRADECENAL

Widder *et al.* (2008)		0.0026

(*E*)-8-TETRADECENAL

Widder *et al.* (2008)		0.000 009

(*Z*)-8-TETRADECENAL

Widder *et al.* (2008)		0.00124

(*Z*)-9-TETRADECENAL

Widder *et al.* (2008)		0.00144

1-TETRADECENE *[1120-36-1]*

Tamura *et al.* (1995)		0.06

4,5,6,7-TETRAHYDROBENZOTHIAZOLE (cyclohexathiazole) *[4433-49-2]*

Pelosi *et al.* (1983)	d	0.22

3a,4,7,7a-TETRAHYDRO-4,7-METHANOINDENE (dicyclopentadiene) *[77-73-6]*

Ventura *et al.* (1997)		0.000 01 - 0.000 025

1,2,3,4-TETRAHYDRONAPHTHALENE (tetralin) *[119-64-2]*

Rosen *et al.* (1963)		0.018

1,2,3,4-TETRAHYDROPYRIDINE (1-piperideine) *[505-18-0]*

Amoore *et al.* (1975b)	d	0.302

1,2,3,6-TETRAHYDROPYRIDINE (3-piperideine) *[694-05-3]*

Amoore *et al.* (1975b)	d	226

1,2,3,4-TETRAHYDRO-1,1,6-TRIMETHYLNAPHTHALENE *[475-03-6]*

Buttery *et al.* (1990b)	d	0.002

tetrahydronootkatone⇒8-ISOPROPYL-5,6-DIMETHYLBICYCLO[4.4.0]-3-DECANONE
tetralin ⇒1,2,3,4-TETRAHYDRONAPHTHALENE

1,2,7,7-TETRAMETHYL-*exo*-BICYCLO[2.2.1]-2-HEPTANOL (2-methylisoborneol, MIB) *[2371-42-8]*

Medsker *et al.* (1969)		0.000 1
Rosen *et al.* (1970)		0.000 1
Zoeteman & Piet (1973)		0.000 01
Persson & York (1978)	r	0.000 018
Persson & York (1978)	r	0.000 020
Persson (1979a)	r	0.000 042
Persson (1979a)	r	0.000 035
Sugiura *et al.* (1983)		0.000 007
Persson (1983)		0.000 038
Persson (1983)		0.000 008
Ito *et al.* (1988)	d	0.000 004 - 0.000 005
Ito *et al.* (1988)	r	0.000 008 - 0.000 033
Sano (1988)	d	0.000 004
Sano (1988)	d	0.000 012
Tomita *et al.* (1988)		0.000 002 - 0.000 02
Belitz & Grosch (1992g)		0.000 03
Vitzthum *et al.* (1994)		0.000 0025
Young *et al.* (1996)		0.000 015
Rashash *et al.* (1997)		0.000 006 - 0.000 01
Piriou *et al.* (2009)	d	0.000 0012 - 0.000 0018

(1*R*)-1,2,7,7-TETRAMETHYL-*exo*-BICYCLO[2.2.1]-2-HEPTANOL ((-)-2-methylisoborneol)

Blank & Grosch (2002)		0.000 002 – 0.000 003

4-(1,1,3,3-TETRAMETHYLBUTYL)PHENOL *[140-66-9]*

Dietz & Traud (1978)		2.7

cis-1,2,2,6-TETRAMETHYLCYCLOHEXANOL *[139895-91-3]*
Finato *et al.* (1992) d 0.000 08

trans-1,2,2,6-TETRAMETHYLCYCLOHEXANOL *[139895-92-4]*
Finato *et al.* (1992) d 0.0015

(+)-4-((1*S*,5*R*)-2,5,6,6-TETRAMETHYL-2-CYCLOHEXEN-1-YL)-*cis*-3-BUTEN-2-ONE ((+)-*cis*-α-irone)
Brenna *et al.* (1999) d 100

(-)-4-((1*R*,5*S*)-2,5,6,6-TETRAMETHYL-2-CYCLOHEXEN-1-YL)-*cis*-3-BUTEN-2-ONE ((-)-*cis*-α-irone)
Brenna *et al.* (1999) d 10

2,6,6,9-TETRAMETHYL-1,4,8-CYCLOUNDECATRIENE (α-humulene, humulene) *[6753-98-6]*
Guadagni *et al.* (1966); Guadagni (1969,1970c);
Buttery *et al.* (1987b) d 0.12 - 0.16
Tamura *et al.* (2001); Boonbumrung *et al.* (2001) d 0.39
Pino & Mesa (2006) d 0.16

3aβ,6,6,9aβ-TETRAMETHYL-1,2,3a,4,5,5aβ,6,7,8,9a,9bβ-DODECAHYDRONAPHTHO[2,1-b]FURAN (5β-ambrox)
Escher *et al.* (1990) d 0.011

(3a*R*)-3a,6,6,9a-TETRAMETHYL-3aα,5aβ,9aα,9bβ-DODECAHYDRONAPHTHO[2,1-b]FURAN (ambrox) *[6790-58-5]*
Pickenhagen (1989) 0.000 6

(+)-(3a*R*)-3a,6,6,9a-TETRAMETHYL-3aα,5aβ,9aα,9bβ-DODECAHYDRONAPHTHO[2,1-b]FURAN ((*d*)-ambrox)
Ohloff *et al.* (1985); Pickenhagen (1989) 0.0024 - 0.0026

(3a*S*)-3a,6,6,9a-TETRAMETHYL-3aα,5aα,9aβ,9bα-DODECAHYDRONAPHTHO-[2,1-b]-FURAN (isoambrox) *[68365-88-8]*
Ohloff *et al.* (1985) 0.034

(-)-(3a*S*)-3a,6,6,9a-TETRAMETHYL-3aα,5aβ,9aα,9bβ-DODECAHYDRONAPHTHO[2,1-b]FURAN ((*l*)-ambrox)
Ohloff *et al.* (1985); Pickenhagen (1989) 0.000 3

(3a*S*)-3a,6,6,9a-TETRAMETHYL-3aα,5aβ,9aα,9bβ-DODECAHYDRONAPHTHO[2,1-b]FURAN ((*l*)-9-epiambrox)
Ohloff *et al.* (1985) 0.000 15

1,5,9,9-TETRAMETHYL-[1R-(1R*,4E,7E,11R*)]-12-0XABICYCLO[9.1.0]DODECA-4,7-DIENE (humulene epoxide I) *[19888-33-6]*
Tressl *et al.* (1979a) 0.01

2,6,10,10-TETRAMETHYL-1-OXASPIRO[4,5]DECAN-6-OL (dihydro-6-hydroxytheaspirane, 6-hydroxydihydrotheaspirane) *[65620-50-0]*
Ohloff (1977,1978) 0.000 2

TETRAMETHYLPYRAZINE *[1124-11-4]*
Koehler *et al.* (1971) d 10
Calabretta (1975,1978) d 1.0
Giri *et al.* (2010) d 2.52502

2,4,5,7-TETRATHIAOCTANE (bis(methylthiomethyl) disulphide)
Kubota *et al.* (1994) d 0.0035

2-thenyl mercaptan⇒2-THENYLTHIOL

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

2-THENYLTHIOL (2-thenyl mercaptan) *[6258-63-5]*

Schieberle & Hofmann (1996a, 1996b)		0.000 04

thialdine⇒ DIHYDRO-2,4,6-TRIMETHYL-1,3,5(4*H*)-DITHIAZINE

THIAMINE *[59-43-8]*

Comrey *et al.* (1958)		0.024 - 6,250

THIAMINE HYDROCHLORIDE *[67-03-8]*

Kurkela & Uutela (1972)	d	0.0098 - 0.156

4-thiapentanal⇒3-(METHYLTHIO)PROPANAL

THIAZOLE *[288-47-1]*

Teranishi *et al.* (1974)	d	3.1
Marchand *et al.* (2000)	d	0.038

thibetolid⇒15-HYDROXYPENTADECANOIC ACID LACTONE

2-THIENYLACETIC ACID *[1918-77-0]*

Davídek *et al.* (1979)		22.6

3-(2-THIENYL)PROPANOIC ACID *[5928-51-8]*

Davídek *et al.* (1979)		13.2

2-thienylthiol⇒2-THIOPHENETHIOL
γ-thiodecalactone⇒5-HEXYLDIHYDRO-2(3*H*)-THIOPHENONE
δ-thiodecalactone⇒6-PENTYLTETRAHYDRO-2*H*-THIOPYRAN-2-ONE

THIOPHENE *[110-02-1]*

Henkin & Bartter (1966); Henkin (1967a)	d	0.084
Henkin & Bartter (1966)	r	84

2-THIOPHENECARBOXALDEHYDE *[98-03-3]*

Buttery *et al.* (1994a)	d	5

2-THIOPHENEMETHANOL *[636-72-6]*

Buttery *et al.* (1994a)	d	15

2-THIOPHENETHIOL (2-thienylthiol) *[7774-74-5]*

Marchand *et al.* (2000)	d	0.000 8
Burdack-Freitag & Schieberle (2010)		0.000 6

thiophenol⇒BENZENETHIOL
α/β-thujone⇒1-ISOPROPYL-4-METHYLBICYCLO[3.1.0]-3-HEXANONE
thymol⇒2-ISOPROPYL-5-METHYLPHENOL
tiglaldehyde⇒(*E*)-2-METHYL-2-BUTENAL
tiglic aldehyde⇒(*E*)-2-METHYL-2-BUTENAL

TOLUENE (methylbenzene) *[108-88-3]*

Zoeteman *et al.* (1971)		1
Alexander *et al.* (1982)	d	0.024
Sarzynski (1982)		5
Granzer *et al.* (1986)		1-20
Schirack *et al.* (2006)	d	0.527

α-toluenenitrile⇒BENZYL CYANIDE
4-tolyl acetate⇒4-METHYLPHENYL ACETATE
tonalide⇒7-ACETYL-1,1,3,4,4,6-HEXAMETHYL-1,2,3,4-TETRAHYDRONAPHTHALENE

2,3,6-TRIBROMOANISOLE

Diaz *et al.* (2004b, 2005)		0.000 010 - 0.000 011

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

2,4,6-TRIBROMOANISOLE *[607-99-8]*

Saxby (1985)		0.000 000 008
UKWIR (1996)		0.000 000 03
Whitfield *et al.* (1997)	d	0.000 000 02
Diaz *et al.* (2004b, 2005)		0.000 010 - 0.000 012

TRIBROMOMETHANE (bromoform) *[75-25-2]*

De Grunt (1975); De Grunt *et al.* (1977)	d	0.3
CIRSEE in Mackey *et al.* (2004)		0.005
McDonald *et al.* (2009)		0.16

2,4,6-TRIBROMOPHENOL *[118-79-6]*

Piriou *et al.* (2007)	d	0.029 - 0.03

TRICHLORAMINE (nitrogen trichloride) *[10025-85-1]*

Krasner & Barrett (1984)		0.03

2,4,6-TRICHLOROANILINE *[634-93-5]*

Mändli (1990)		0.050

2,3,6-TRICHLOROANISOLE *[50375-10-5]*

Curtis *et al.* (1972)		0.000 000 000 3
Griffiths & Fenwick (1977)	d	0.000 004
Guadagni & Buttery (1978)	d	0.000 0074 - 0.000 0245
Diaz *et al.* (2005)		0.000 005

2,4,6-TRICHLOROANISOLE *[87-40-1]*

Curtis *et al.* (1972)		0.000 000 03
Nijssen (1988)	d	0.000 000 076
Young *et al.* (1996)		0.000 000 9
Diaz *et al.* (2005)		0.000 030

1,2,3-TRICHLOROBENZENE *[87-61-6]*

Kölle *et al.* (1972)		0.01
De Grunt *et al.* (1977)	d	0.02

1,2,4-TRICHLOROBENZENE *[120-82-1]*

Kölle *et al.* (1972)		0.005
Zoeteman (1978)	d	0.005

1,3,5-TRICHLOROBENZENE *[108-70-3]*

Kölle *et al.* (1972)		0.05

1,1,1-TRICHLORO-2,2-BIS(4-CHLOROPHENYL)ETHANE (DDT) *[50-29-3]*

Sigworth (1965)	d	0.35

1,1,1-TRICHLORO-2,2-BIS(4-METHOXYPHENYL)ETHANE (methoxychlor) *[72-43-5]*

Sigworth (1965)	d	4.7

3,4,5-trichlorocatechol⇒ 3,4,5-TRICHLORO-1,2-DIHDROXYBENZENE

3,4,5-TRICHLORO-1,2-DIHYDROXYBENZENE (3,4,5-trichlorocatechol) *[56961-20-7]*

Kovacs *et al.* (1984)	d	> 3.5

1,1,1-TRICHLOROETHANE (methylchloroform) *[71-55-6]*

Alexander *et al.* (1982)	d	0.47
Young *et al*. (1996)		20

TRICHLOROETHENE *[79-01-6]*

Cherkinski (1961)	d	0.5

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Alexander *et al.* (1982)	d	0.56
Sarzynski (1982)		29

3,4,5-trichloroguaiacol⇒2-METHOXY-3,4,5-TRICHLOROPHENOL
4,5,6-trichloroguaiacol⇒2-METHOXY-4,5,6-TRICHLOROPHENOL

TRICHLOROMETHANE (chloroform) *[67-66-3]*

De Grunt (1975); De Grunt *et al.* (1977)	d	0.1
Alexander *et al.* (1982)	d	1
Skvortsov *et al.* (1983)		18.03
Young *et al.* (1996)		30
Khiari *et al.* (2002)		0.09
McDonald *et al.* (2009)		0.12

2,3,6-TRICHLOROPHENOL *[933-75-5]*

Dietz & Traud (1978)		0.3

2,4,5-TRICHLOROPHENOL *[95-95-4]*

Hoak (1956)		0.011
Dietz & Traud (1978)		0.2
Young *et al.* (1996)		0.35

2,4,6-TRICHLOROPHENOL *[88-06-2]*

Hoak (1956)		0.1
Burttschell *et al.* (1959)		> 1
Dietz & Traud (1978)		0.3
Kovacs *et al.* (1984)	d	0.94

2,4,5-TRICHLOROPHENOXYACETIC ACID *[93-76-5]*

Sigworth (1965)	d	2.92

2-(2,4,5-TRICHLOROPHENOXY)PROPANOIC ACID *[93-72-1]*

Sigworth (1965)	d	0.78

(*d*)-tricycloketone⇒[1a*R*-(1aα,5β,5aβ,6aα)]-5,5a-DIMETHYL-6a-(1-METHYLETHYL)-1a,4,5,5a,6,6a-HEXAHYDROCYCLOPROP[a]INDEN-3(1*H*)-ONE
(*l*)-tricycloketone⇒[1a*S*-(1aα,5β,5aβ,6aα)]-5,5a-DIMETHYL-6a-(1-METHYLETHYL)-1a,4,5,5a,6,6a-HEXAHYDROCYCLOPROP[a]INDEN-3(1*H*)-ONE

TRIDECANAL *[10486-19-8]*

Von Ranson & Belitz (1992)	r	0.008
Yang *et al.* (1992); Tamura *et al.* (1993)	d	0.07
Tamura *et al.* (1995)		0.01

TRIDECANOIC ACID *[638-53-9]*

Cherkinski (1961)	d	10.0

TRIETHANOLAMINE *[102-71-6]*

Alexander *et al.* (1982)	d	160

TRIETHYLAMINE *[121-44-8]*

Amoore & Forrester (1976)	d	2.21

triethylene glycol⇒3,6-DIOXA-1,8-OCTANEDIOL

2',3',4'-TRIHYDROXYACETOPHENONE *[528-21-2]*

Dietz & Traud (1978)		2

1,2,3-TRIHYDROXYBENZENE (1,2,3-benzenetriol, pyrogallol) *[87-66-1]*

Dietz & Traub (1978)		2

1,2,4-TRIHYDROXYBENZENE (1,2,4-benzenetriol, hydroxyquinol) *[533-73-3]*

Dietz & Traub (1978)		> 10

1,3,5-TRIHYDROXYBENZENE (1,3,5-benzenetriol, phloroglucinol) *[108-73-6]*

Dietz & Traub (1978)		> 10

2,4,4'-TRIHYDROXYBENZOPHENONE *[1470-79-7]*

Dietz & Traud (1978)		6

TRIIODOMETHANE (iodoform) *[75-47-8]*

Bruchet *et al.* (1989)		0.000 020
Cancho *et al.* (2001)		0.000 03 - 0.000 1

trimethylacetic acid⇒2,2-DIMETHYLPROPANOIC ACID

TRIMETHYLAMINE *[75-50-3]*

Kauffmann (1907)	r	0.5
Le Magnen (1952)		10 - 20
Baker (1963)		1.7
Hobson-Frohock *et al.* (1973);		
Griffiths *et al.* (1979)	d	0.000 5
Amoore *et al.* (1975b)	d	0.000 367
Amoore & Forrester (1976)	d	0.000 467
Kikuchi *et al.* (1976)		0.6
Trubko & Teplyakova (1981a)		0.02
Tuorila *et al.* (1982)	d	0.008 - 0.023

2,3,6-TRIMETHYLANISOLE

Griffiths & Fenwick (1977)	d	0.009

2,4,6-TRIMETHYLANISOLE *[4028-66-4]*

Griffiths & Fenwick (1977)	d	0.000 8

TRIMETHYLARSINE *[593-88-4]*

Whitfield *et al.* (1983)		0.000 002

1,2,4-TRIMETHYLBENZENE (pseudocumene) *[95-63-6]*

Zoeteman *et al.* (1971)		0.5
Jüttner *et al.* (1995)	r	0.26

1,3,5-TRIMETHYLBENZENE (mesitylene) *[108-67-8]*

Baker (1963)		0.027
Zoeteman *et al.* (1971)		0.5
De Grunt (1975); De Grunt *et al.* (1977)	d	0.003
Jüttner *et al.* (1995)	r	0.7

(1*R*,2*S*)-(+)-*endo*-1,7,7,-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANOL (borneol, (1*R*,2*S*-(+)-borneol) *[507-70-0]*

Yang *et al.* (1992); Tamura *et al.* (1993)	d	0.14
Padrayuttawat *et al.* (1997)	d	0.18

(1*S*,2*R*)-(-)-*endo*-1,7,7,-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANOL ((1*S*,2*R*)-(-)-borneol) *[464-45-9]*

Padrayuttawat *et al.* (1997)	d	0.08

exo-1,7,7-TRIMETHYLBICYCLO[2.2,1]-2-HEPTANOL (isoborneol) *[124-76-5]*

Müller & Heller (1977)		0.01 - 0.001
De Grunt *et al.* (1977)	d	0.0025
Zoeteman *et al.* (1978)	d	0.015
Köster *et al.* (1981)	d	0.0085 - 0.016

1,3,3-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANONE (fenchone) *[1195-79-5]*

Yang *et al.* (1992); Tamura *et al.* (1993)	d	1.2

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Zeller & Rychlik (2006)	r	0.44

(1*S*)-(+)-1,3,3-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANONE ((*d*)-fenchone, (+)-fenchone) *[4695-62-9]*

Pelosi & Pisanelli (1981)	d	0.51
Padrayuttawat *et al.* (1997)	d	0.24

(1*R*)-(-)-1,3,3-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANONE ((*l*)-fenchone, (-)-fenchone) *[7787-20-4]*

Pelosi & Pisanelli (1981)	d	0.35
Padrayuttawat *et al.* (1997)	d	0.44
Czerny *et al.* (2008)	d	0.11
Czerny *et al.* (2008)	r	0.44

1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANONE (camphor) *[76-22-2]*

Le Magnen (1952)		0.25 - 0.83
Sugisawa *et al.* (1991); Yang *et al.* (1992); Tamura *et al.* (1993)	d	4.6

(1*R*)-(+)-1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANONE ((*d*)-camphor, (1*R*)-(+)-camphor) *[464-49-3]*

Lillard & Powers (1975)	d	1.29
Pelosi & Pisanelli (1981)	d	1.0
Padrayuttawat *et al.* (1997)	d	1.36

(1*S*)-(-)-1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANONE ((1*S*)-(-)-camphor) *[464-48-2]*

Padrayuttawat *et al.* (1997)	d	0.52
Zeller & Rychlik (2006)	r	1.47

2,6,6-TRIMETHYLBICYCLO[3.1.1]-2-HEPTENE (α-pinene) *[80-56-8]*

Guadagni *et al.* (1966); Guadagni (1969); Buttery *et al.* (1974,1987b); Takeoka (2001)	d	0.006
Appell (1969)		0.5
De Grunt (1975); De Grunt *et al.* (1977)	d	0.0025 - 0.003
Ahmed *et al.* (1978a)	d	0.0095
Pino *et al.* (1986)	d	0.062
Tamura *et al.* (1996)	d	2.08
Tamura *et al.* (1996)	r	2.082
Tamura *et al.* (2001)	d	0.19
Pino & Mesa (2006)	d	0.006
Czerny *et al.* (2008)	d	0.014
Czerny *et al.* (2008)	r	0.041

(1*S*,5*S*)-(-)-2,6,6-TRIMETHYLBICYCLO[3.1.1]-2-HEPTENE ((1*S*,5*S*)-(-)-α-pinene) *[7785-26-4]*

Padrayuttawat *et al.* (1997); Boonbumrung *et al.* (2001); Mookdasanit *et al.* (2003)	d	0.10

(1*R*,5*R*)-(+)-2,6,6-TRIMETHYLBICYCLO[3.1.1]-2-HEPTENE ((1*R*,5*R*)-(+)-α-pinene, (*R*)-α-pinene) *[7785-70-8]*

Padrayuttawat *et al.* (1997)	d	0.02
Buettner & Schieberle (2001a, 2001b)		0.005
Zeller & Rychlik (2006); Czerny *et al.* (2008)	r	0.0046
Czerny *et al.* (2008)	d	0.0022

3,7,7-TRIMETHYLBICYCLO[4.1.0]-2-HEPTENE ((+)-2-carene) *[4497-92-1]*

Buttery *et al.* (1987b)	d	4.0

3,7,7-TRIMETHYLBICYCLO[4.1.0]-3-HEPTENE (δ3-carene, 3-carene) *[13466-78-9]*

Tamura *et al.* (1996)	d	0.77
Tamura *et al.* (1996)	r	2.03
Pino & Mesa (2006)	d	0.77

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

(1*S*)-(+)-3,7,7-TRIMETHYLBICYCLO[4.1.0]-3-HEPTENE ((1*S*)-(+)-3-carene) *[498-15-7]*

Boonbumrung *et al.* (2001)	d	0.044

endo-1,7,7,-TRIMETHYLBICYCLO[2.2.1]-2-HEPTYL ACETATE (bornyl acetate) *[76-49-3]*

Buttery *et al.* (1968); Takeoka (2001)	d	0.075
Yang *et al.* (1992); Tamura *et al.* (1993)	d	1.38

1-(2,6,6-TRIMETHYL-1,3-CYCLOHEXADIEN-1-YL)-(*E*)-2-BUTEN-1-ONE ((*E*)-β-damascenone) *[23726-93-4]*

Ohloff (1977,1978)		0.01
Ohloff (1978)		0.000 009
Meilgaard (1978)		0.5
Braell (1984)	d	0.000 002
Buttery *et al.* (1988b,1989,1990a); Buttery (1993)	d	0.000 002
Schieberle (1991)	d	0.000 004
Semmelroch *et al.* (1995)		0.000 000 75
Ong & Acree (1999)		0.000 01
Boonbumrung *et al.* (2001)	d	0.000 0013
Plotto *et al.* (2006)	d	0.000 00358 - 0.000 00834 (perceivers)
Plotto *et al.* (2006)	d	0.000 0382 - 0.000 13 (nonperceivers)
Plotto *et al.* (2006)	r	0.000 0113 (perceivers)
Pino & Mesa (2006)	d	0.01
Schuh & Schieberle (2006)	d	0.000 004
Czerny *et al.* (2007c)	r	0.000 43
Czerny *et al.* (2008)	d	0.000 013
Czerny *et al.* (2008)	r	0.000 056
Averbeck & Schierberle (2009, 2011)	d	0.000 002

4-(2,6,6-TRIMETHYL-1,3-CYCLOHEXADIEN-1-YL)-3-BUTEN-2-ONE (dihydro-β-ionone) *[17283-81-7]*

Meilgaard (1978)	0.001	

2,2,6-TRIMETHYLCYCLOHEXANONE *[2408-37-9]*

Persson & Jüttner (1983)	r	0.310
Takeoka *et al.* (1990a)	d	0.1

2,6,6-TRIMETHYL-1-CYCLOHEXENE-1-CARBOXALDEHYDE (β-cyclocitral) *[432-25-7]*

Persson & Jüttner (1983)	r	0.0193
Buttery *et al.* (1989,1990a); Takeoka *et al.* (1990a); Hansen *et al.* (1992); Buttery (1993)	d	0.005
Rashash *et al.* (1997)		0.003

1,3,3-TRIMETHYLCYCLOHEXENE *[68555-95-3]*

Persson & Jüttner (1983)	r	7.1

α,α,4-trimethyl-3-cyclohexene-1-methanethiol⇒4-ISOPROPYL-8-MERCAPTO-1-METHYL-1-CYCLOHEXENE

2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-ONE *[20013-73-4]*

Persson & Jüttner (1983)	r	0.420
Buttery *et al.* (1990b)	d	0.15

3,5,5-TRIMETHYL-2-CYCLOHEXEN-1-ONE (isophorone) *[78-59-1]*

Pelosi & Viti (1978)	d	11.9
Pelosi & Pisanelli (1981)	d	11

(*E*)-1-(2,6,6-TRIMETHYL-1-CYCLOHEXEN-1-YL)-2-BUTEN-1-ONE (β-damascone , (*E*)-β-damascone) *[23726-91-2]*

Pickenhagen & Furrer (1987,1990)	0.000 09
Belitz & Grosch (2001)	0.000 002

(*R*)-(+)-1-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-2-BUTEN-1-ONE ((*d*)-α-damascone) *[28102-28-5]*

Pickenhagen (1989, 2001); Fehr & Galindo (1991)	0.1

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

(*S*)-(-)-1-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-2-BUTEN-1-ONE ((*l*)-α-damascone) *[116497-11-1]*

Pickenhagen (1989, 2001); Fehr & Galindo (1991)		0.0015

4-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE (α-ionone) *[127-41-3]*

Appell (1969)		0.06
Larsen & Poll (1990)	d	0.001 - 0.01
Buttery & Ling (1995)	d	0.000 6
Buttery *et al.* (1997)	d	0.000 4
Tamura *et al.* (1995)		0.05
Plotto *et al.* (2006)	d	0.00378
Plotto *et al.* (2006)	r	0.0106

(*R*)-(+)-4-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE ((*d*)-α-ionone) *[24190-29-2]*

Polak *et al.* (1989)		0.000 03 - 0.0059
Polak *et al.* (1989)		0.000 64 - 0.328
Polak *et al.* (1989)		0.000 04 - 0.036

(*S*)-(-)-4-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE ((*l*)-α-ionone) *[14398-36-8]*

Polak *et al.* (1989)		0.000 1 - 0.021
Polak *et al.* (1989)		0.000 64 - 0.656
Polak *et al.* (1989)		0.000 8 - 0.487

4-(2,6,6-TRIMETHYL-1-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE (β-ionone) *[14901-07-6]*

Guadagni (1966,1968,1969,1970c); Buttery *et al.* (1971,1987a, 1989,1990a); Teranishi *et al.* (1974); Buttery (1981,1993); Hansen *et al.* (1992);		
Buttery & Ling (1995)	d	0.000 007
Larsen & Poll (1990)	d	0.000 1 - 0.001
Sugisawa *et al.* (1991); Yang *et al.* (1992);		
Tamura *et al.* (1993)	d	0.006 - 0.205
Tamura *et al.* (1995)		0.06
Tandon *et al.* (2000)	d	0.023
Buttery *et al.* (1997)	d	0.000 01
Tamura *et al.* (2001); Boonbumrung *et al.* (2001)	d	0.0059
Plotto *et al.* (2006)	d	0.000 135 - 0.000 168 (perceivers)
Plotto *et al.* (2006)	d	0.823 - 1.31 (nonperceivers)
Plotto *et al.* (2006)	r	0.000 521 (perceivers)
Plotto *et al.* (2006)	r	1.78 (nonperceivers)
Schuh & Schieberle (2006)	d	0.000 2
Czerny *et al.* (2007c)		0.000 2
Czerny *et al.* (2008)	d	0.0035
Czerny *et al.* (2008)	r	0.0084

(*E*)-4-(2,6,6-TRIMETHYL-1-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE ((*E*)-β-ionone) *[79-77-6]*

Pino & Mesa (2006)	d	0.000 007

cis-1-(2,6,6-TRIMETHYLCYCLOHEXYL)-3-HEXANOL

Schulte-Elte *et al.* (1987)	d	0.240

trans-1-(2,6,6-TRIMETHYLCYCLOHEXYL)-3-HEXANOL

Schulte-Elte *et al.* (1987)	d	0.032

3a,6,9a-TRIMETHYL-3aα,5aβ,9aα,9bα-DODECAHYDRONAPHTHO[2,1-b]FURAN (9-epi-19-norambrox)

Ohloff *et al.* (1985)		0.075

(3a*R*)-3a,6,9a-TRIMETHYL-3aα,5aβ,6α,9aα,9bα-DODECAHYDRONAPHTHO[2,1-b]FURAN ((*d*)-9-epi-18-norambrox)

Ohloff *et al.* (1985)		0.012

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

2,6,10-TRIMETHYL-2,6,9,11-DODECATETRAENAL (α-sinensal) *[17909-77-2]*

Guadagni (1969,1970c); Teranishi *et al.* (1981)	d	0.000 05
Sugisawa *et al.* (1991)	d	0.22

3,7,11-TRIMETHYL-2,6,10-DODECATRIEN-1-OL (farnesol) *[4602-84-0]*

Ohloff (1977,1978); Pickenhagen & Furrer (1990)		0.02 - 0.2
Sugisawa *et al.* (1991)	d	1

3,7,11-TRIMETHYL-1,6,10-DODECATRIEN-3-OL (nerolidol) *[7212-44-4]*

Pickenhagen & Furrer (1990)		0.12
Larsen & Poll (1992)	d	0.01 - 0.1
Sugisawa *et al.* (1991); Yang *et al.* (1992); Tamura *et al.* (1993)	d	2.25 - 10

(*E*)-3,7,11-TRIMETHYL-1,6,10-DODECATRIEN-3-OL ((*E*)-nerolidol) *[40716-66-3]*

Pino & Mesa (2006)	d	0.25

trimethylenediamine⇒1,3-PROPANEDIAMINE
trimethyleneimine⇒AZETIDINE

4,11,11-TRIMETHYL-8-METHYLENEBICYCLO[7.2.0]-4-UNDECENE (caryophyllene, β-caryophyllene) *[87-44-5]*

Guadagni *et al.* (1966); Guadagni (1969); Buttery *et al.* (1978b,1987b)	d	0.064 - 0.090
Tamura *et al.* (2001)	d	0.15
Boonbumrung *et al.* (2001)	d	1.54
Pino & Mesa (2006)	d	0.064

[1*R*-(1*R**,4*R**,6*R**,10*S**)]-4,12,12-TRIMETHYL-9-METHYLENE-5-OXATRICYCLO[8.2.0^{4,6}]DODECANE (caryophyllene oxide, caryophyllene epoxide) *[1139-30-6]*

Buttery *et al.* (1987b)	d	0.2
Sugisawa *et al.* (1991); Yang *et al.* (1992); Tamura *et al.* (1993)	d	5.5
Tamura *et al.* (2001)	d	0.41

(3aα,5aβ,6β,9aα,9bβ)-3a,6,9a-TRIMETHYLNAPHTHO[2,1-b]FURAN (19-norambrox)

Ohloff *et al.* (1985)		0.003

[3a*R*-(3aα,5aβ,6α,9aα,9bβ)]-3a,6,9a-TRIMETHYLNAPHTHO[2,1-b]FURAN ((*l*)-18-norambrox)

Ohloff *et al.* (1985)		0.0014

[3a*R*-(3aα ,5aβ,6α,9aβ,9bα)]-3a,6,9a-TRIMETHYLNAPHTHO[2,1-b]FURAN ((*d*)-18-norisoambrox)

Ohloff *et al.* (1985)		0.17

[3a*S*-(3aα,5aβ,6β,9aβ,9bα)]-3a,6,9a-TRIMETHYLNAPHTHO[2,1-b]FURAN ((*l*)-19-norisoambrox)

Ohloff *et al.* (1985)		0.18

4-(2,2,6-TRIMETHYL-7-OXABICYCLO[4.1.0]HEPT-1-YL)-3-BUTEN-2-ONE (epoxy-β-ionone) *[36340-49-5]*

Buttery *et al.* (1989); Takeoka *et al.* (1990a); Buttery (1993)	d	0.1

1,3,3-TRIMETHYL-2-OXABICYCLO[2.2.2]OCTANE (1,8-cineole, eucalyptol) *[470-82-6]*

Le Magnen (1952)		0.013 - 0.04
Amoore & Venstrom (1966); Amoore (1969)	d	0.012
Appell (1969)		1
Buttery *et al.* (1974,1987b); Takeoka (2001)	d	0.0013
Pelosi & Pisanelli (1981)	d	0.021
Sugisawa *et al.* (1991); Yang *et al.* (1992); Tamura *et al.* (1993)	d	0.023 - 0.064

WATER detection (d) and recogniton (r) odour threshold values in mg/kg

Padrayuttawat *et al.* (1997)	d	0.003
Pino & Mesa (2006)	d	0.0013
Zeller & Rychlik (2006); Czerny *et al.* (2008)	r	0.0046
Czerny *et al.* (2008)	d	0.0011

2,4,5-TRIMETHYLOXAZOLE *[20662-84-4]*

Marchand *et al.* (2000)	d	0.017

2,2,4-TRIMETHYL-1-PENTANOL *[123-44-4]*

Schnabel *et al.* (1988)	d	0.42 - 0.82

2,2,4-TRIMETHYL-3-PENTANOL *[5162-48-1]*

Pelosi & Pisanelli (1981)	d	0.035
Schnabel *et al.* (1988)	d	0.0078 - 0.042

2,3,5-TRIMETHYLPHENOL (isopseudocumenol) *[697-82-5]*

Dietz & Traud (1978)		2.3

2,4,5-TRIMETHYLPHENOL *[496-78-6]*

Dietz & Traud (1978)		2.3

2,4,6-TRIMETHYLPHENOL (mesitol) *[527-60-6]*

Dietz & Traud (1978)		0.5

3,4,5-TRIMETHYLPHENOL *[527-54-8]*

Dietz & Traud (1978)		> 10

(2*R*,4*S*,6*R*)-2,4,6-TRIMETHYL-4-PHENYL-1,3-DIOXANE

Kappey *et al.* (2000)		0.000 85

(2*R*,4*R*,6*R*)-2,4,6-TRIMETHYL-4-PHENYL-1,3-DIOXANE

Kappey *et al.* (2000)		0.332

TRIMETHYLPYRAZINE *[14667-55-1]*

Koehler *et al.* (1971)	d	9
Calabretta (1973,1975,1978)	d	0.4
Czerny & Wagner (1995)		0.091
Buttery *et al.* (1997,1999); Buttery & Ling (1998)	d	0.023
Giri *et al*. (2010)	d	0.35012

2,4,5-TRIMETHYL-1,3-THIAZOLE *[13623-11-5]*

Giri *et al*. (2010)	d	0.045

cis-α,α,5-TRIMETHYL-5-VINYLTETRAHYDROFURFURYL ALCOHOL (*cis*-linalool oxide) *[5989-33-3]*

Yang *et al.* (1992); Tamura *et al.* (1993)	d	0.32
Boonbumrung *et al.* (2001)	d	0.10

trans-α,α,5-TRIMETHYL-5-VINYLTETRAHYDROFURFURYL ALCOHOL (*trans*-linalool oxide) *[34995-77-2]*

Yang *et al.* (1992); Tamura *et al.* (1993)	d	0.32
Boonbumrung *et al.* (2001)	d	0.19

2,4,6-TRINITROPHENOL (picric acid) *[88-89-1]*

Dietz & Traud (1978)		4

TRIPROPYLAMINE *[102-69-2]*

Amoore & Forrester (1976)	d	5.14

2,3,5-trithiahexane⇒METHYL(METHYLTHIO)DITHIOMETHANE
2,3,4-trithiapentane⇒METHYLTRITHIOMETHANE

1,2,4-TRITHIOLANE *[289-16-7]*

Rödel & Kruse (1982)		0.00766

L-tyrosine⇒(*S*)-(-)--2-AMINO-3-(4-HYDROXYPHENYL)PROPANOIC ACID

U

umbellulone⇒1-ISOPROPYL-4-METHYLBICYCLO[3.1.0]-3-HEXEN-2-ONE
γ-undecalactone⇒4-HYDROXYUNDECANOIC ACID LACTONE
(*R*)-(+)-δ-undecalactone⇒(*R*)-(+)-5-HYDROXYUNDECANOIC ACID LACTONE
(*S*)-(-)-δ-undecalactone⇒(*S*)-(-)-5-HYDROXYUNDECANOIC ACID LACTONE

2,4-UNDECADIENAL *[13162-46-4]*

Tressl *et al.* (1980)		0.000 01

UNDECANAL *[112-44-7]*

Guadagni *et al.* (1963); Teranishi *et al.* (1974)	d	0.005
Schnabel *et al.* (1988)	d	0.000 41 - 0.0083
Sugisawa *et al.* (1991); Yang *et al.* (1992);		
Tamura *et al.* (1993)	d	0.04 - 0.1
Padrayuttawat *et al.* (1997)	d	0.01
Fabrellas *et al.* (2004)		0.000 03 - 0.000 09
Pino & Mesa (2006)	d	0.0125

UNDECANE *[1120-21-4]*

Zoeteman *et al.* (1971)		10

UNDECANOIC ACID *[112-37-8]*

Cherkinski (1961)	d	10.0

1-UNDECANOL *[112-42-5]*

Schnabel *et al.* (1988)	d	0.086 - 0.41
Tamura *et al.* (1995)		0.7

2-UNDECANOL *[1653-30-1]*

Schnabel *et al.* (1988)	d	0.0086 - 0.041

3-UNDECANOL *[6929-08-4]*

Schnabel *et al.* (1988)	d	0.0086 - 0.041

4-UNDECANOL *[4272-06-4]*

Schnabel *et al.* (1988)	d	0.041 - 0.083

5-UNDECANOL *[37493-70-2]*

Schnabel *et al.* (1988)	d	0.41 - 0.83

6-UNDECANOL *[23708-56-7]*

Schnabel *et al.* (1988)	d	0.0086 - 0.041

2-UNDECANONE (methyl nonyl ketone) *[112-12-9]*

Guadagni *et al.* (1966); Teranishi *et al.* (1974);		
Buttery *et al.* (1988a); Buttery & Ling (1995)	d	0.007
Schnabel *et al.* (1988)	d	0.041 - 0.082
Giri *et al*. (2010)	d	0.0055

3-UNDECANONE *[2216-87-7]*

Schnabel *et al.* (1988)	d	0.0083 - 0.041

4-UNDECANONE *[14476-37-0]*

Schnabel *et al.* (1988)	d	0.041 - 0.083

5-UNDECANONE *[33083-83-9]*
Schnabel *et al.* (1988) d 0.041 - 0.083

6-UNDECANONE (diamyl ketone) *[927-49-1]*
Schnabel *et al.* (1988) d 0.085 - 0.41

(*E*,*Z*)-1,3,5-UNDECATRIENE *[51447-08-6]*
Steinhaus & Schieberle (2005) d 0.000 02

(6*E*/*Z*,8*E*)-6,8,10-UNDECATRIEN-2-ONE
Miyazawa *et al.* (2010) 0.000 7

(6*E*/*Z*,8*E*)-6,8,10-UNDECATRIEN-3-ONE
Miyazawa *et al.* (2010) 0.000 01

(6*E*/*Z*,8*E*)-6,8,10-UNDECATRIEN-4-ONE
Miyazawa *et al.* (2010) 0.000 072

(*E*)-2-UNDECENAL *[53448-07-0]*
Czerny *et al*. (2008) d 0.000 78
Czerny *et al*. (2008) r 0.0014

UNDECYL ACETATE *[1731-81-3]*
Yang *et al.* (1992); Tamura *et al.* (1993) d 0.25

UNDECYL CYANIDE (dodecanenitrile) *[2437-25-4]*
Fabrellas *et al.* (2004) 0.000 09 - 0.000 32

uric acid⇒7,9-DIHYDRO-1H-PURINE-2,6,8(3*H)*-TRIONE

V

valeraldehyde⇒PENTANAL
valeric acid⇒PENTANOIC ACID
valeronitrile⇒BUTYL CYANIDE
vanillin⇒4-HYDROXY-3-METHOXYBENZALDEHYDE
versalide⇒1-(3-ETHYL-5,6,7,8-TETRAHYDRO-5,5,8,8-TETRAMETHYL-2-NAPHTHALENYL)ETHANONE
α-vetivone⇒8-ISOPROPYLIDENE-5,6-DIMETHYLBICYCLO[4.4.0]-1-DECEN-3-ONE
vinylbenzene⇒ETHENYLBENZENE
vinyl cyanide⇒ETHENYL CYANIDE
p-vinylguaiacol⇒4-ETHENYL-2-METHOXYPHENOL
4-vinylguaiacol⇒4-ETHENYL-2-METHOXYPHENOL
vinylidene chloride⇒1,1-DICHLOROETHENE
4-vinylphenol⇒4-ETHENYLPHENOL
vinyl propanoate⇒ETHENYL PROPANOATE
vinyl propionate⇒ETHENYL PROPANOATE
vinylpyrazine⇒ETHENYLPYRAZINE

X

xanthine⇒3,7-DIHYDRO-1H-PURINE-2,6-DIONE
m-xylene⇒1,3-DIMETHYLBENZENE
o-xylene⇒1,2-DIMETHYLBENZENE
p-xylene⇒1,4-DIMETHYLBENZENE
2,3-xylenol⇒2,3-DIMETHYLPHENOL
2,4-xylenol⇒2,4-DIMETHYLPHENOL
2,5-xylenol⇒2,5-DIMETHYLPHENOL
2,6-xylenol⇒2,6-DIMETHYLPHENOL
3,4-xylenol⇒3,4-DIMETHYLPHENOL
3,5-xylenol⇒3,5-DIMETHYLPHENOL

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